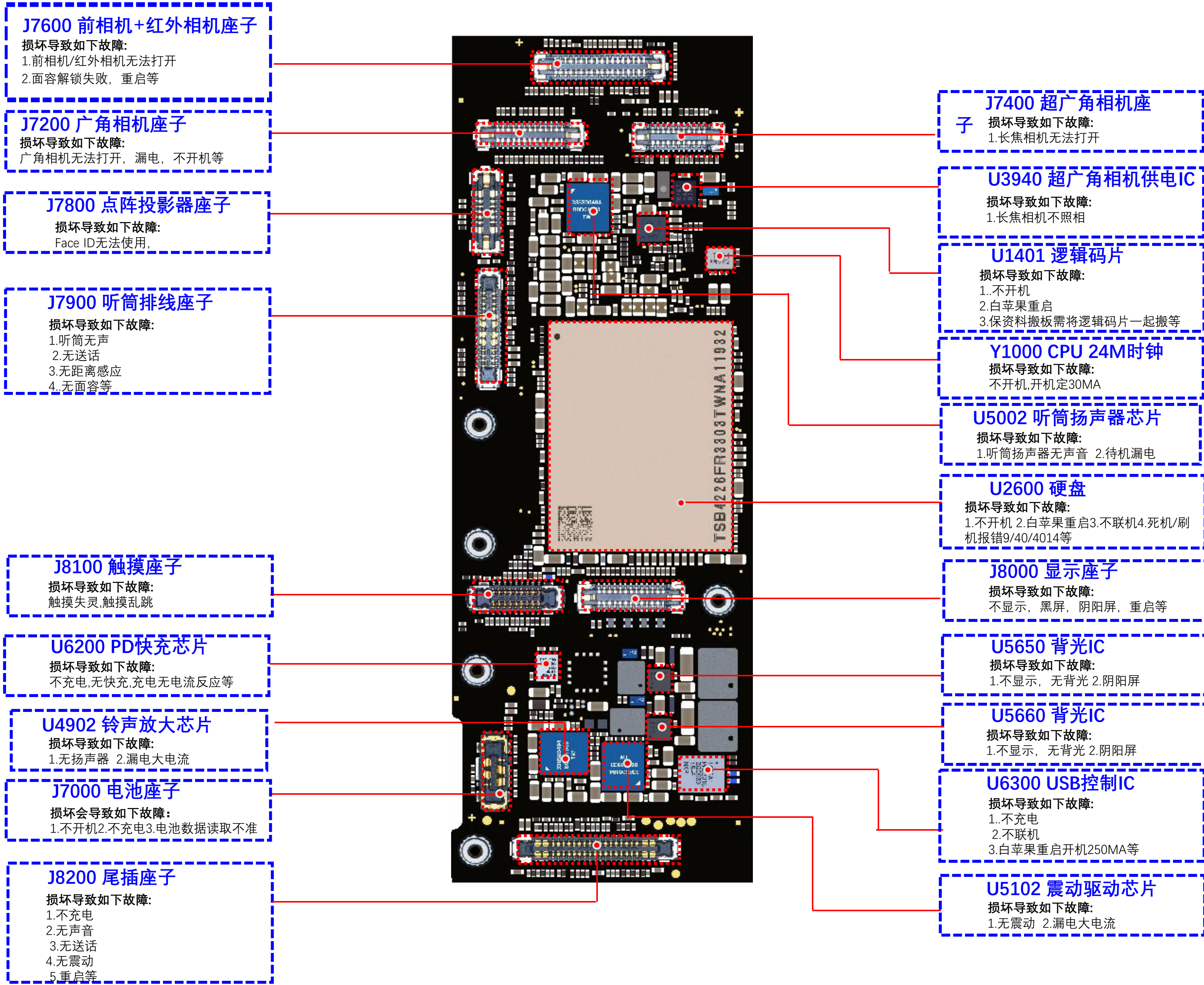
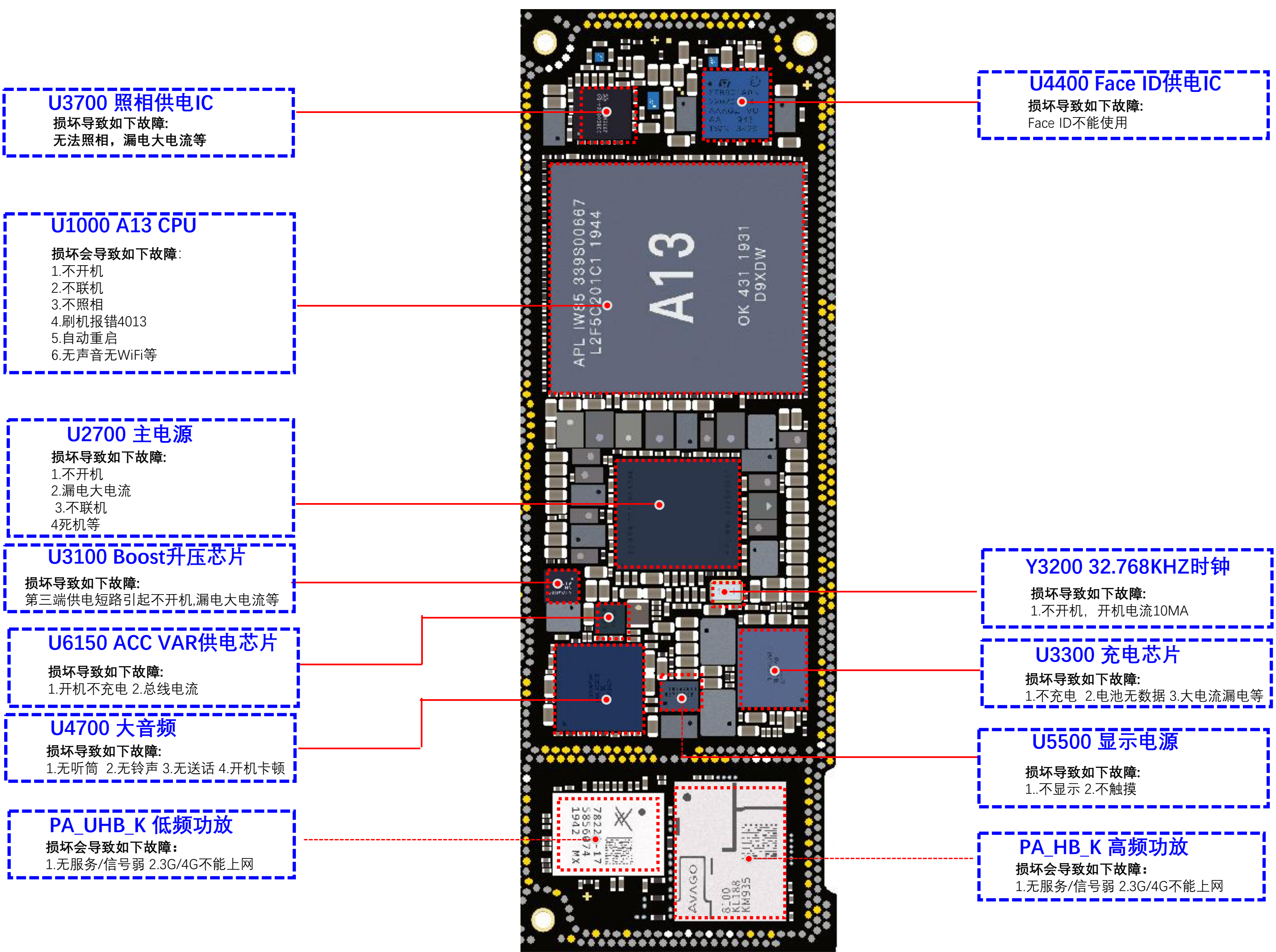


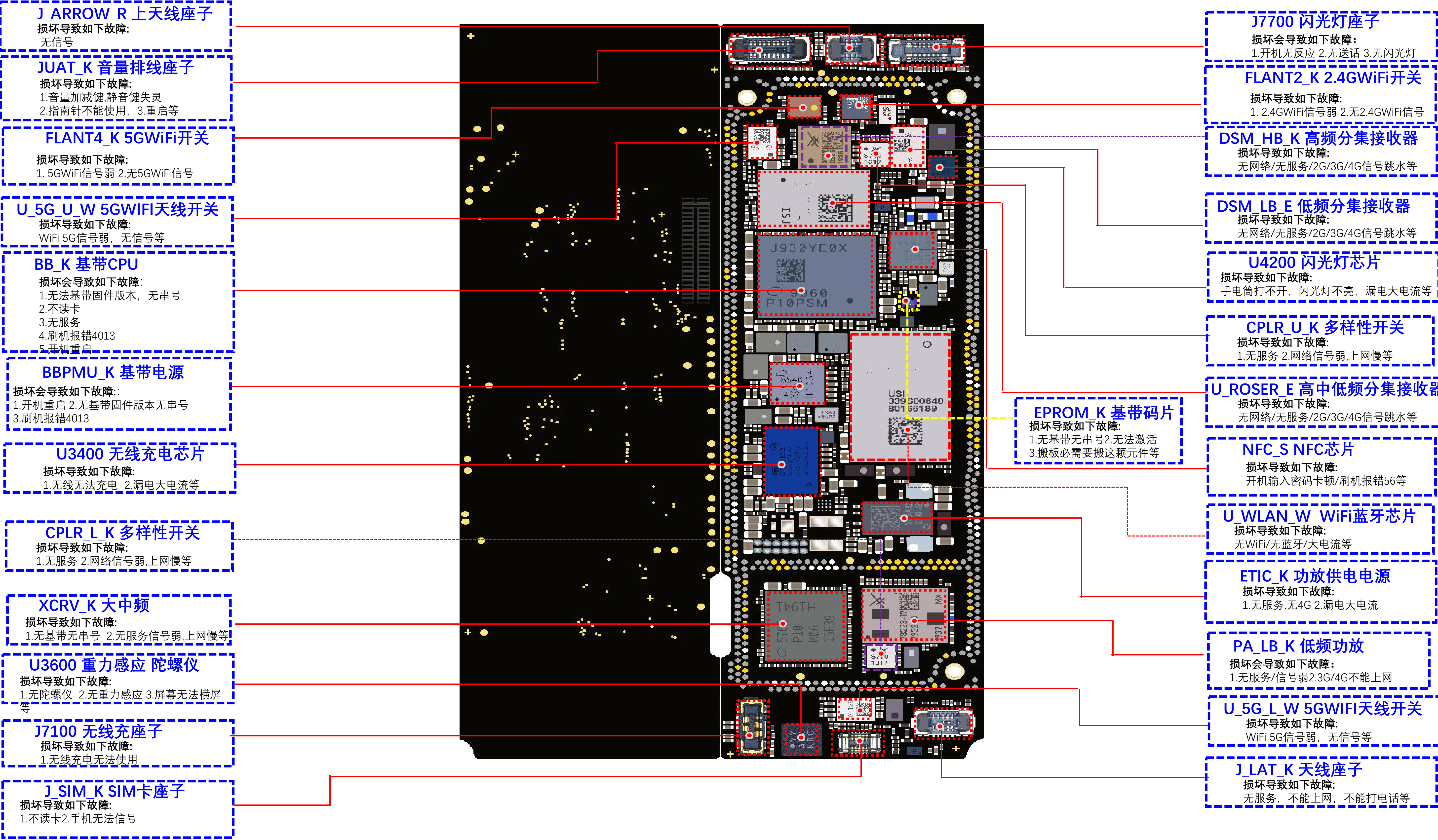
11逻辑板故障彩图A



11逻辑板故障彩图B



11射频板故障彩图

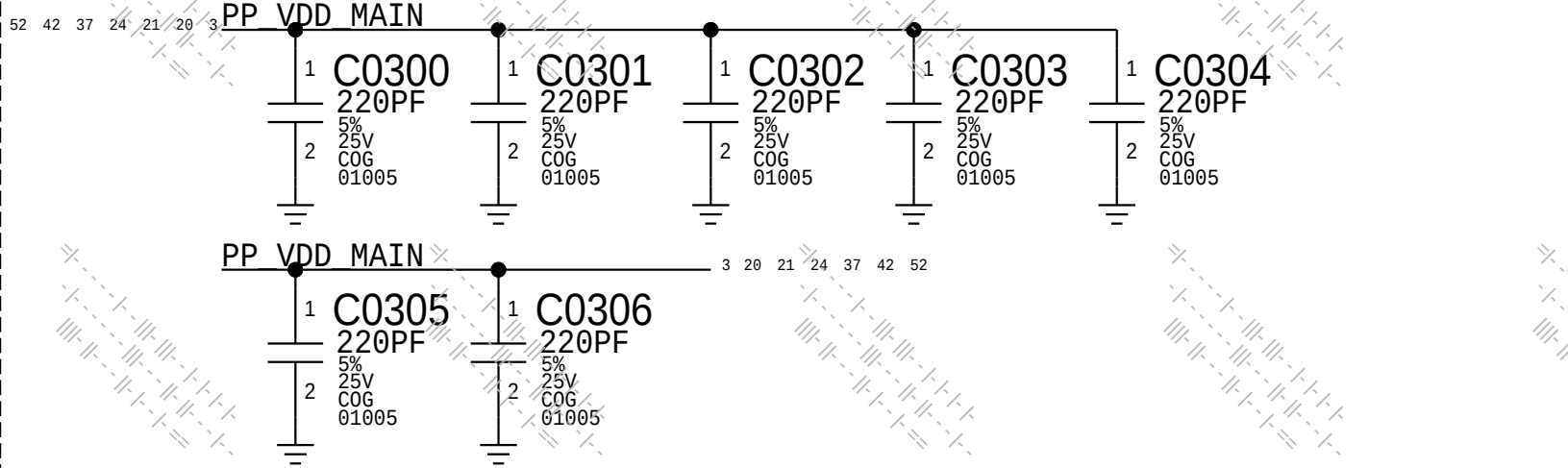


EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB_TOP, 639-86359, ULTIMATE)	EEEE_KF2F	CRITICAL	ULTIMATE
825-7691	1	EEEE FOR (MLB_TOP, 639-86359, SUPREME)	EEEE_KF2R	CRITICAL	SUPREME
825-7691	1	EEEE FOR (MLB_TOP, 639-86360, EXTREME)	EEEE_KF34	CRITICAL	EXTREME
825-7691	1	EEEE FOR (MLB_TOP, 639-86949, ULTIMATE)	EEEE_L6R1	CRITICAL	ULTIMATE
825-7691	1	EEEE FOR (MLB_TOP, 639-86951, SUPREME)	EEEE_L6RQ	CRITICAL	SUPREME
825-7691	1	EEEE FOR (MLB_TOP, 639-86952, EXTREME)	EEEE_L6T3	CRITICAL	EXTREME

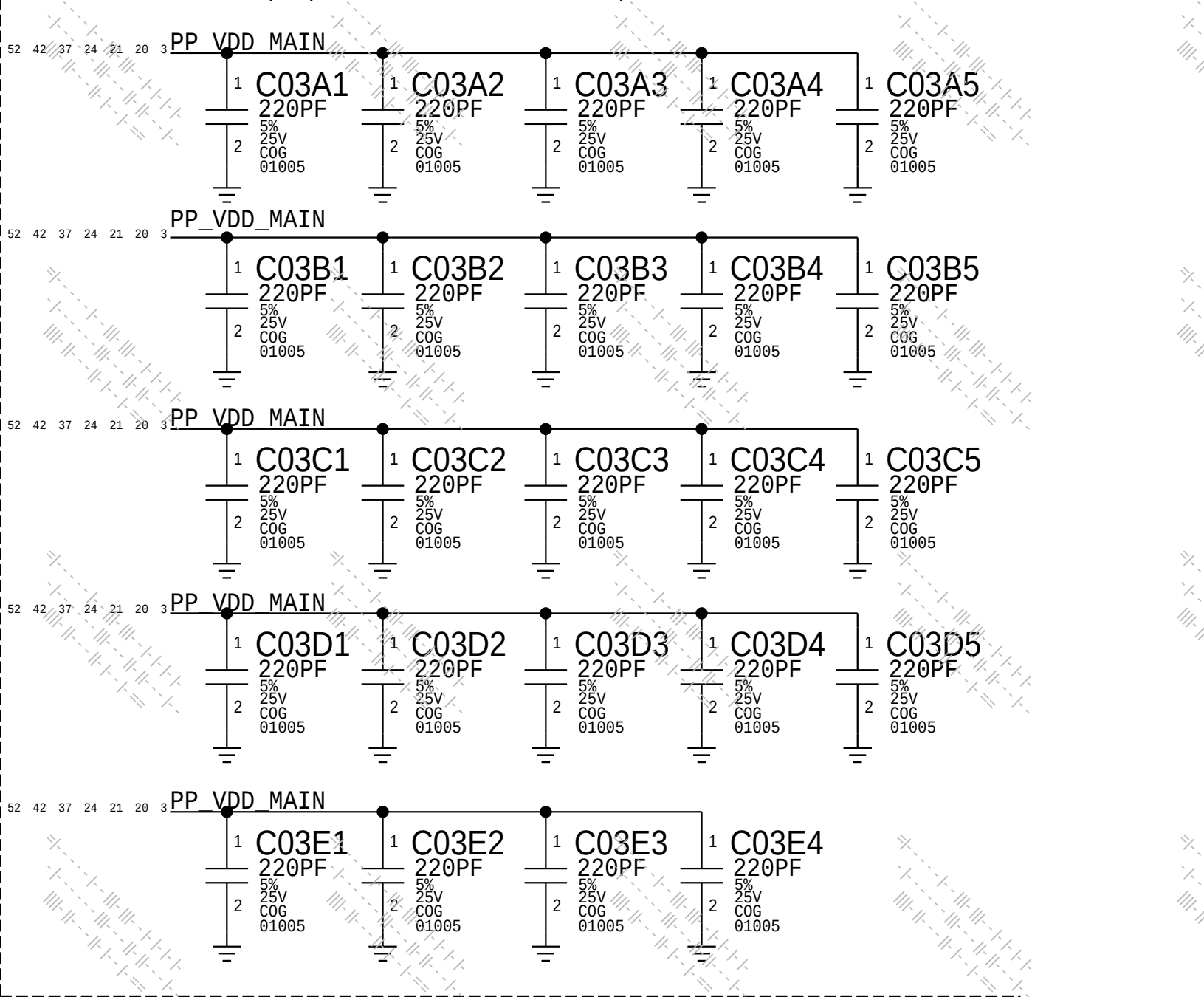
AC Return Steering - 7X

Caps provide return current path for FCAM LPDP & JU MIPI

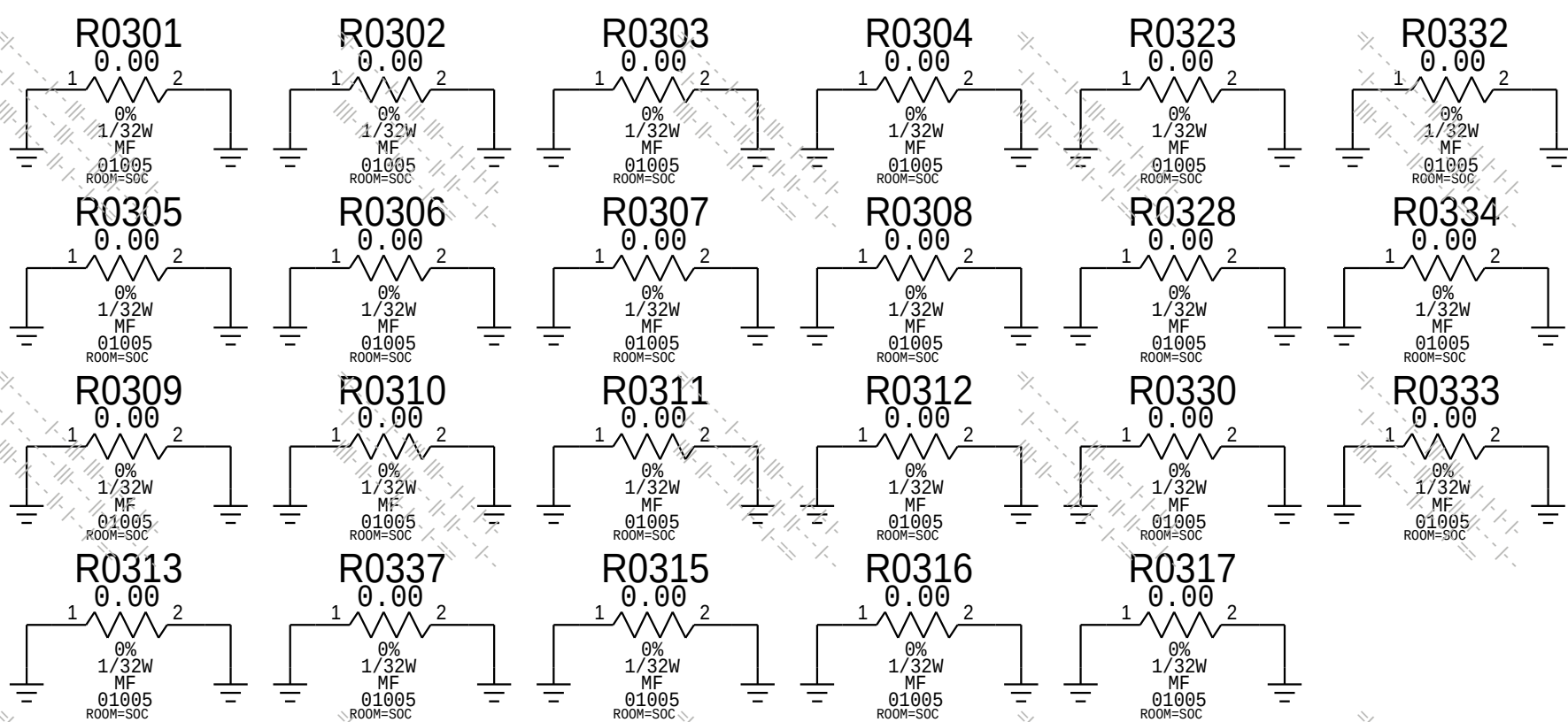


AC Return Steering - 24X

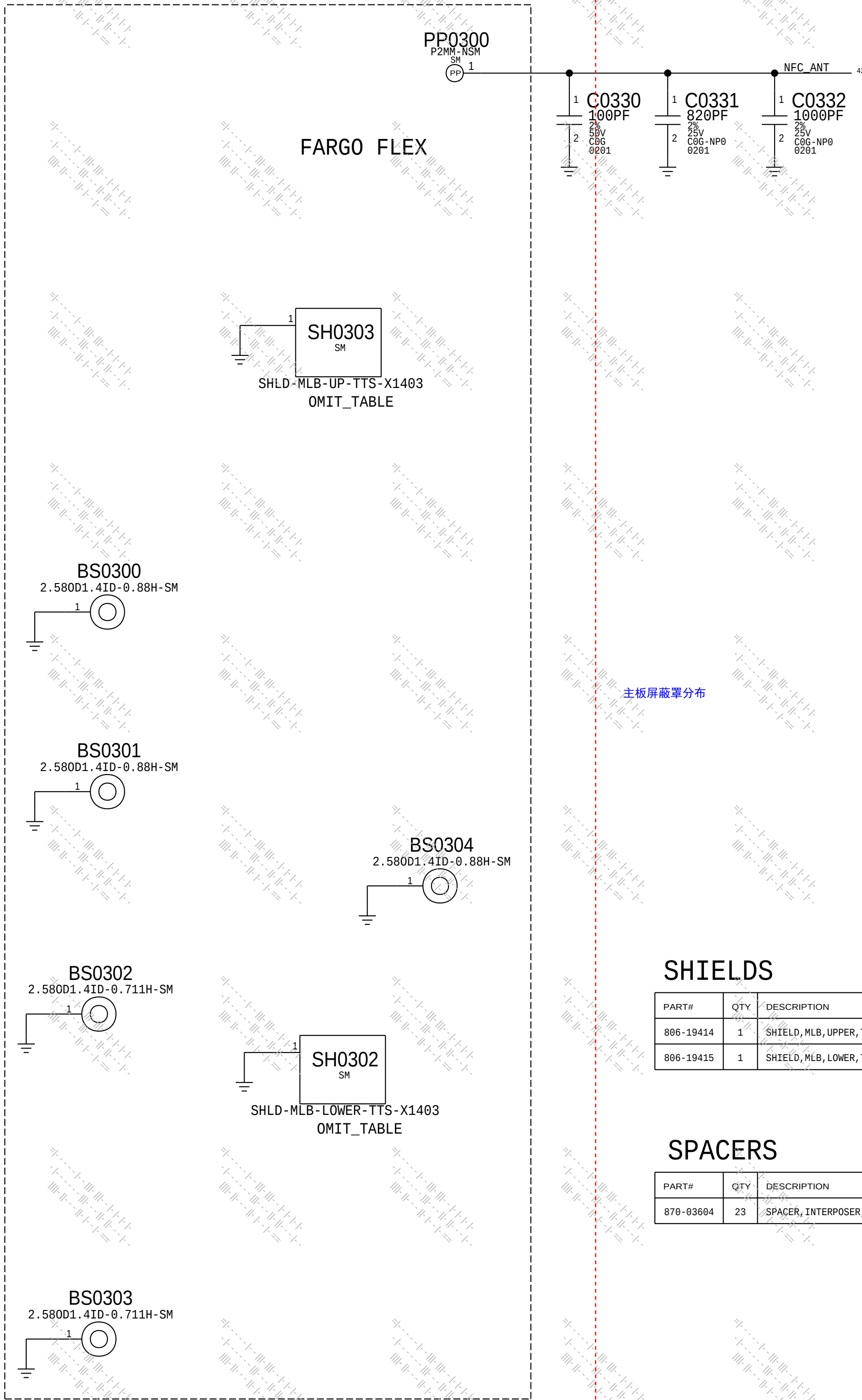
Caps provide return current path for RCAM LPDP



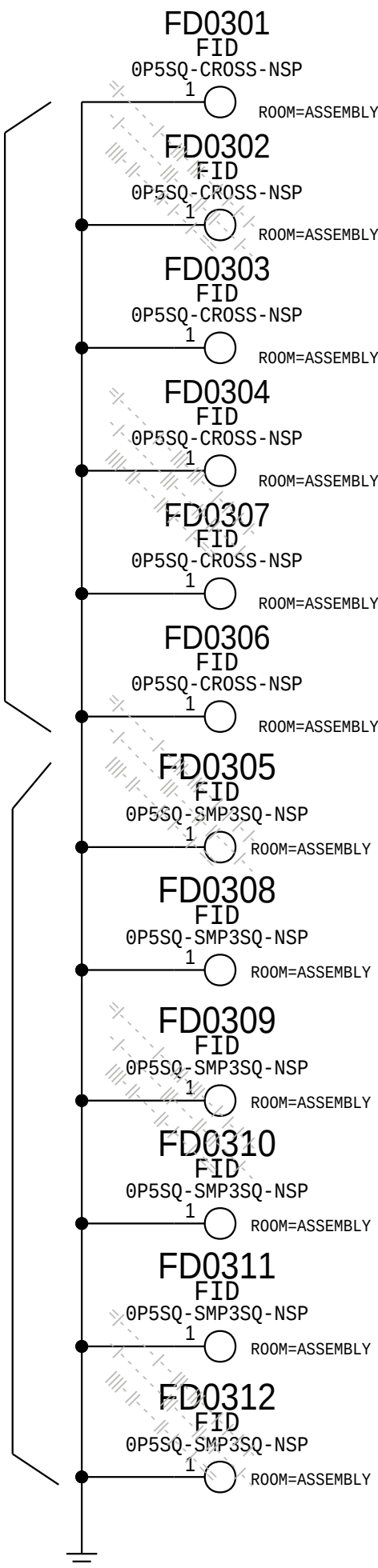
UNDERFILL DAM



FARGO FLEX



FIDUCIALS



Crosses

Squares

SHIELDS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
886-19414	1	SHIELD, MLB, UPPER, TTS, P2, N184	SH0303	CRITICAL	?
886-19415	1	SHIELD, MLB, LOWER, TTS, P2, N184	SH0302	CRITICAL	?

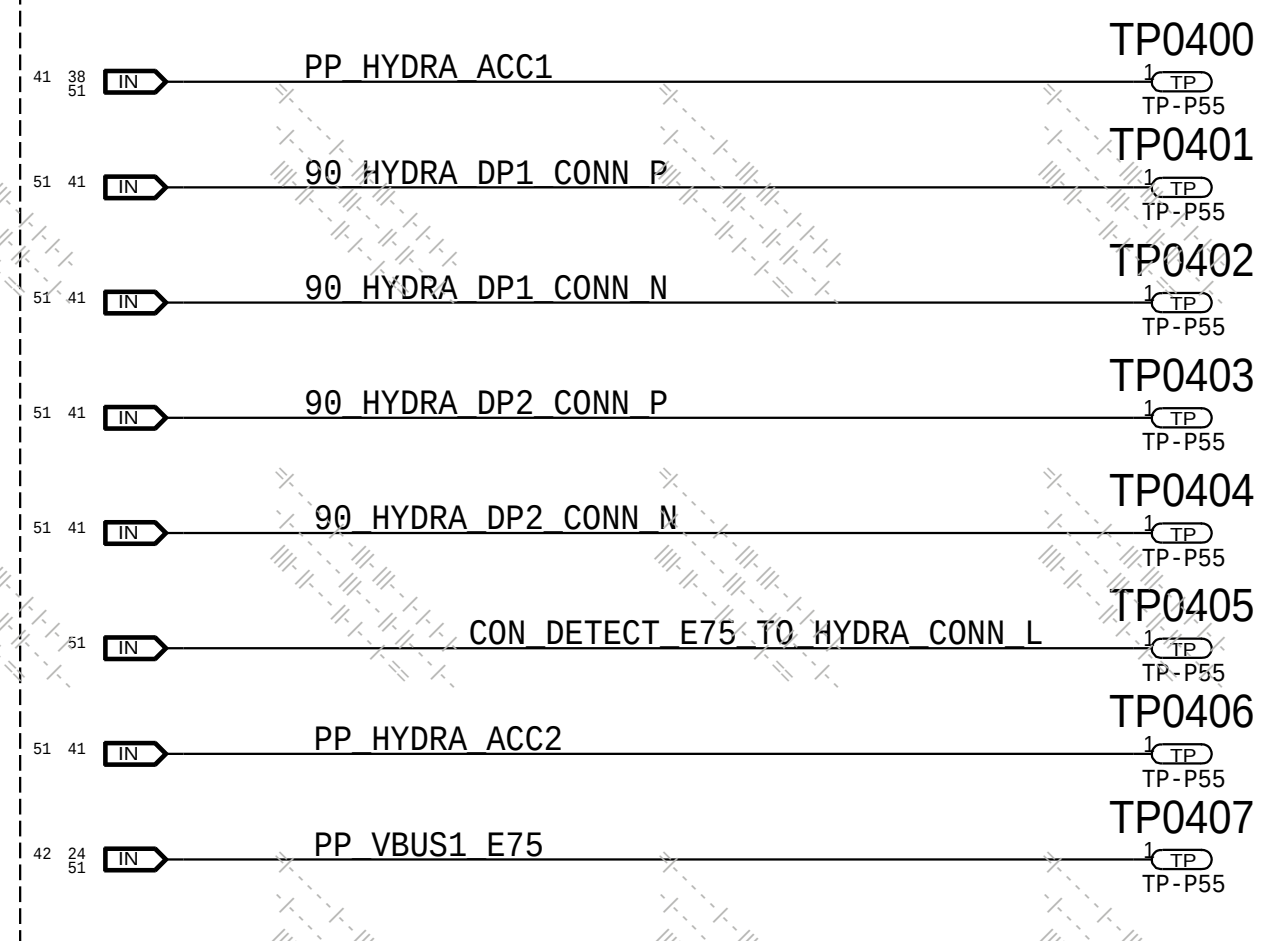
SPACERS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
870-93604	23	SPACER, INTERPOSER, SMALL, X891	SP1-SP26	CRITICAL	?

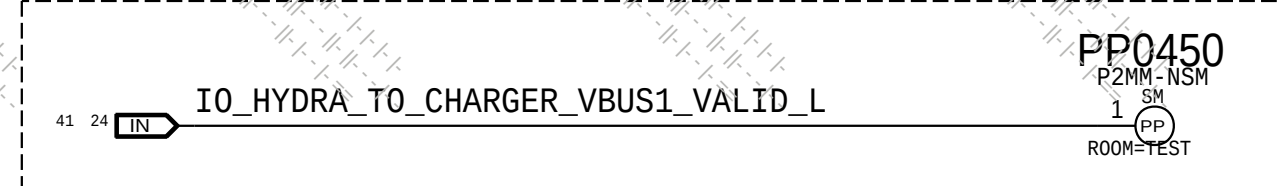
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SYSTEM: FF BOM Specific & Mechanical		SYNG MASTER=	
		DRAWING NUMBER	051-04047
		REVISION	6.0.0
		BRANCH	
		PAGE	3 OF 150
		SHEET	3 OF 60
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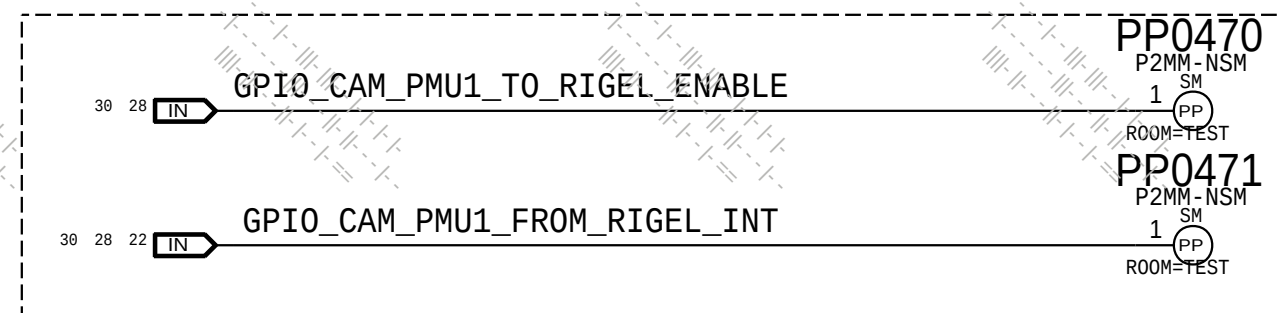
TEST POINTS



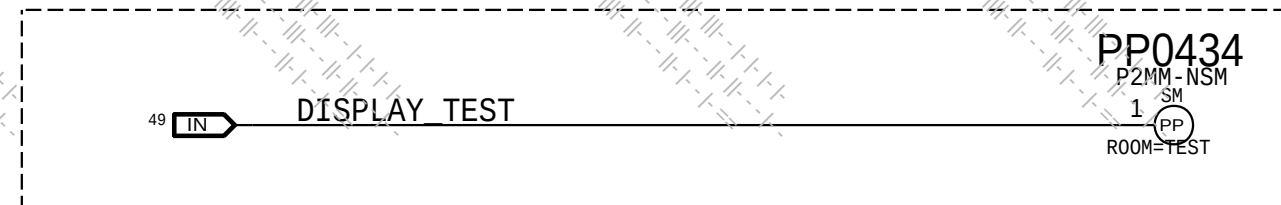
Hydra VBUS



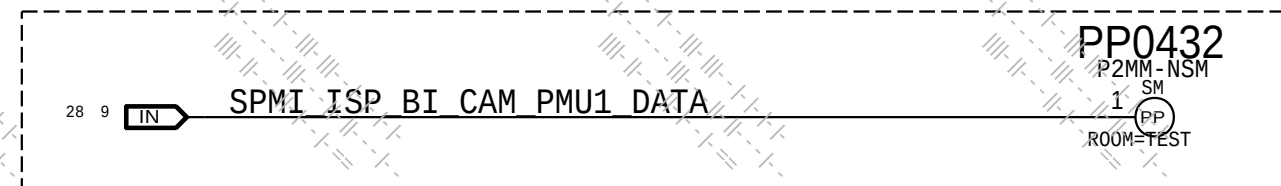
Rigel



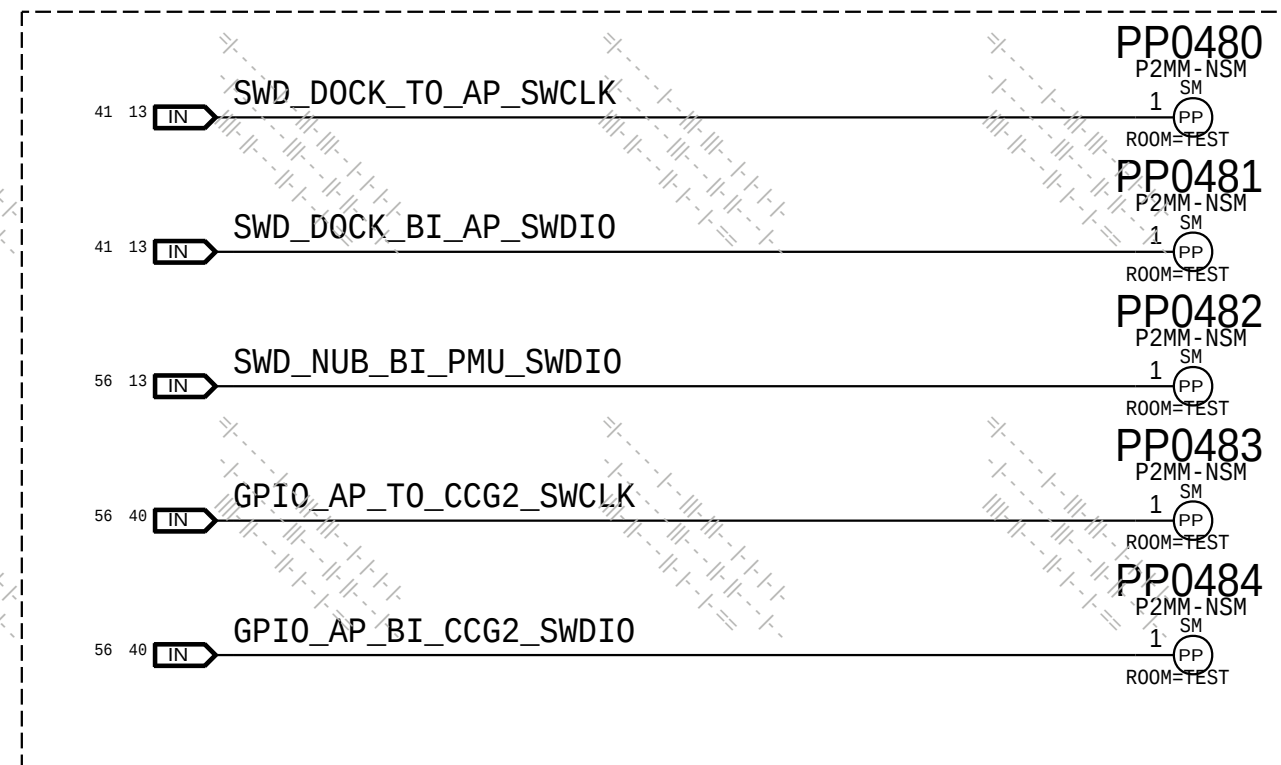
DISPLAY



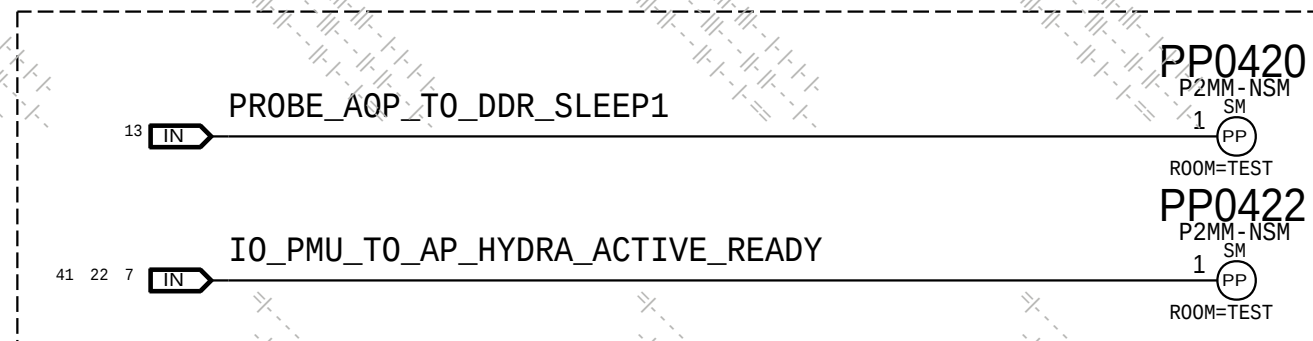
SPMI DATA



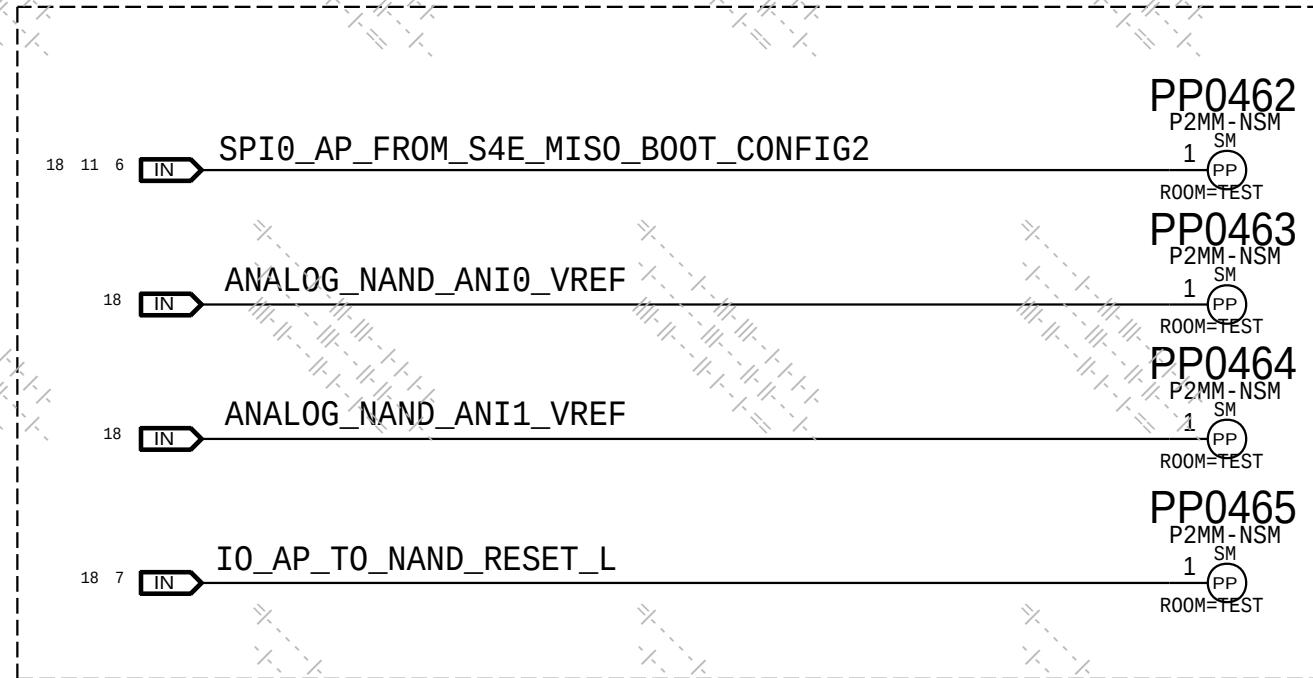
SWD



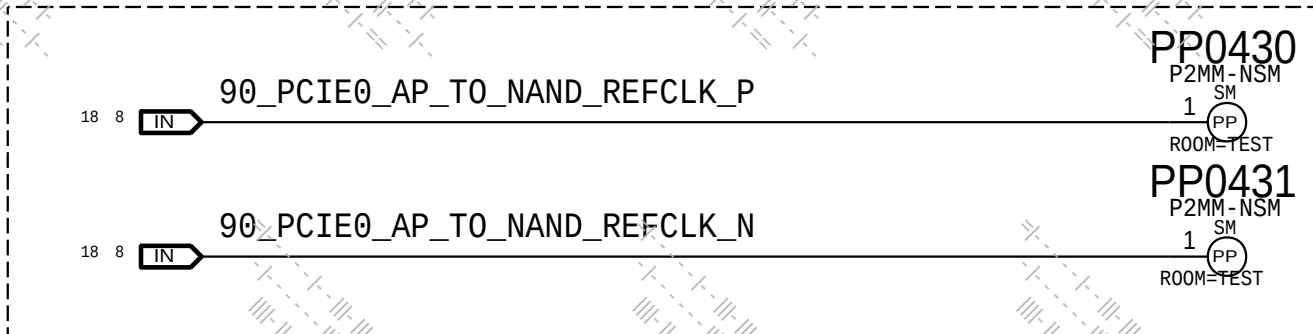
PMU



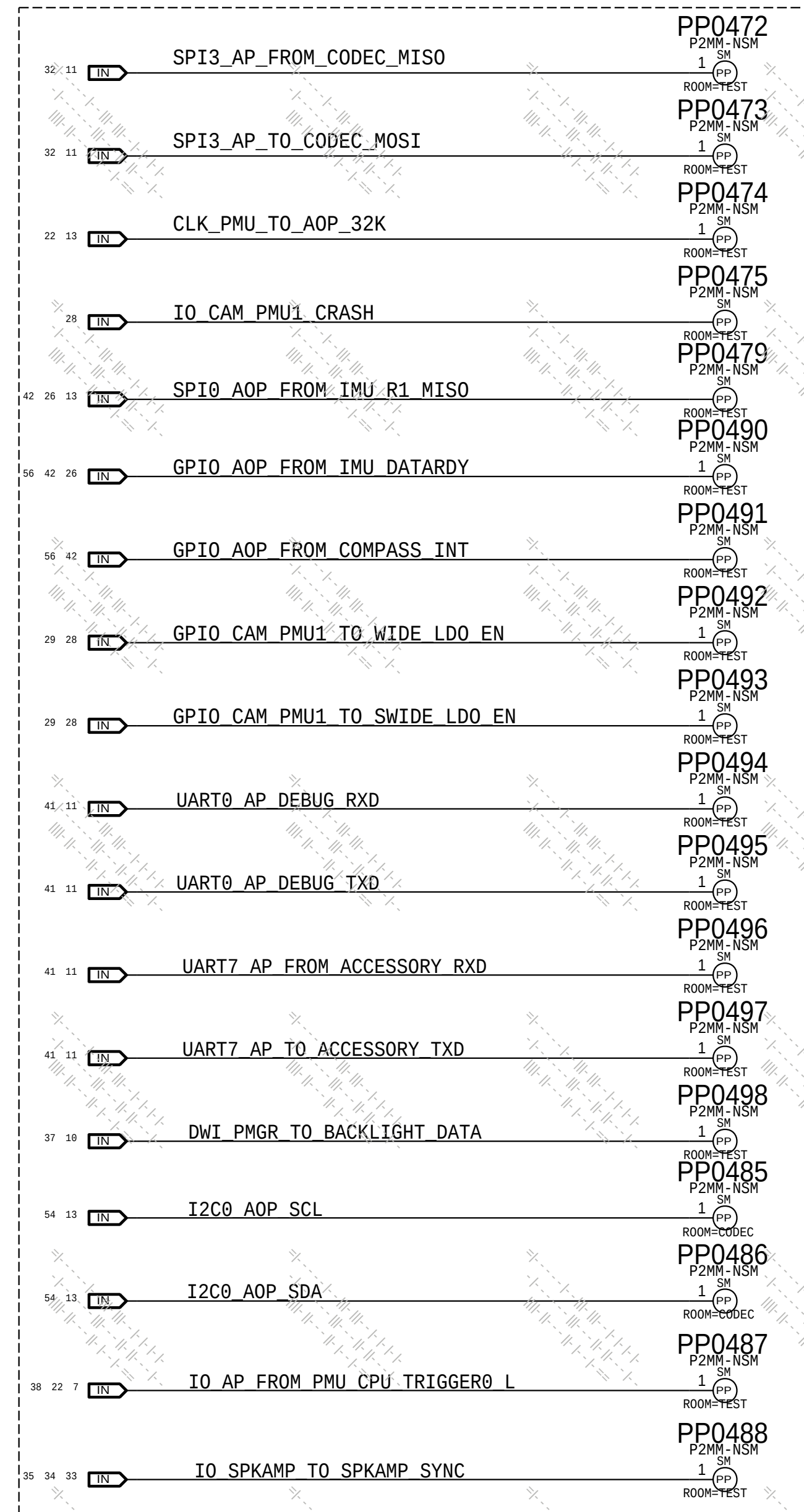
NAND



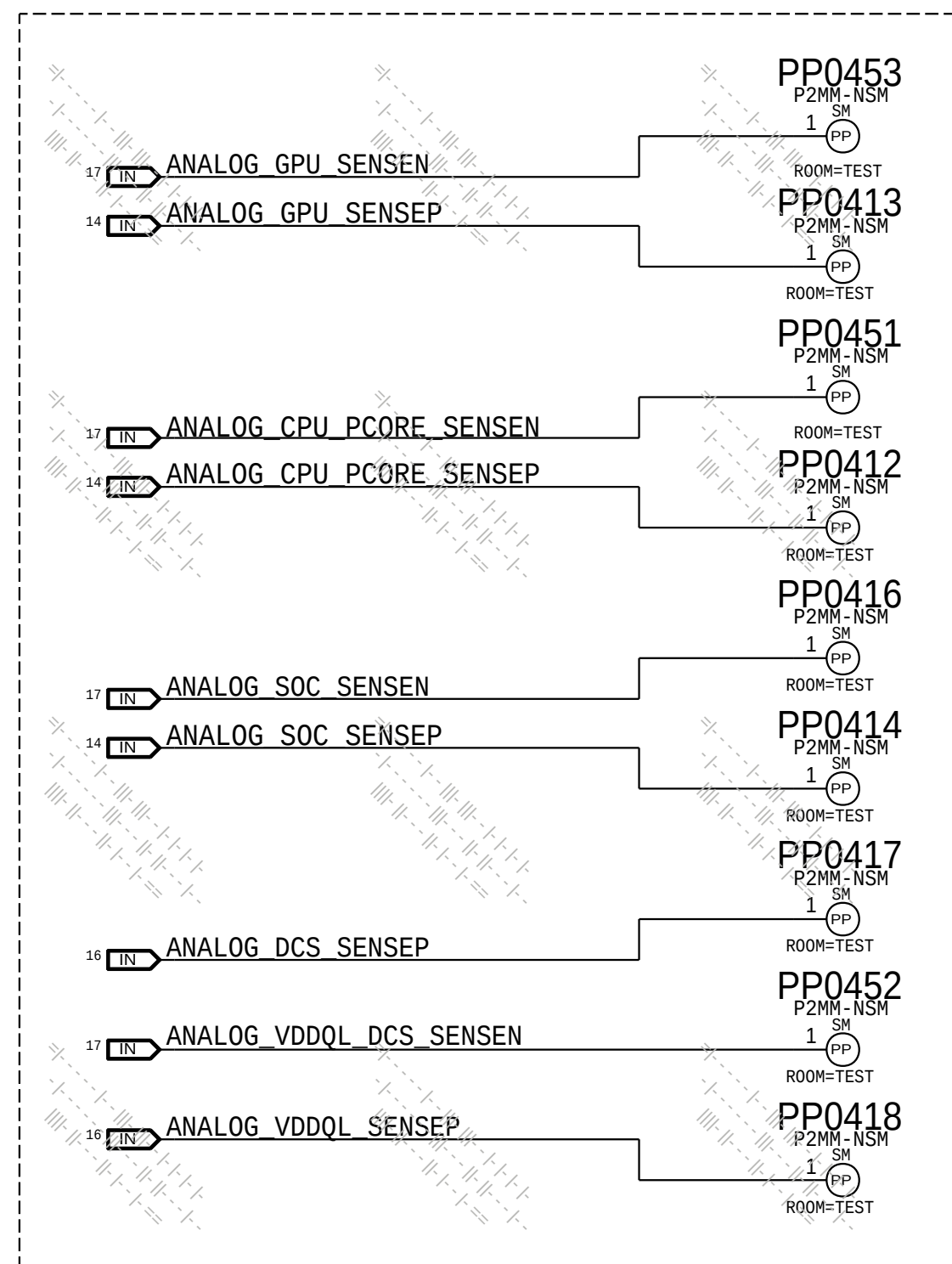
PCIE Refclk



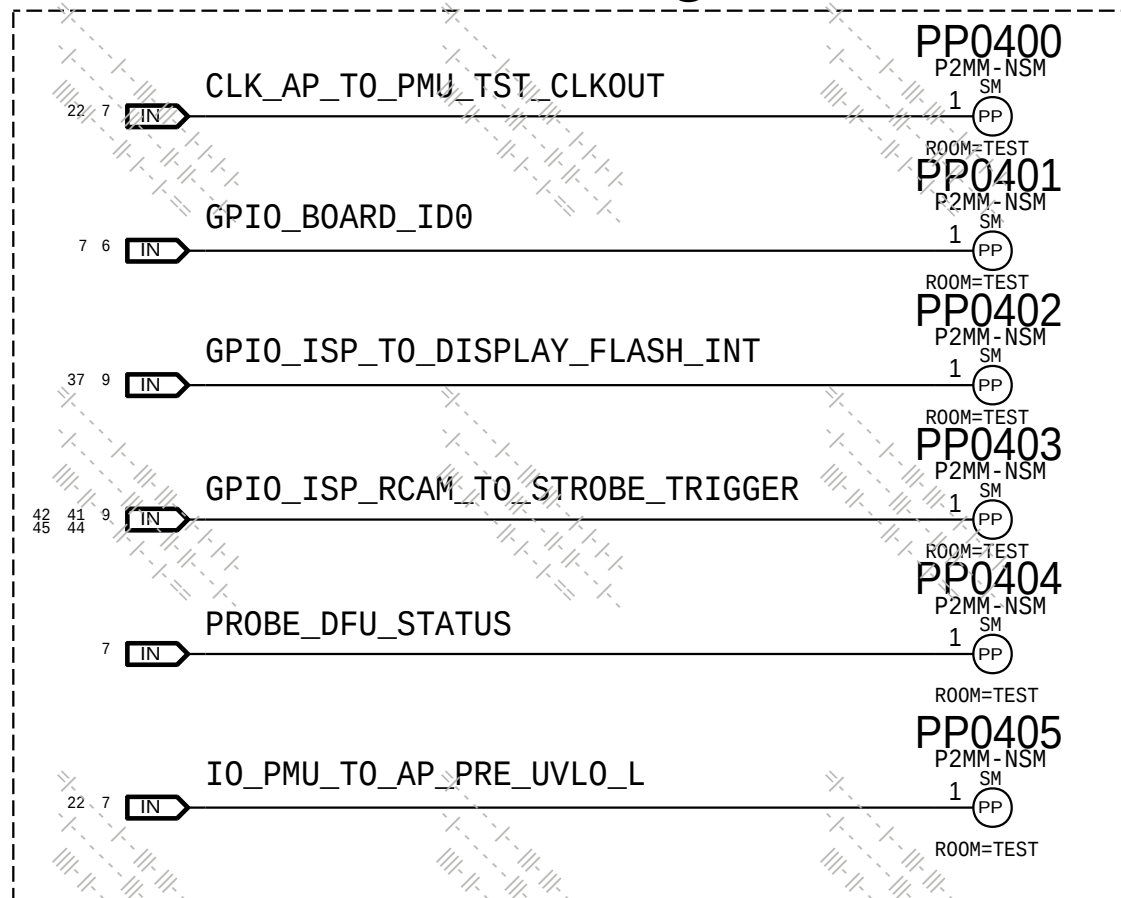
VALIDATION



BUMP SENSE



SOC Debug

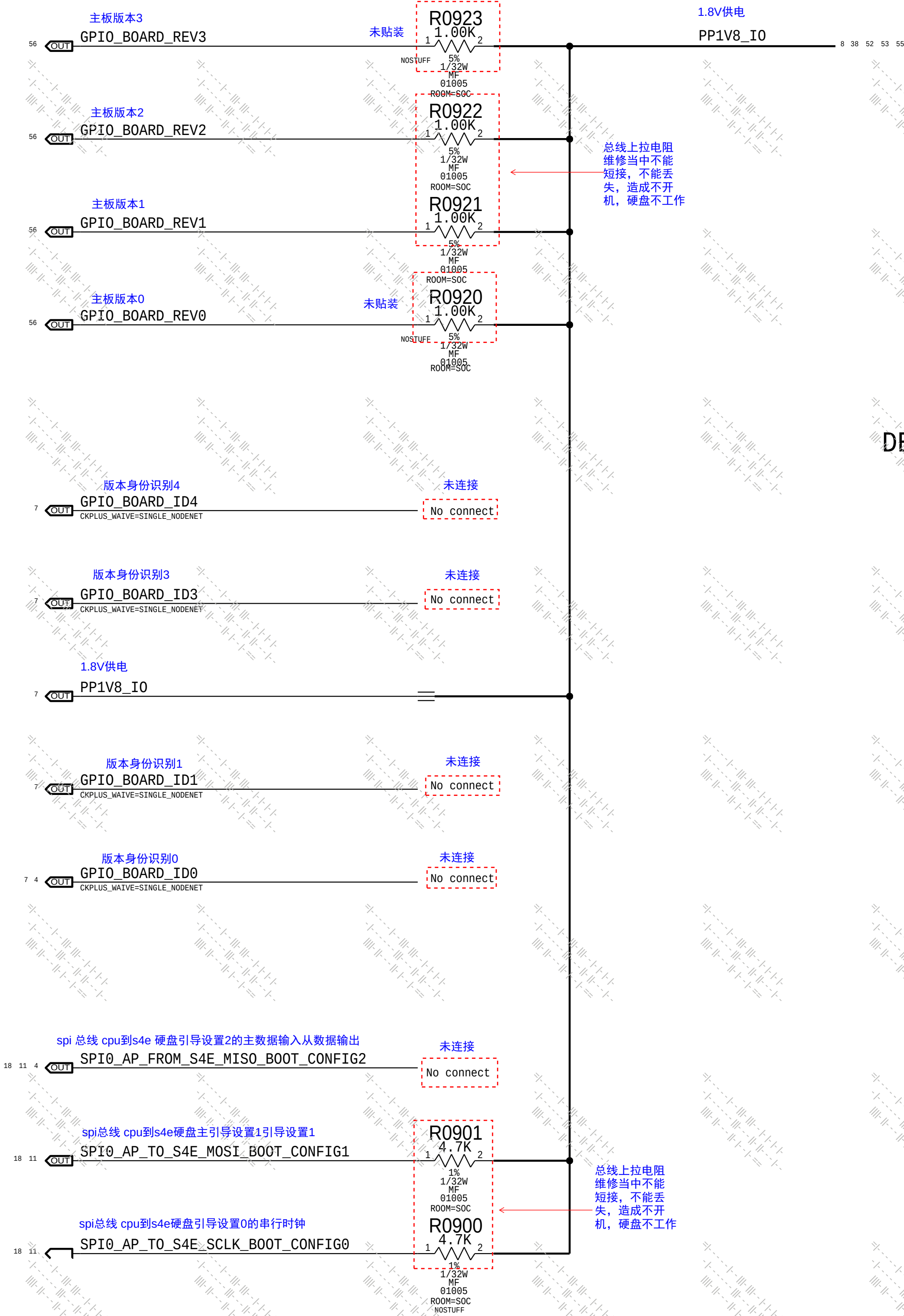


PAGE TITLE		
SYSTEM: Testpoints (Top)		
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	SHEET	
	4 OF 60	

BOOTSTRAPPING:
BOARD REV

BOARD ID

BOOT CONFIG



SELECTED -->

DEFAULT -->

POR -->

Proto Builds -->

主板配置对照表

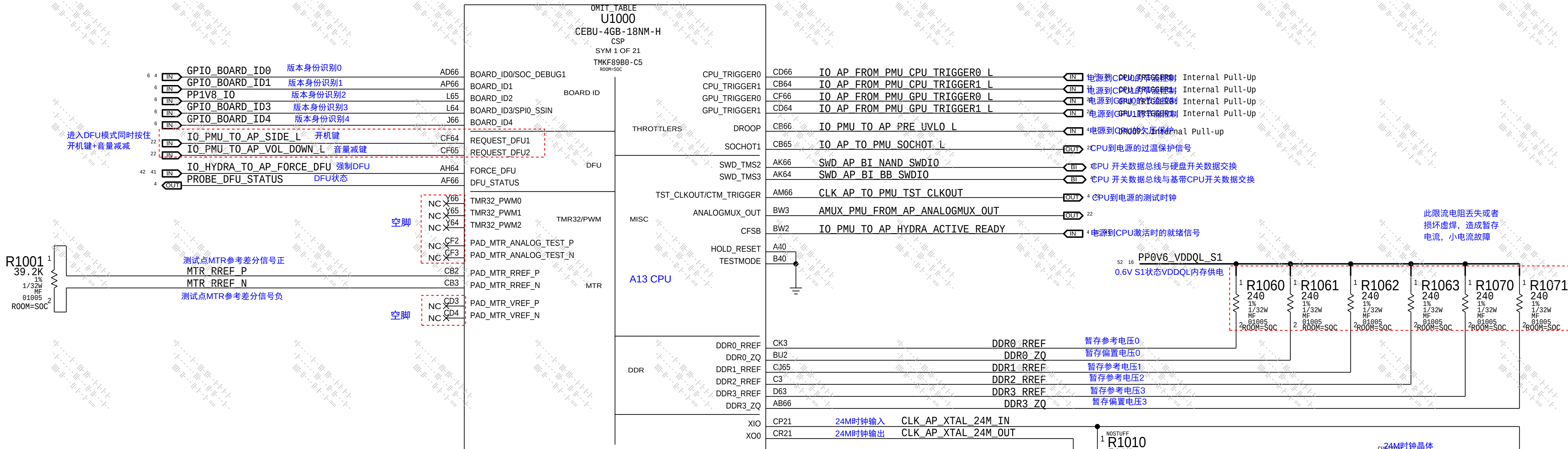
Board Rev[3:0]				
Float=Low PU=High				
	3	2	1	0
PreProto	1	1	1	1
Proto 1	1	1	1	0
(Spare)	1	1	0	1
Proto 2	1	1	0	0
P2+ B0/B0 4G	1	0	1	1
EVT	1	0	1	0
(Spare)	0	1	1	1
Carrier	0	1	1	0
(Spare)	0	0	1	1
DVT	0	0	1	0
(Spare)	0	0	0	1
PVT	0	0	0	0

Board ID[4:0]						
Float=Low PU=High						
	BoardID[4:0]	4	3	2	1	0
		Unused	0 = ICE, 1 = MAV	00 = None 01 = x1367/9 11 = x1364/8 10 = x1403	0 = MLB, 1 = DEV	
MLB x1403	5'b00100	0	0	1	0	0
DEV x1403	5'b00101	0	0	1	0	1
MLB x1364	5'b01110	0	1	1	1	0
DEV x1364	5'b01111	0	1	1	1	1
MLB X1368	5'b00110	0	0	1	1	0
DEV X1368	5'b00111	0	0	1	1	1
MLB x1367	5'b01010	0	1	0	1	0
DEV x1367	5'b01011	0	1	0	1	1
MLB x1369	5'b00010	0	0	0	1	0
DEV x1369	5'b00011	0	0	0	1	1

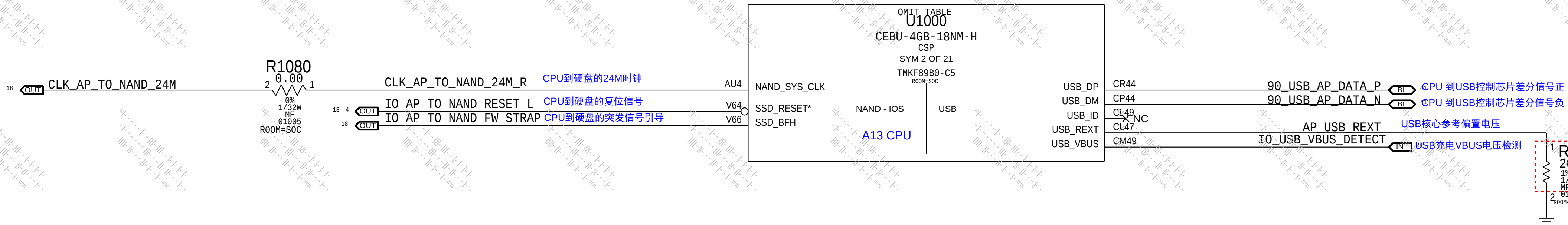
Boot Config [2:0]			
Float = Low PU = High			
	2	1	0
SPI NOR on SPI0 12 MHz	0	0	0
SPI NOR on SPI0 12 MHz Test	0	0	1
SPI NAND on SPI0 12 MHz	0	1	0
SPI NAND on SPI0 12 MHz Test	0	1	1
SPI NOR on SPI0 40 MHz	1	0	0
SPI NOR on SPI0 40 MHz Test	1	0	1
SPI NOR on SPI0 6 MHz	1	1	0
SPI NOR on SPI0 6 MHz Test	1	1	1

PAGE TITLE		
BOOTSTRAPPING		
	DRAWING NUMBER	051-04047
	REVISION	6.0.0
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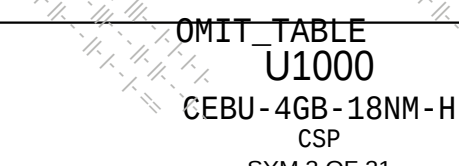
SOC: Misc



SOC: NAND + USB



PAGE TITLE		
SOC: NAND & USB & Misc		
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BRANCH		
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SOC: ISP

来自前置相头到CPU LPDP总线的数据接收1差分信号正
来自前置相头到CPU LPDP总线的数据接收1差分信号负
来自前置相头到CPU LPDP总线的数据接收0差分信号正
来自前置相头到CPU LPDP总线的数据接收0差分信号负
LPDP LANE SWAPPING
OPTIMIZES ROUTING

来自后置超广角相头到CPU LPDP总线的数据接收0差分信号正
来自后置超广角相头到CPU LPDP总线的数据接收0差分信号负
来自后置超广角相头到CPU LPDP总线的数据接收1差分信号正
来自后置超广角相头到CPU LPDP总线的数据接收1差分信号负
来自后置超广角相头到CPU LPDP总线的数据接收2差分信号正
来自后置超广角相头到CPU LPDP总线的数据接收2差分信号负

来自后置广角相头到CPU LPDP总线的数据接收0差分信号正
来自后置广角相头到CPU LPDP总线的数据接收0差分信号负

来自后置广角相头到CPU LPDP总线的数据接收2差分信号正
来自后置广角相头到CPU LPDP总线的数据接收2差分信号负
来自后置广角相头到CPU LPDP总线的数据接收1差分信号正
来自后置广角相头到CPU LPDP总线的数据接收1差分信号负

90 LPDP ISP FROM FCAM_RX_D1_P
90 LPDP ISP FROM FCAM_RX_D1_N
90 LPDP ISP FROM FCAM_RX_D0_P
90 LPDP ISP FROM FCAM_RX_D0_N

90 LPDP ISP FROM SWIDE_RX_D0_N
90 LPDP ISP FROM SWIDE_RX_D0_P
90 LPDP ISP FROM SWIDE_RX_D1_N
90 LPDP ISP FROM SWIDE_RX_D1_P
90 LPDP ISP FROM SWIDE_RX_D2_N
90 LPDP ISP FROM SWIDE_RX_D2_P

90 LPDP ISP FROM WIDE_RX_D0_N
90 LPDP ISP FROM WIDE_RX_D0_P

90 LPDP ISP FROM WIDE_RX_D2_N
90 LPDP ISP FROM WIDE_RX_D2_P
90 LPDP ISP FROM WIDE_RX_D1_N
90 LPDP ISP FROM WIDE_RX_D1_P

LPDPRX_RX_D0_P
LPDPRX_RX_D0_N
LPDPRX_RX_D1_P
LPDPRX_RX_D1_N

LPDPRX_RX_D2_P
LPDPRX_RX_D2_N
LPDPRX_RX_D3_P
LPDPRX_RX_D3_N
LPDPRX_RX_D4_P
LPDPRX_RX_D4_N

LPDPRX_RX_D5_P
LPDPRX_RX_D5_N
LPDPRX_RX_D6_P
LPDPRX_RX_D6_N

LPDPRX_RX_D7_P
LPDPRX_RX_D7_N
LPDPRX_RX_D8_P
LPDPRX_RX_D8_N

LPDPRX0_RCAL_P
LPDPRX0_RCAL_N
LPDPRX1_RCAL_P
LPDPRX1_RCAL_N

来自前置相头到CPU LPDP总线的辅助信号
来自前置相头到CPU LPDP总线的辅助信号
来自前置相头到CPU LPDP总线的辅助信号
来自前置相头到CPU LPDP总线的辅助信号

来自后置超广角相头到CPU LPDP总线的辅助信号
来自后置超广角相头到CPU LPDP总线的辅助信号

LPDPRX_AUX_D0_P
LPDPRX_AUX_D1_P
LPDPRX_AUX_D2_P
LPDPRX_AUX_D3_P
LPDPRX_AUX_D4_P
LPDPRX_AUX_D5_P
LPDPRX_AUX_D6_P
LPDPRX_AUX_D7_P
LPDPRX_AUX_D8_P
LPDPRX_AUX_D9_P
LPDPRX_AUX_D10_P
LPDPRX_AUX_D11_P

OMIT_TABLE
U1000
CEBU-4GB-18NM-H
CSP
SYM 4 OF 21
TMKF89B0-C5
ROOM=SOC

A13 CPU

MIPI0C_DPCLK
MIPI0C_DNCLK
MIPI0C_DPDATA0
MIPI0C_DNDATA0
MIPI0C_DPDATA1
MIPI0C_DNDATA1
MIPI0C_REXT

ISP_I2C0_SCL/ISP_GPIO_15
ISP_I2C0_SDA/ISP_GPIO_14
ISP_I2C1_SCL/ISP_GPIO_13
ISP_I2C1_SDA/ISP_GPIO_12
ISP_I2C2_SCL/ISP_GPIO_11
ISP_I2C2_SDA/ISP_GPIO_10
ISP_I2C3_SCL/ISP_GPIO_9
ISP_I2C3_SDA/ISP_GPIO_8

ISP_GPIO0/SIO_LEAP_MAD_IN/AOP_LEAP_MAD_IN
ISP_GPIO1/SIO_LEAP_MAD_OUT/AOP_LEAP_MAD_OUT
ISP_GPIO2/SOC_DEBUG2/MTR_ADC_DOUT
ISP_GPIO3/SOC_DEBUG3
ISP_GPIO4/ISP_SPML_SDATA/MTR_ADC_CLKOUT
ISP_GPIO5/ISP_SPML_SCLK
ISP_GPIO6/ISP_FCAM_SPML_SDATA/PLL_DIG0BS_0
ISP_GPIO7/ISP_FCAM_SPML_SCLK/PLL_DIG0BS_1

SENSOR0_CLK

SENSOR1_CLK

SENSOR2_CLK

SENSOR3_CLK

90 MIPI ISP FROM IRCAM_CLK_P
90 MIPI ISP FROM IRCAM_CLK_N
90 MIPI ISP FROM IRCAM_D0_N
90 MIPI ISP FROM IRCAM_D0_P
90 MIPI ISP FROM IRCAM_D1_N
90 MIPI ISP FROM IRCAM_D1_P

MIPI0C_ISP_REXT

I2C0_ISP_SCL
I2C0_ISP_SDA

I2C3_ISP_SCL
I2C3_ISP_SDA

GPIO_ISP_TO_CAM_PMU1_RESET_L
GPIO_ISP_TO_DISPLAY_FLASH_INT

GPIO_ISP_RCAM_TO_STROBE_TRIGGER
SPMI_ISP_BI_CAM_PMU1_DATA

SPMI_ISP_TO_CAM_PMU1_CLK_R
SPMI_ISP_TO_CAM_PMU1_CLK_L

CLK_ISP_TO_CAM_PMU1_24M_R

CLK_ISP_TO_RIGEL_12M_R

来自前置红外相头到CPU MIPI总线的数据时钟差分信号正
来自前置红外相头到CPU MIPI总线的数据时钟差分信号负
来自前置红外相头到CPU MIPI总线的数据接收1差分信号正
来自前置红外相头到CPU MIPI总线的数据接收1差分信号负
来自前置红外相头到CPU MIPI总线的数据接收0差分信号正
来自前置红外相头到CPU MIPI总线的数据接收0差分信号负

MIPI0C_ISP_REXT

CPU的闪光灯 I2C0总线串行时钟
CPU的闪光灯 I2C0总线串行数据地址

CPU的闪光灯 I2C0总线串行时钟
CPU的闪光灯 I2C0总线串行数据地址

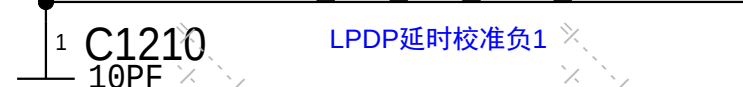
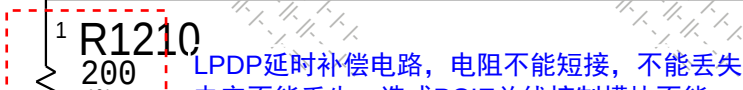
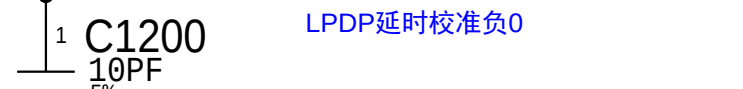
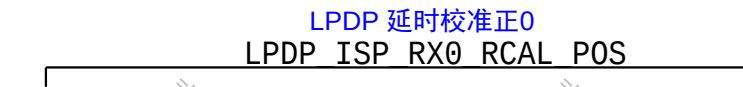
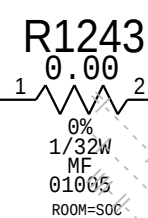
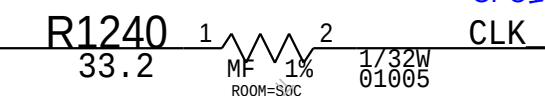
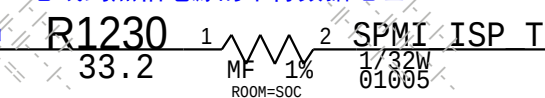
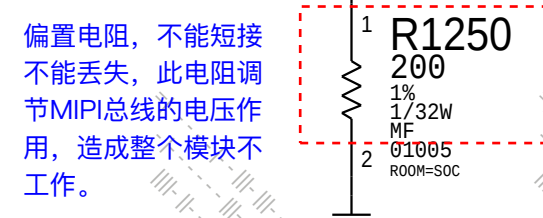
CPU到照相电源的复位信号
GPIO_ISP_TO_DISPLAY_FLASH_INT

CPU SPMI总线到照相电源的串行数据地址
SPMI_ISP_TO_CAM_PMU1_CLK_R

CPU SPMI总线到照相电源的串行时钟
SPMI_ISP_TO_CAM_PMU1_CLK_L

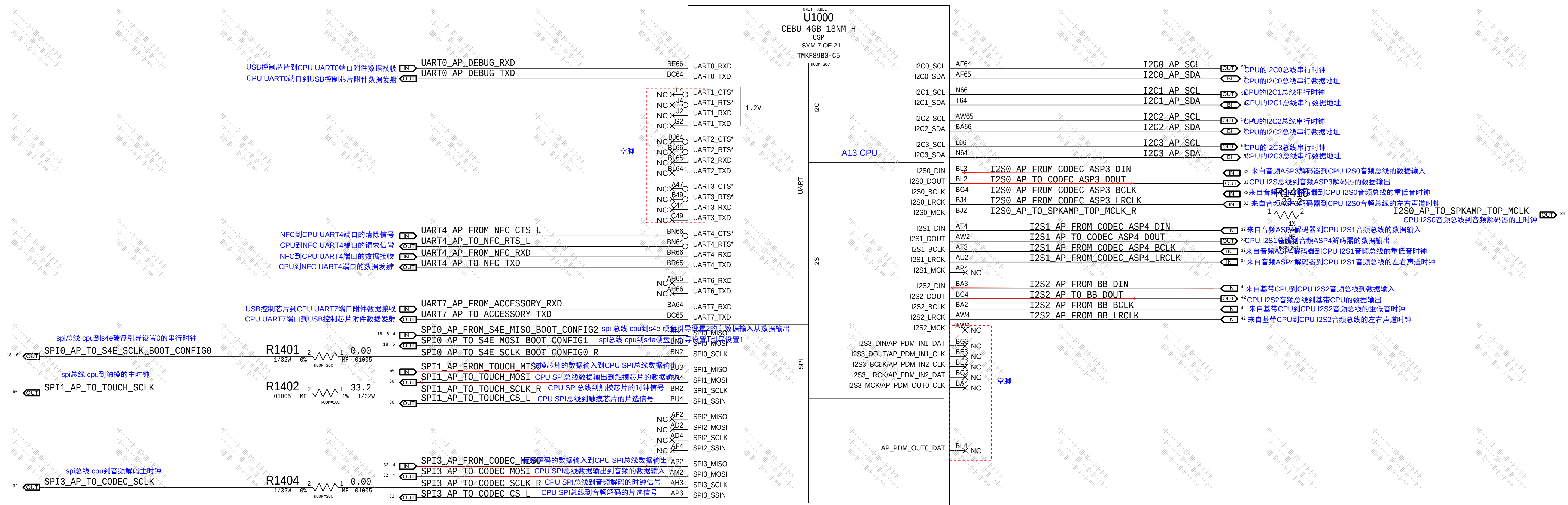
CPU到照相电源的24M时钟
CLK_ISP_TO_CAM_PMU1_24M_R

CPU到点阵的12M时钟
CLK_ISP_TO_RIGEL_12M_R



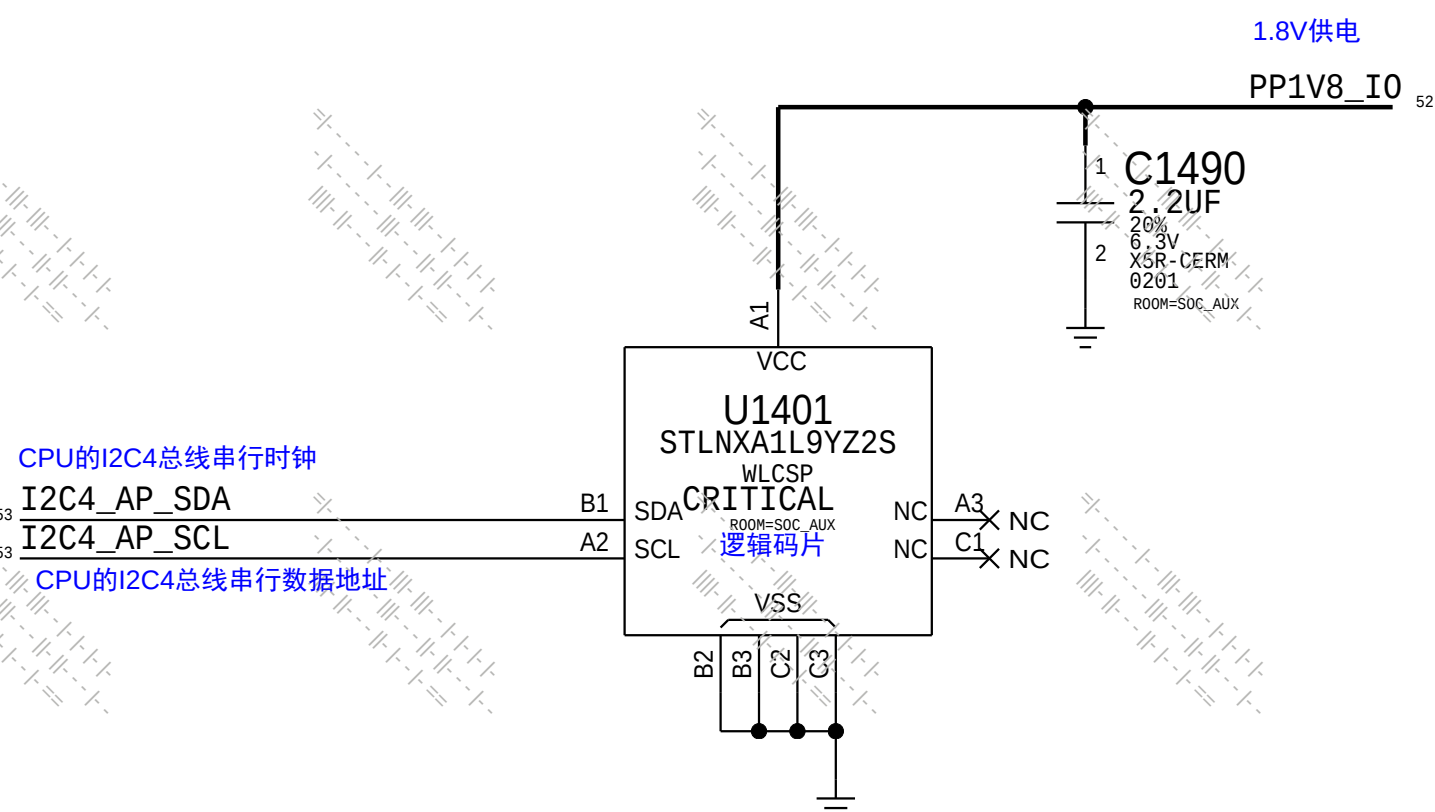
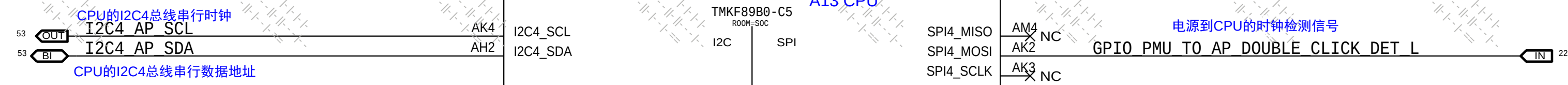
PAGE TITLE			DRAWING NUMBER			SIZE		
S0C: ISP			051-04047			D		
REVISION			6.0.0			BRANCH		
PAGE			12 OF 150			SHEET		
9 OF 60								

SOC: AP Serial



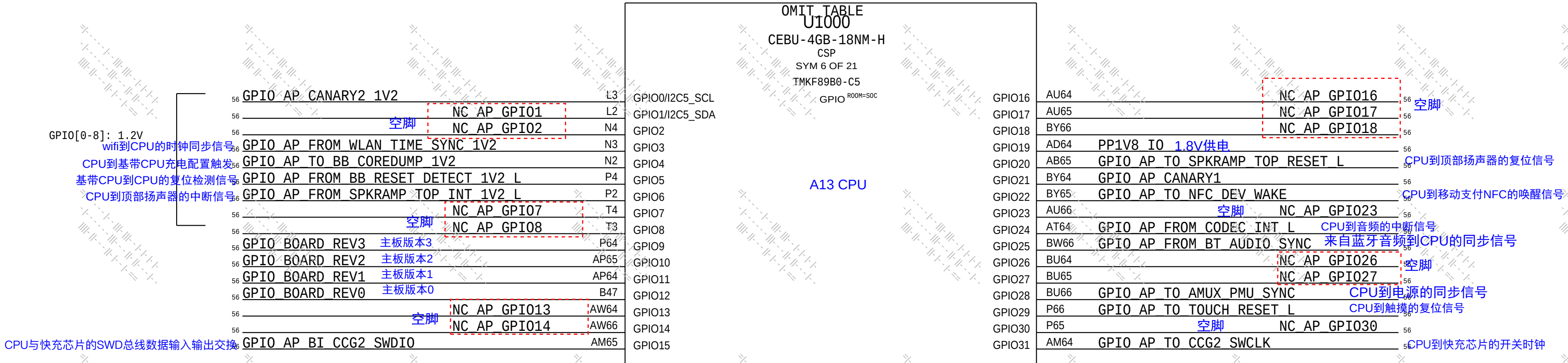
SPI: Route as Daisy-Chain. No T's Allowed

Place series terminations close to SoC Pins

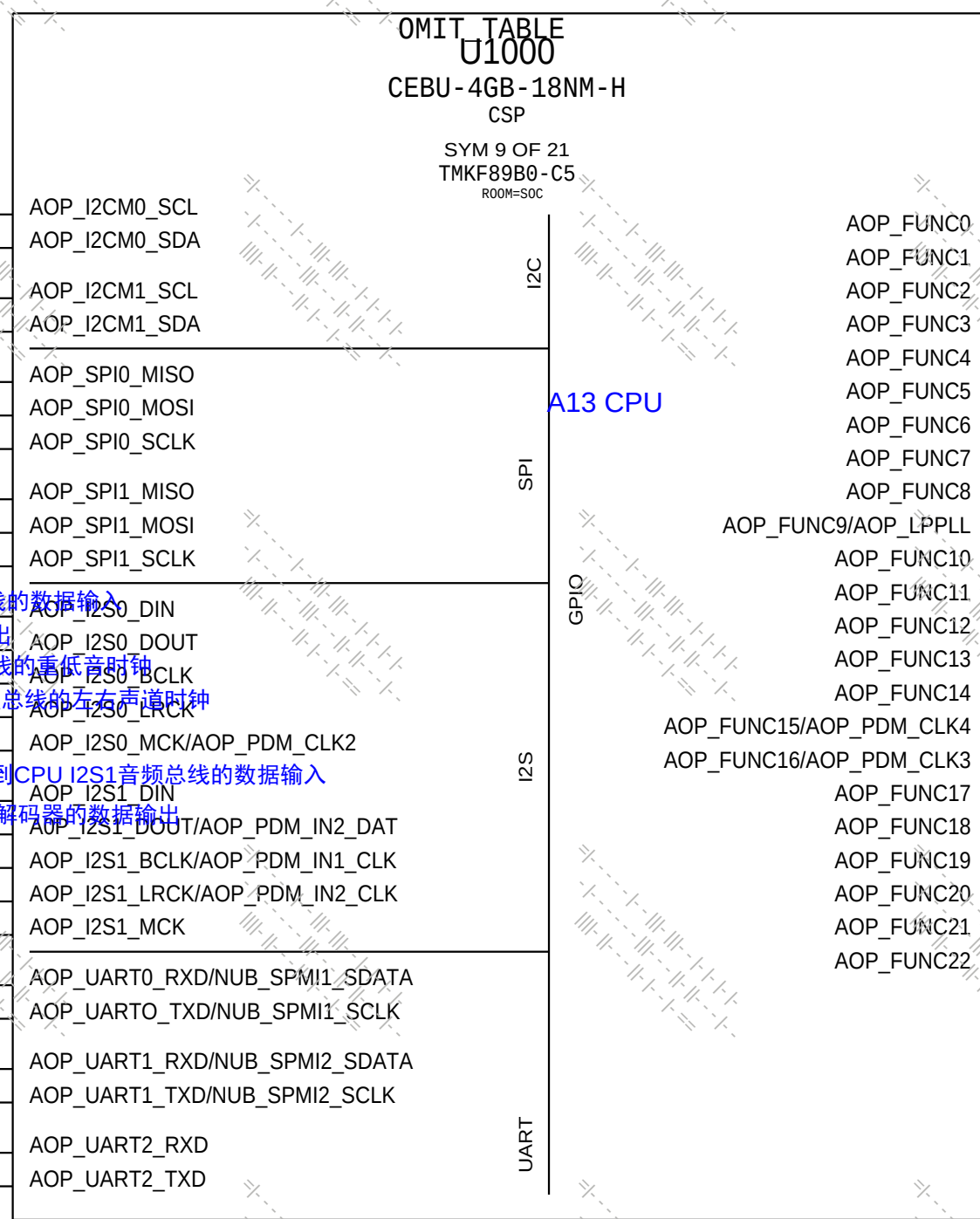
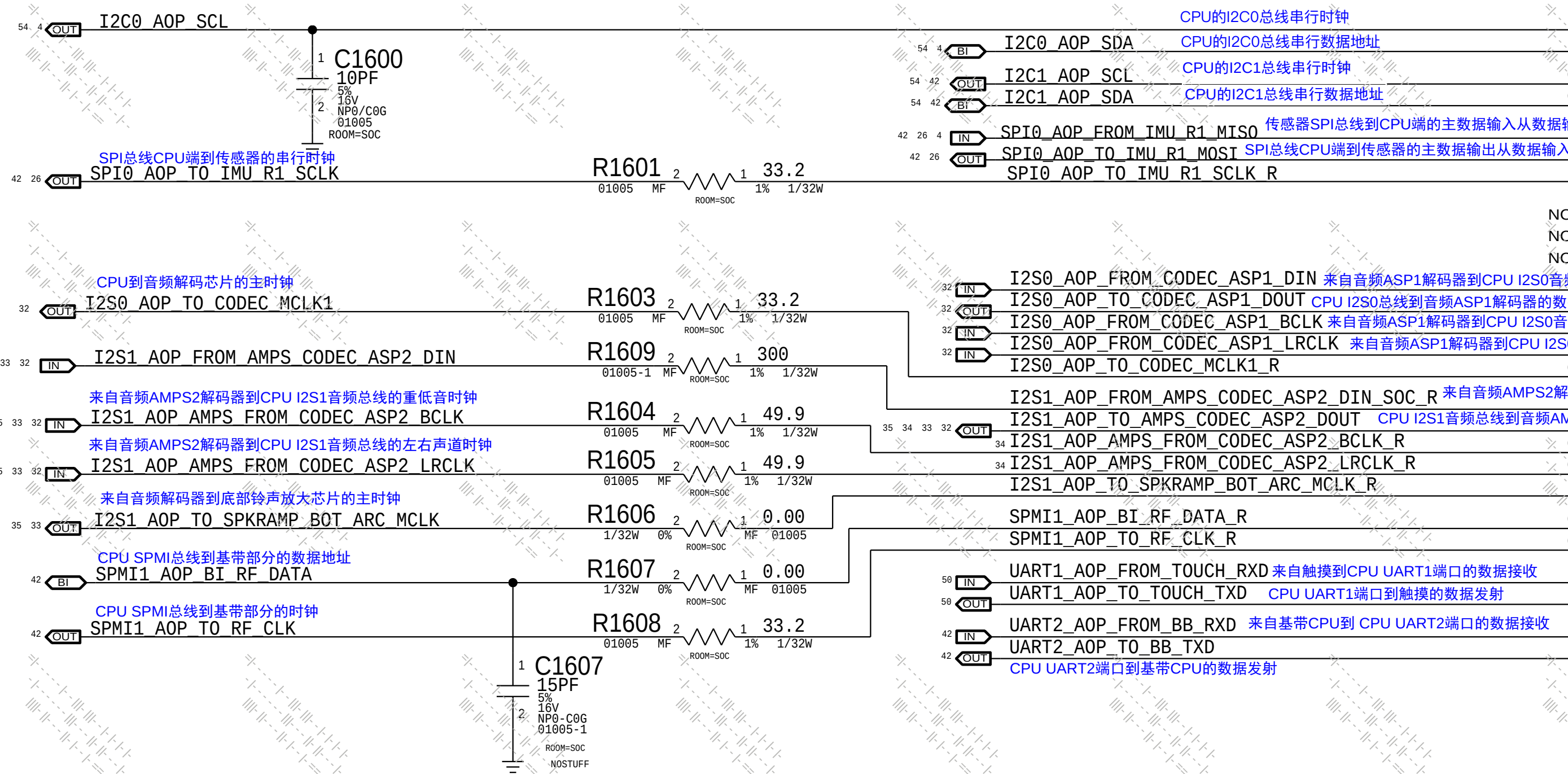


PAGE TITLE		
SOC: SERIAL		
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	SHEET	11 OF 60

SOC: AP GPIO

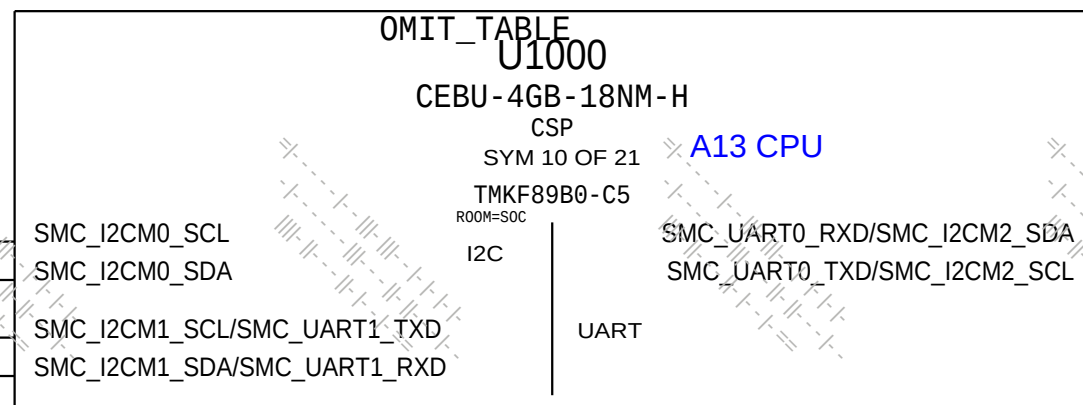
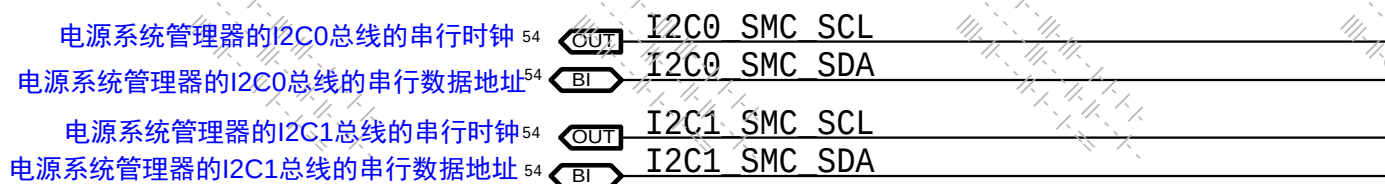


SOC: AOP

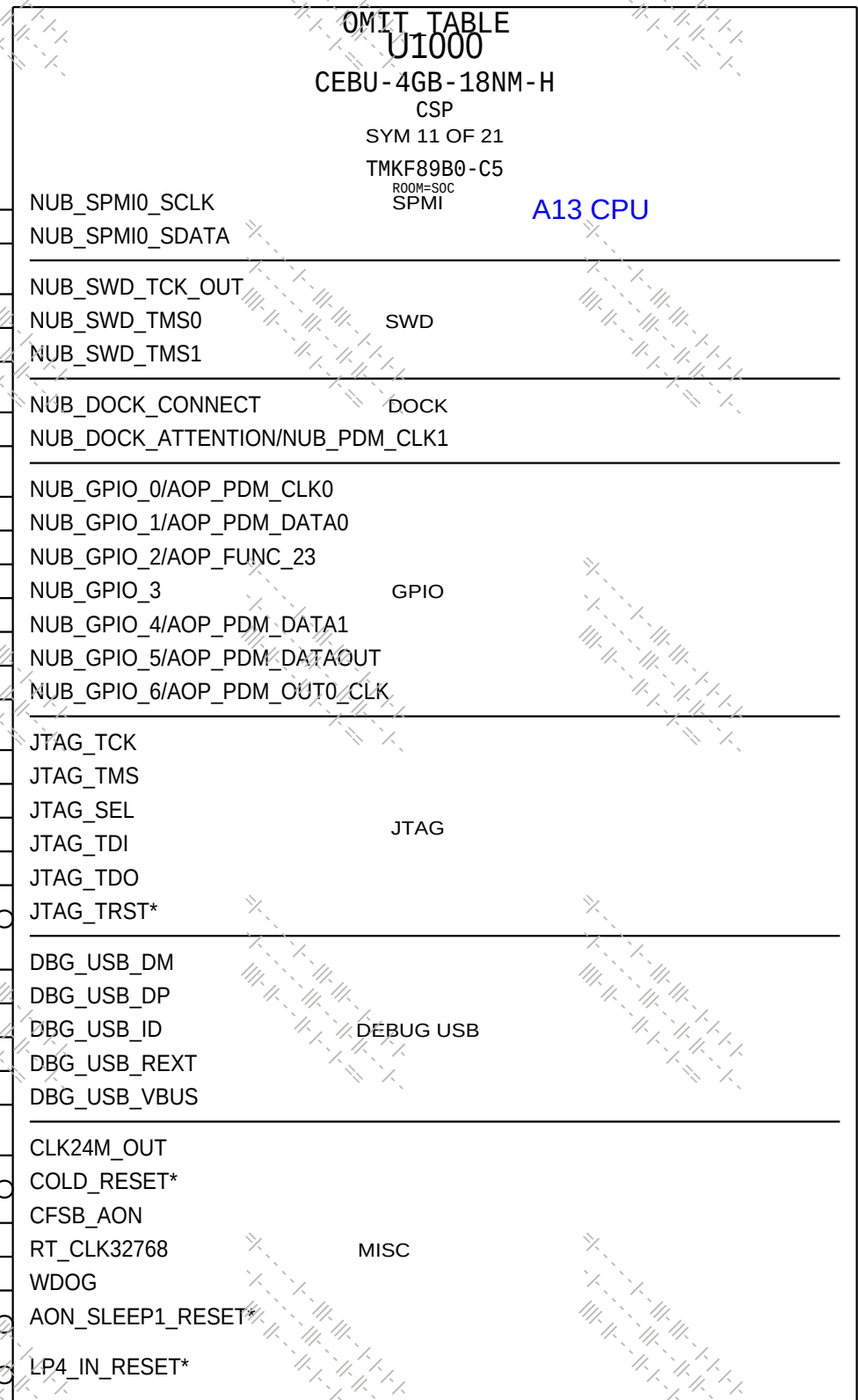
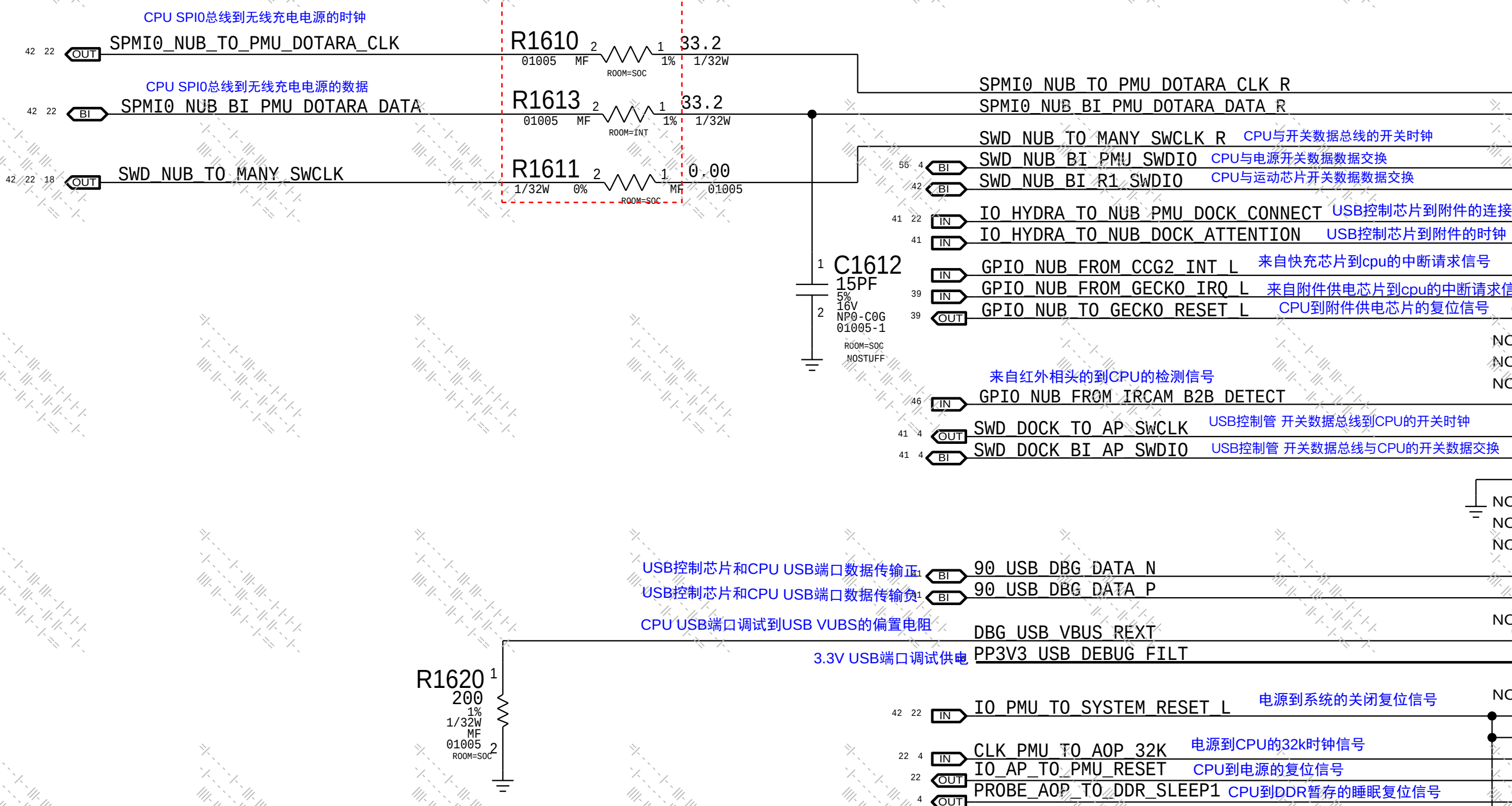


CL62	GPIO_AOP_FROM_IMU_DATARDY	加速器陀螺仪到CPU中断请求信号
CL34	GPIO_AOP_TO_R1_COREDUMP_TRIGGER	CPU到运动芯片的复位信号
CM63	GPIO_AOP_TO_CODEC_CLP_EN	CPU到音频解码的控制开启
CJ34	GPIO_AOP_TO_R1_SPI_CS_L	CPU SPI总线到运动芯片的片选信号
CK34	GPIO_AOP_FROM_R1_INT	CPU到运动芯片的中断信号
CM55	GPIO_AOP_TO_CODEC_RESET_L	CPU到音频解码的复位信号
CR38	GPIO_AOP_FROM_NFC_HQST_WAKE	CPU到NFC的热启动唤醒信号
CM53	GPIO_AOP_FROM_SPKAMP_BOT_ARC_INT_L	底部铃声放大芯片到CPU的中断信号
CP38	GPIO_AOP_TO_WLAN_CONTEXT_A	CPU到wifi的Context接口程序配置A
CM51	GPIO_AOP_TO_IMU_SPI_CS_L	CPU到加速器, 陀螺仪SPI总线片选信号
CM36	GPIO_AOP_TO_R1_TIME_SYNC	CPU到运动芯片的时间同步信号
CM47	NC_AOP_GPIO11	空脚
CL36	GPIO_AOP_FROM_ROMEO_B2B_DETECT	点阵到CPU的B2B连接检测信号
CL64	CPU到触摸芯片的24M时钟信号	CLK_AP_TO_TOUCH_SCAN
CJ36	空脚	NC_AOP_GPIO14
CK57	GPIO_AOP_BI_PROX_INT_L	感光到CPU的中断信号
CK36	GPIO_AOP_FROM_ALS_INT_L	感光到CPU的中断信号
CM64	GPIO_AOP_FROM_K2_INT_L	K2到CPU的中断信号
CM38	GPIO_AOP_FROM_COMPASS_INT	指南针到CPU的中断信号
CM59	NC_AOP_GPIO19	空脚
CL38	GPIO_AOP_FROM_TOUCH_CTS	来自触摸到CPU的清除信号
CJ61	GPIO_AOP_FROM_TOUCH_INT_L	触摸到CPU的中断信号
CJ38	GPIO_AOP_TO_WLAN_CONTEXT_B	CPU到wifi的Context接口程序配置B

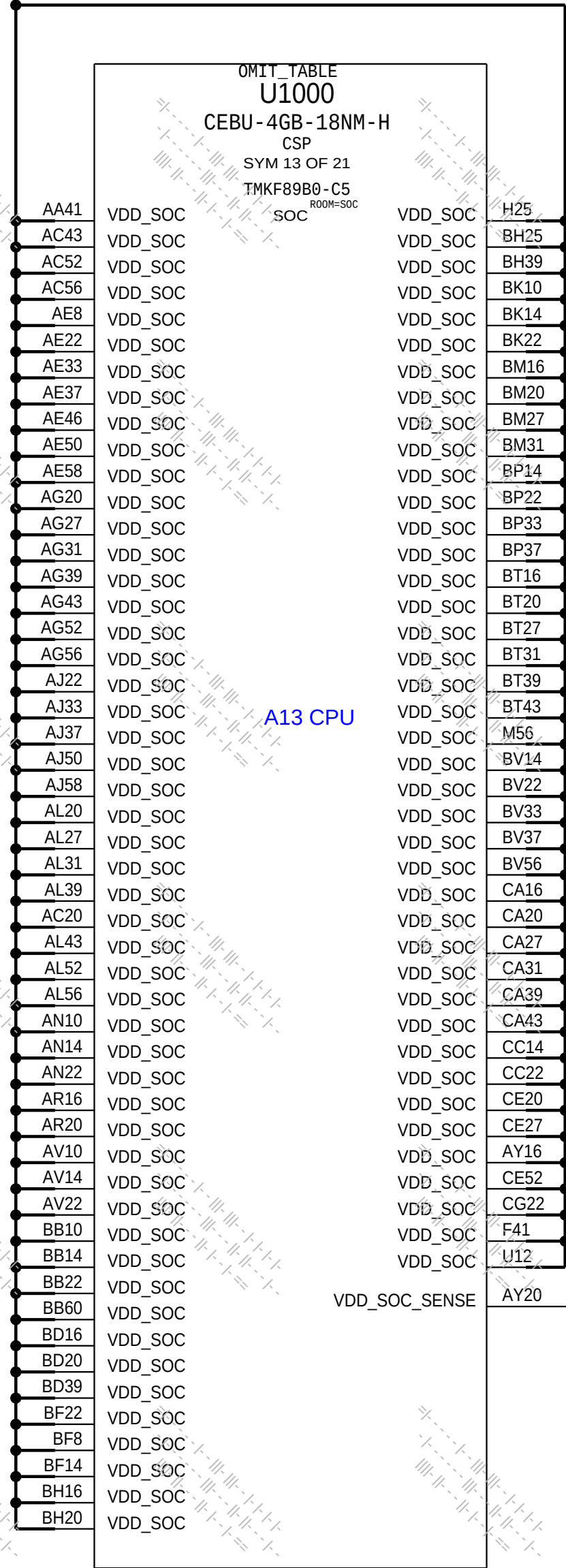
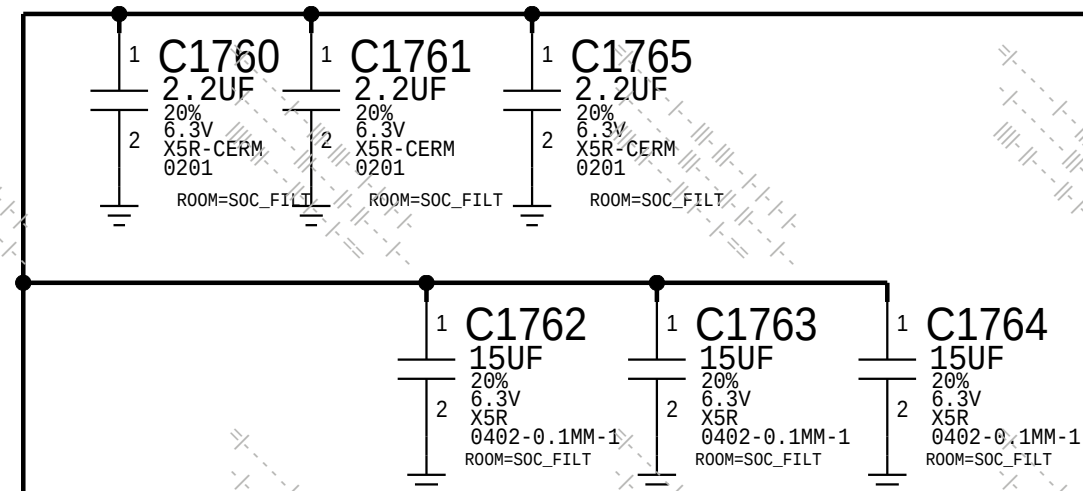
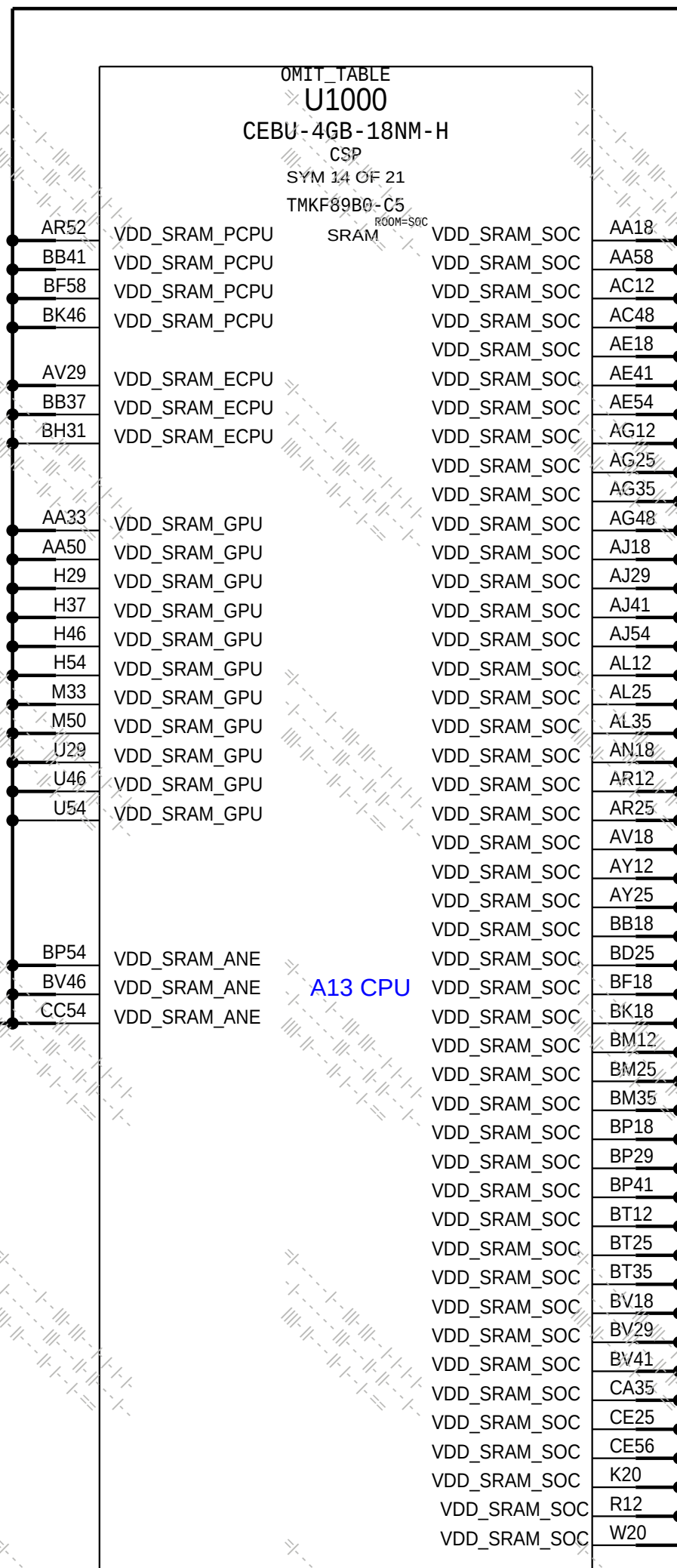
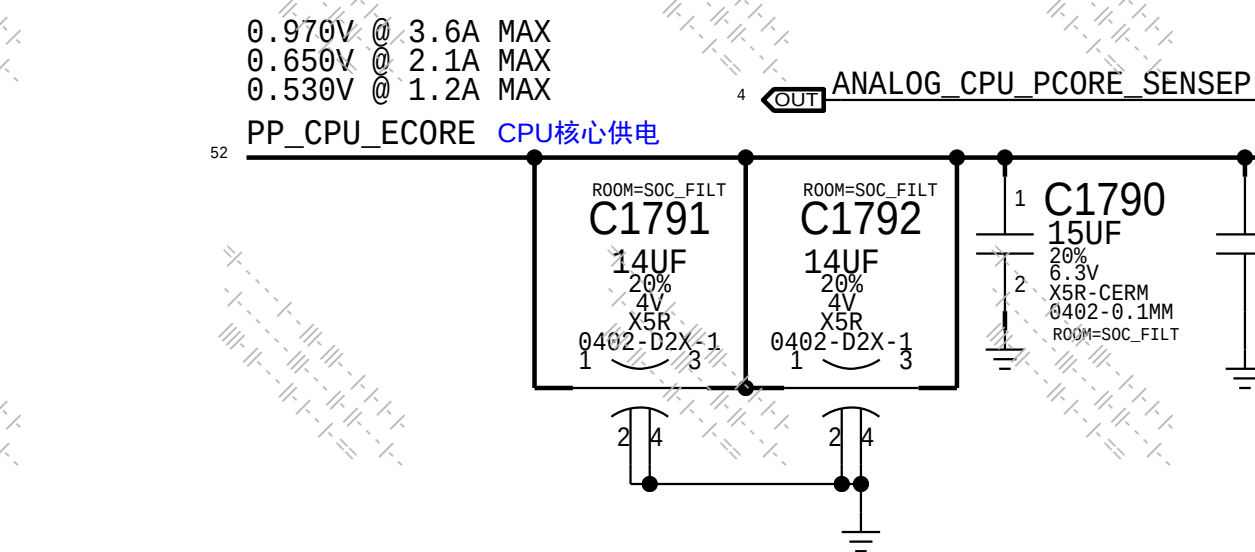
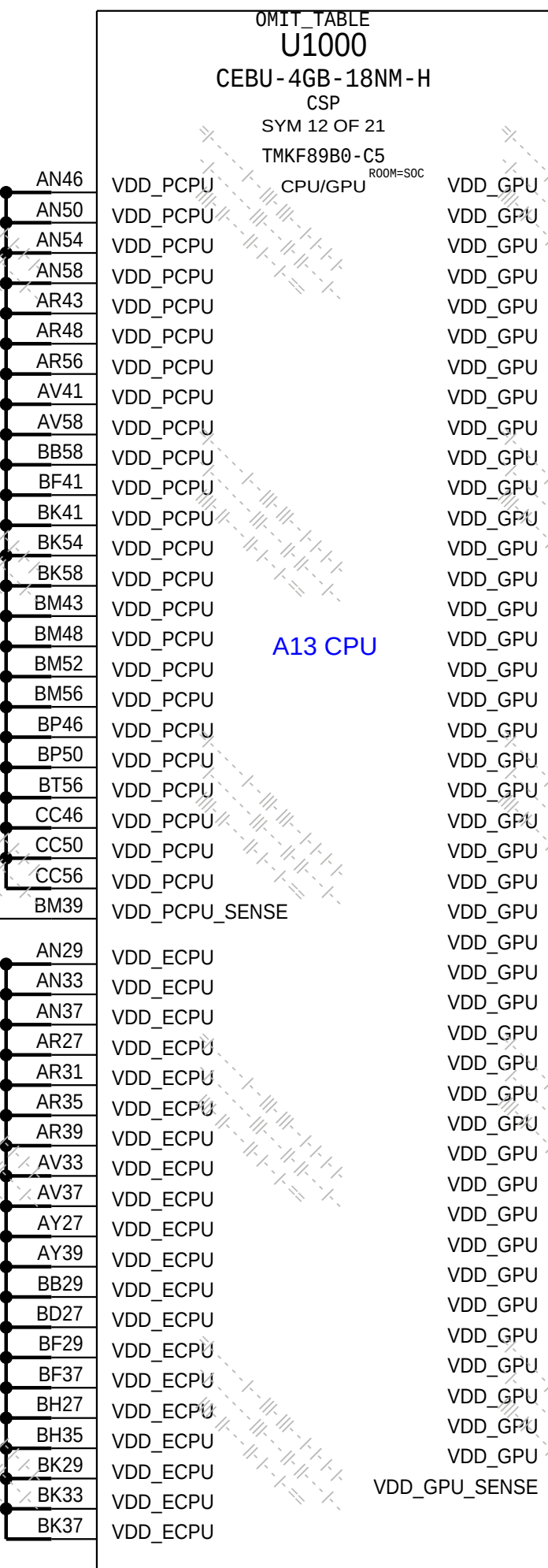
SOC: SMC



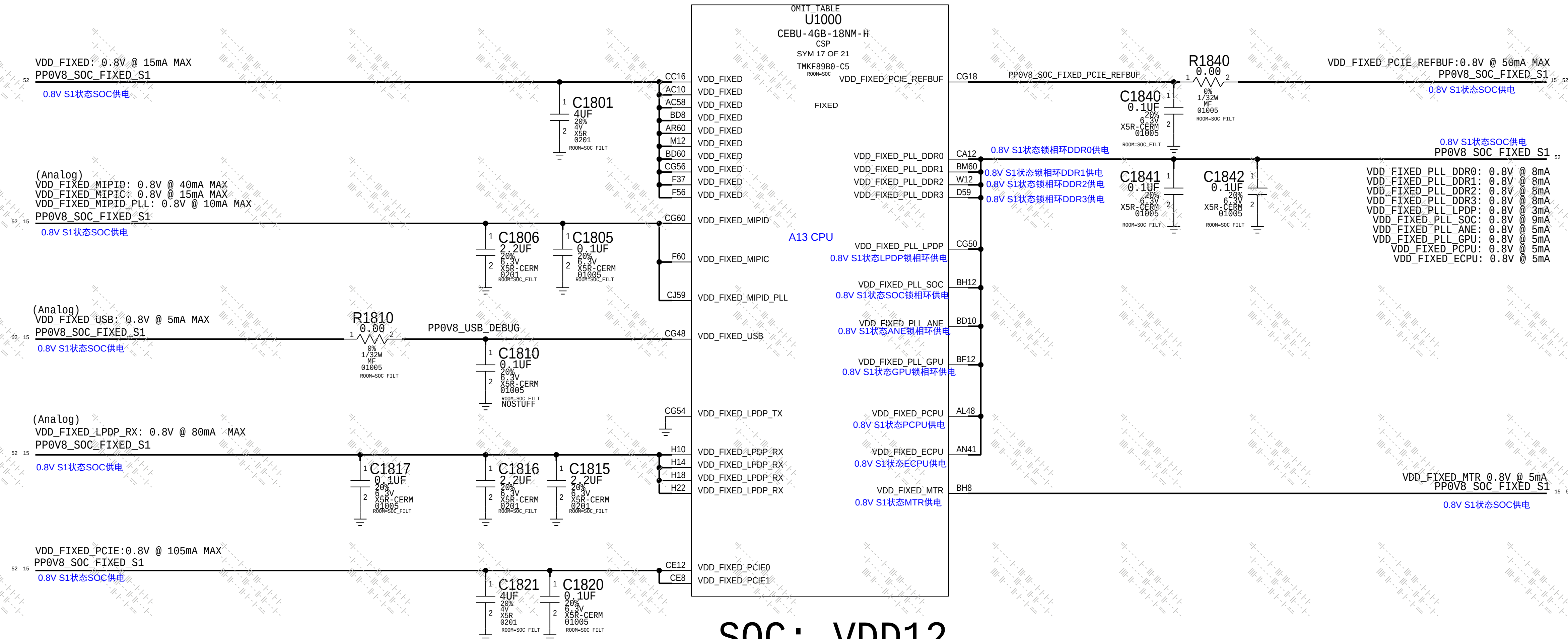
SOC: NUB



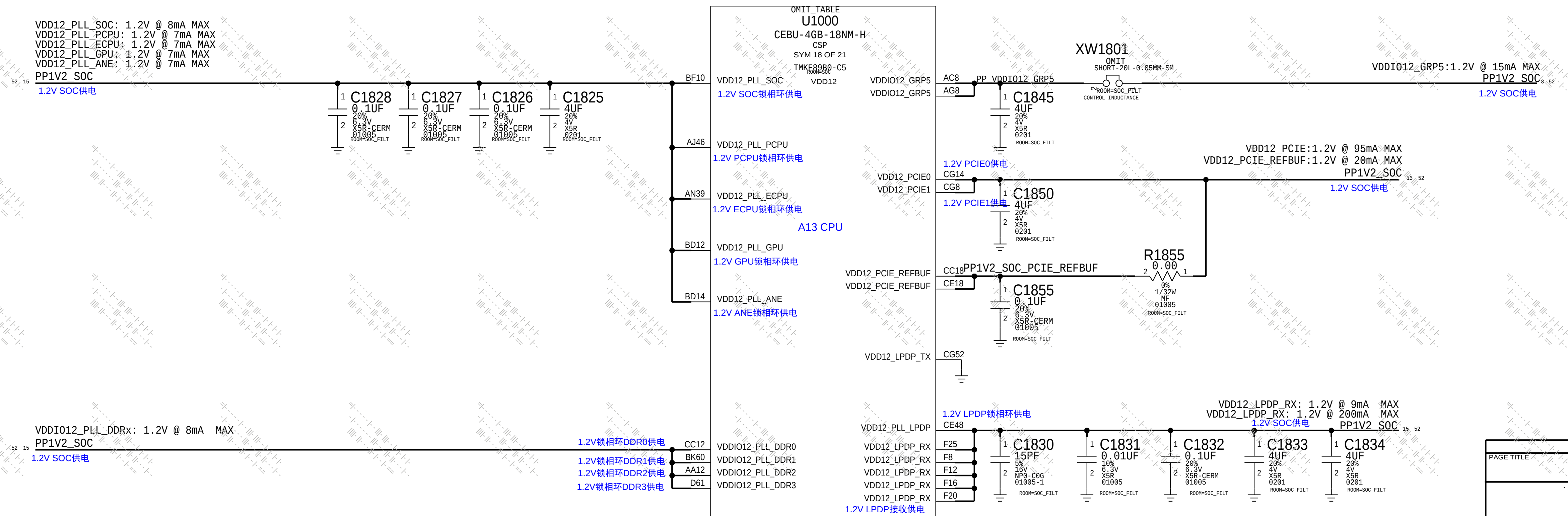
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		6.0.0		
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		PAGE		
		SHEET		13 OF 60



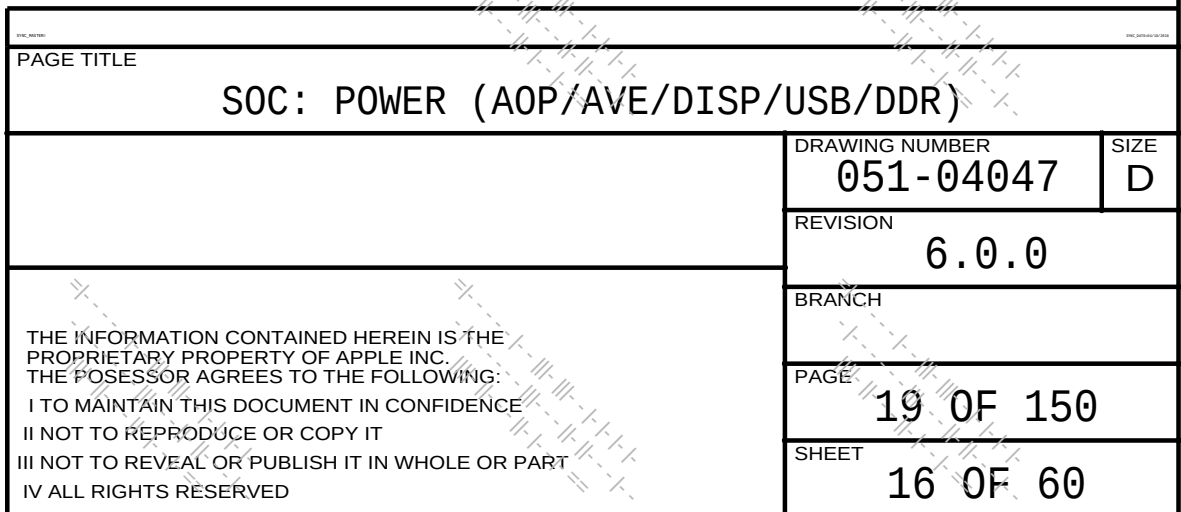
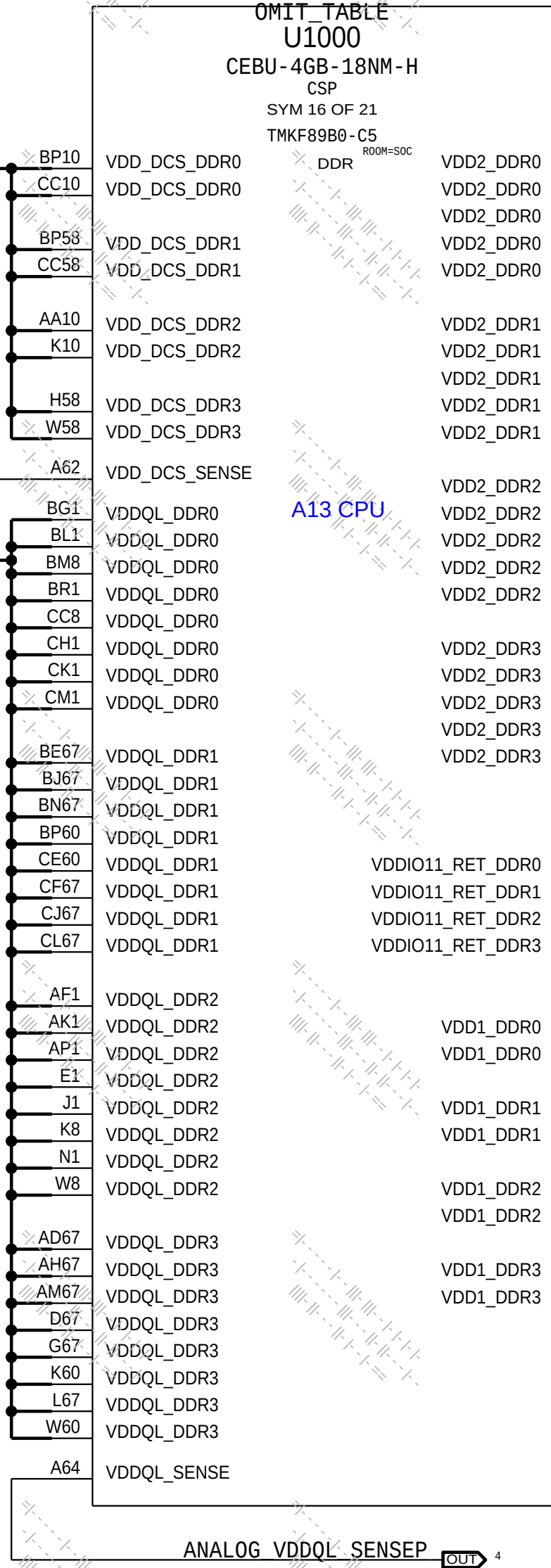
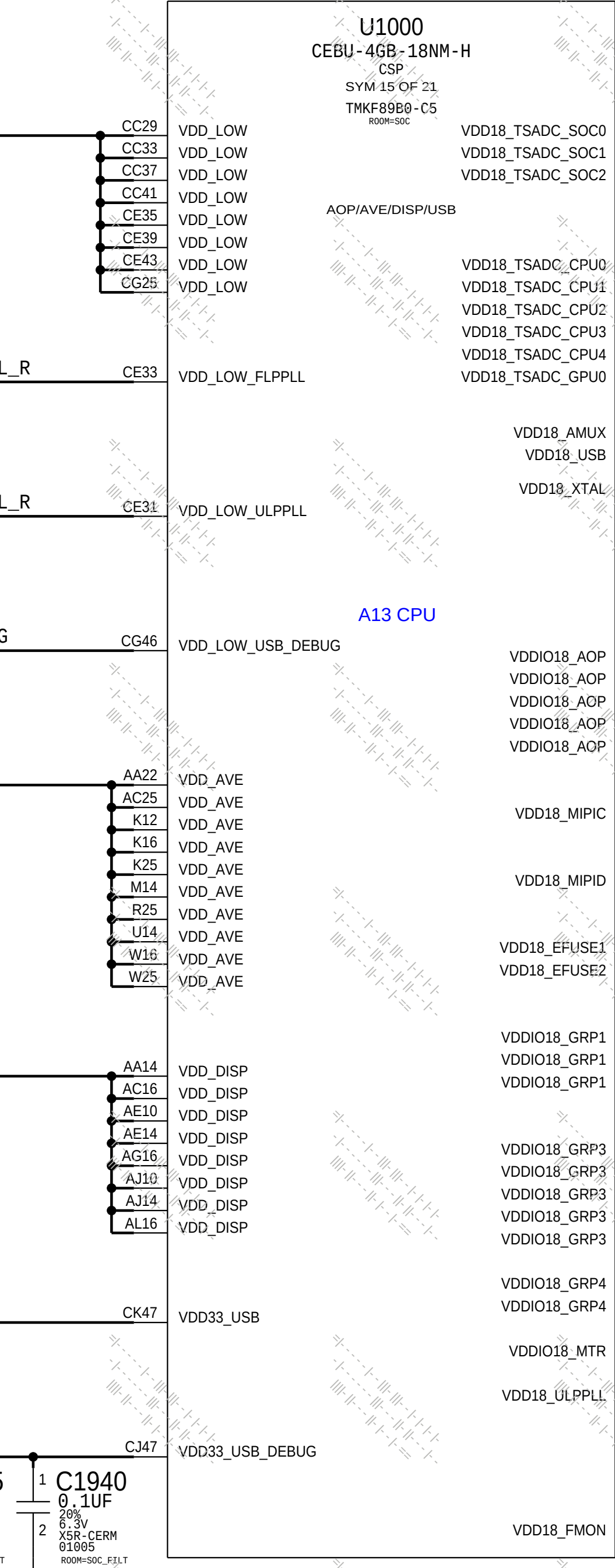
SOC: FIXED



SOC: VDD12



PAGE TITLE			SYNC DATE=04/18/2018 SYNC MASTER=	
SOC: POWER (Fixed/1v2)			DRAWING NUMBER	SIZE
			051-04047	D
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				6.0.0
			BRANCH	
			PAGE	18 OF 150
			SHEET	
				15 OF 60



SOC : GND

OMIT_TABLE
U1000
CEBU-4GB-18NM-H
CSP
SYM 19 OF 21
TMKF89B0-C5
ROOM=SOC

OMIT_TABLE
U1000
CEBU-4GB-18NM-H
CSP
SYM 20 OF 21
TMKF89B0-C5
ROOM=SOC

OMIT_TABLE
U1000
CEBU-4GB-18NM-H
CSP
SYM 21 OF 21
TMKF89B0-C5
ROOM=S0C

F10	SS
F14	SS
F18	SS
F22	SS
F29	SS
F33	SS
F50	SS
F54	SS
G1	SS
G65	SS
H12	SS
H16	SS
H20	SS
H27	SS
H31	SS
H35	SS
H39	SS
H48	SS
H52	SS
H56	SS
H60	SS
H8	SS
J3	SS
J67	SS
K14	SS
K18	SS
K22	SS
K29	SS
K33	SS
K37	SS
K46	SS
K41	SS
K50	SS
K54	SS
K58	SS
L1	SS
M25	SS
M27	SS
M31	SS
M35	SS
M48	SS
M52	SS
N65	SS
N67	SS
P3	SS
R19	SS
R24	SS
R33	SS
R46	SS
R50	SS
R54	SS
T1	SS
T67	SS
U25	SS
U27	SS
U31	SS
U35	SS
U48	SS
U52	SS
U56	SS
V65	SS
W10	SS
W14	SS
W18	SS
W22	SS
W29	SS
W33	SS
W37	SS
W41	SS
W46	SS
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W54	SS
Y1	SS
Y3	SS
Y67	SS

```
<-- Corner Ball Test Pins. System connects to GND.
```

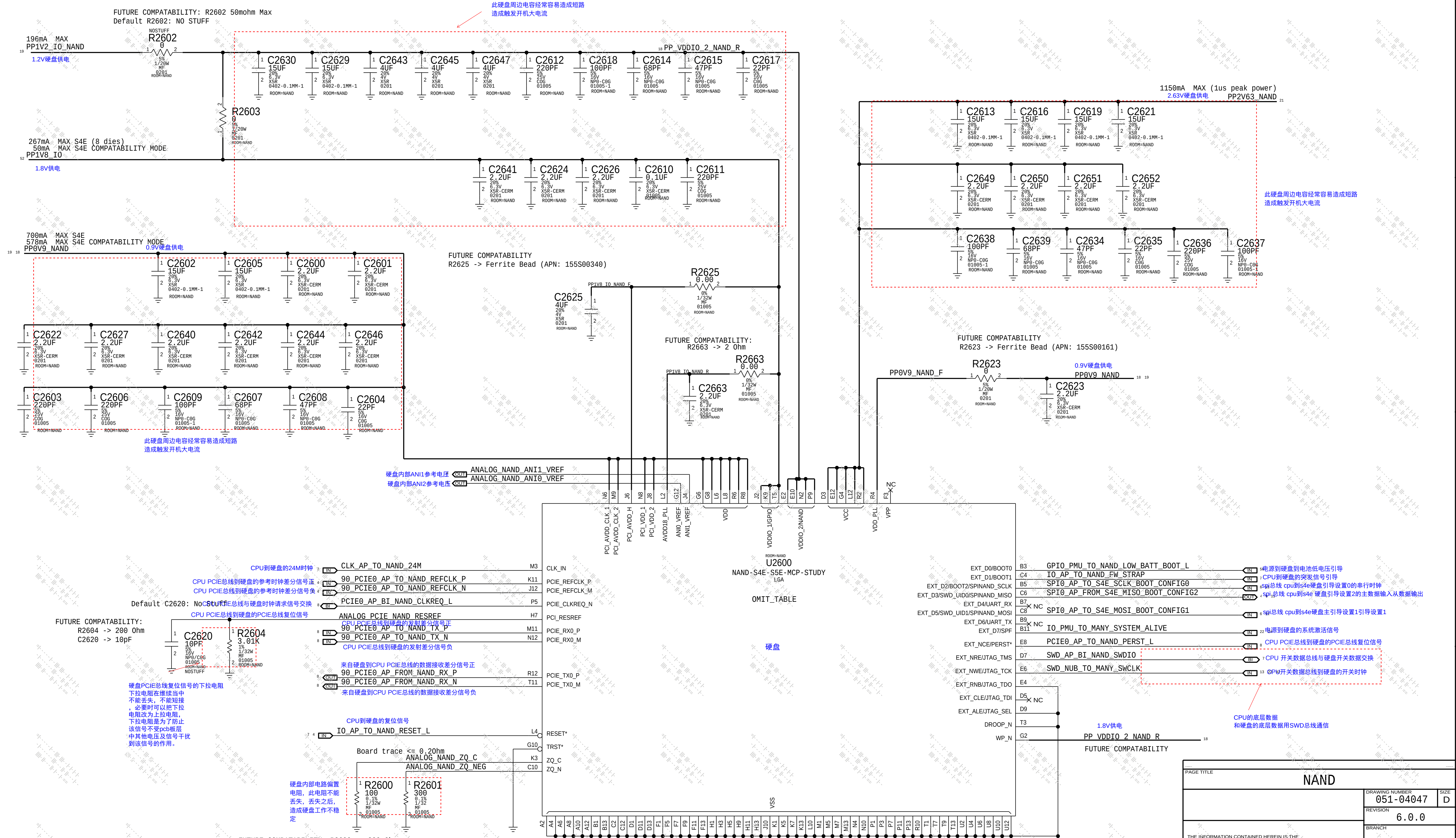
VSS_4	A63	VDDQL_DCS 供电检测测	
VSS_DDR Sense	A63	ANALOG VDDQL_DCS SENSEN	GPU
VSS_GPU Sense	AE29	GPU供电检测测	ANALOG GPU SENSEN
VSS_PCPU Sense	BP39	ANALOG CPU PCORE SENSEN	CPU核
VSS Sense	AY18	ANALOG SOC SENSEN	SOC
		SOC供电检测测	

SOC: POWER (GND)

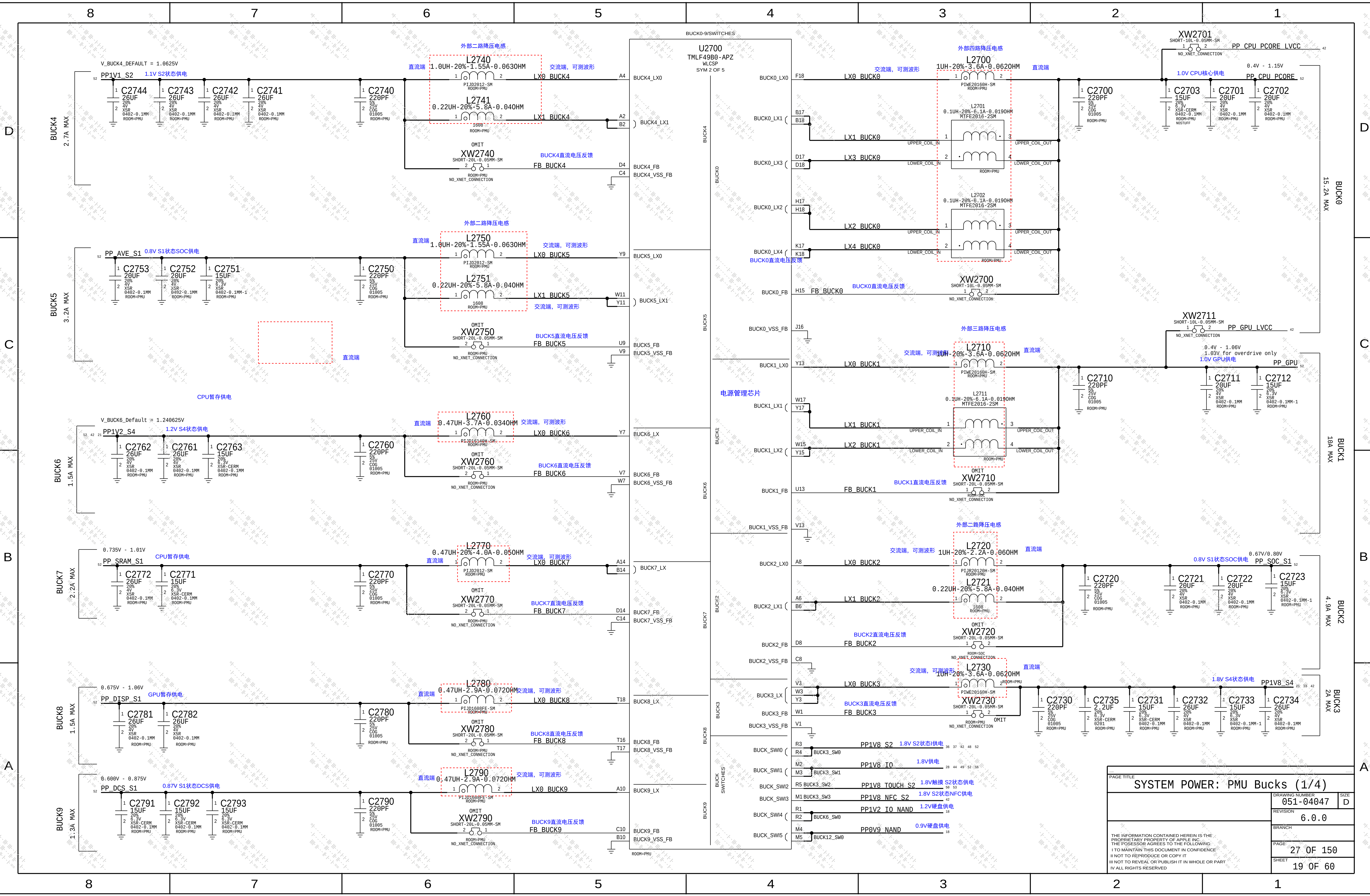
DRAWING NUMBER 051-04047		SIZE D
REVISION 6.0.0		
BRANCH		
PAGE 20 OF 150		
SHEET 17 OF 60		

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S4E NAND

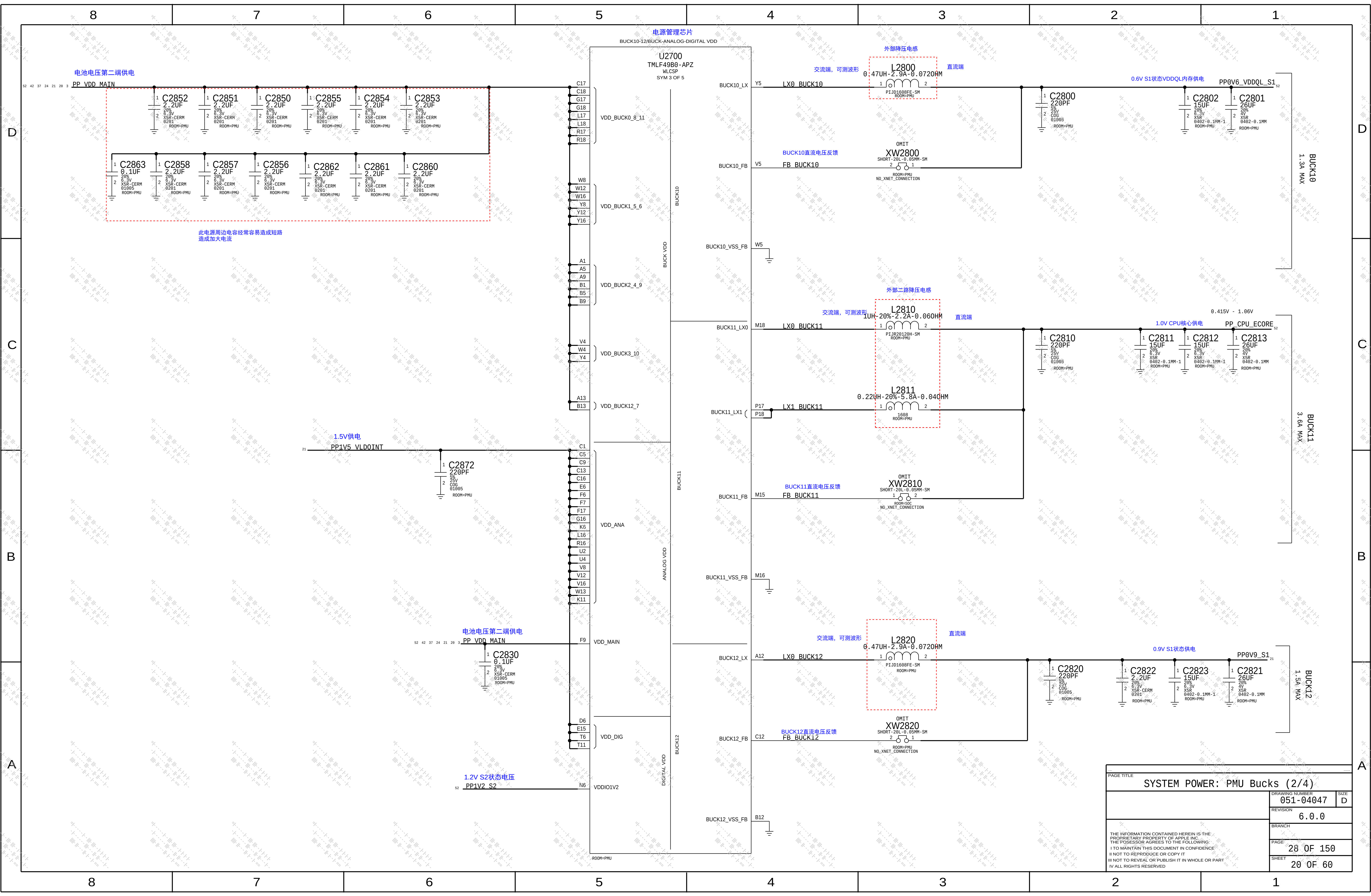


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	PAGE		26 OF 150		
	SHEET		18 OF 60		

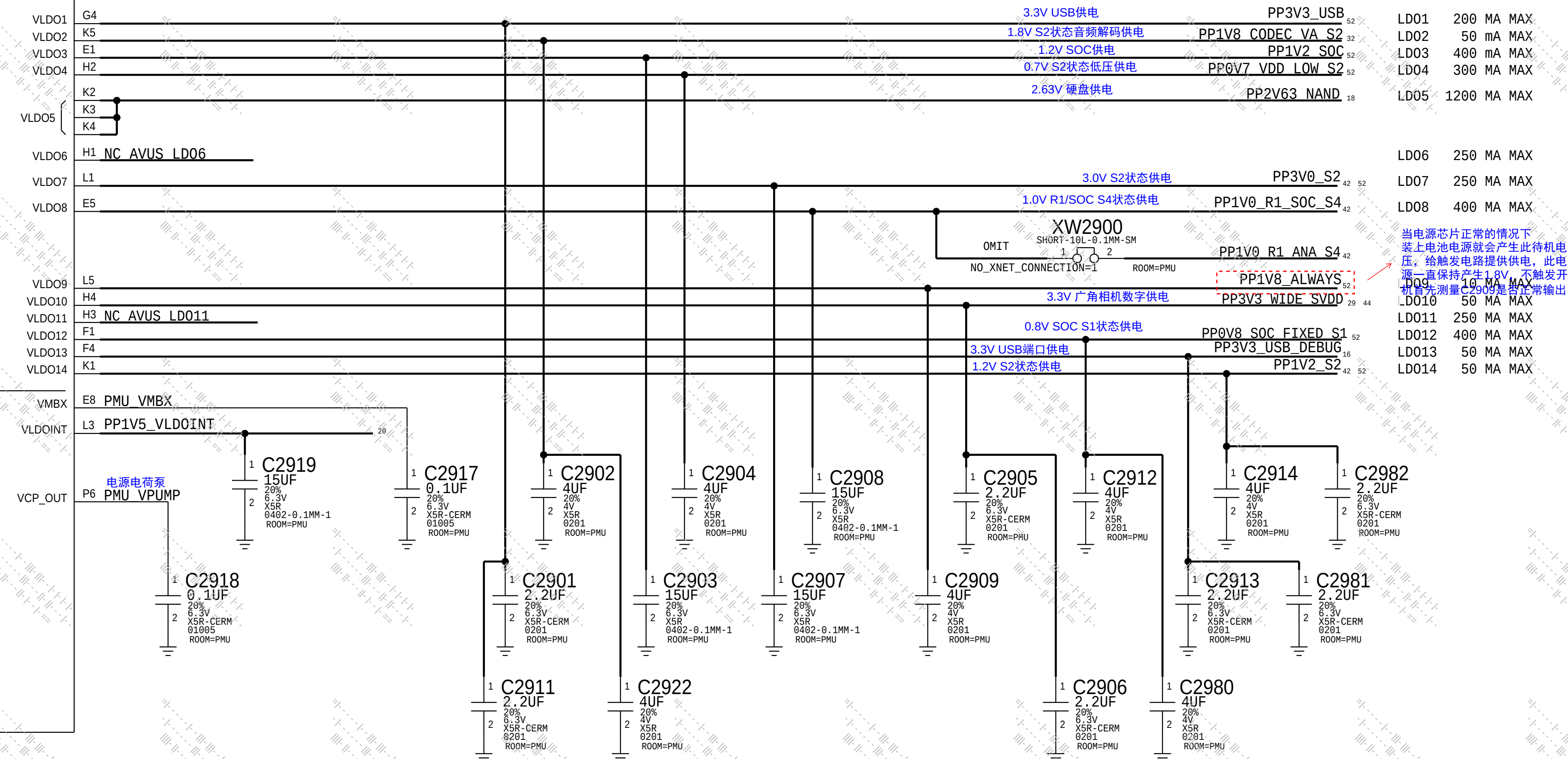
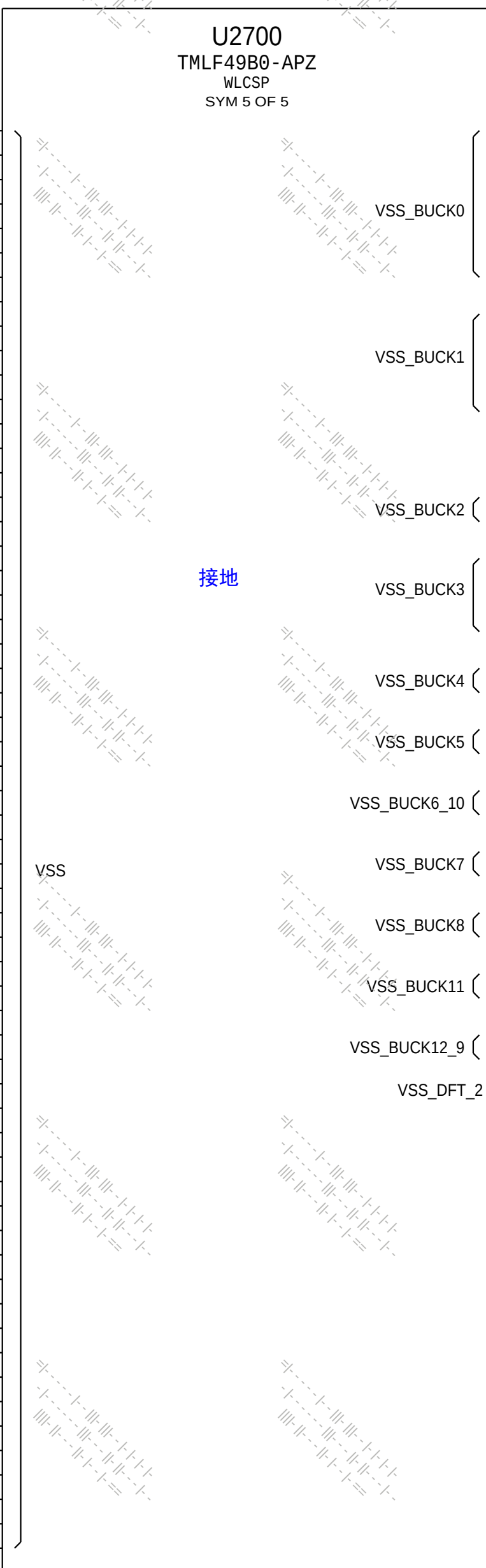
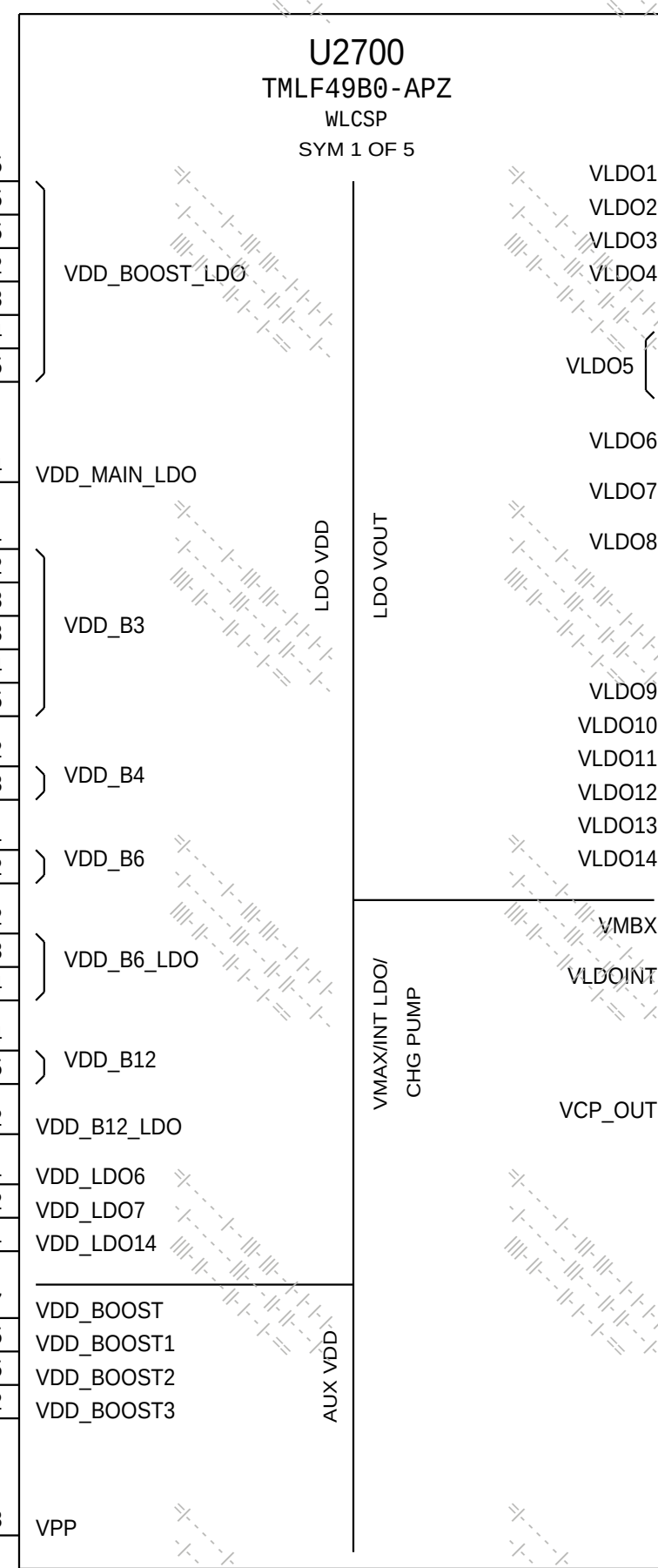
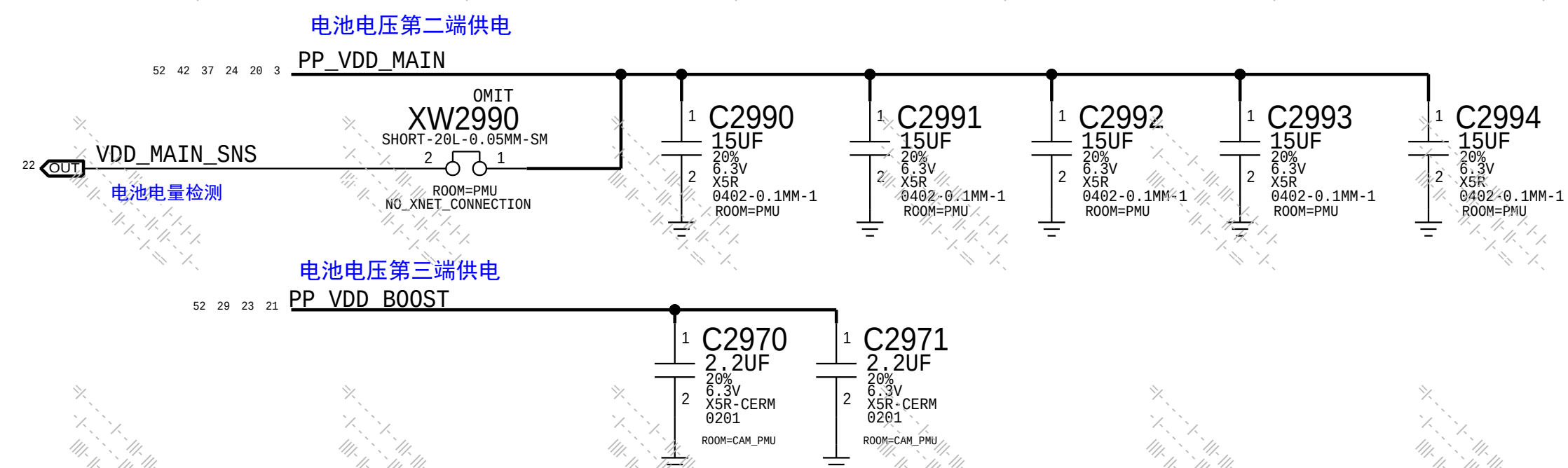


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DRAWING NUMBER	051-04047	SIZE
REVISION	6.0.0	D
BRANCH		
PAGE	27 OF 150	
SHEET	19 OF 60	

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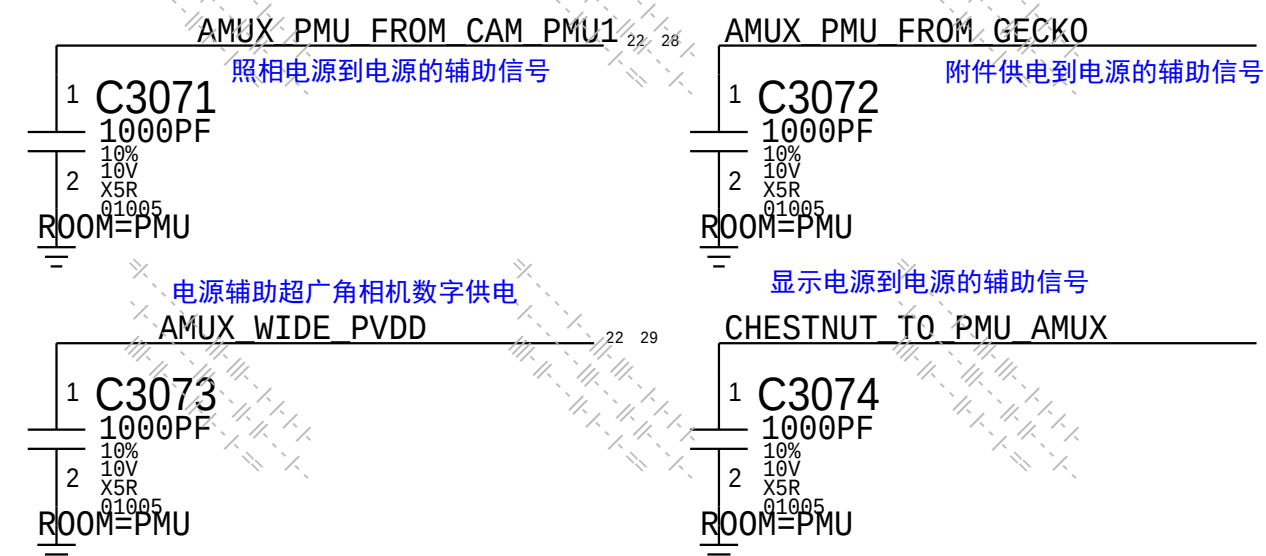
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		PAGE	28 OF 150
		SHEET	20 OF 60



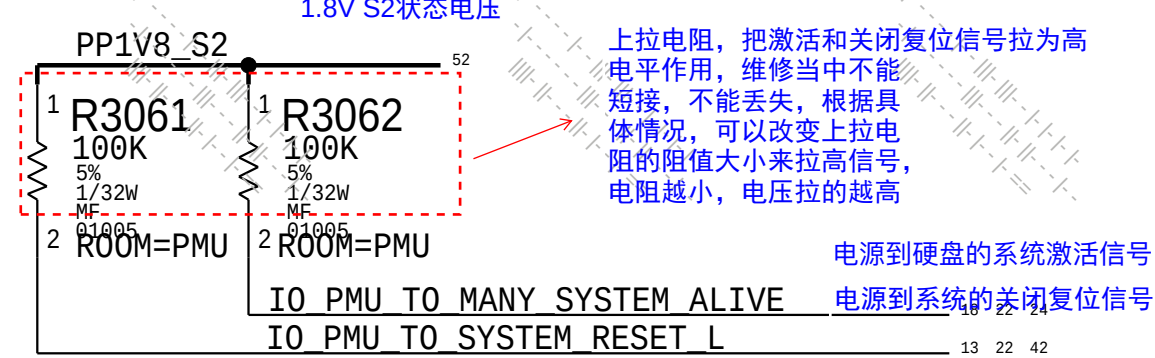
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	PAGE	29 OF 150
	SHEET	21 OF 60

PMU - GPIOs

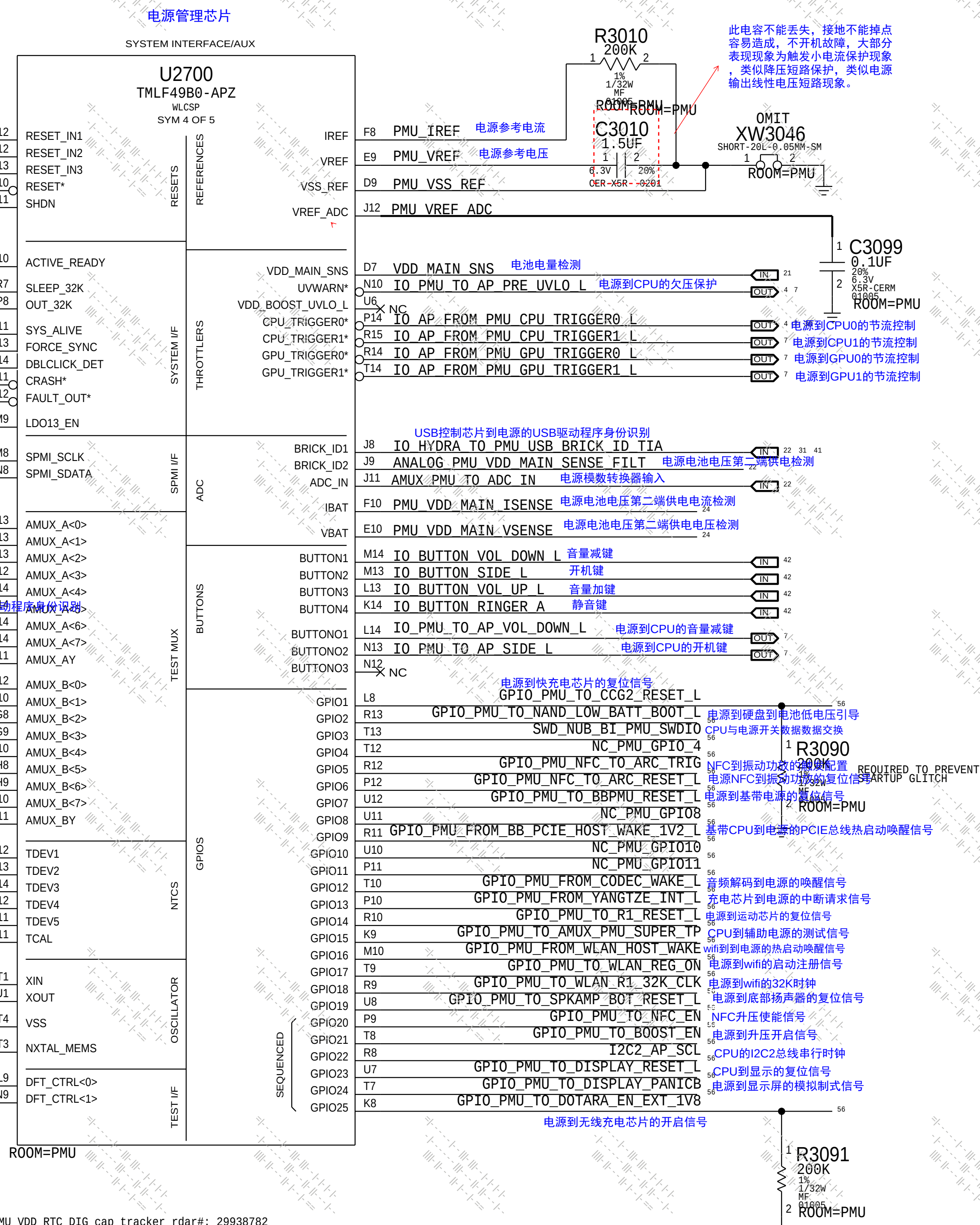
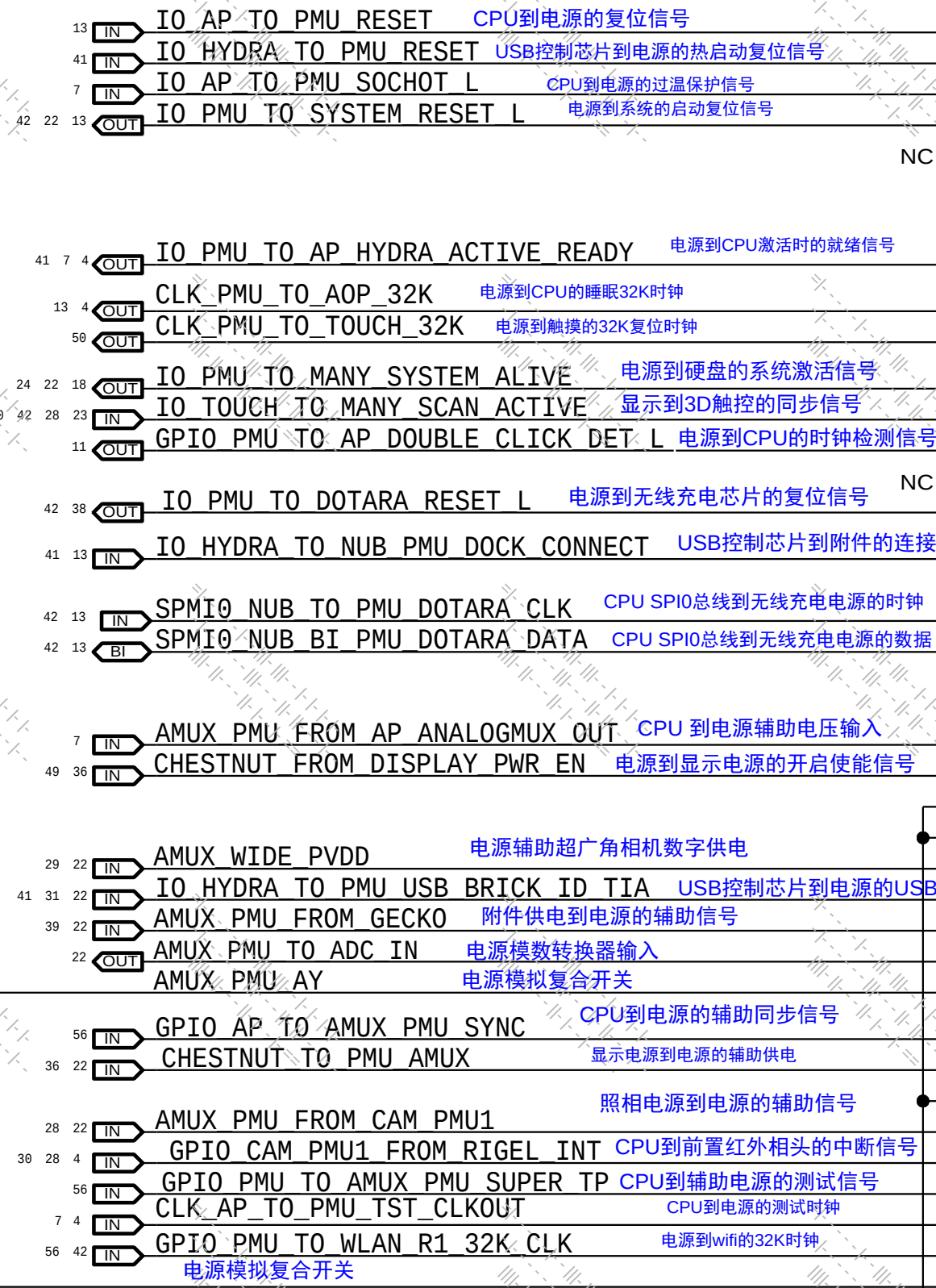
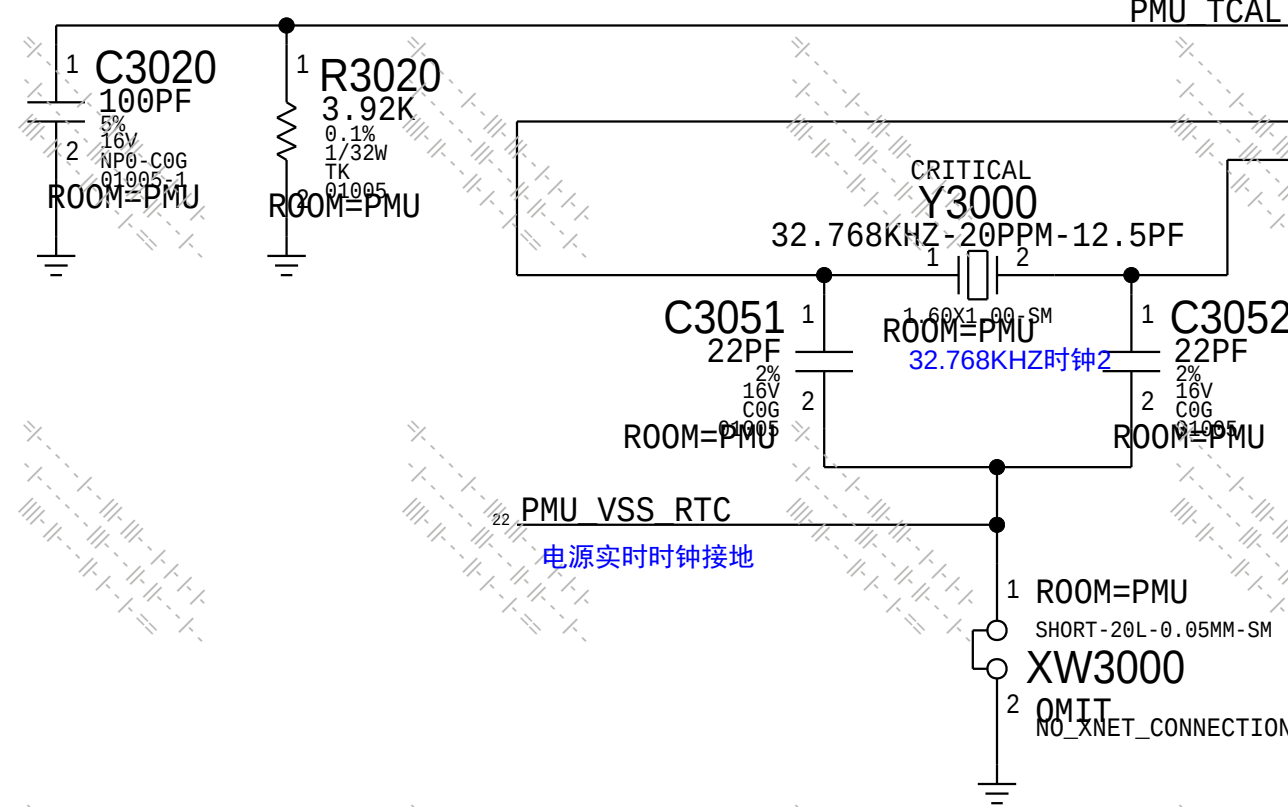
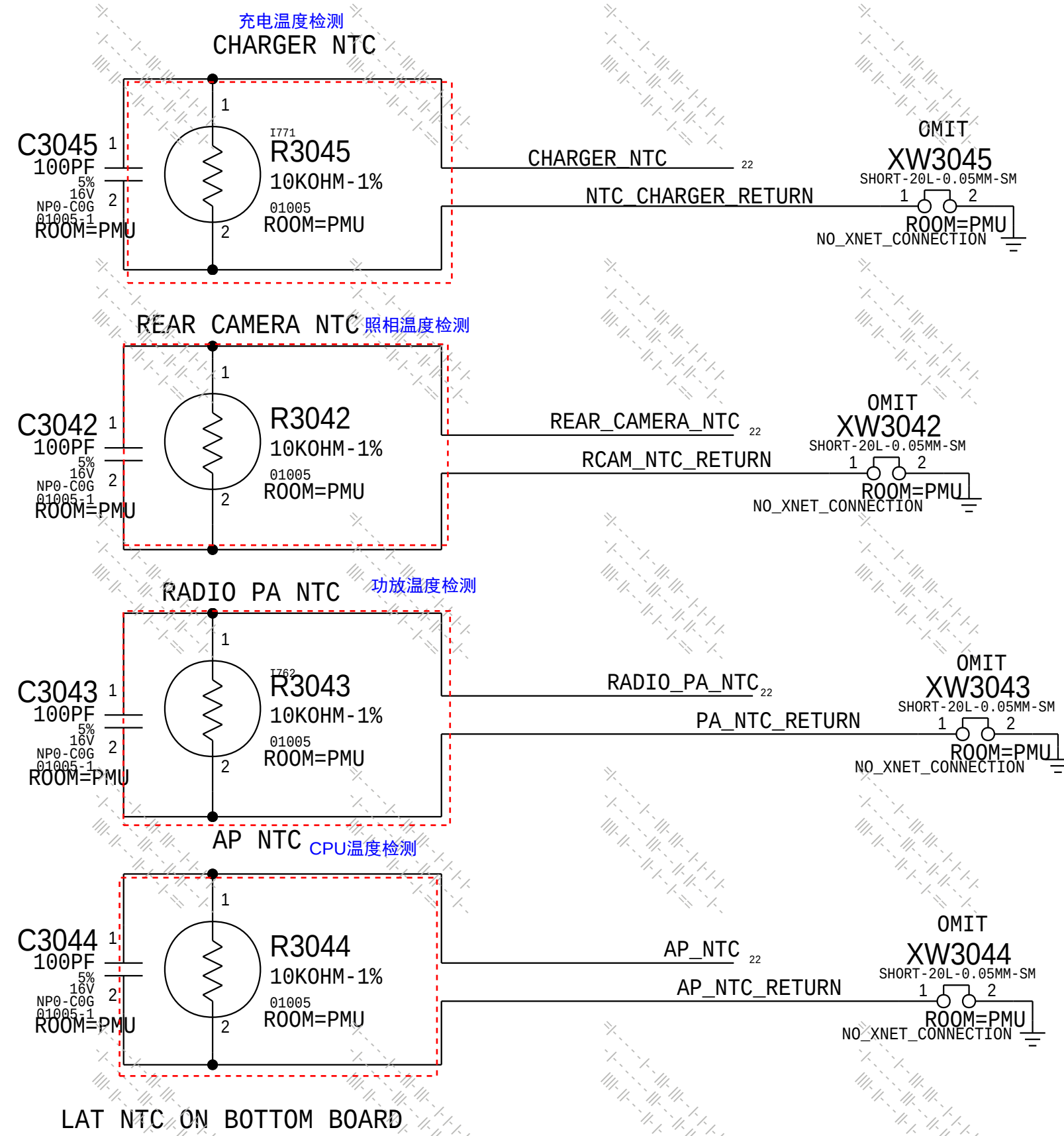
AMUX CAPS



COLD_RESET & SYSTEM_ALIVE



NTCs



TODO: Update
CONTROL PIN NOTES:

NOTE (1): INPUT PULL-DOWN 100-300k

NOTE (2): INPUT PULL-DOWN 1M

NOTE (3): INPUT PULL-UP OR DOWN $\times 100k-300k$

NOTE (4): OUTPUT OPEN-DRAIN, REQUIRES PULL-UP

PAGE TITLE		
SYSTEM POWER: PMU (4/4)		
	DRAWING NUMBER	SIZE
	051-04047	D
	REVISION	
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	PAGE	30 OF 150
	SHEET	22 OF 60

YANGTZE CHARGER
(Veridian Charger Control)

Yangtze Alt Ind

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00070	138S00187	ALT_PARTS	ALL	TAIYO ALT
138S00240	138S00187	ALT_PARTS	ALL	TAIYO ALT

152S01021 152S01019 ALT_PARTS ALL TAIYO ALT

SYSTEM POWER: Charger

DRAWING NUMBER 051-04047 SIZE D

REVISION 6.0.0

BRANCH

PAGE 33 OF 150

SHEET 24 OF 60

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YANGTZE CHARGER
(Veridian Charger Control)

NON-ZRB Alts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00070	138S00187	ALT_PARTS		注意功率电压，此电压当插入充电器输入电压时，才能产生，当插入充电器电压时测量此电压是否
138S00240	138S00187	ALT_PARTS		来大数判断充电器是否工作，此一堆滤波电容如果漏电容者短路，会造成适配器输入电压偏低，造成不充电，没充电图标
CRITICAL PART#	COMMENT			
138S00187	CAP, XSR, 4.7UF, 20%, 25V, 0402			

Yangtze Alt Ind

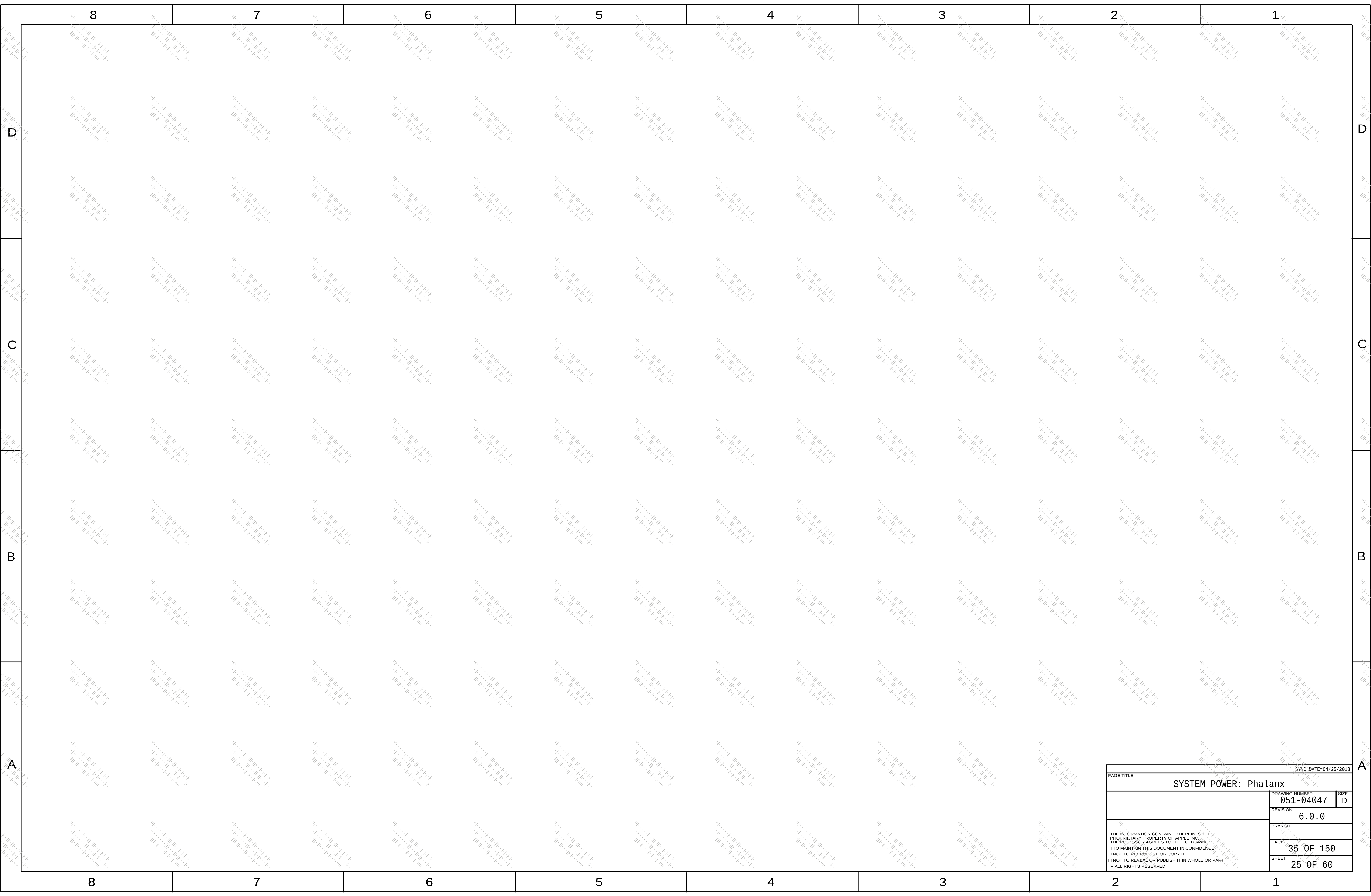
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152S01021	152S01019	ALT_PARTS	ALL	TAIYO ALT

SYSTEM POWER: Charger

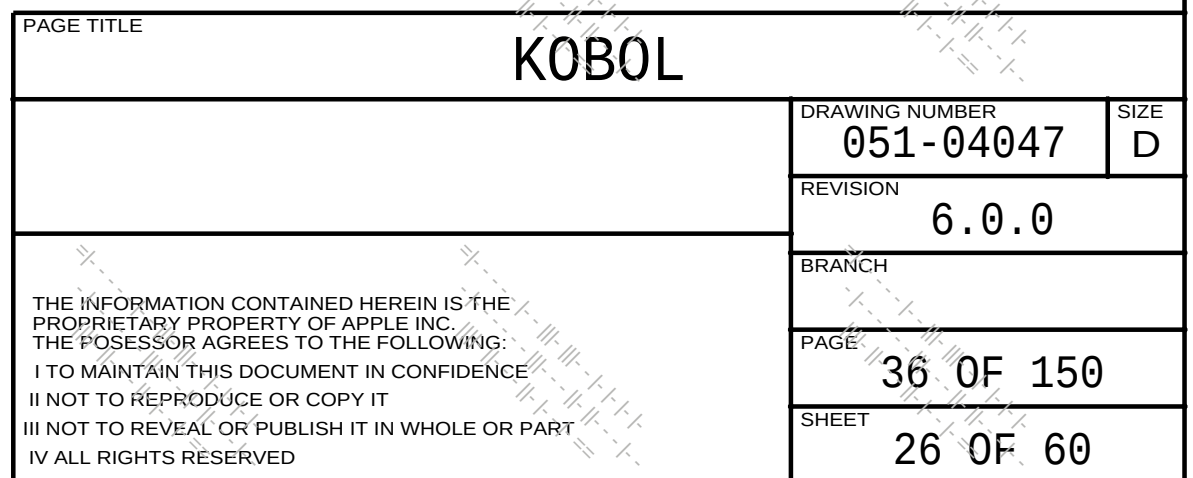
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REVISION: 6.0.0
PAGE: 33 OF 150
SHEET: 24 OF 60

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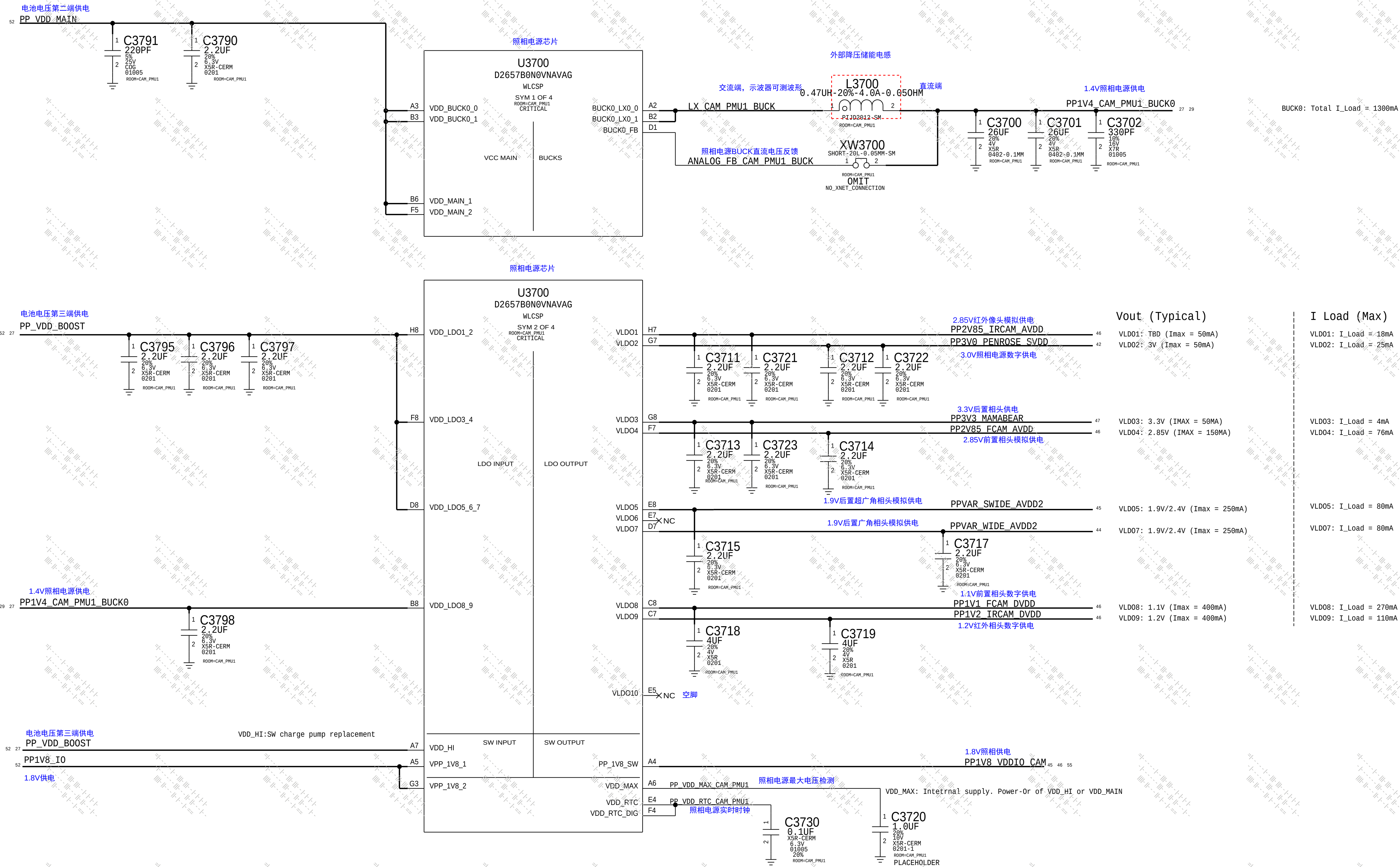
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		SHEET	
		24 OF 60	



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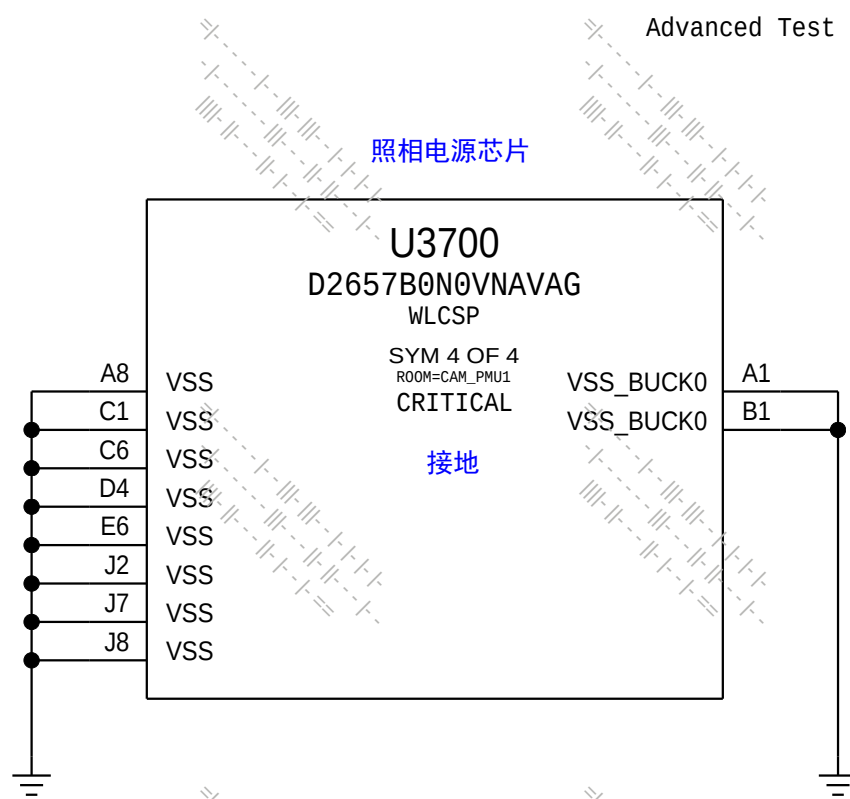
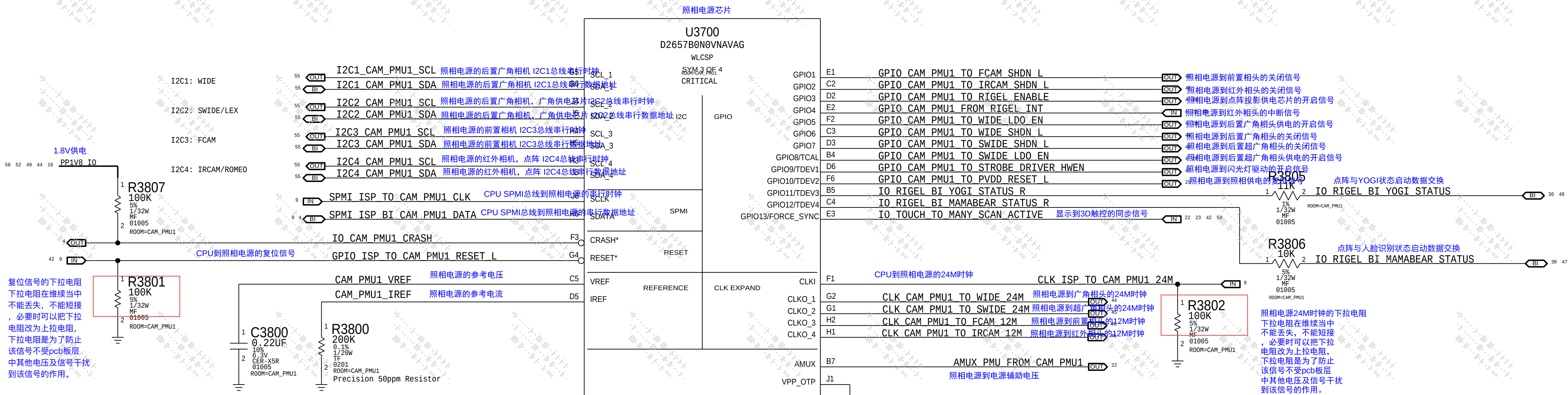


CAMERA PMU1



PAGE TITLE				
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	REVISION		6.0.0	
	BRANCH			
	PAGE		37 OF 150	
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CAMERA PMU1



PAGE TITLE		
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	PAGE	38 OF 150
	SHEET	28 OF 60

WIDE AVDD_ADC LDO

Camera LDO Alts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S01728	353S00015	ALT_PARTS	U3900, U3900	ORIENT ALTERNATE
353S01690	353S01904	ALT_PARTS	U3910, U3910	OLDER ALTERNATE

WIDE DVDD LDO

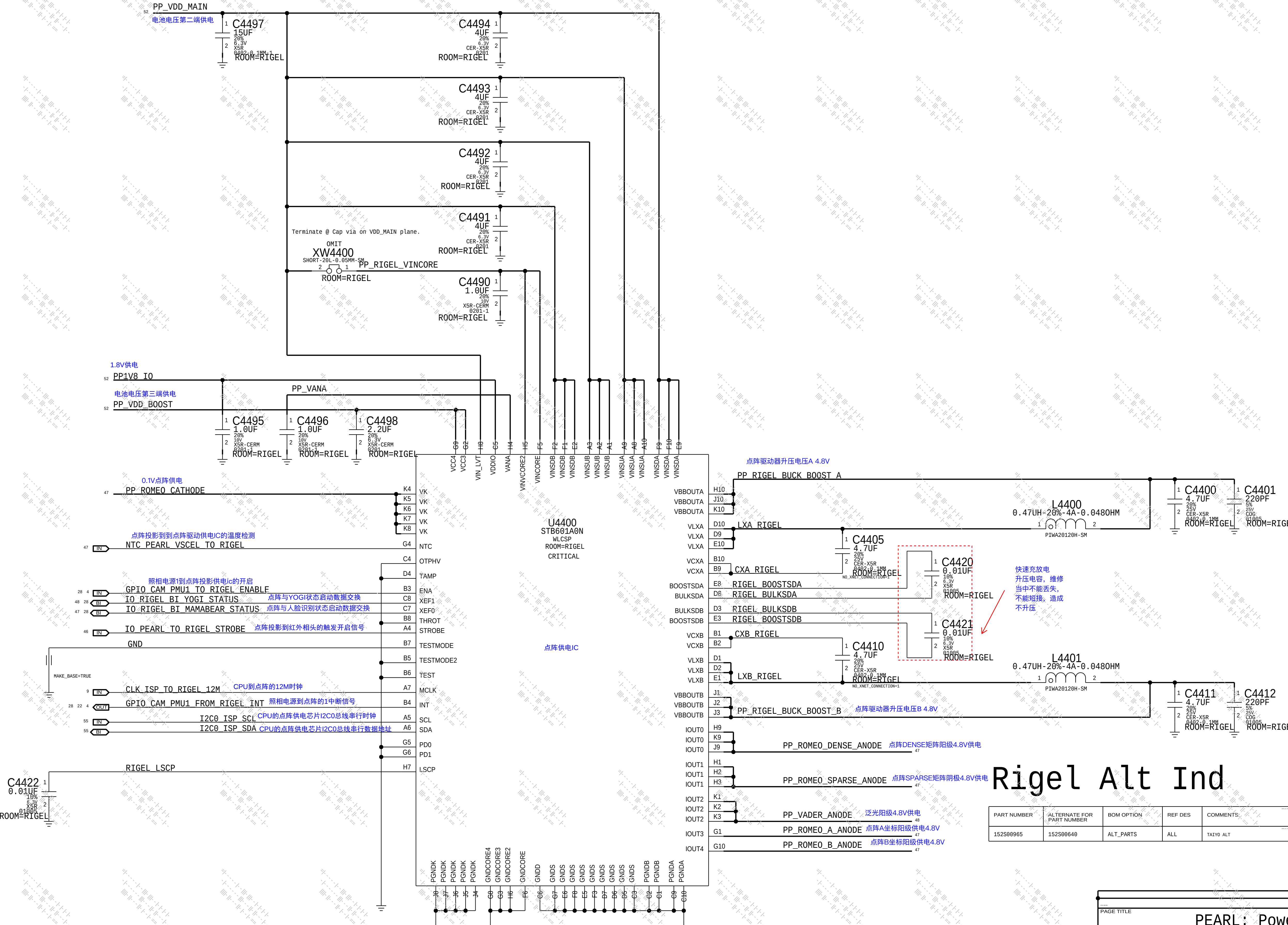
SUPERWIDE VDDL LDO

SUPERWIDE AVDD LDO

LEX

PAGE TITLE			SYNC. DATE=84/18/2918	
CAMERA: Discrete Power				
			DRAWING NUMBER	SIZE
			051-04047	D
			REVISION	6.0.0
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			PAGE	39 OF 150
			SHEET	29 OF 60

Rigel Driver



Rigel Alt Inc

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS
152S00965	152S00640	ALT_PARTS	ALL	TAIYO ALT

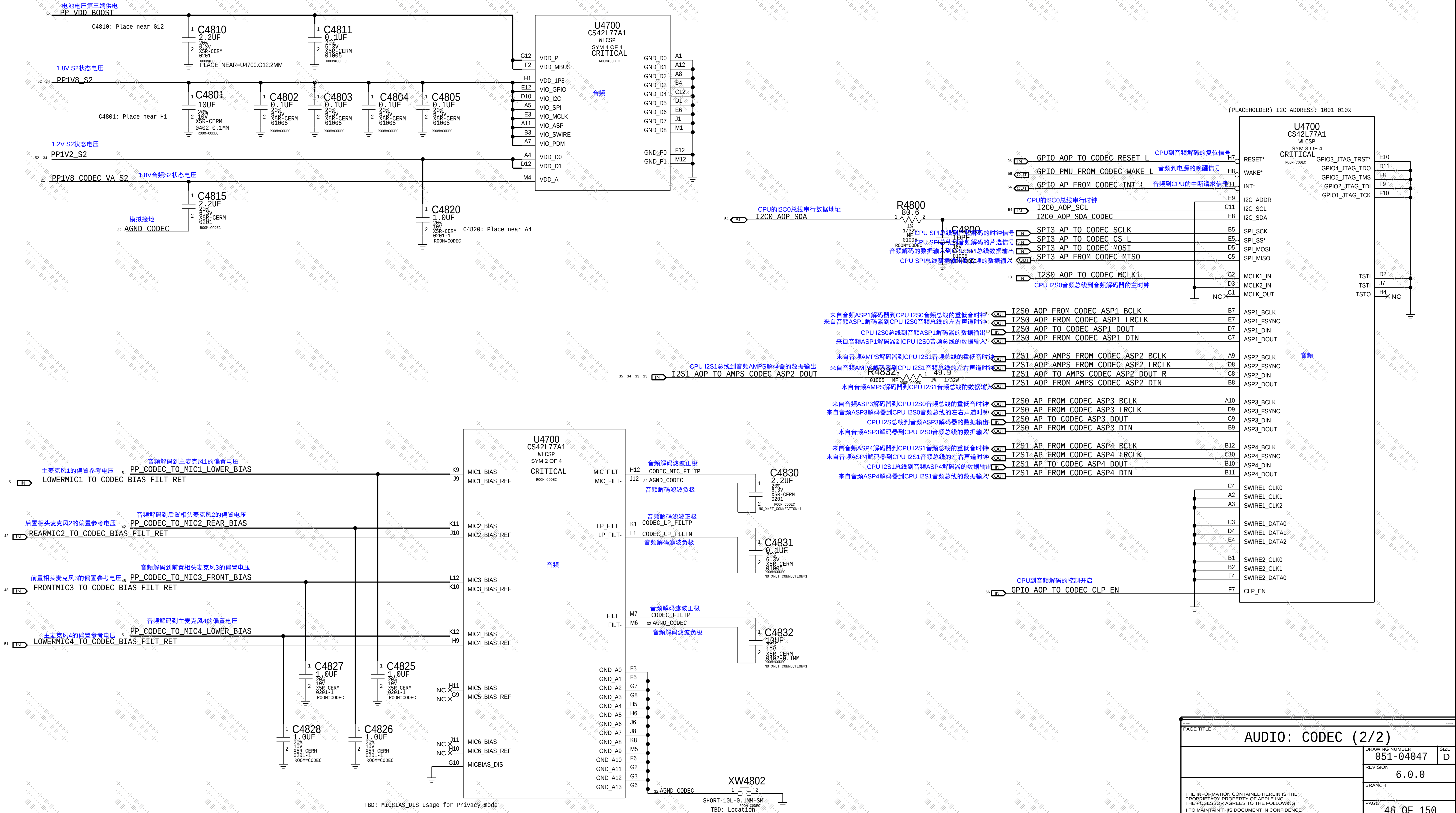
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		PAGE	44 OF 150
		SHEET	30 OF 60

D



A

BRIGHTON AUDIO CODEC (POWER & I/O)

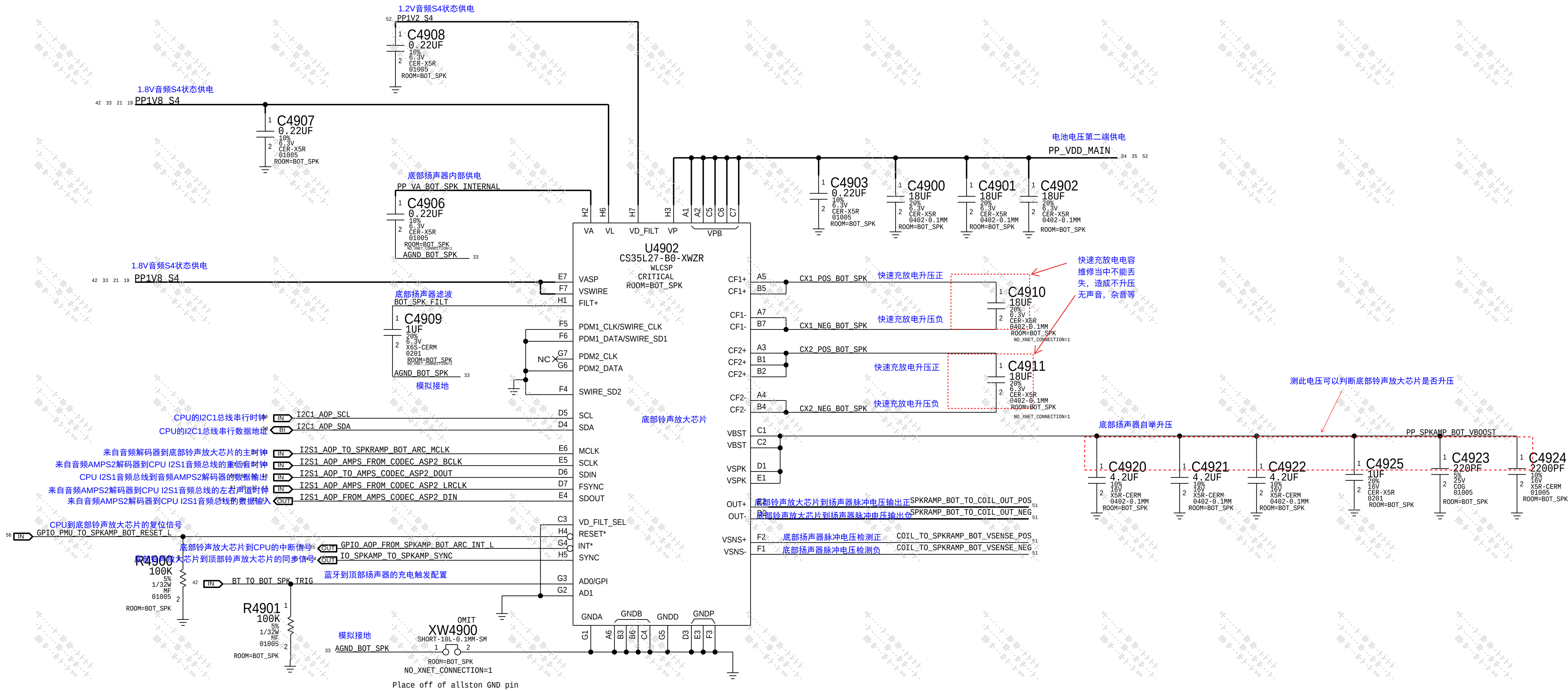


PAGE TITLE			
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DRAWING NUMBER		051-04047	SIZE
REVISION		6.0.0	
BRANCH			
PAGE		48 OF 150	
SHEET		32 OF 60	
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Amp: South Speaker

I2C对照表

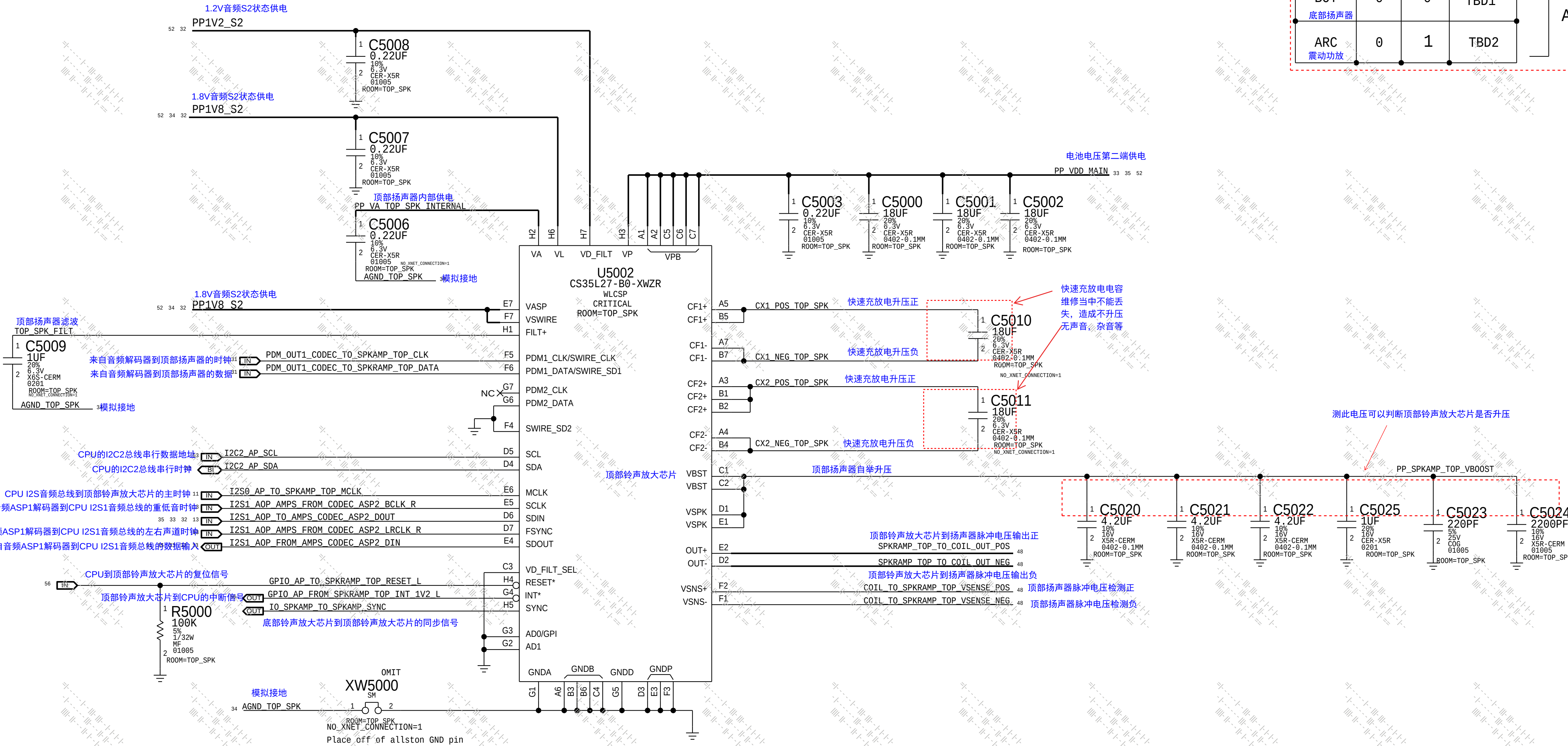
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TOP 顶部扬声器	0	0	TBD1	AP I2C2 AOP I2C1
BOT 底部扬声器	0	0	TBD1	
ARC 震动功放	0	1	TBD2	



Amp: North Speaker

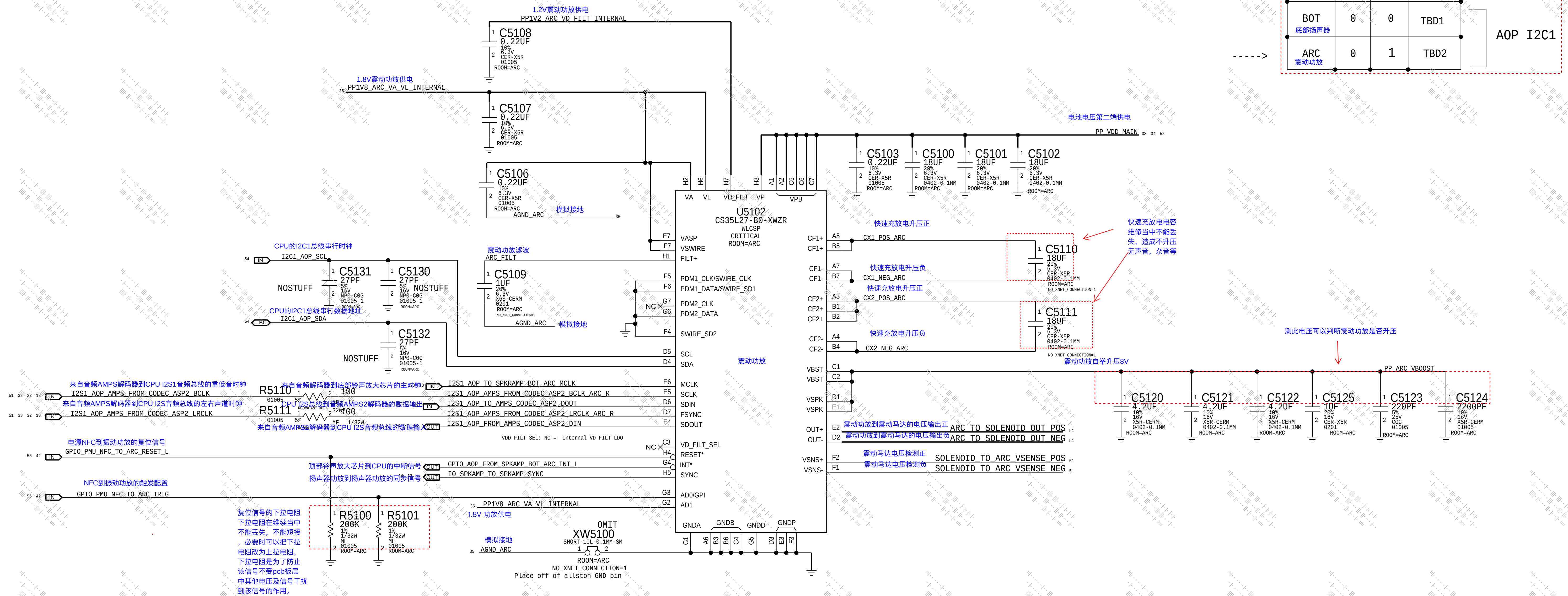
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BOT 底部扬声器	0	0	TBD1	
ARC 震动功放	0	1	TBD2	AOP I2C1



PAGE TITLE			SYNC_DATE=05/25/2018
AUDIO: NORTH SPKAMP			
DRAWING NUMBER		051-04047	SIZE D
REVISION		6.0.0	
BRANCH			
PAGE		50 OF 150	
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Amp: Arc

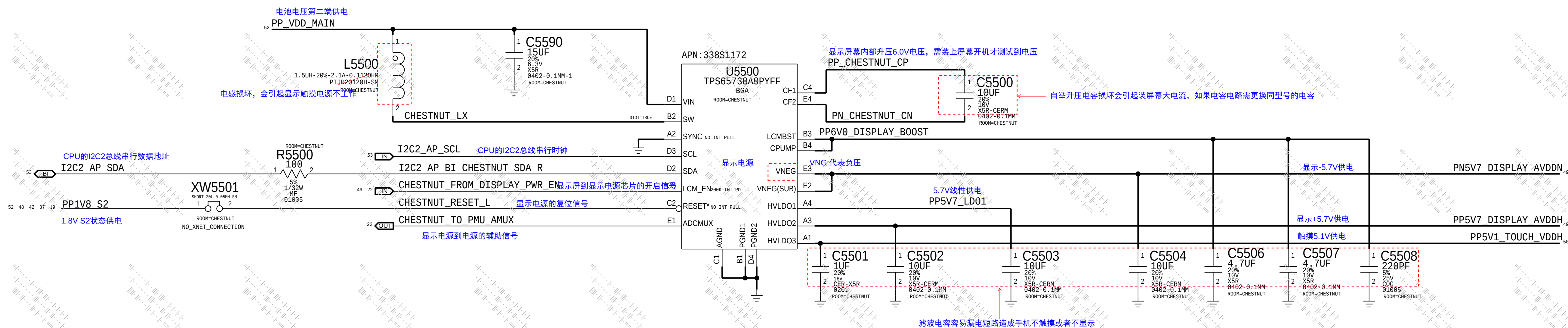


AMP	AD0	AD1	I2C_ADR
TOP 顶部扬声器	0	0	TBD1
BOT 底部扬声器	0	0	TBD1
ARC 震动功放	0	1	TBD2

AP I2C2

AOP I2C1

CHESTNUT DISPLAY PMU



Chestnut Alt Ind

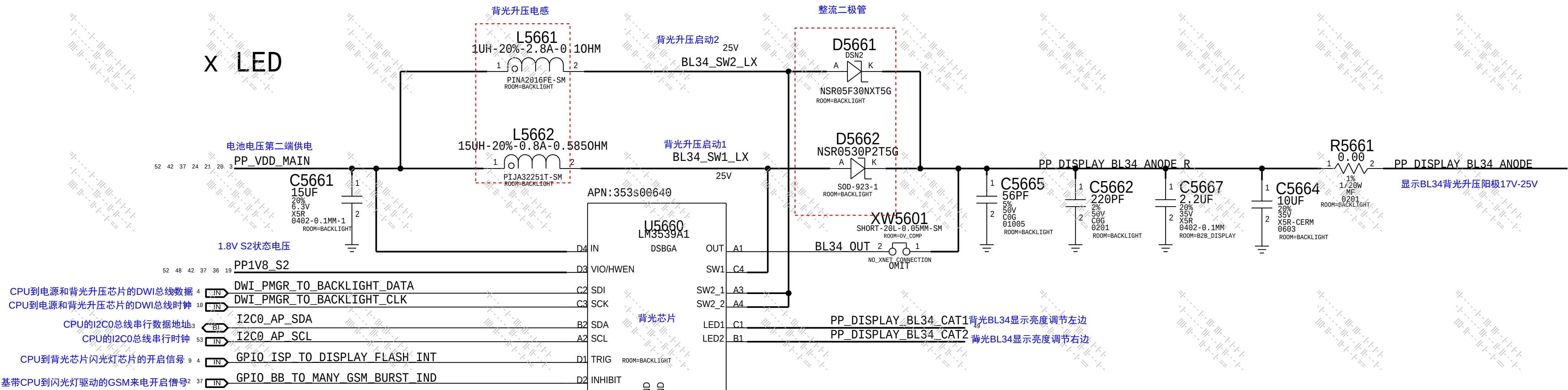
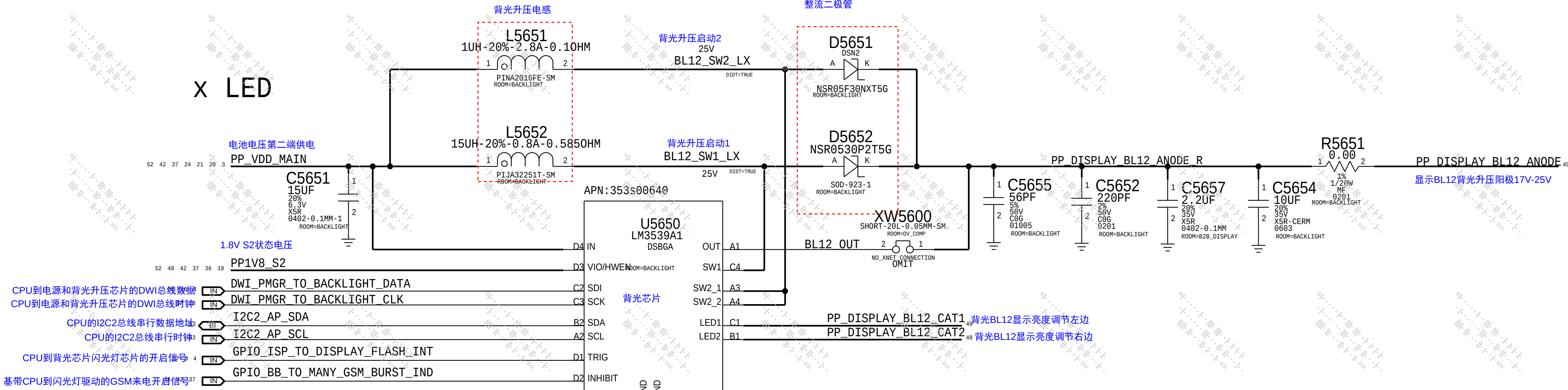
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00913	152S00938	ALT_PARTS	L5500	XTAL, 24M, 1612

PAGE TITLE		
CG: Display PMU (Chestnut)		
	DRAWING NUMBER	051-04047
	REVISION	6.0.0
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	PAGE	55 OF 150
	SHEET	36 OF 60

DWI Clock Pull



MUON - LED BACKLIGHT DRIVERS



Muon Inductor Alt

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00940	152S00939	ALT_PARTS	L5652, L5662	IND, TAIYO, 15UH, 0.8A, 3225

CG: Backlight PMU (Muon)

DRAWING NUMBER
051-04047

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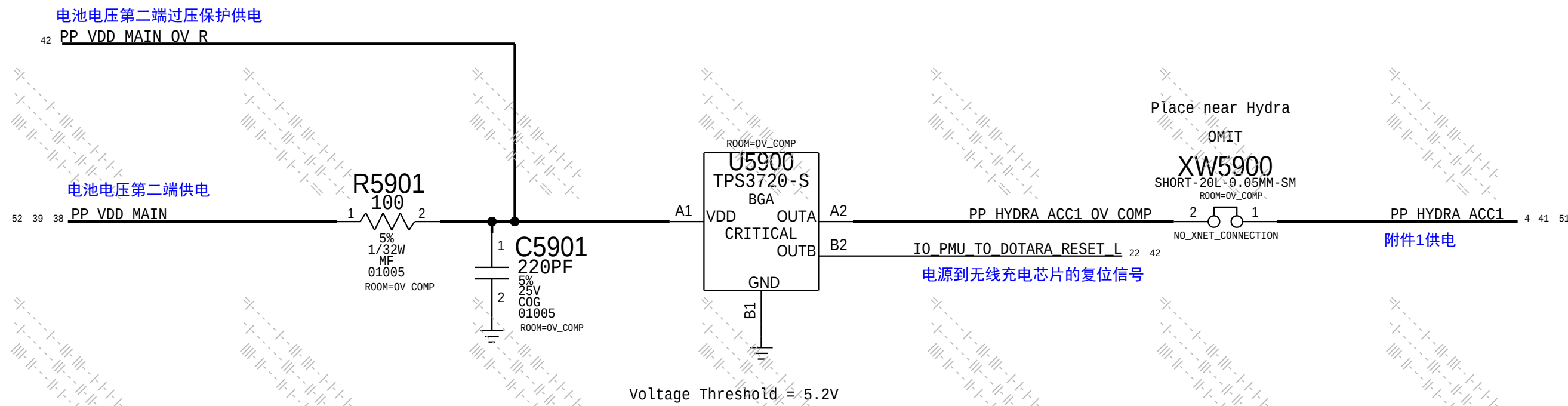
PAGE
56 OF 150

SHEET
37 OF 60

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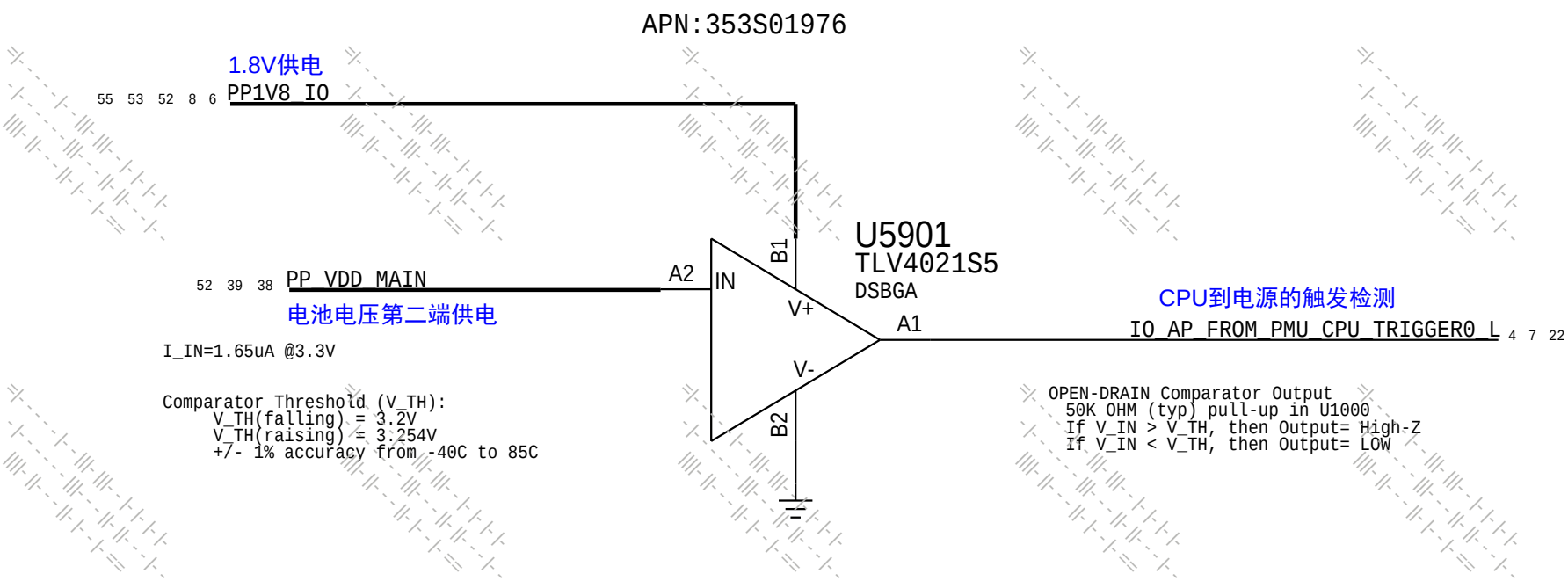
VDD_MAIN COMPARATORS

VDD_MAIN OV CUT-OFF CIRCUIT



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S01375	353S01398	ALT_PARTS	U5900	ON SEMI

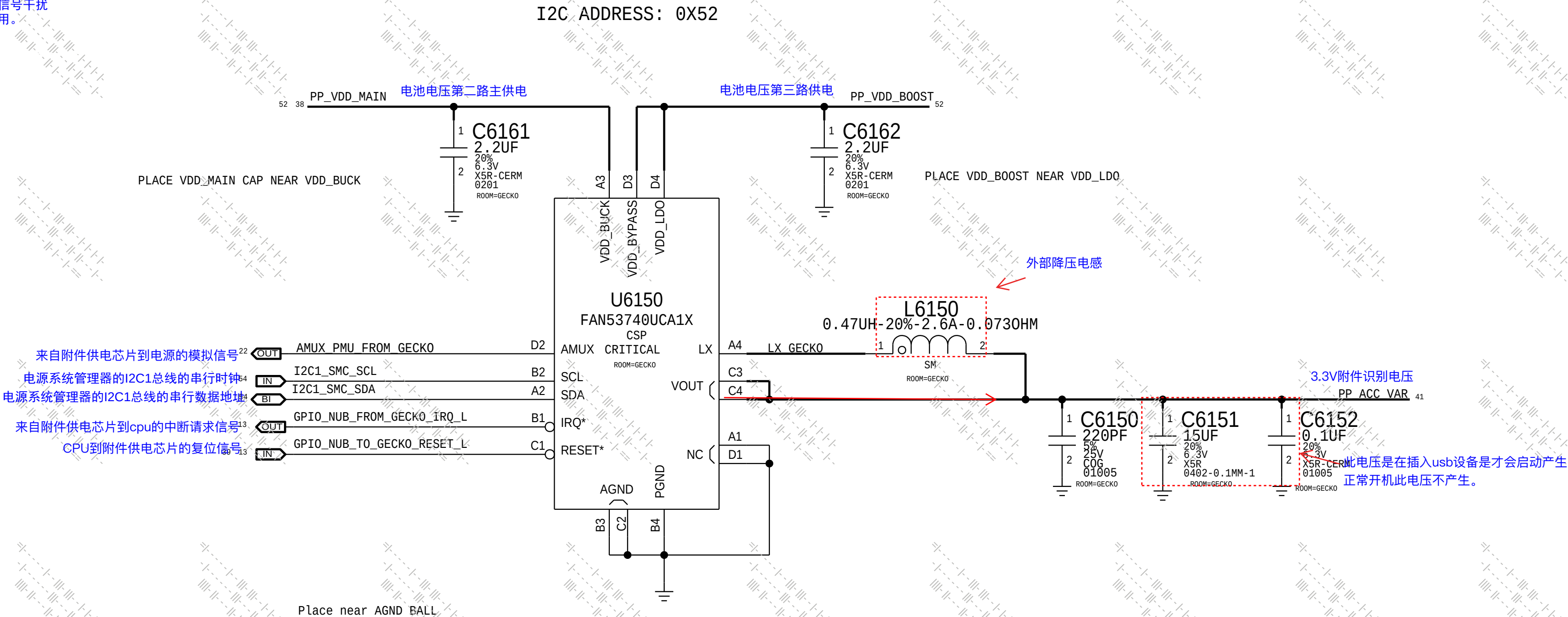
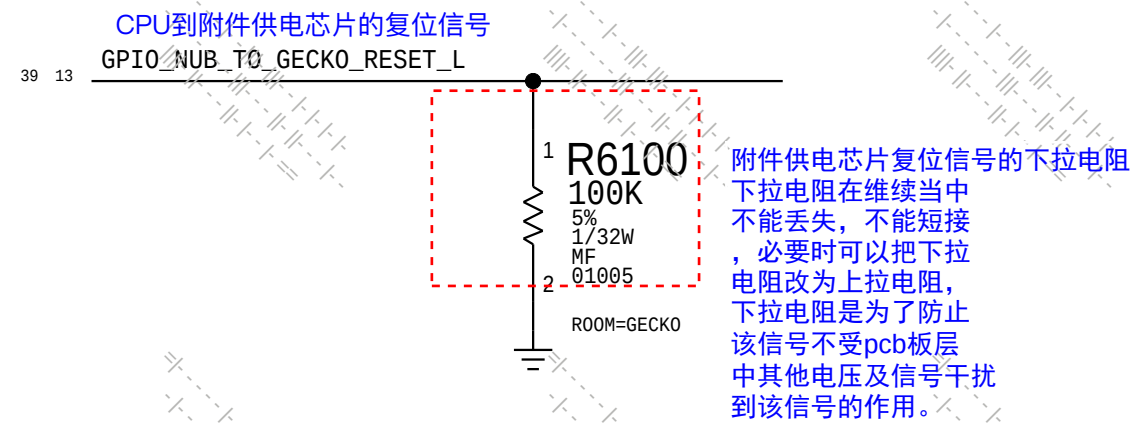
VDD_MAIN PRE-UPO COMPARATOR



PAGE TITLE: VDD_MAIN COMPARATORS			SYNCH DATE=04/18/2018	
PAGE: 59 OF 150			SYNCH MASTER=	
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			REVISION: 6.0.0	
			BRANCH:	
			PAGE: 59 OF 150	
			SHEET: 38 OF 60	

Gecko

GECK0 Reset Pull Down



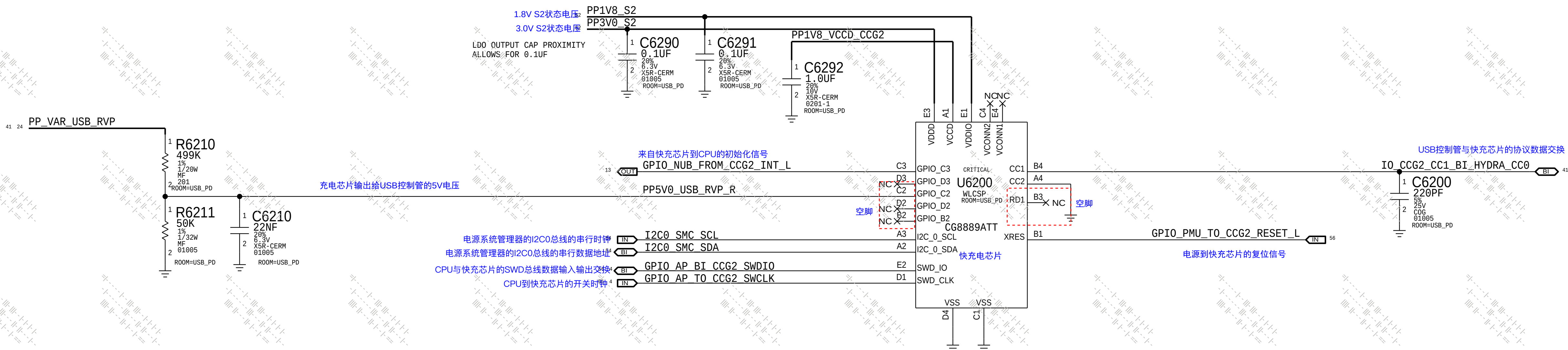
IND Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00853	152S00759	ALT_PARTS	L6150	IND, PWR, 0.47UH, 20%, 2.7A, TY

I/O: Gecko

PAGE TITLE		
	DRAWING NUMBER	051-04047
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	PAGE	61 OF 150
	SHEET	39 OF 60

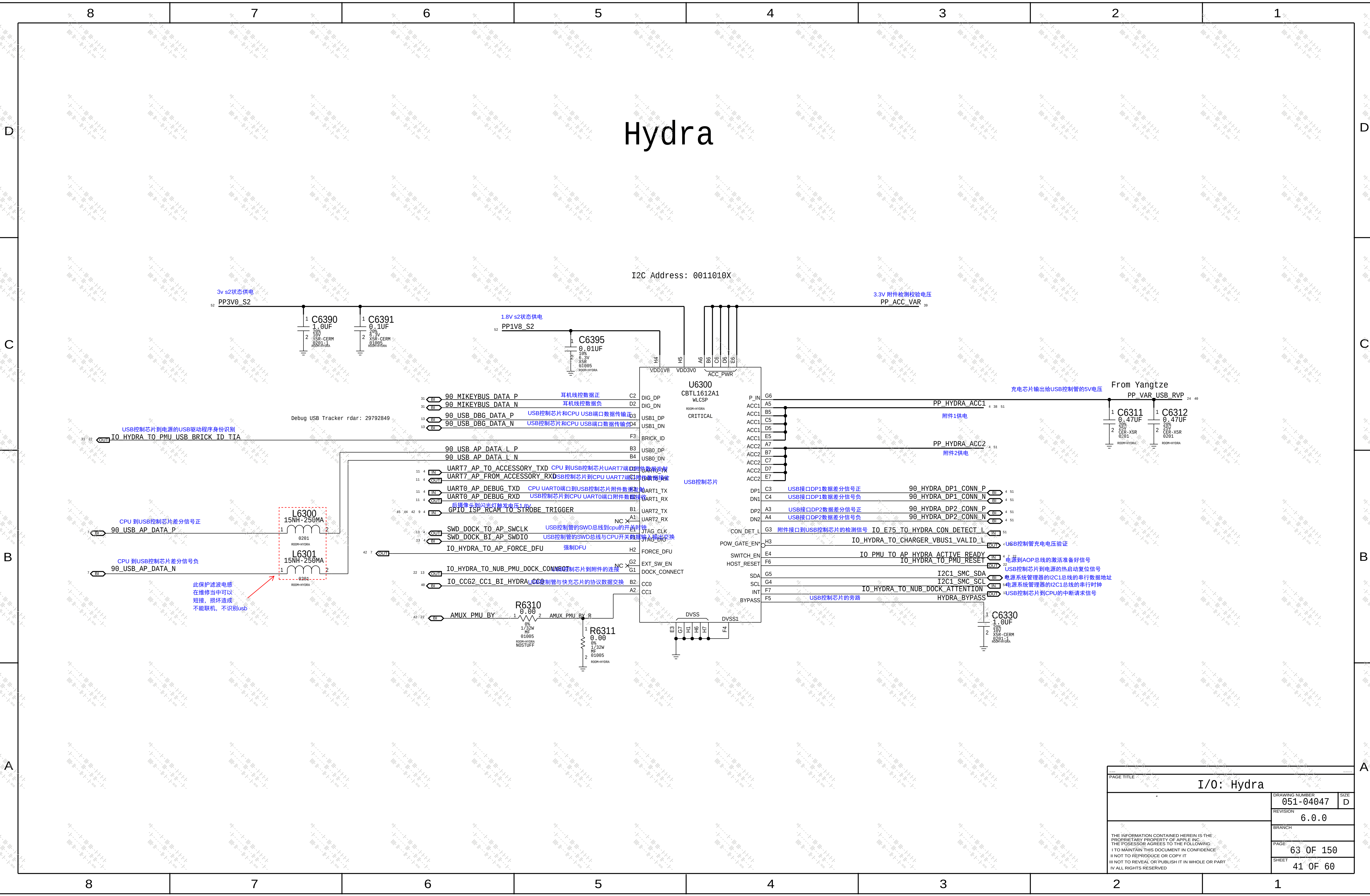
USB-PD



PAGE TITLE			
I/O: USB PD			
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	PAGE	62 OF 150	
	SHEET	40 OF 60	

Hydra

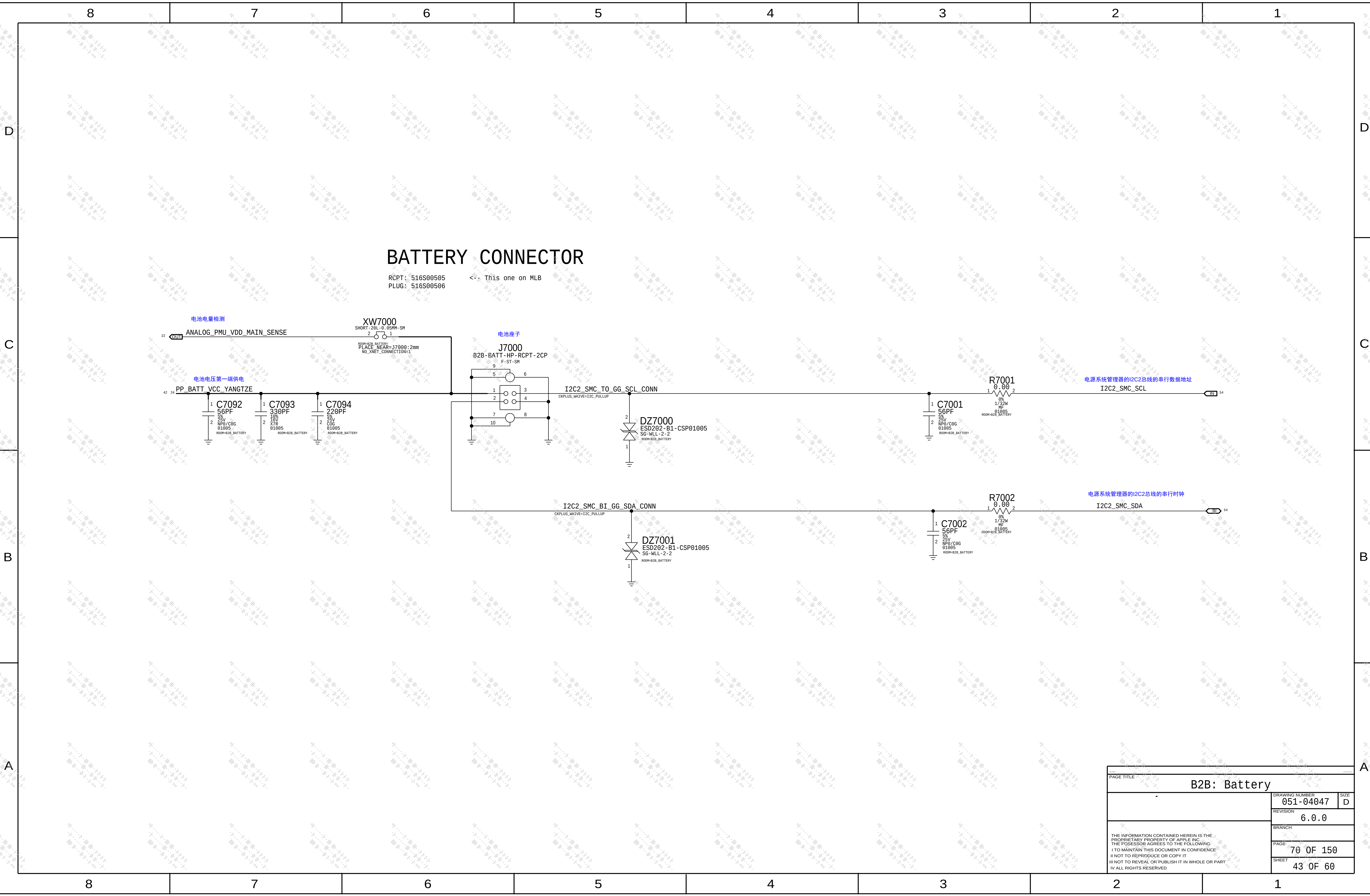
I2C Address: 0011010X



PAGE TITLE			
I/O: Hydra			
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		PAGE	63 OF 150
		SHEET	41 OF 60
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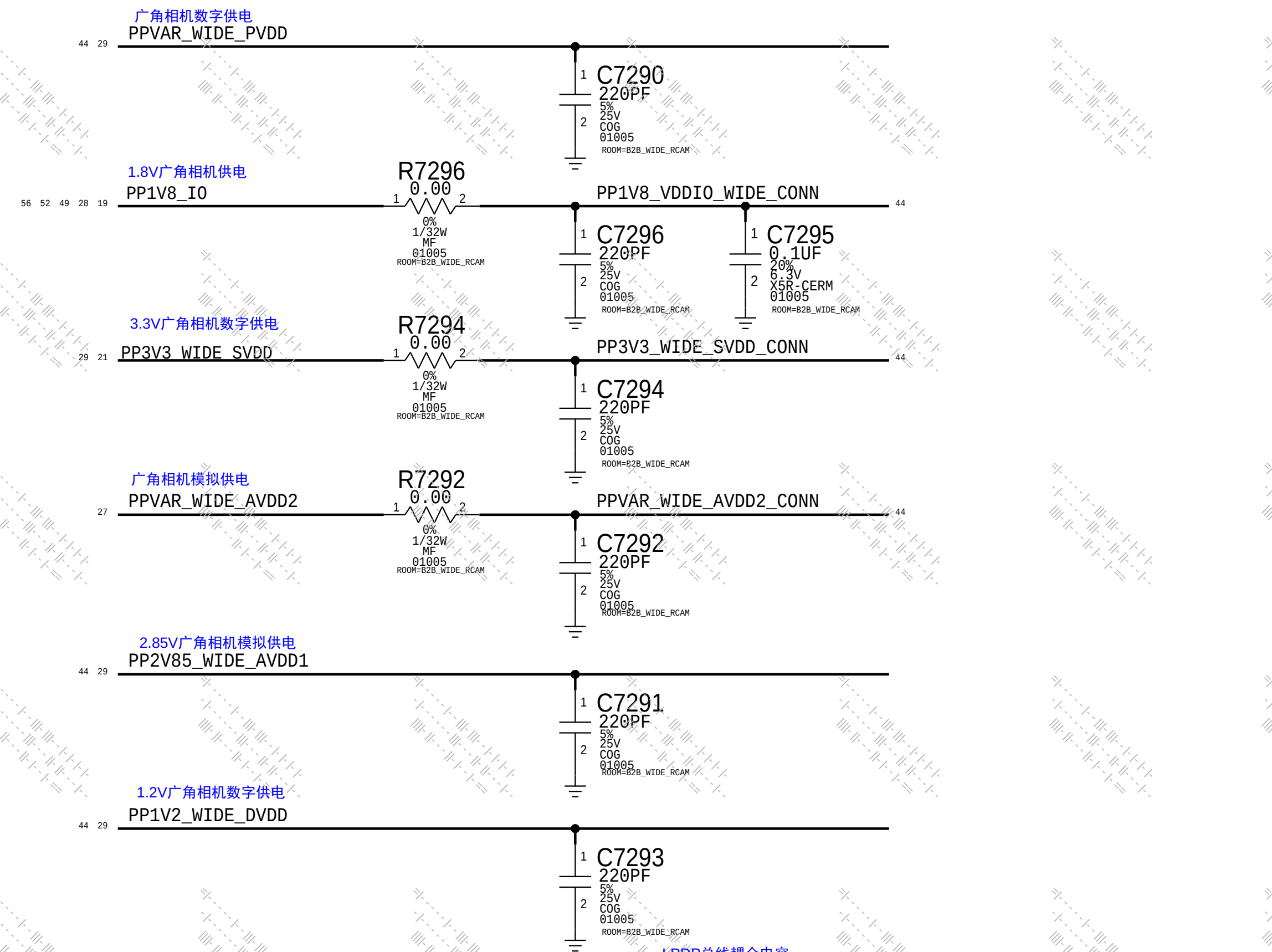
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B2B: Interposer Bot			
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REVISION		6.0.0	
BRANCH			
PAGE		65 OF 150	
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PAGE TITLE		
B2B: Battery		
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	PAGE	70 OF 150
	SHEET	43 OF 60

Power Filtering

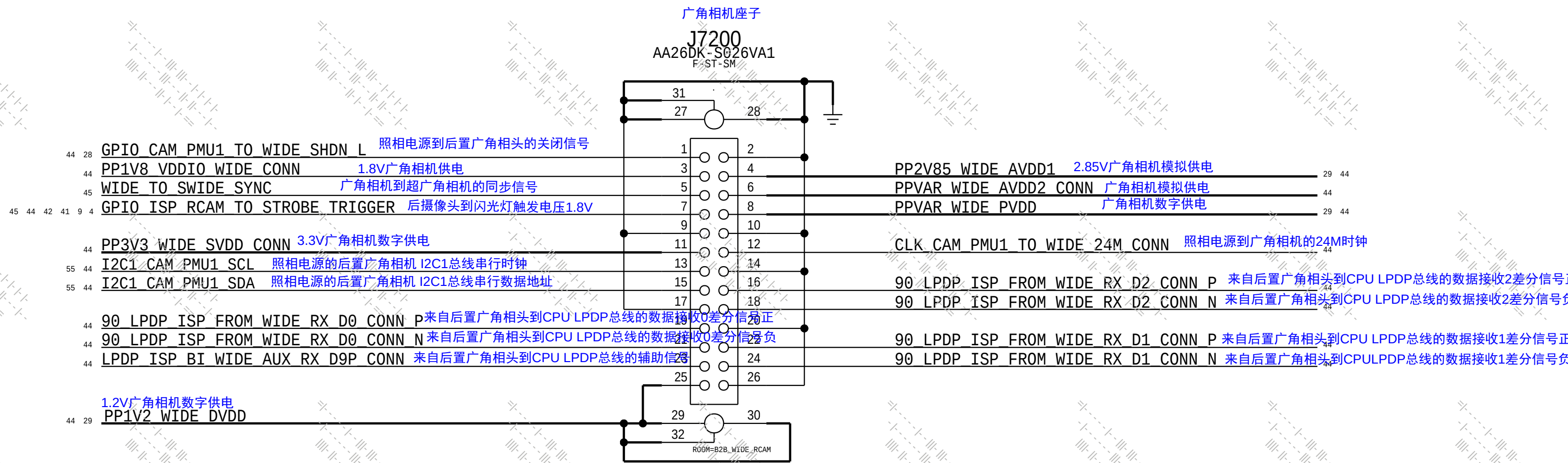


LPDP Filters

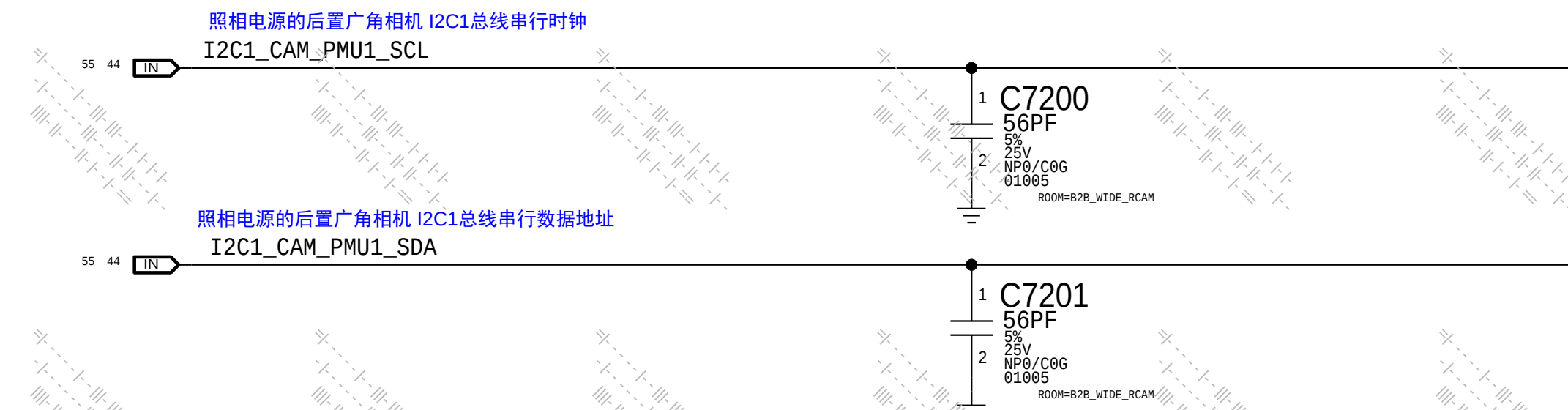


Wide Camera Connector

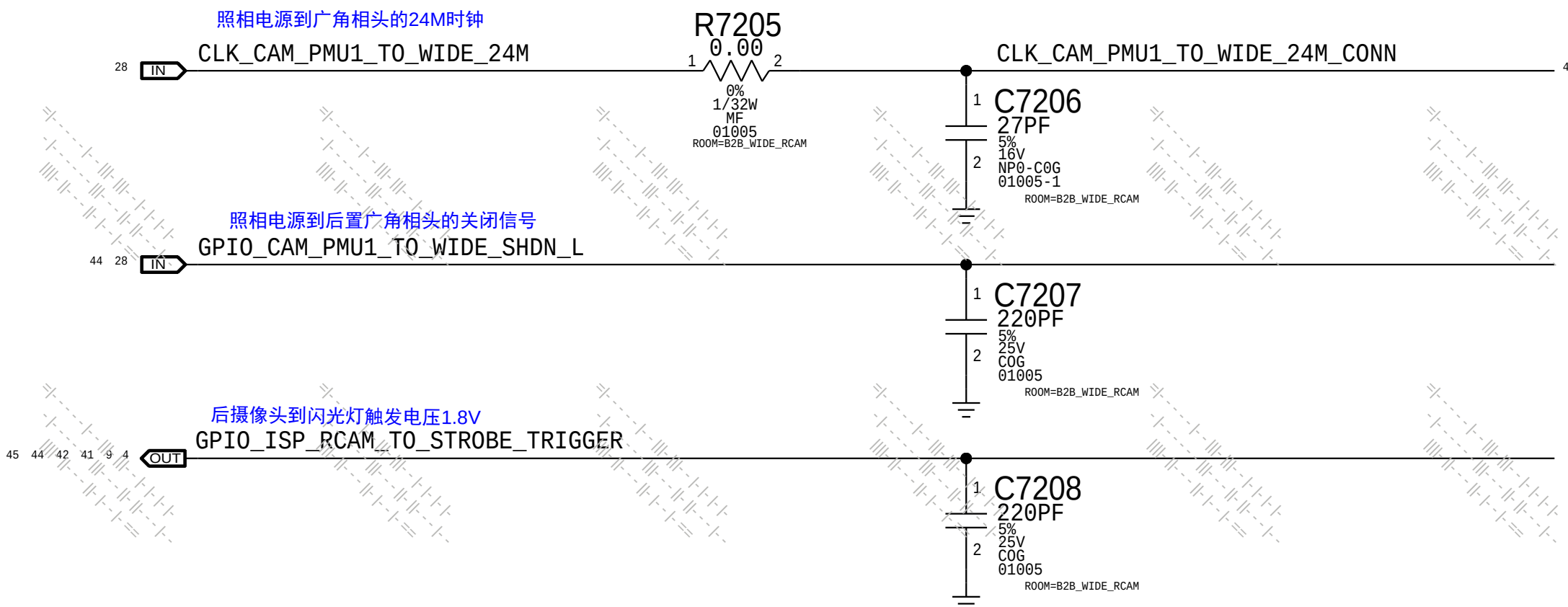
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RCAM I2C



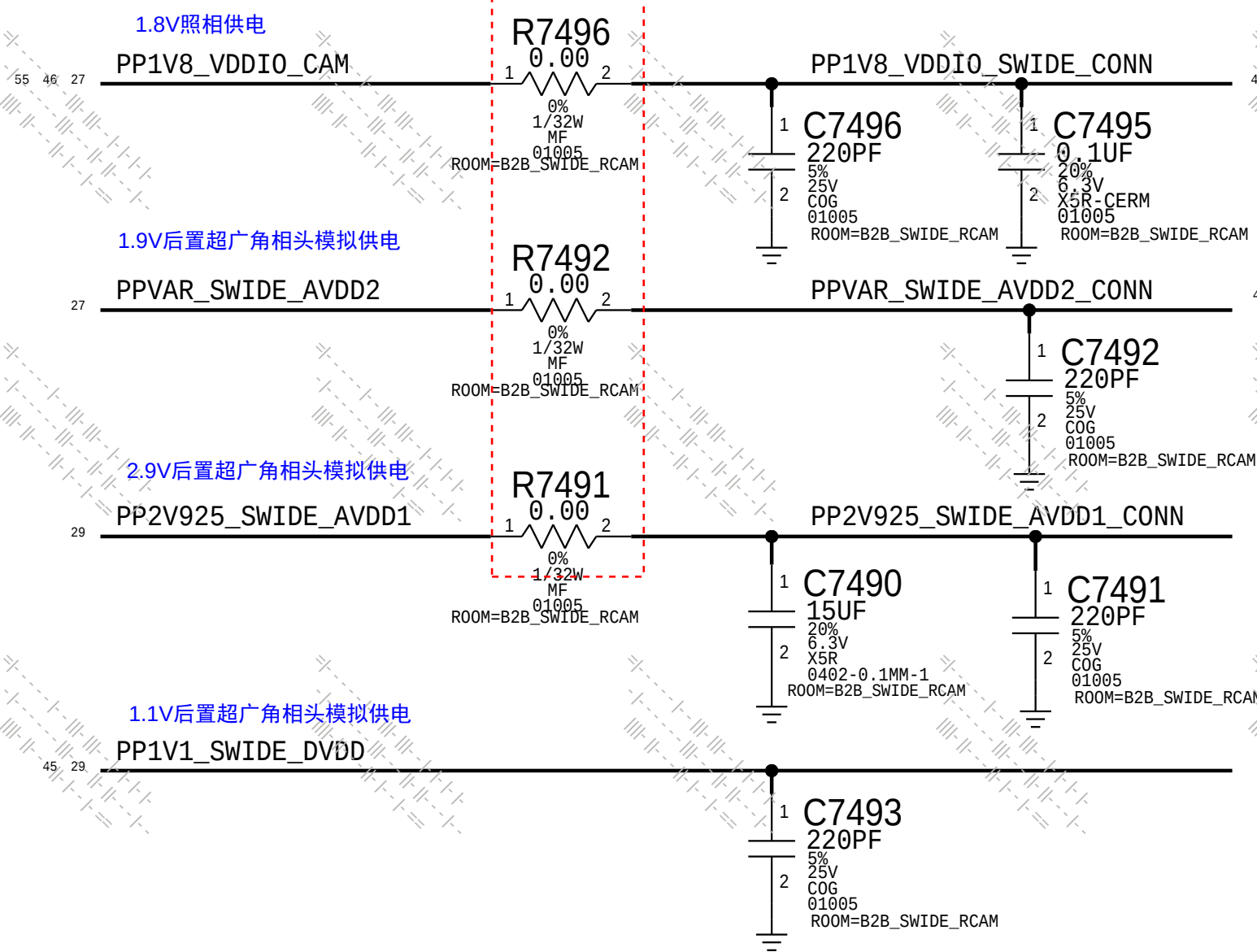
I0 Filters



PAGE TITLE: B2B: Camera Wide (TX)			
	DRAWING NUMBER	051-04047	SIZE
	REVISION	6.0.0	D
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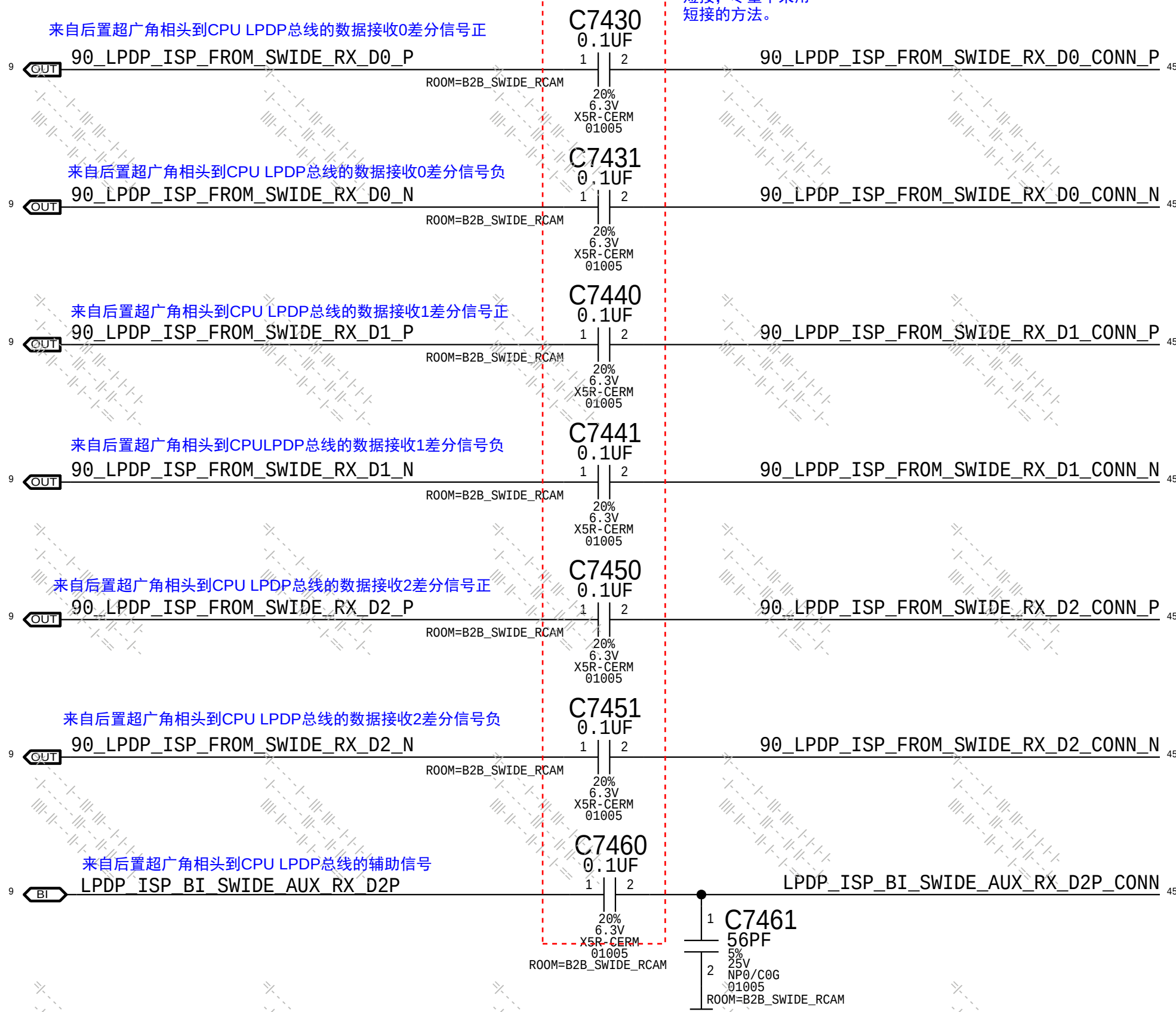
Power Filtering

0欧姆电阻损坏会导致后置超广角相机无法打开
在维修过程中可以短接



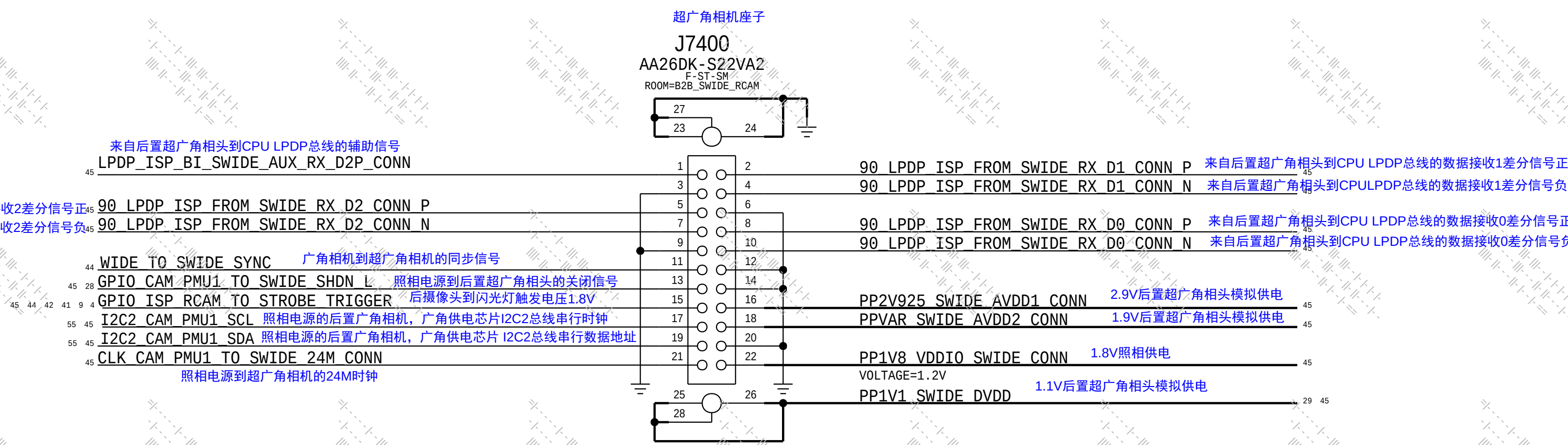
LPDP Filters

LPDP总线耦合电容
通过充电放点的特性，使得信号稳定
无杂波。实际维修
当中，此电容可以
短接，尽量不采用
短接的方法。

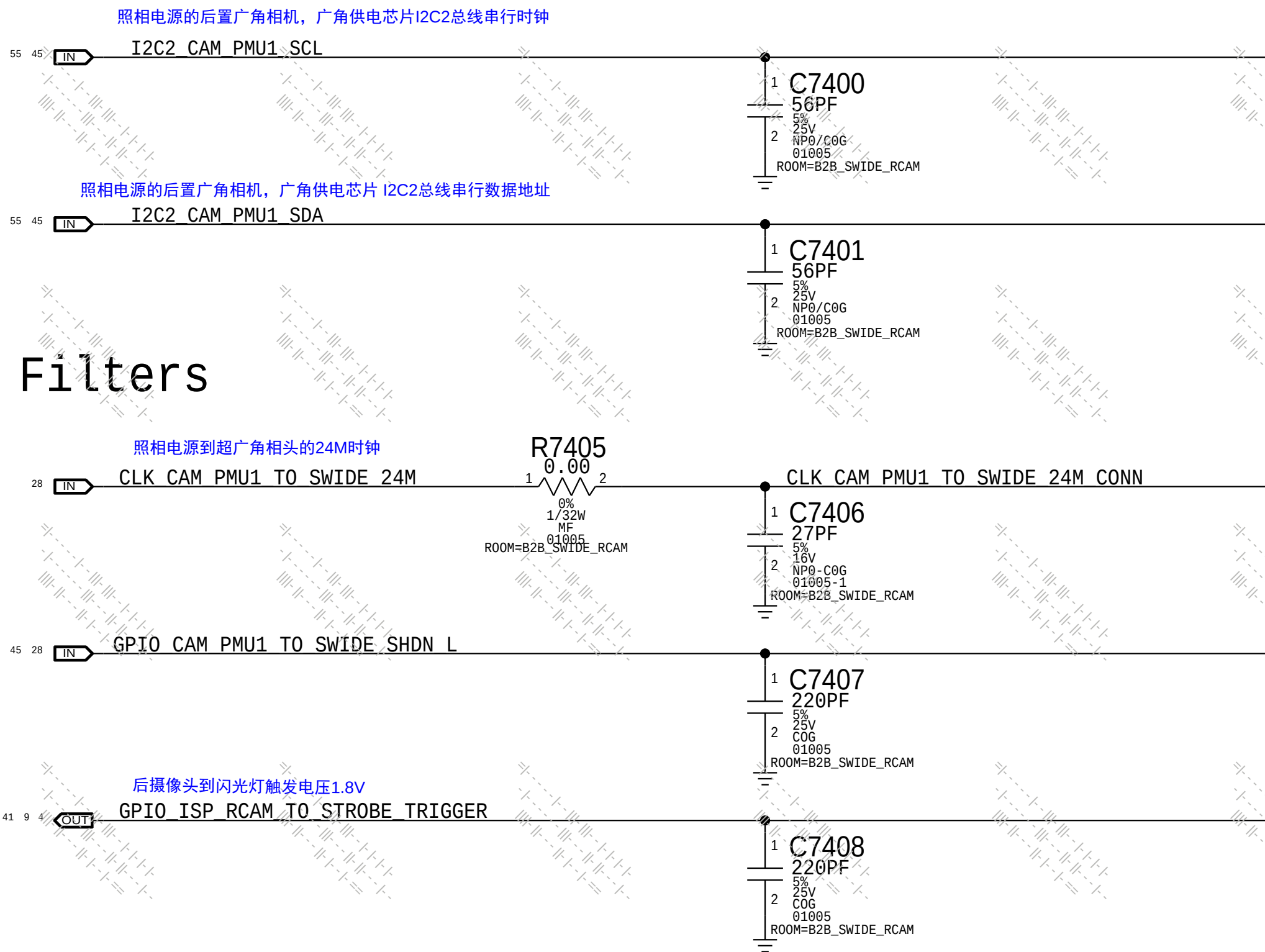


Super Wide Camera Connector

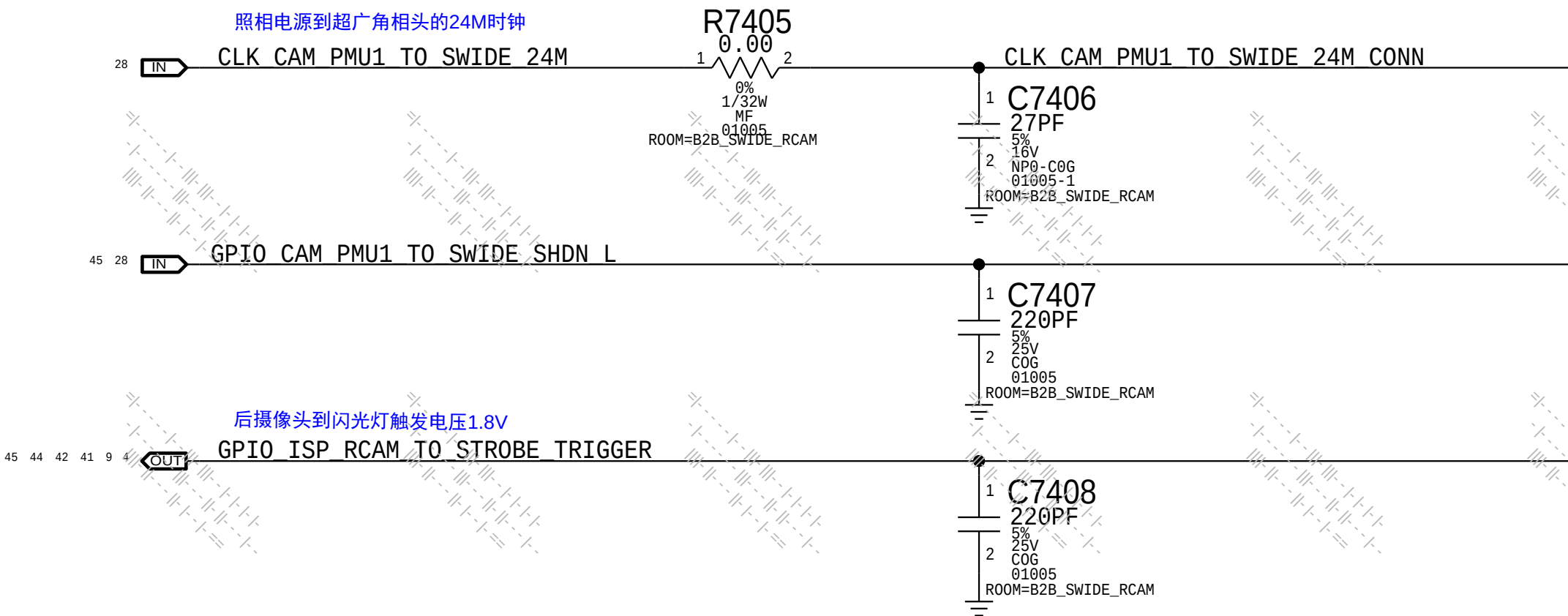
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PLUG: 516S00459
<-- This one on MLB



RCAM I2C

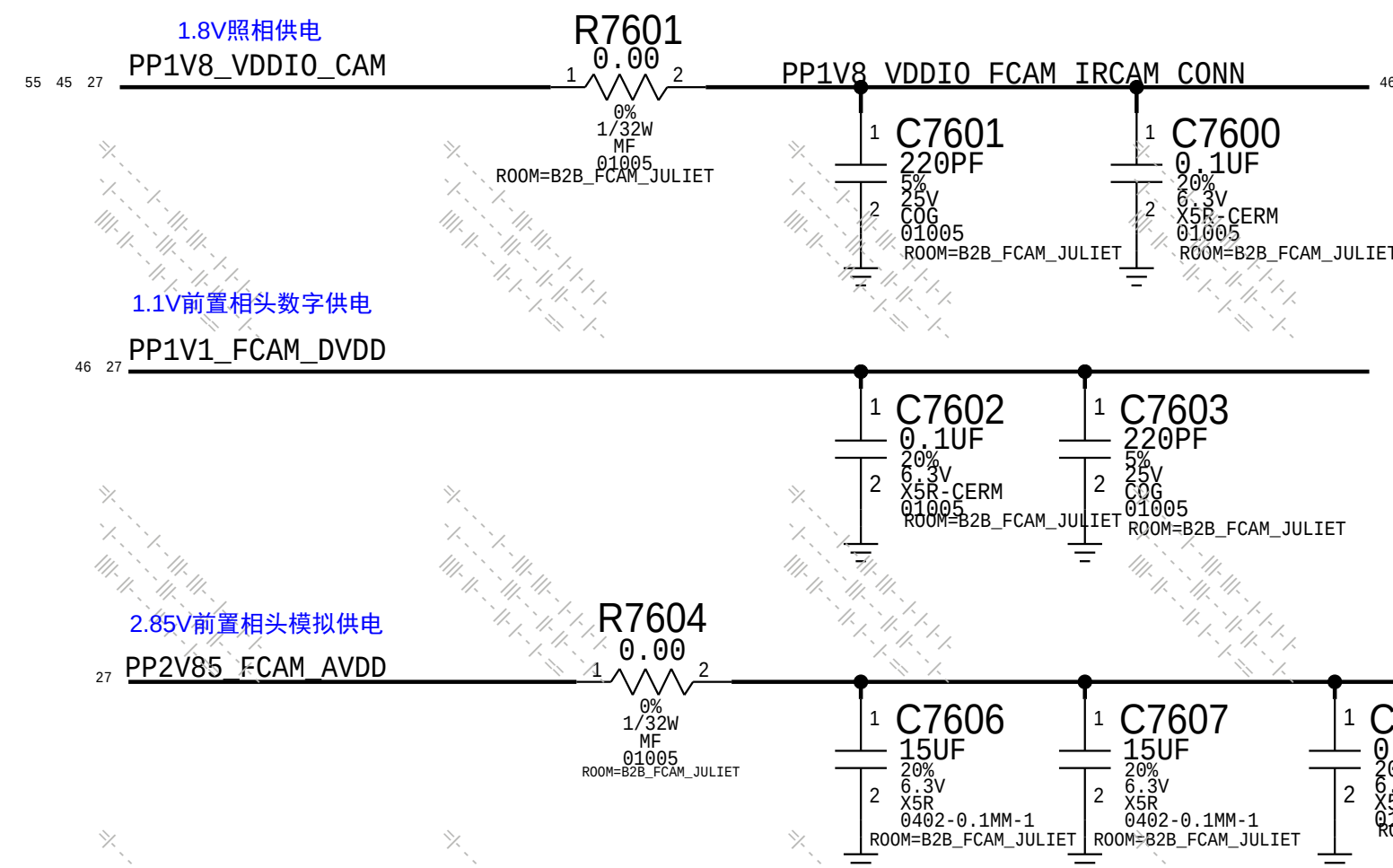


I/O Filters

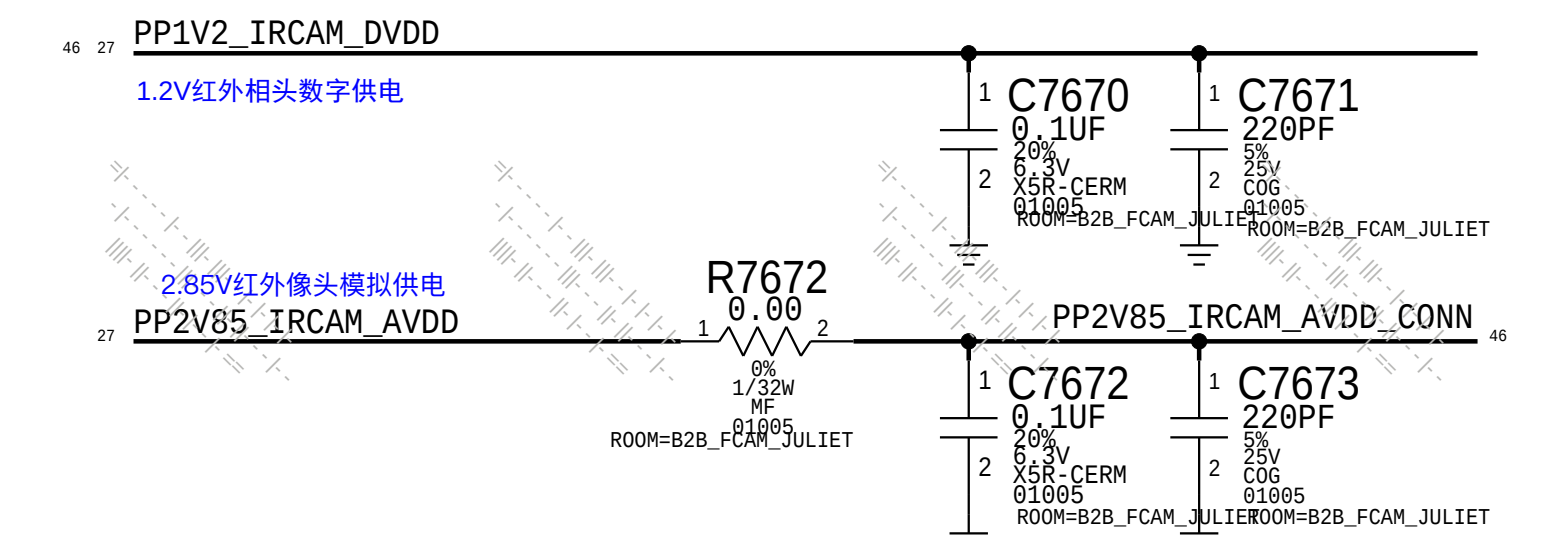


PAGE TITLE			
B2B: Camera Super Wide			
DRAWING NUMBER		051-04047	SIZE
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PAGE		74 OF 150	
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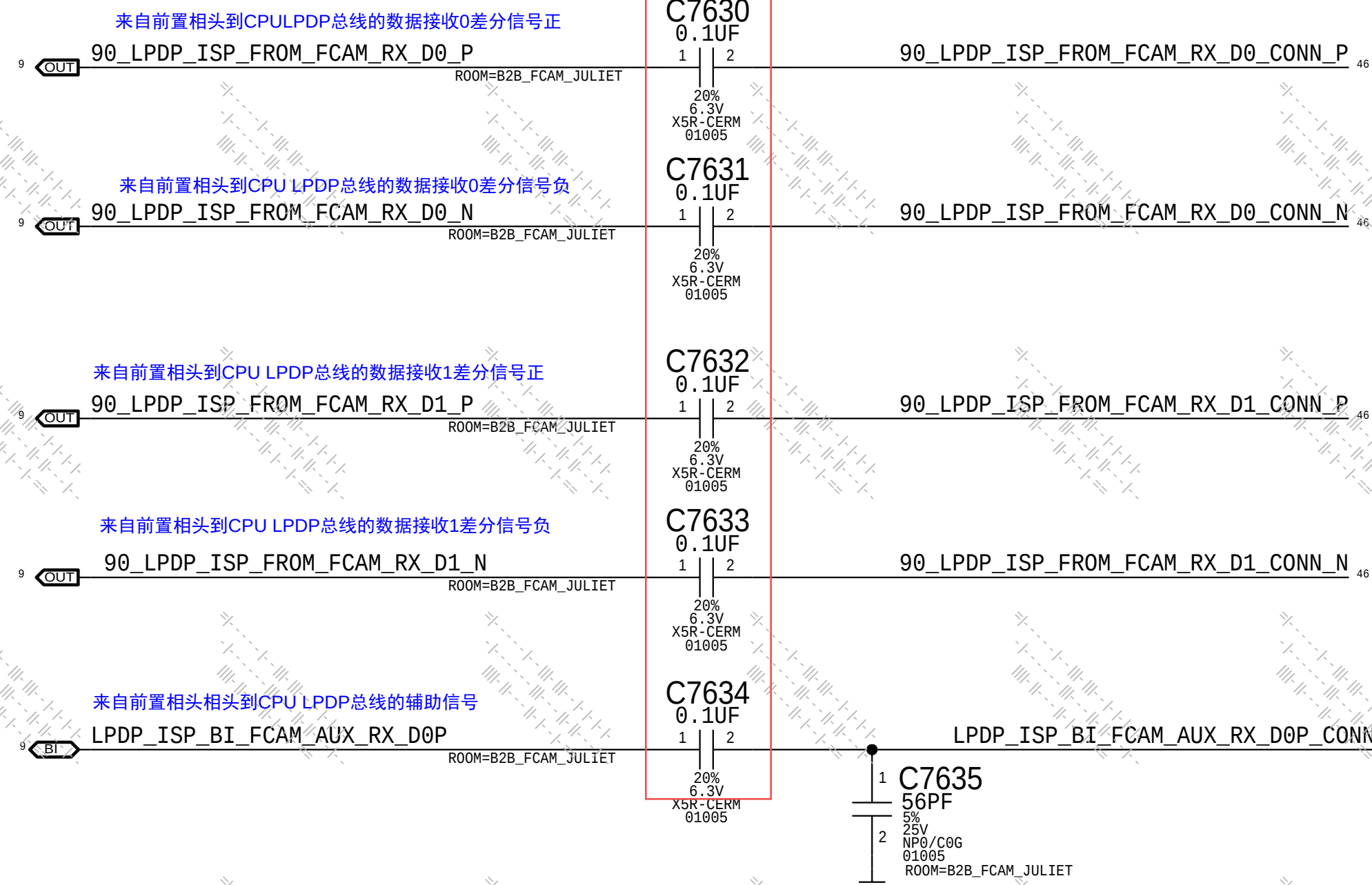
FCAM P0



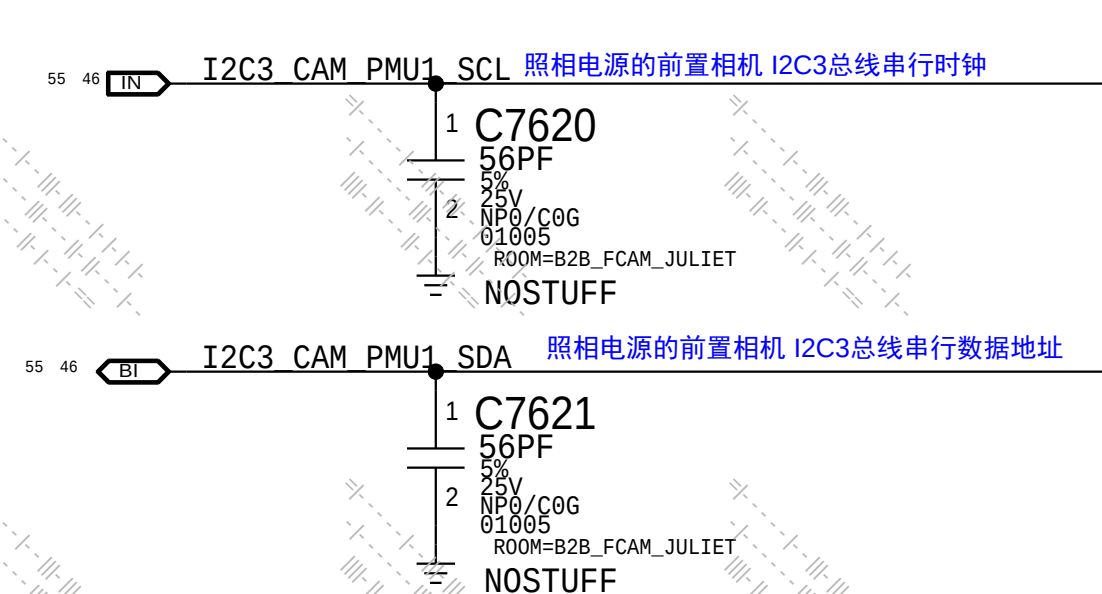
Juliet Power and I/O



LPDP FILTERS

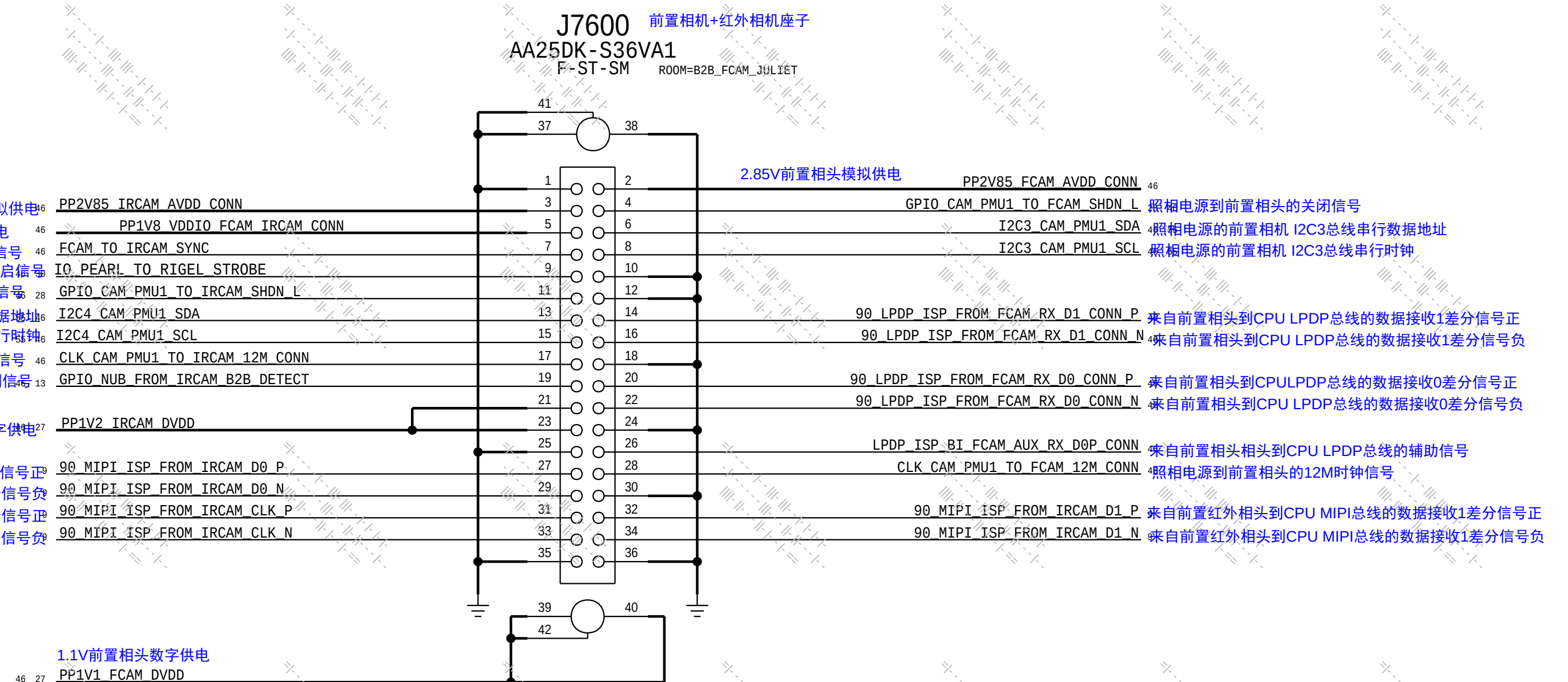


FCAM I2C

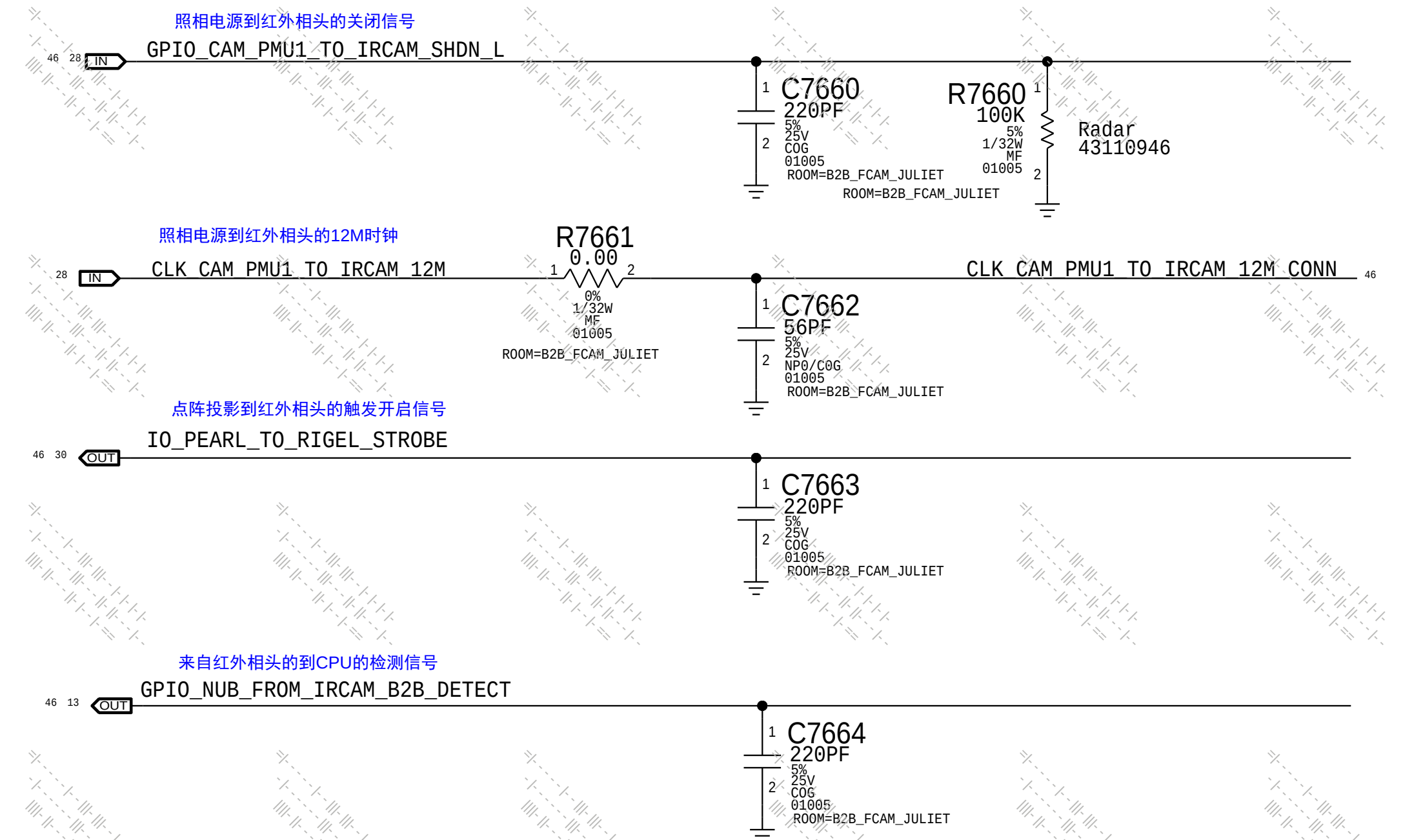


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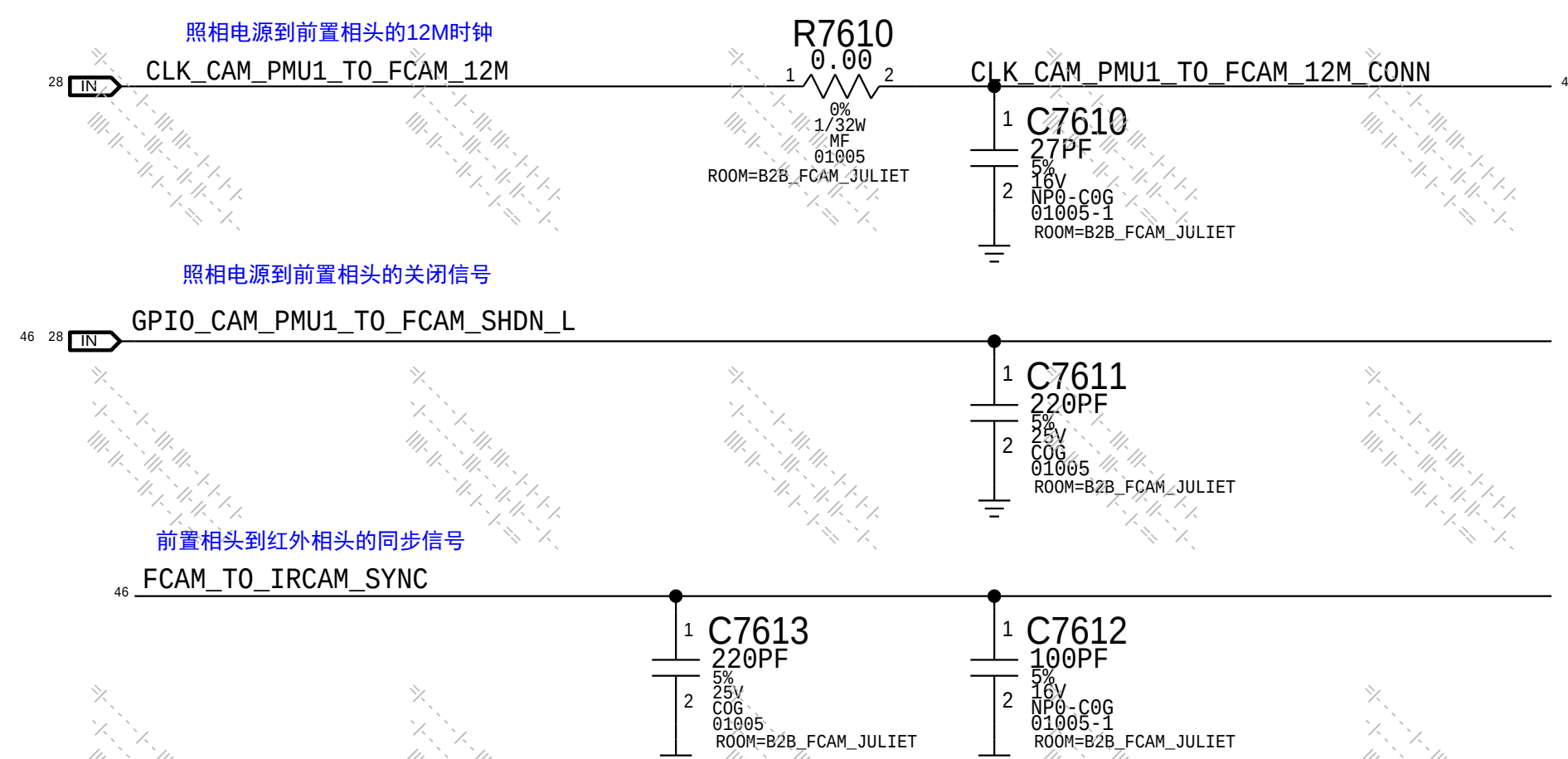
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Rcpt: 516S00521      <-- This one on MLB
Plug: 516S00522
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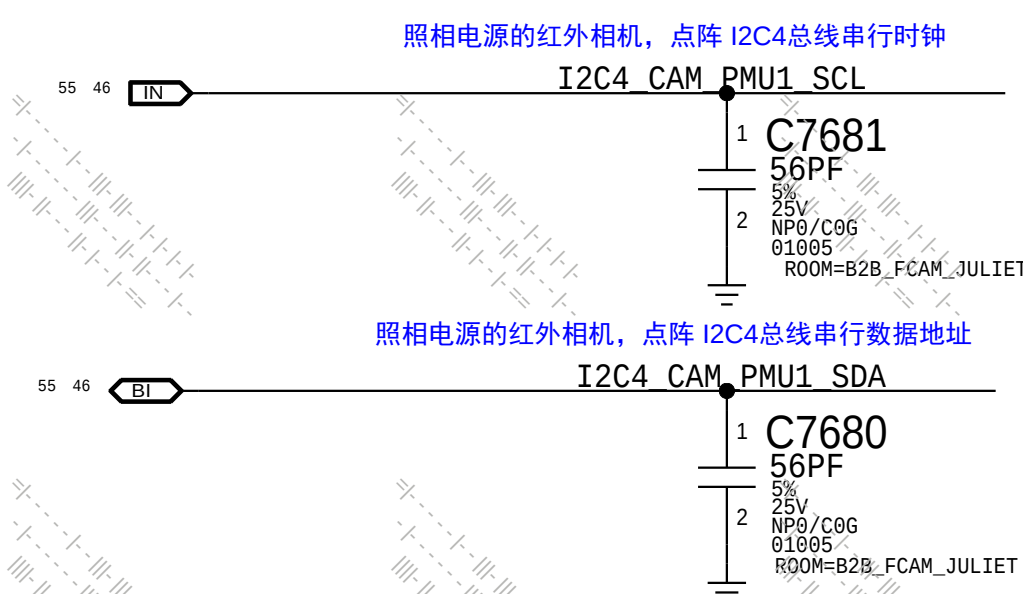
JULIET I/O



FCAM I/O

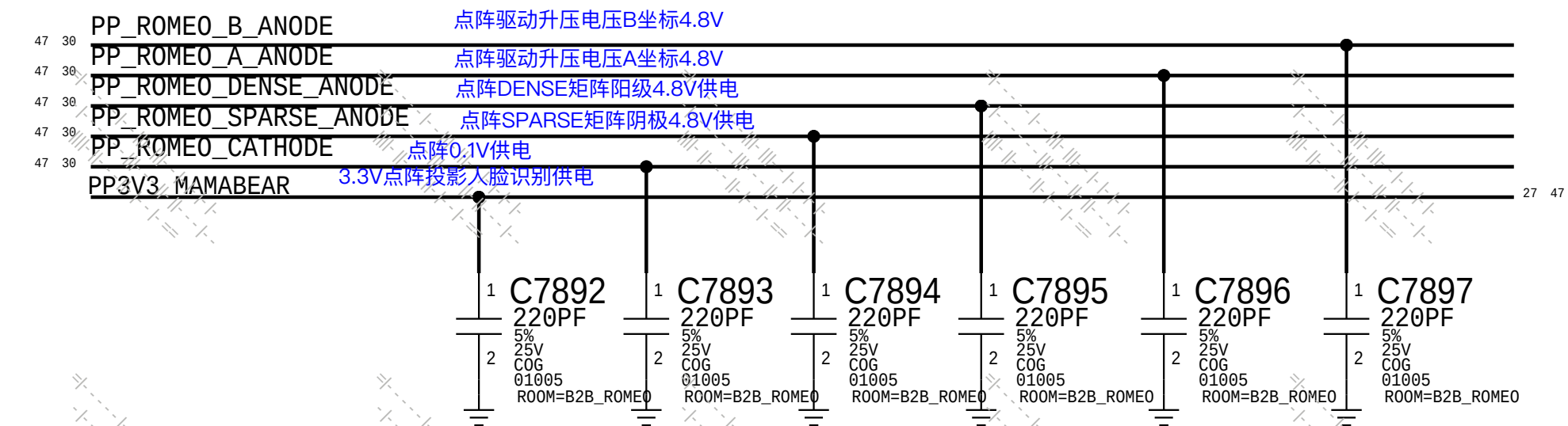


Juliet I2C

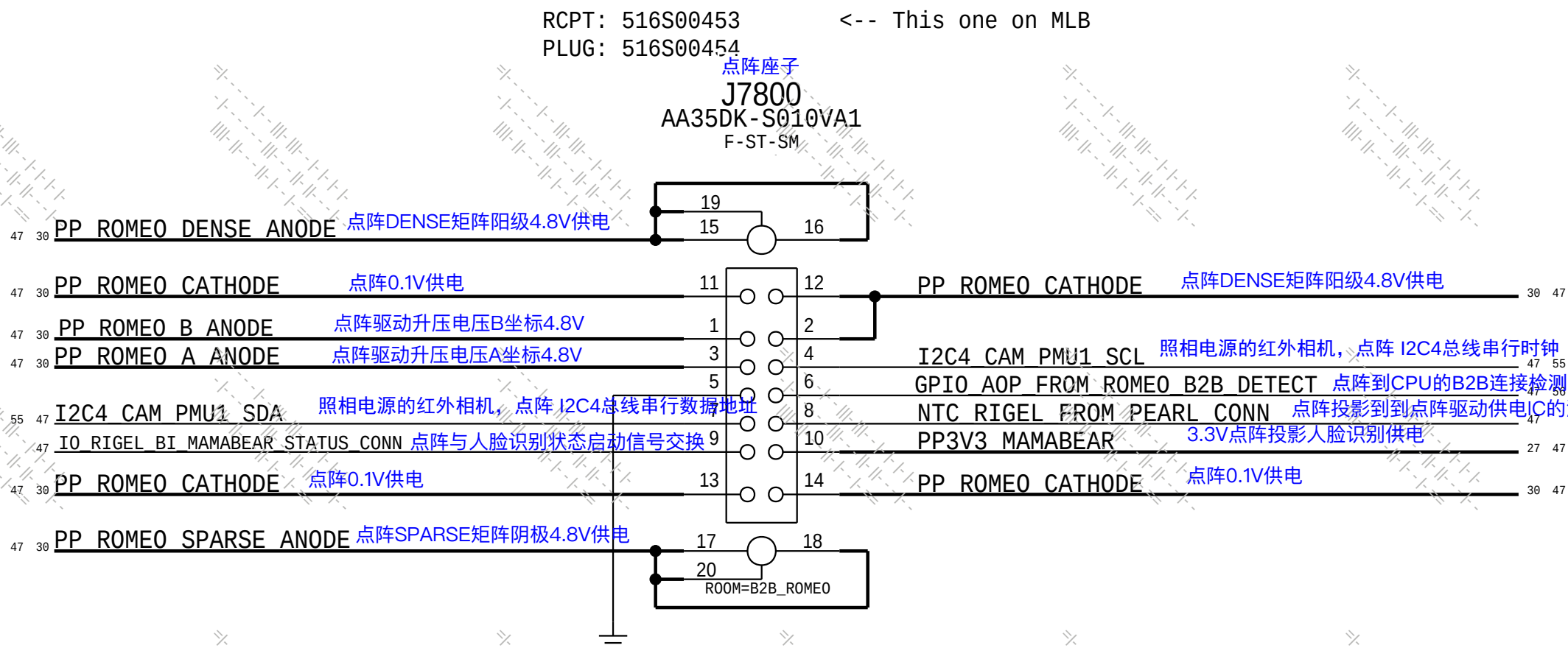


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		46	OF 60

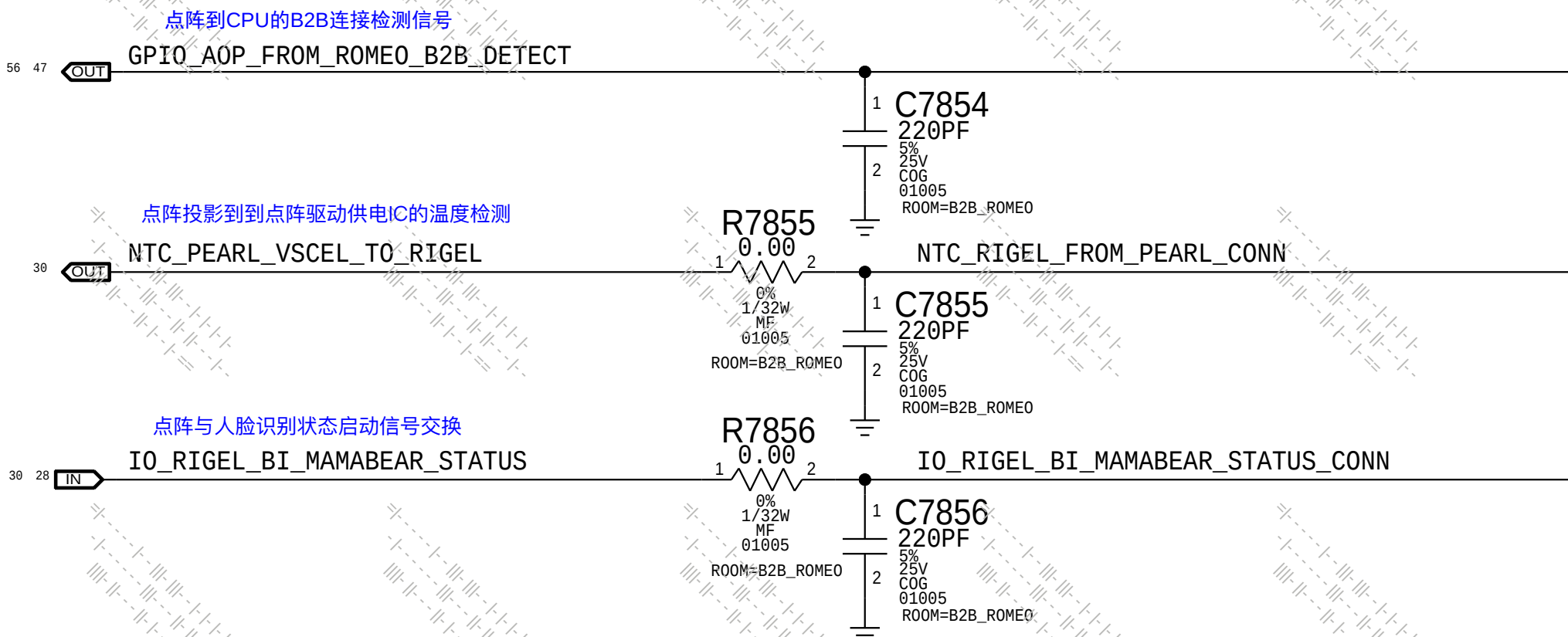
Romeo Power Filtering



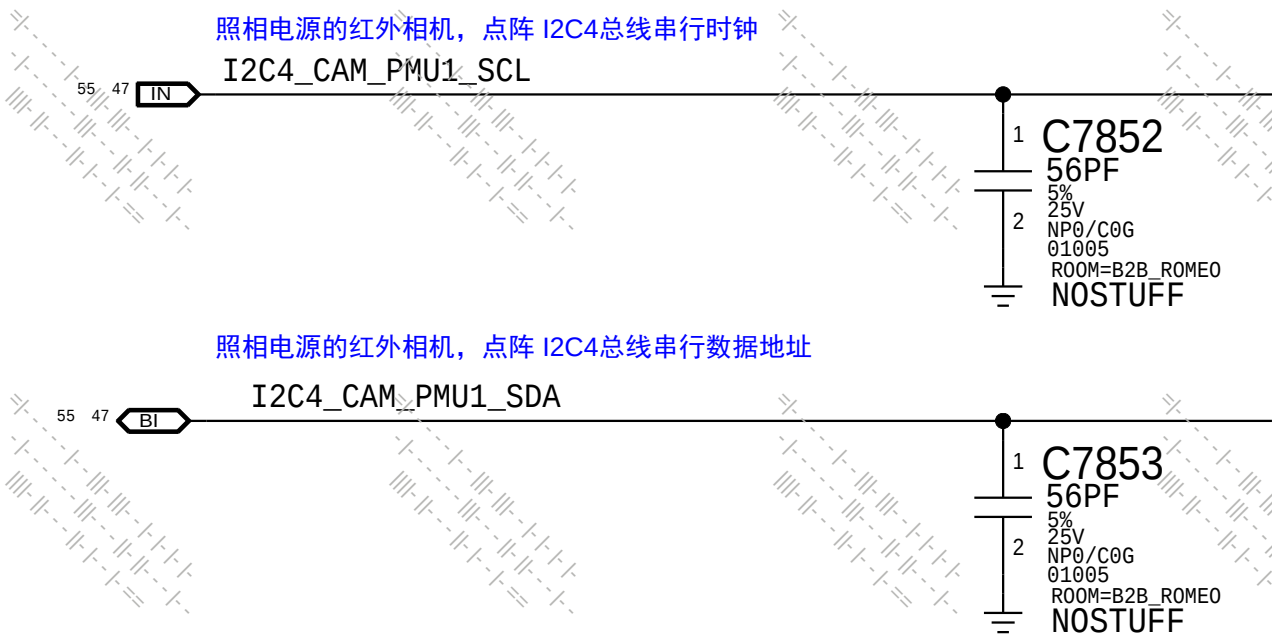
Romeo Connector



Romeo I/O

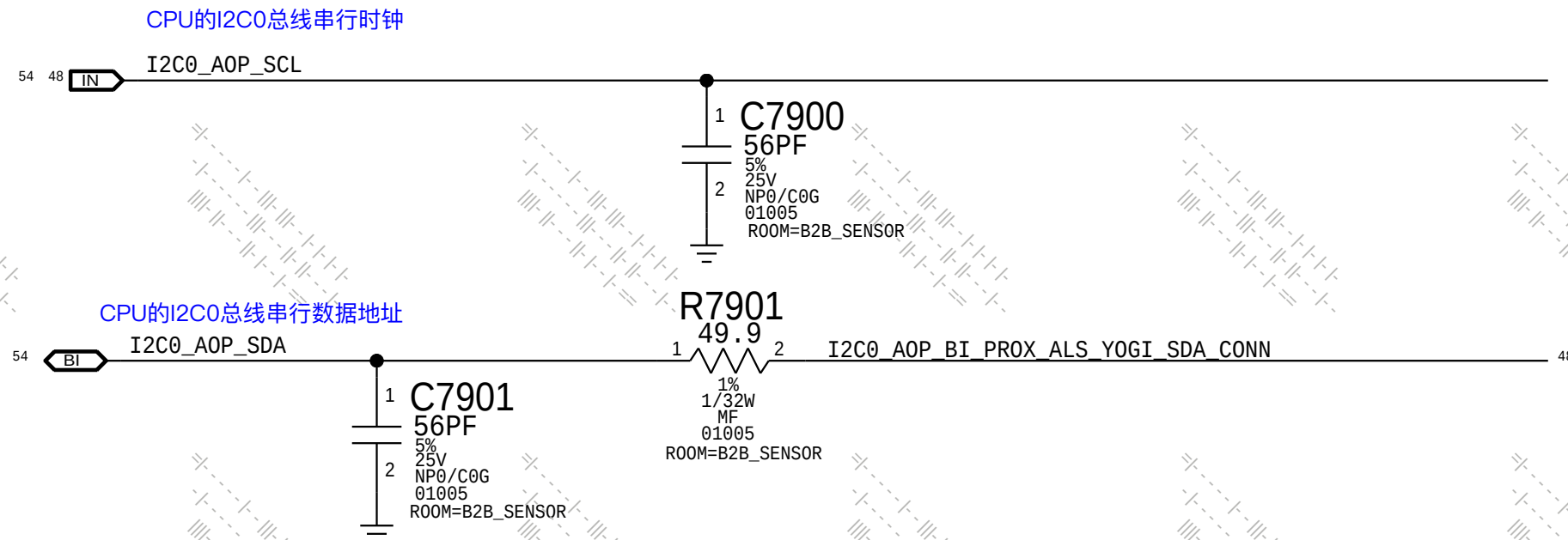


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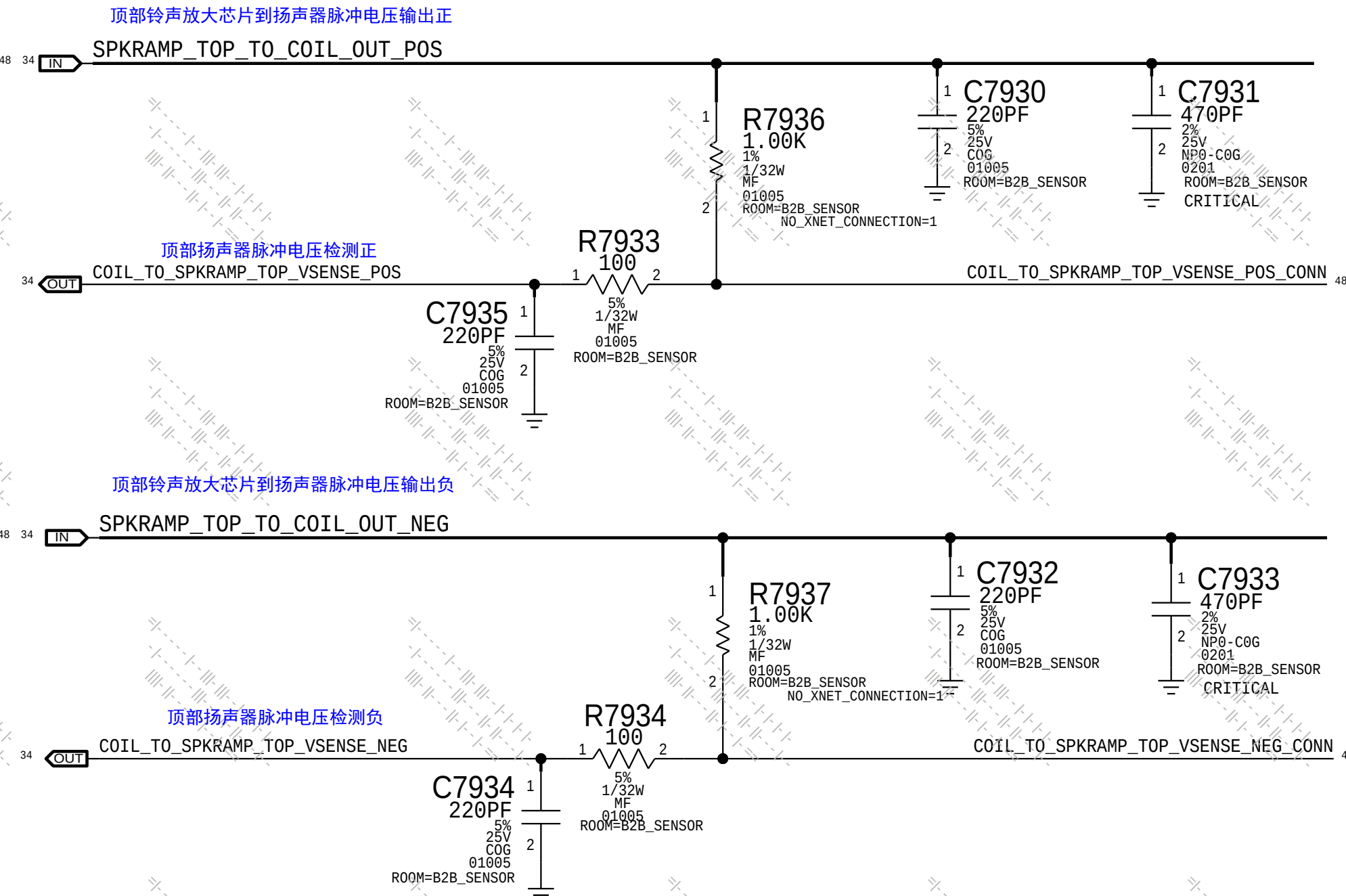


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AOP I2C

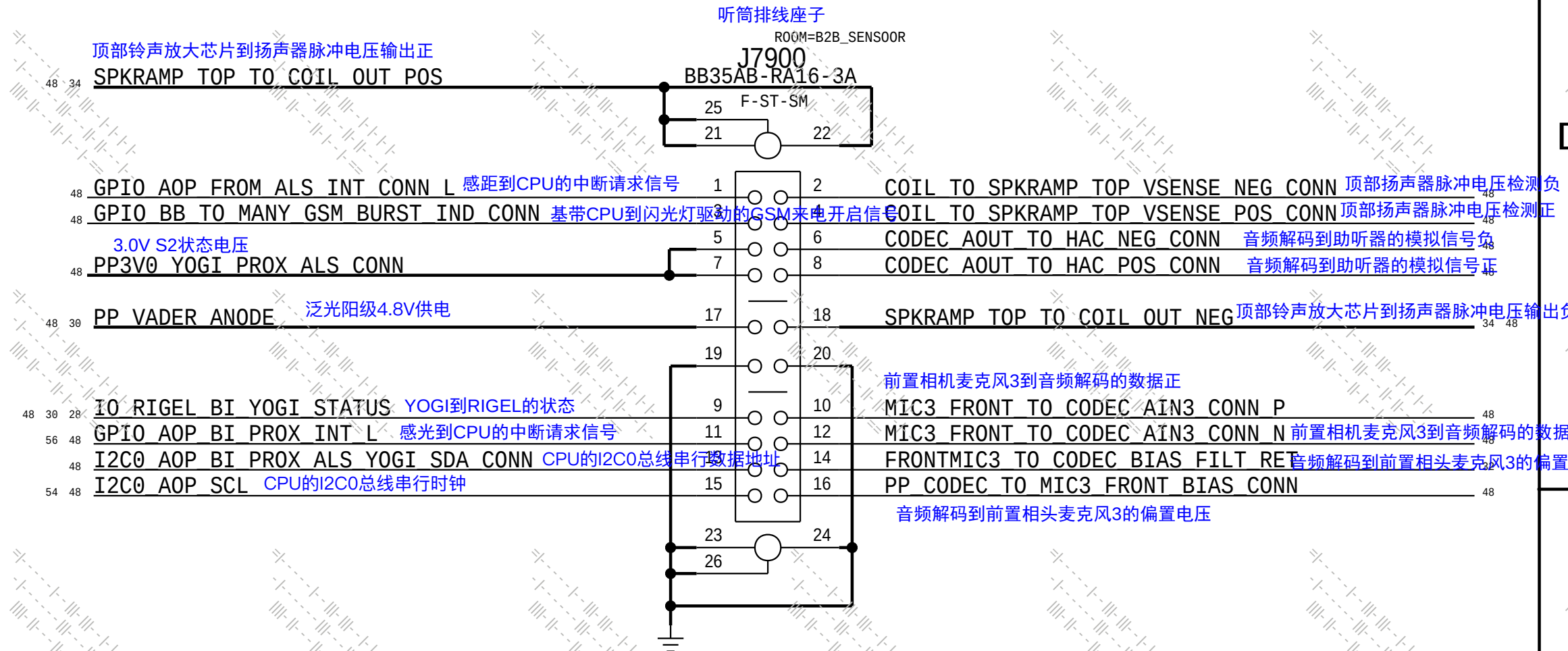


SPEAKER2

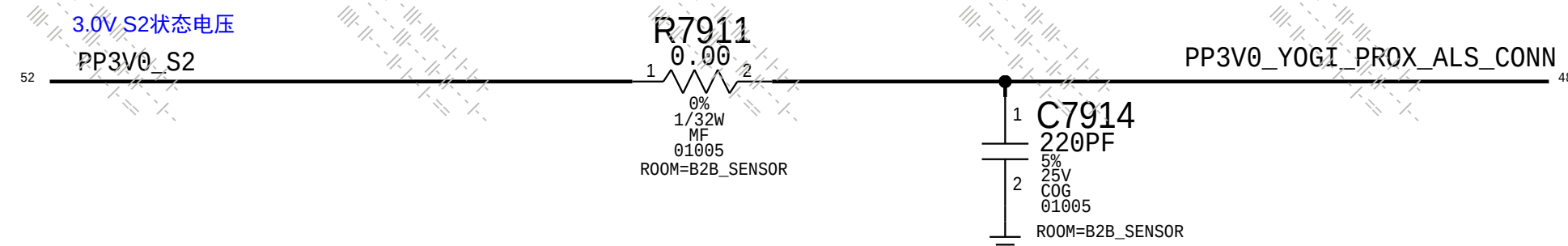


Vader + Sensor Connector

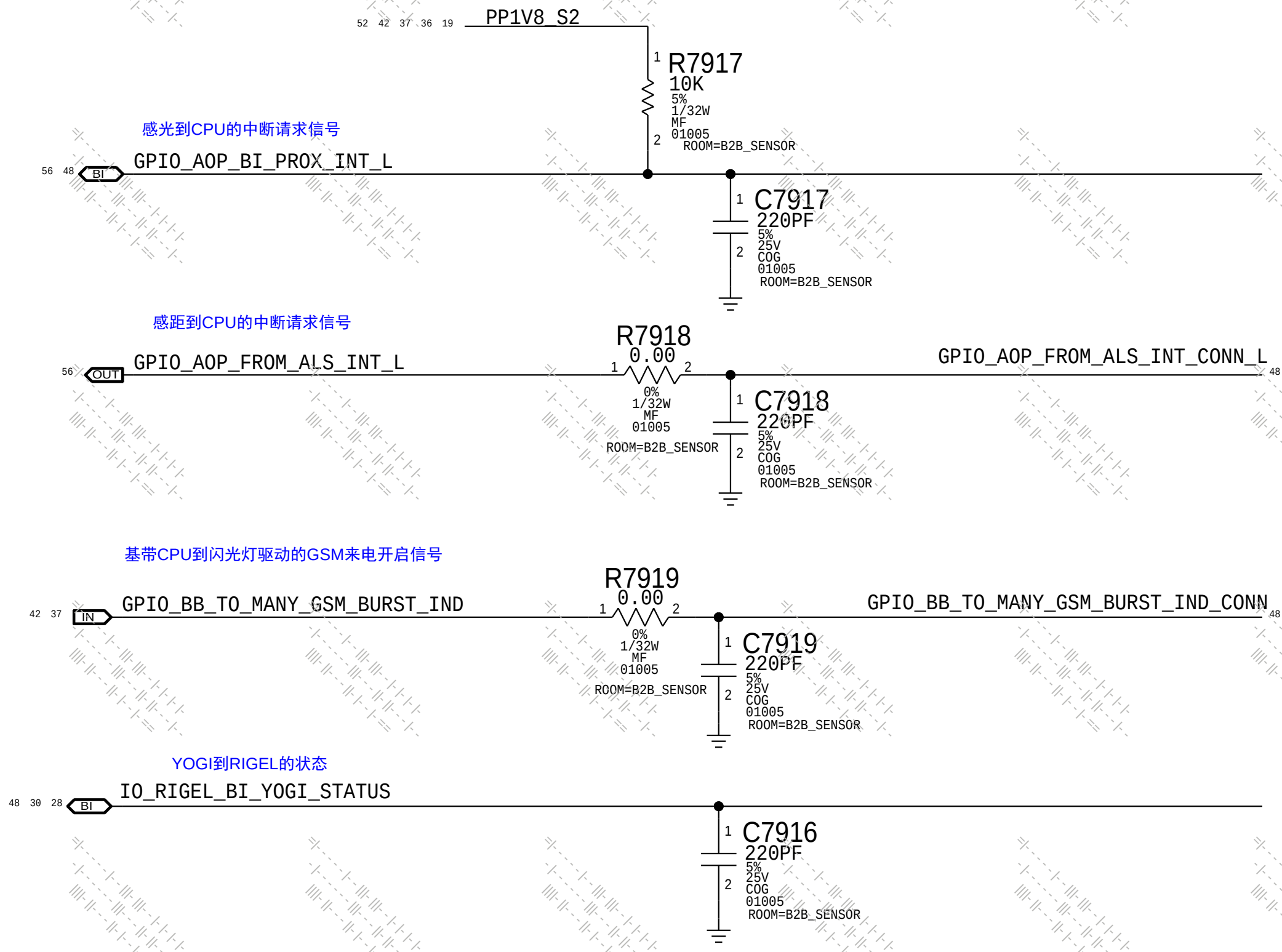
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PLUG: 516S00450



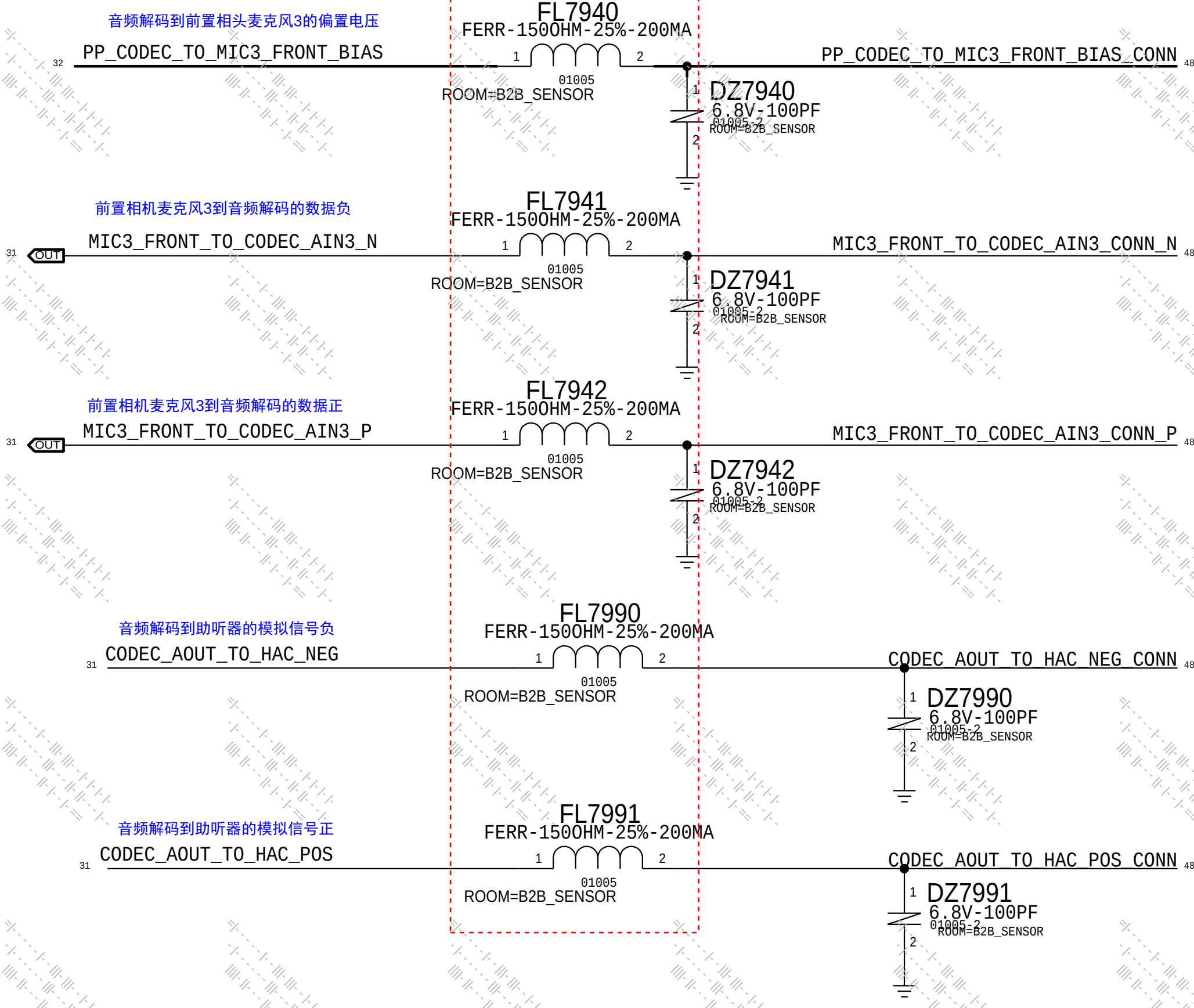
PROX & ALS POWER



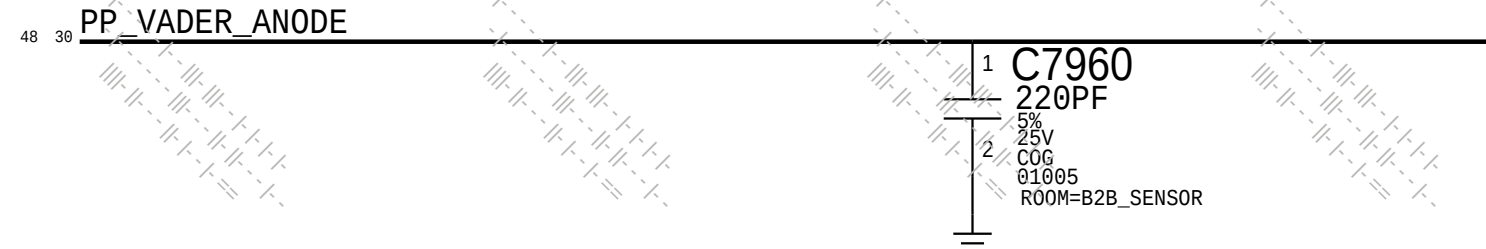
PROX/ALS I/O



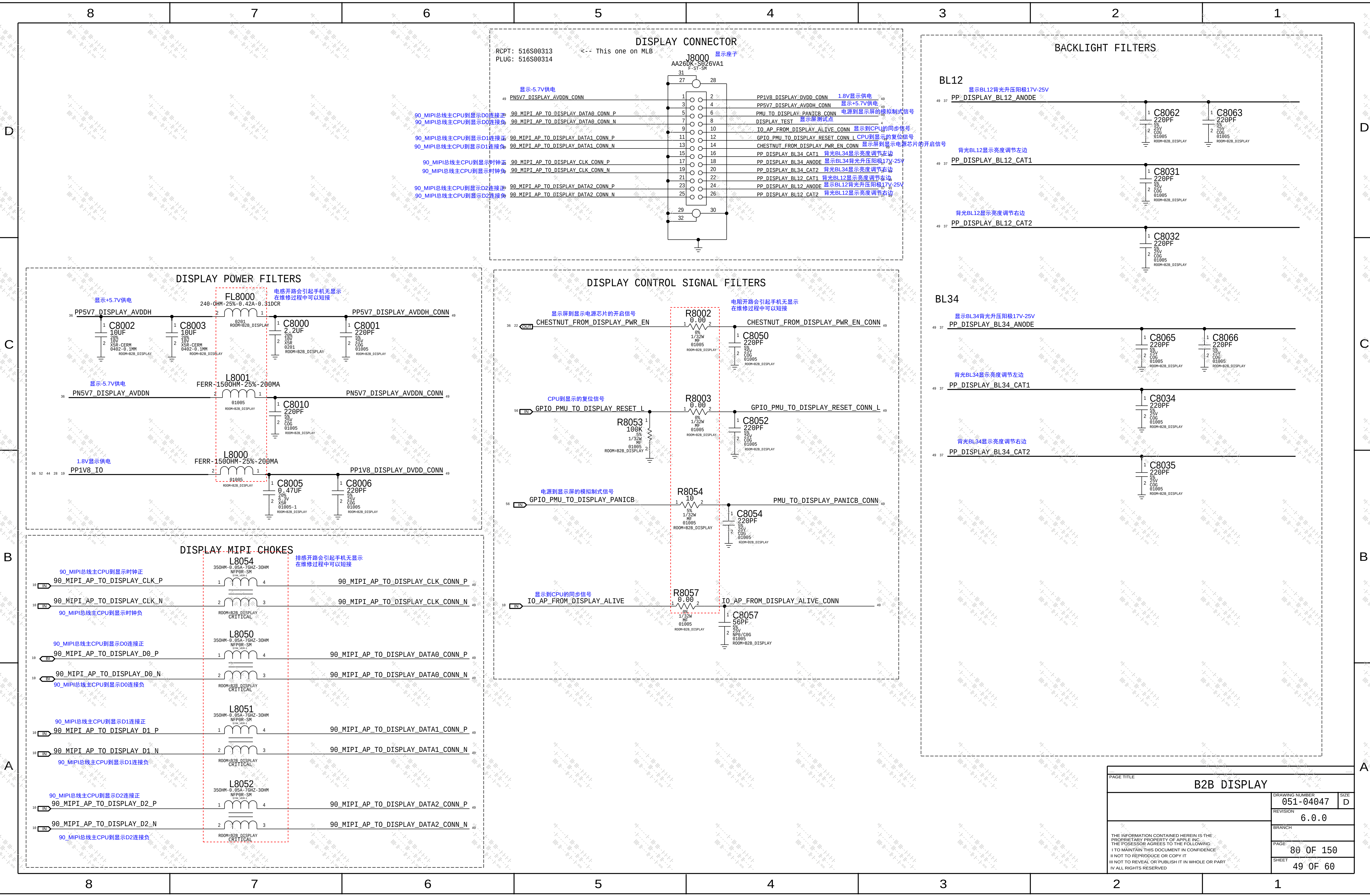
MIC3



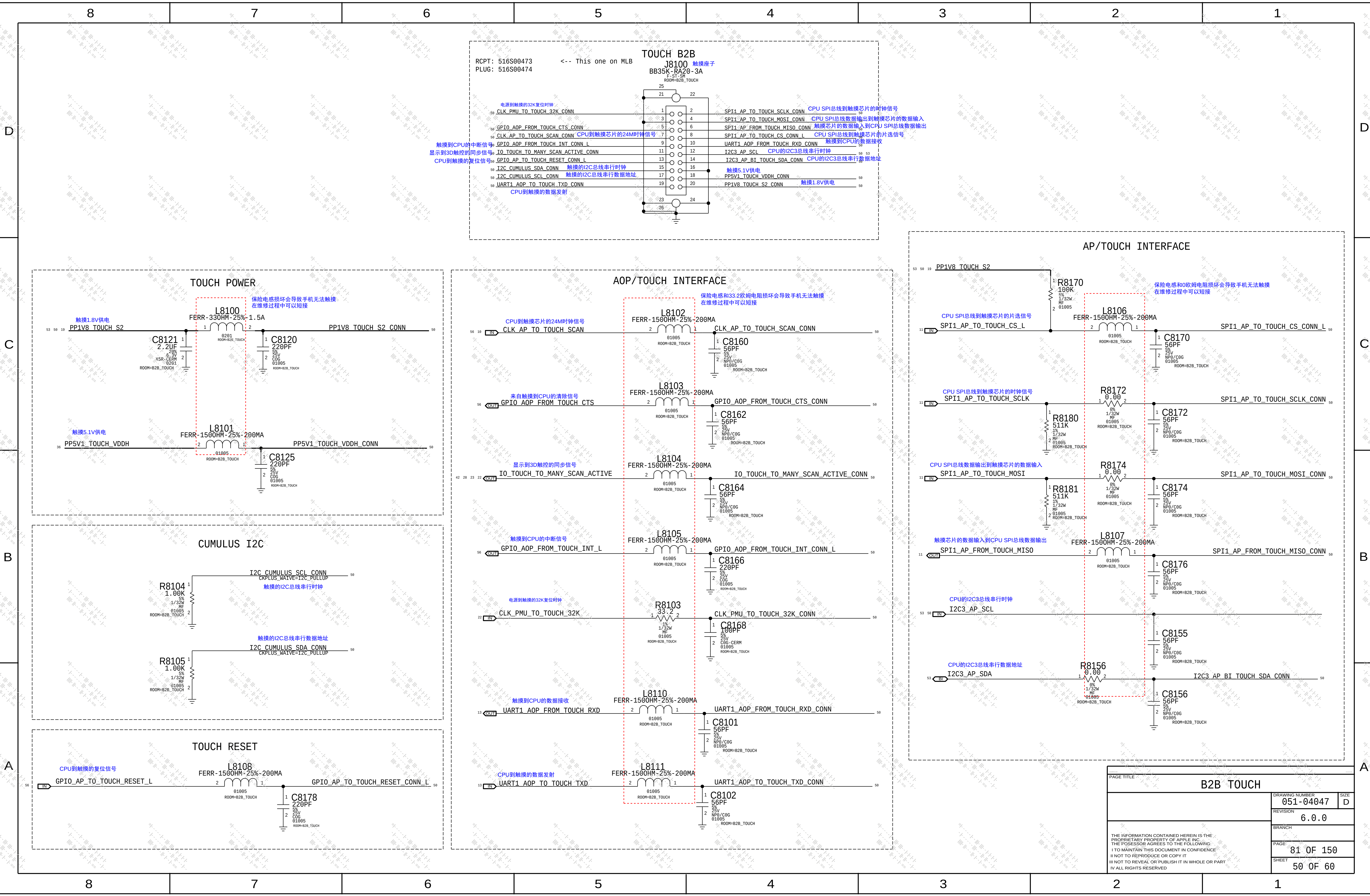
泛光阳极4.8V供电



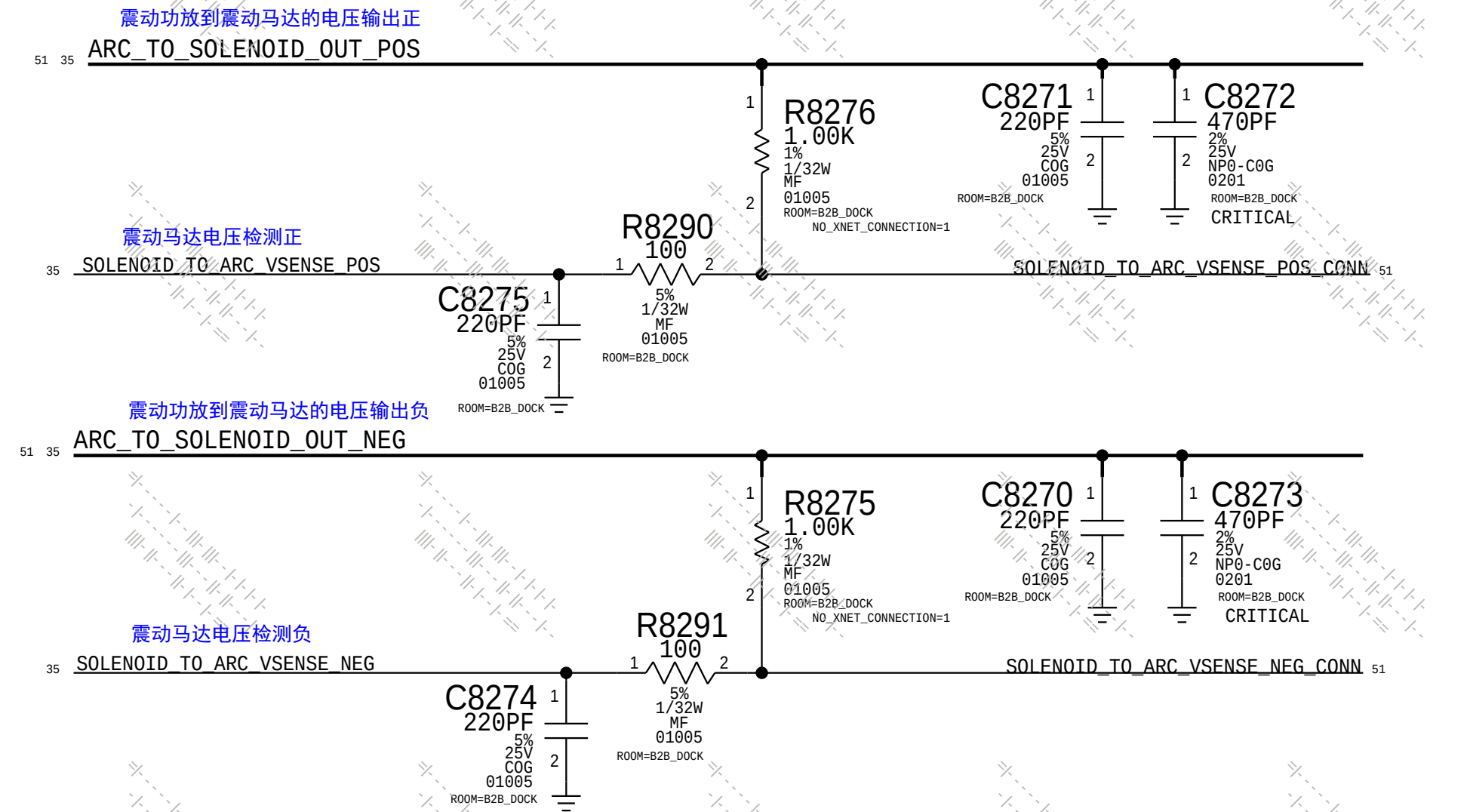
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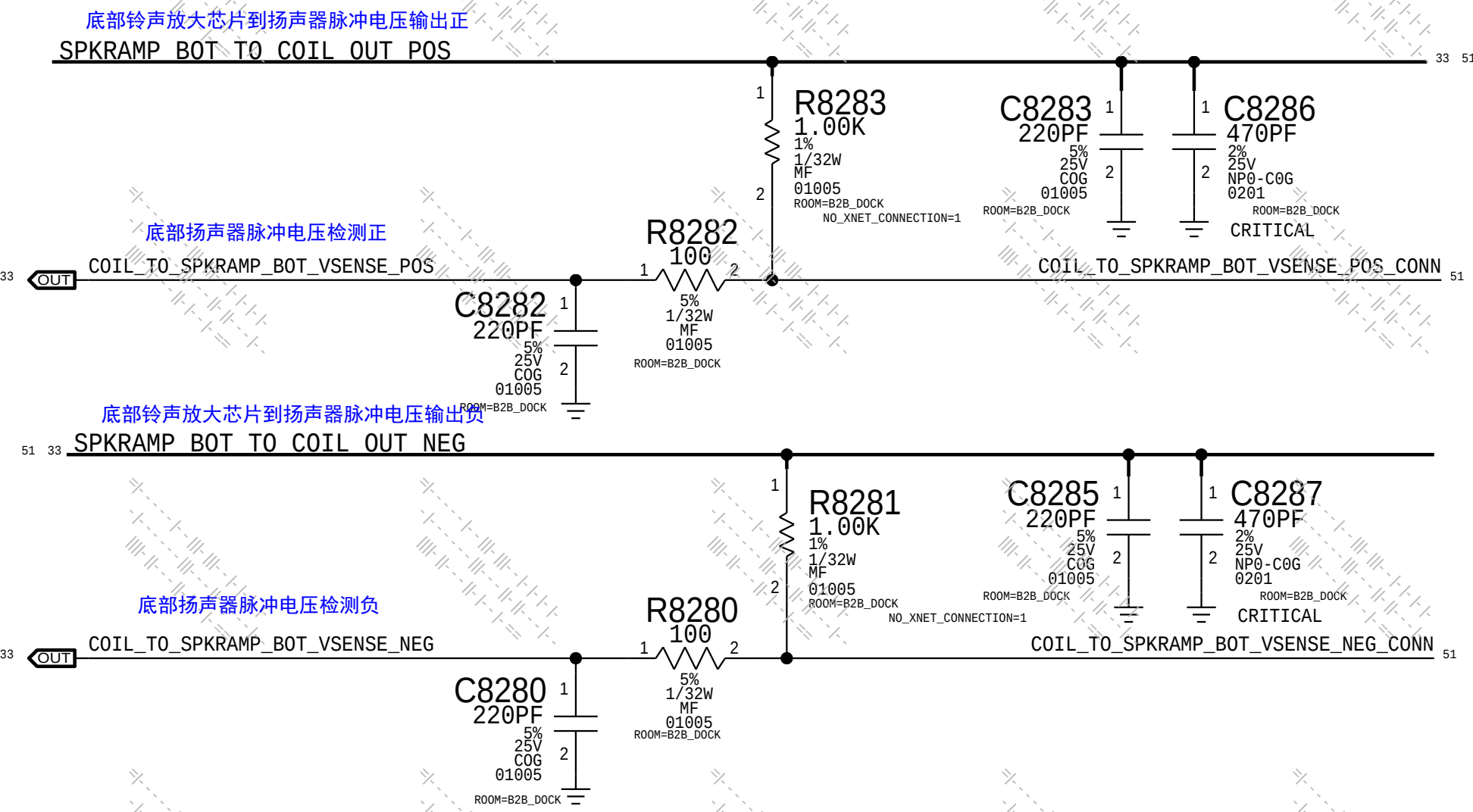
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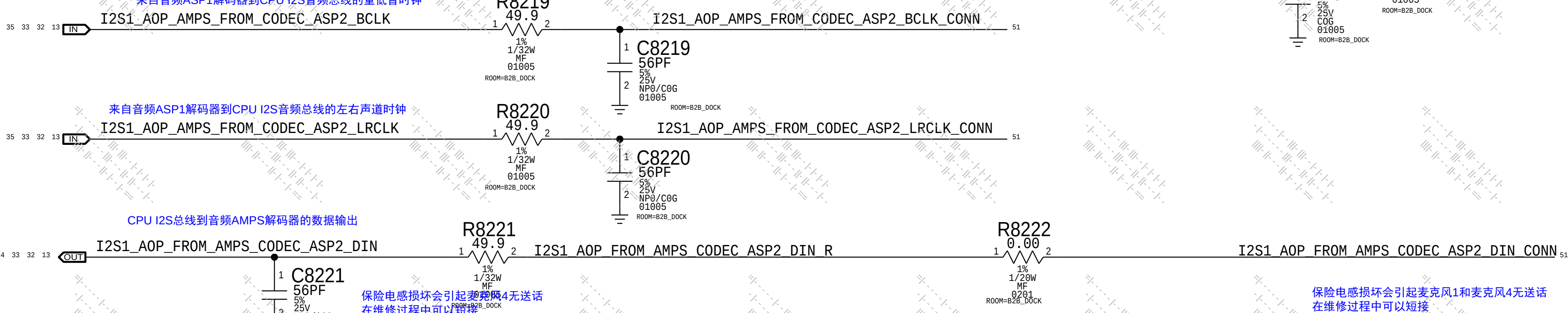
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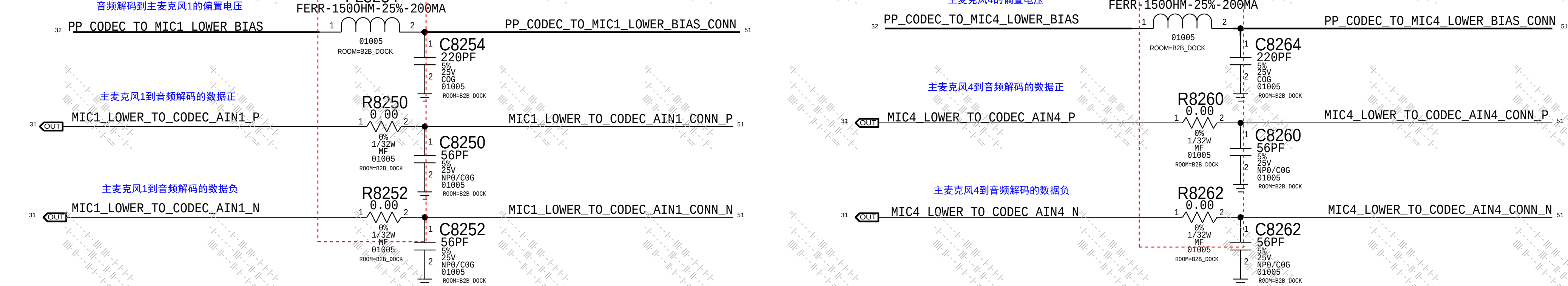
SOUTH SPEAKER



I2S



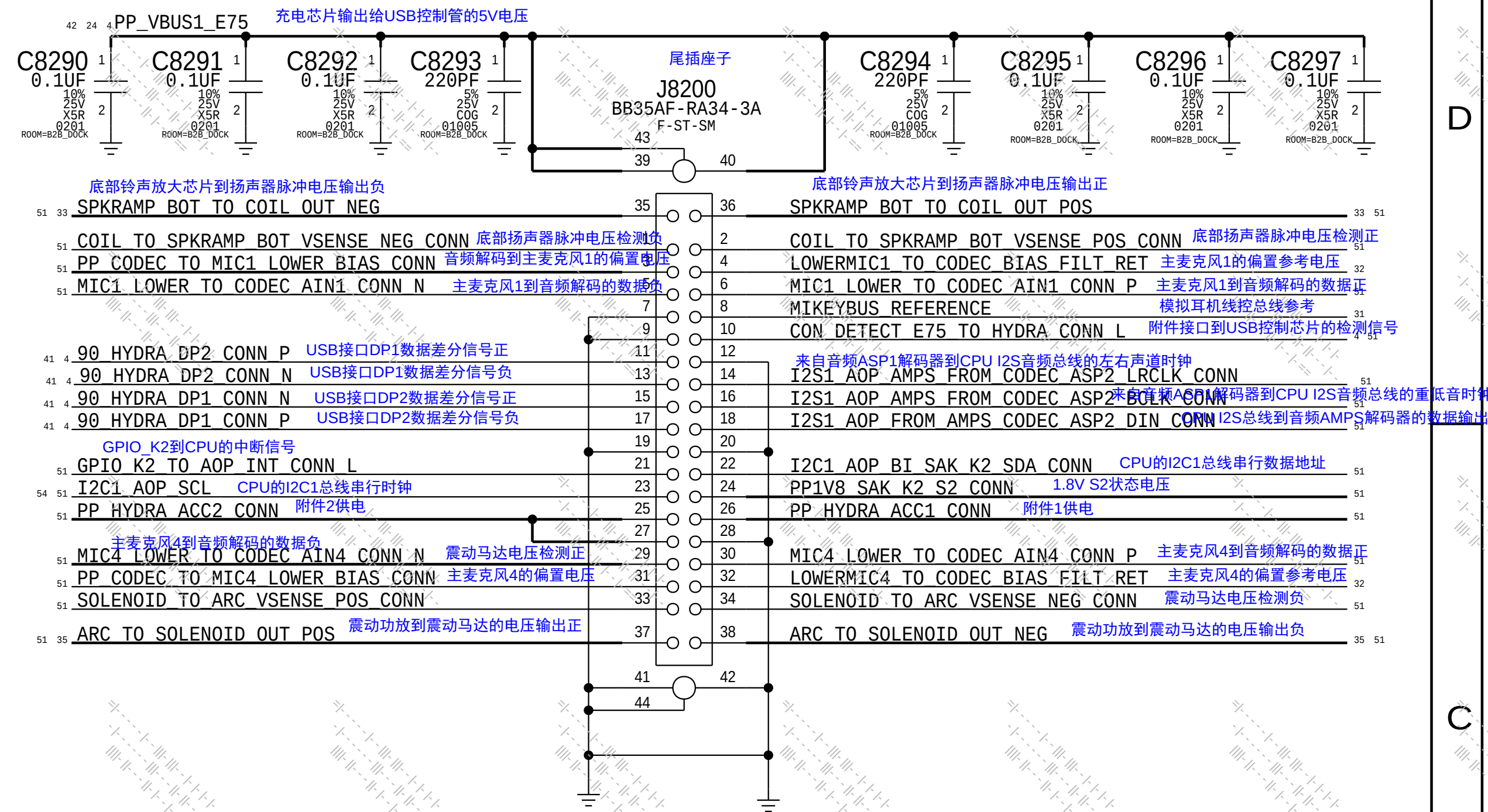
LOWER MIC1 + LOWER MIC4



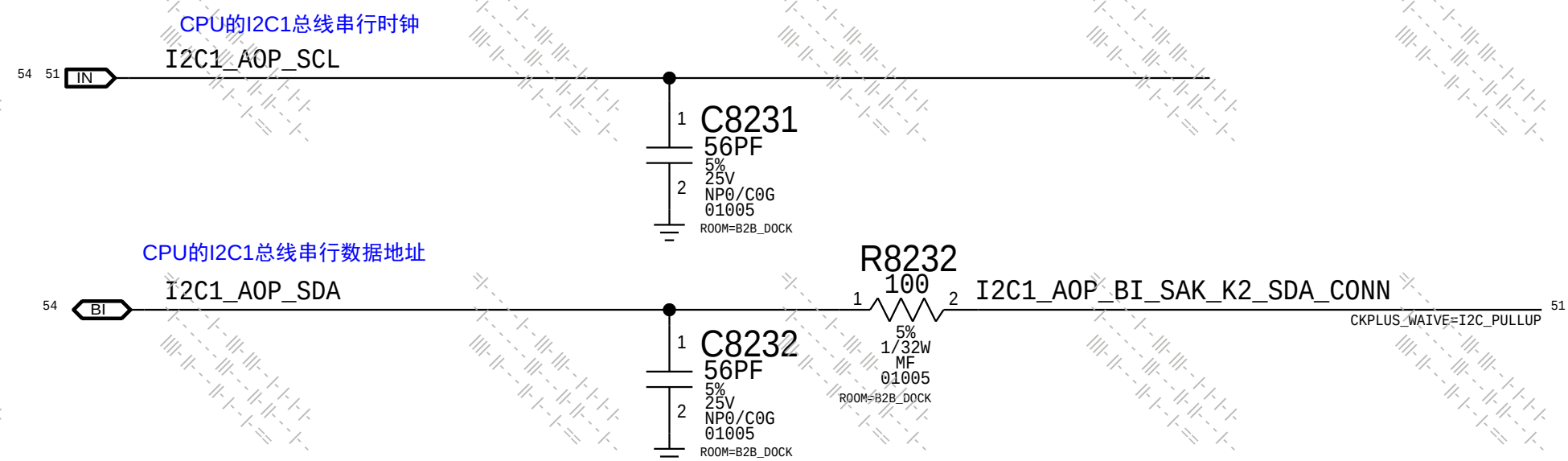
DOCK FLEX CONNECTOR

RCPT: 516S00467
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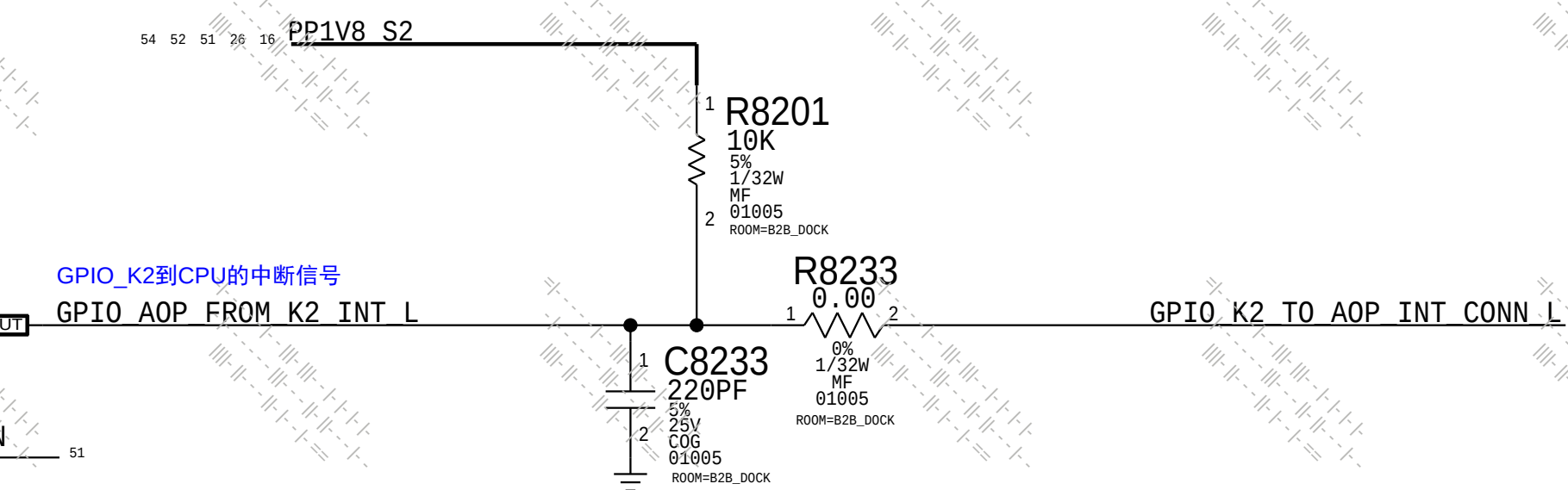
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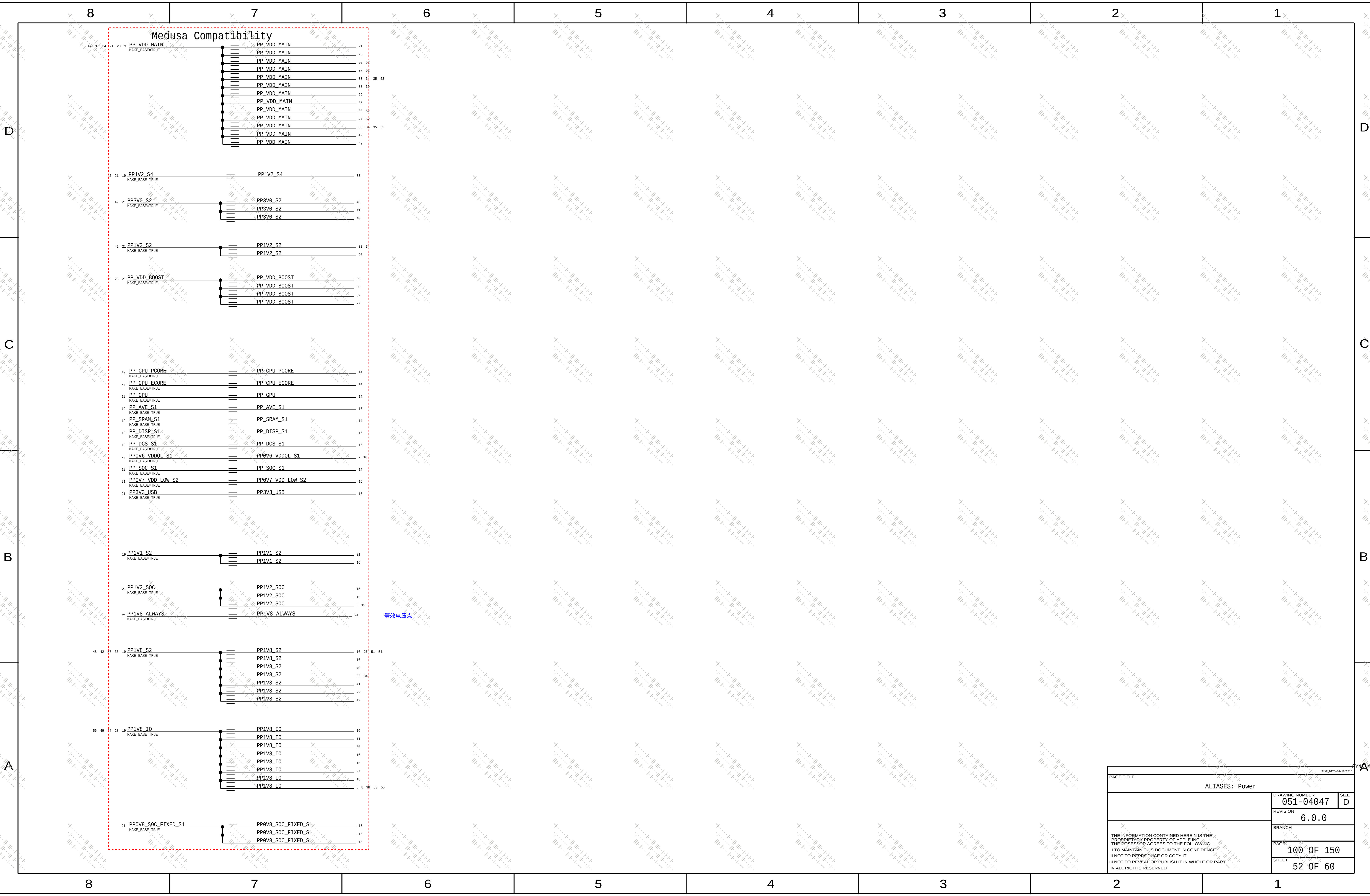
I2C



Potassium Interrupt PU



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AP I2C

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C0	PP1V8_I0	1 MHz	Muon	0x62	1100 010X	0xC4, 0xC5	-	1 MHz	TOP MLB

CPU的I2C1总线对照表

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C1	PP1V8_I0	100 kHz	MIC 麦克风2	0x56	1010 110X	0xAC, 0xAD	-	1 MHz	Strobe Flex

CPU的I2C2总线对照表

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C2	PP1V8_I0	400 kHz	Top Speaker Amp	0x40	1000 000X	0x80, 0x81	-	1 MHz	Top MLB
			Chestnut 显示电源	0x27	0100 111X	0x4E, 0x4F	-	400 kHz	Top MLB
			Muon 背光	0x62	1100 010X	0xC4, 0xC5	-	1 MHz	Top MLB

CPU的I2C3总线对照表

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C3	PP1V8_I0	400 kHz	Touch EEPROM	0x51	1010 001X	0xA2, 0xA3	-	1 MHz	Touch Flex
			Roswell 触摸	0X10	0010 000X	0X20, 0X21	-	400 kHz	Touch Flex

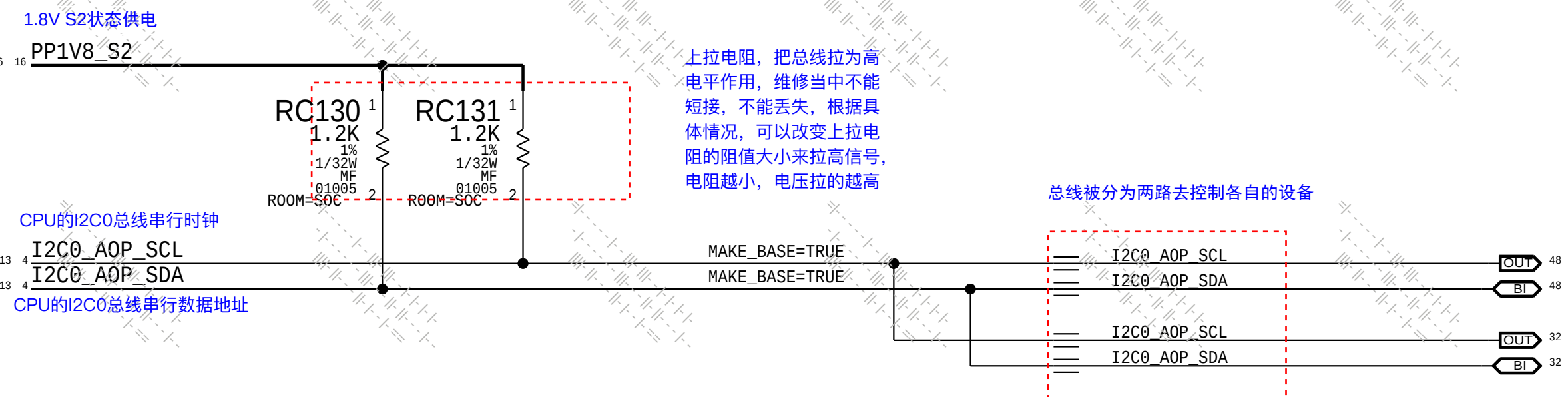
CPU的I2C4总线对照表

Bus Name	Bus Voltage	Bus Speed	Device	Max Speed	Location
AP I2C4	PP1V8_I0	400 kHz	LYNX 逻辑芯片	1 MHz	Top MLB

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AOP/SMC I2C

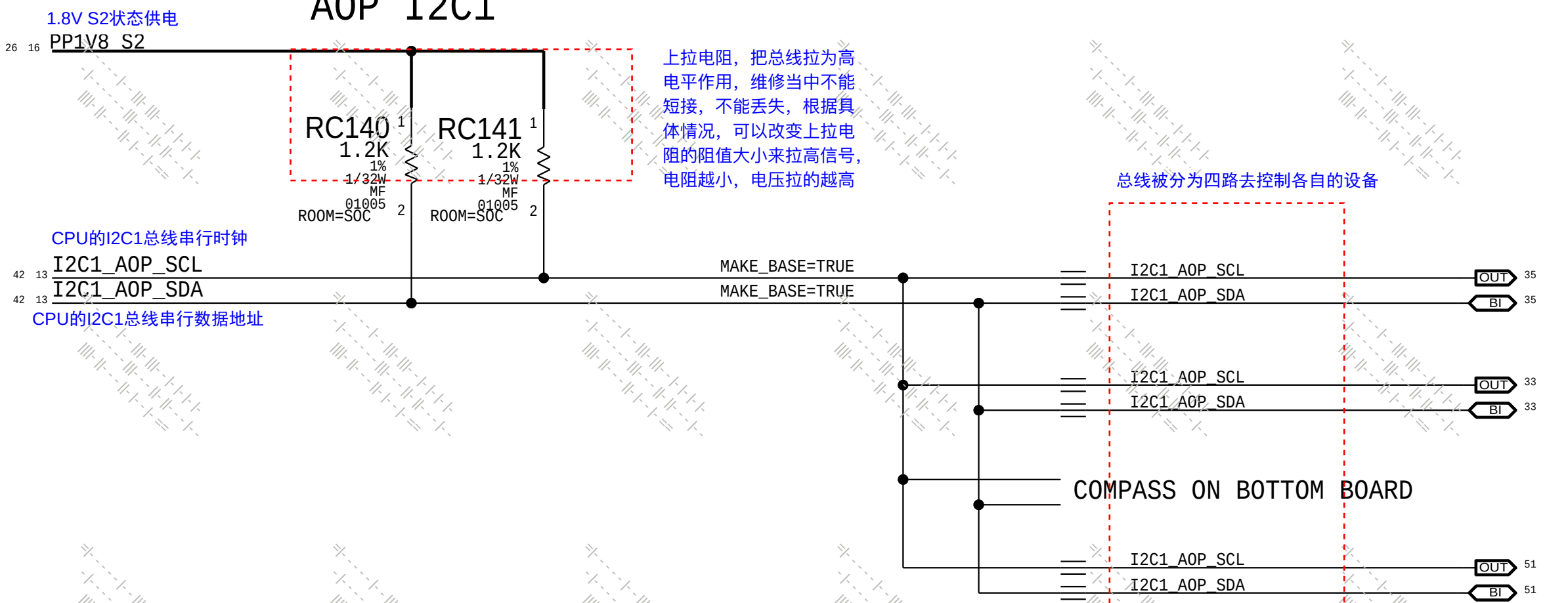
AOP I2C0



CPU的I2C0总线对照表

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AOP I2C0	PP1V8_S2	750 kHz	Doppler	泛光 0x58	1011 000X	0xB0, 0xB1	-	1 MHz	Sensor Flex
			Foxhound	音频 0x29	0101 001X	0x52, 0x53	-	1 MHz	Sensor Flex
			Yogi	人像 0x33	0110 011X	0x66, 0x67	-	1 MHz	Sensor Flex
			Brighton	感光 0x4A	1001 010X	0x94, 0x95	-	1 MHz	Top MLB

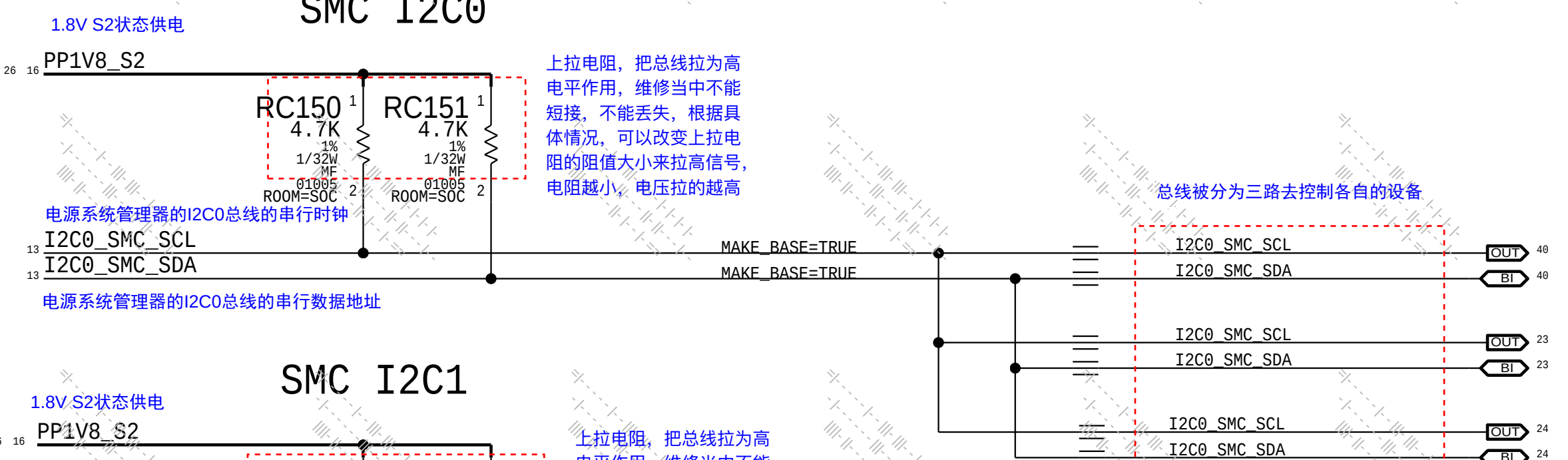
AOP I2C1



CPU的I2C1总线对照表

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AOP I2C1	PP1V8_S2	400 kHz	Arc Speaker Amp	震动功放 0x42	1000 010X	0x84, 0x85	-	1 MHz	Top MLB
			Bot Speaker Amp	底部扬声器 0x40	1000 000X	0x80, 0x81	-	1 MHz	Top MLB
			K2	尾插 0x76	1110 110X	0xEC, 0xED	-	1 MHz	Dock Flex
			Sakonnet	震动 0x08	0001 000X	0x10, 0x11	-	1 MHz	Arc Flex
			Moly	指南针 0x0E	0001 110X	0x1C, 0x1D	-	1 MHz	BTN Combine Flex
			Arc EEPROM	震动码 0x50	1010 000X	0xA0, 0xA1	-	1 MHz	Arc Flex

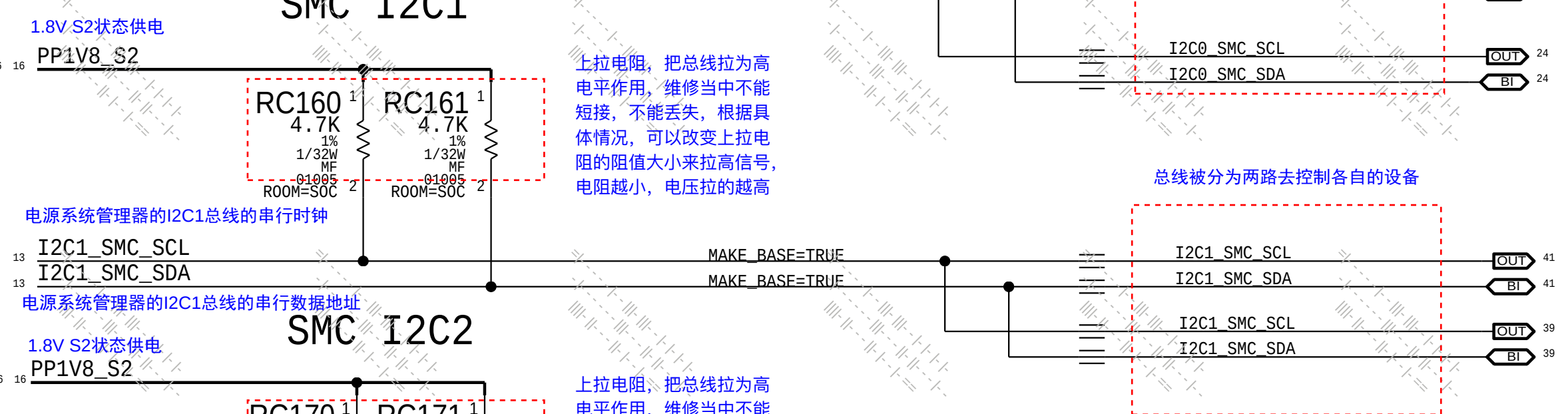
SMC I2C0



电源系统管理器的I2C0总线对照表

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
SMC I2C0	PP1V8_S2	400 kHz	CCG2	快充芯片 0x12	0010 010X	0x24, 0x25	-	1 MHz	Top Board
			Boost	第三端供电芯片 0x75	1110 101X	0xEA, 0xEB	-	1 MHz	Top Board
			Yangtze	充电芯片 0x71	1110 001X	0xE2, 0xE3	-	400 KHz	Top Board

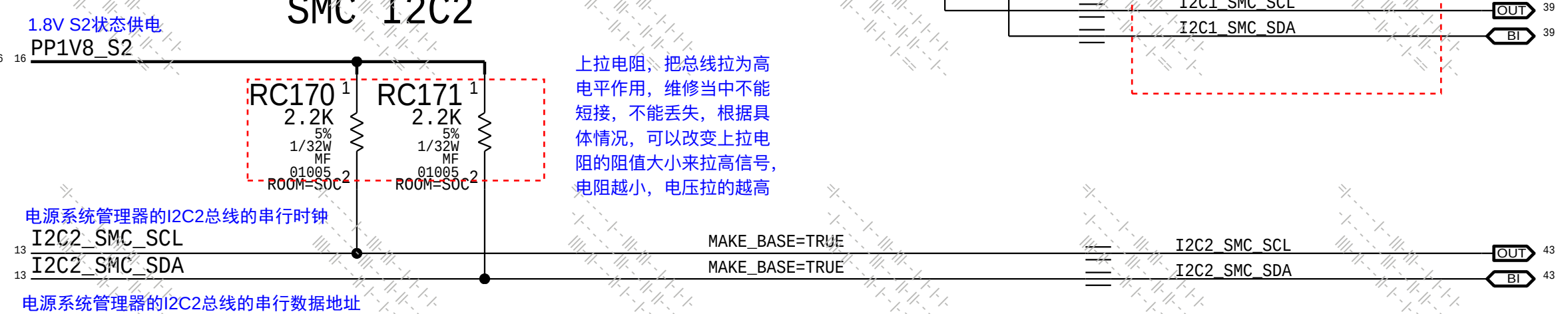
SMC I2C1



电源系统管理器的I2C1总线对照表

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
SMC I2C1	PP1V8_S2	400 kHz	Hydra	USB控制芯片 0x1A	0011 010X	0x34, 0x35	-	400 KHz	Top Board
			Gecko	附件供电芯片 0x52	1010 010X	0xA4, 0xA5	-	400 KHz	Top Board

SMC I2C2



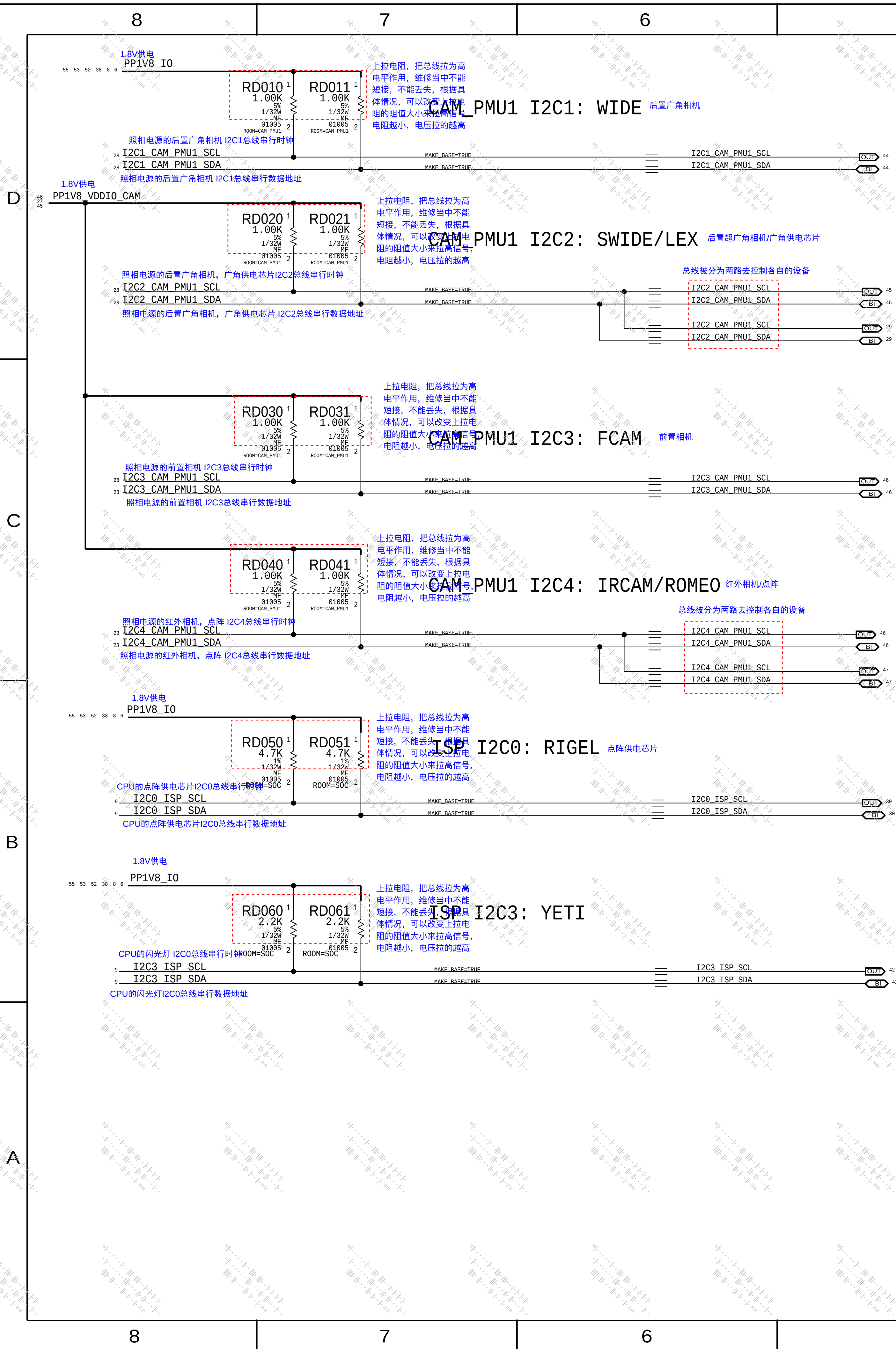
电源系统管理器的I2C2总线对照表

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Location
SMC i2c2	PP1V8_S2	400 kHz	Veridian	电池 0x0B	BMU Flex

ALIASES: I2C - AOP/SMC

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Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
I2C1	PP1V8_IO	1 MHz	Toronto	0X10	0010 000X	0x20, 0x21	-	1 MHz	Wide Cam
			Raman	0X3C	0111 100X	0x78, 0x79	-	1 MHz	Wide Cam

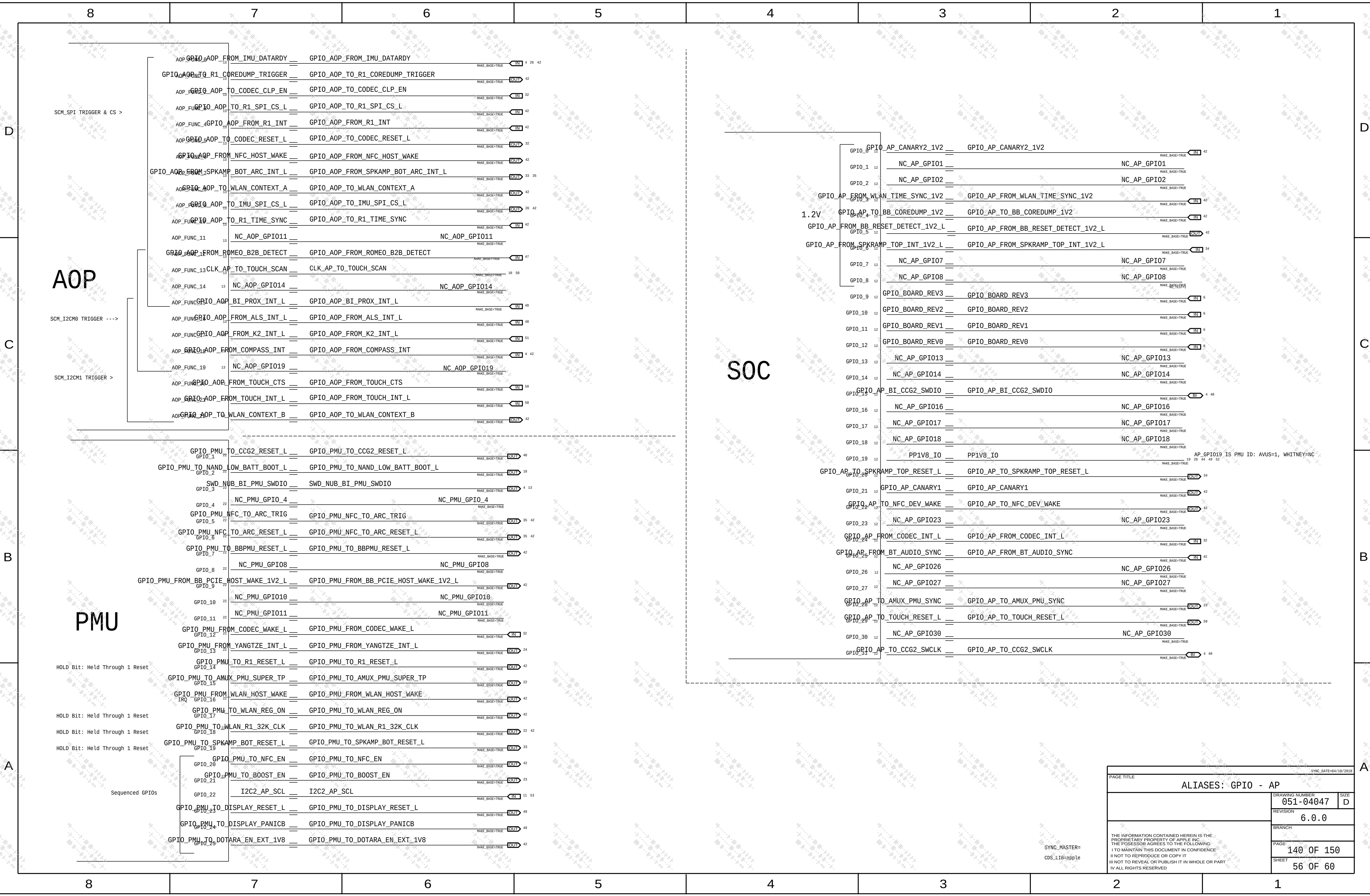
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
I2C2	PP1V8_I0_CAM	1 MHz	Manitoba	0x20	0100 000X	0x40, 0x41	-	1 MHz	SWIDE Cam
			Lex	0x75	1110 101X	0xEA, 0xEB	-	1 MHz	MLB_TOP

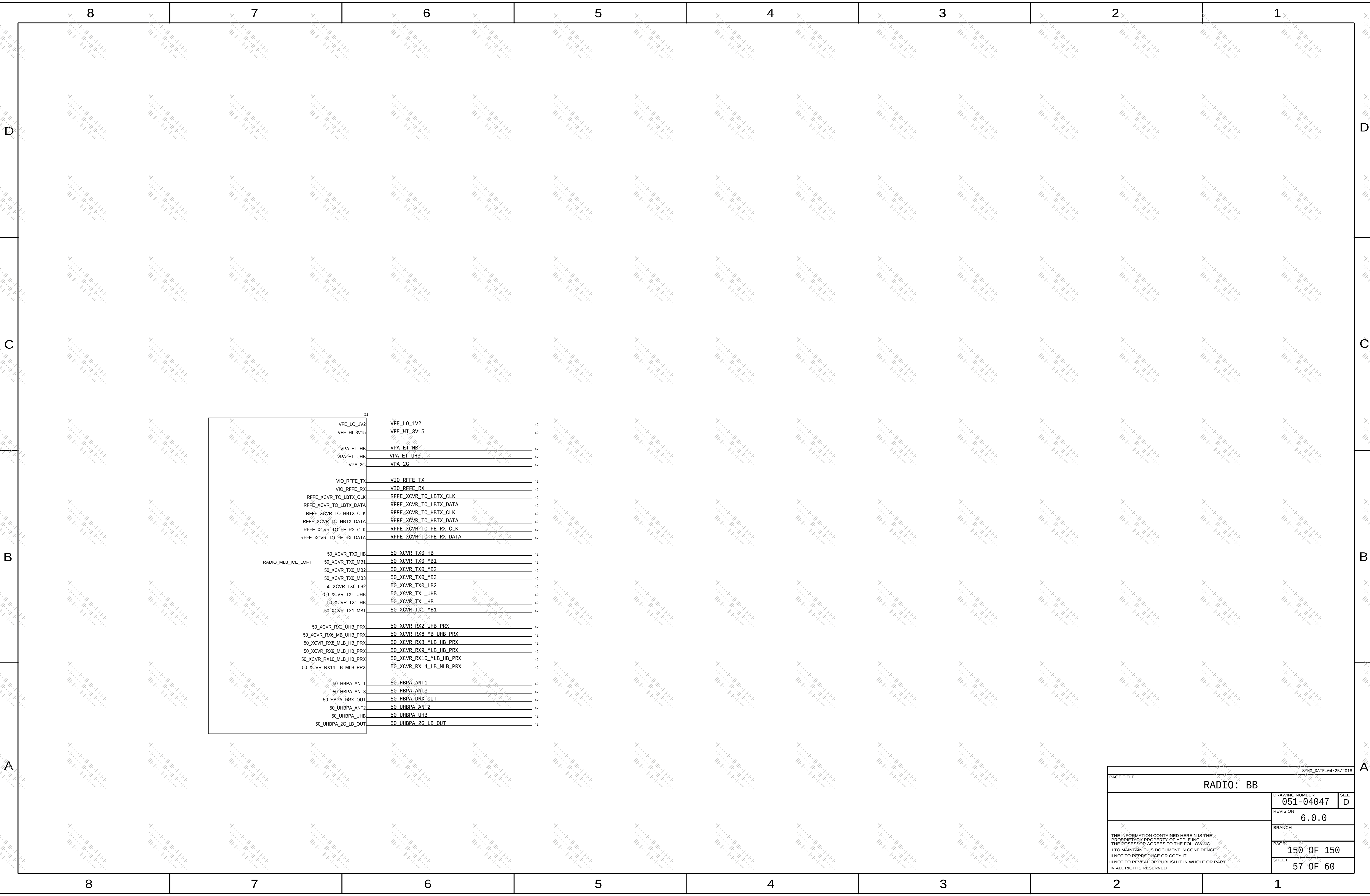
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
I2C3	PP1V8_I0_CAM	1 MHz	Maryland	0x10	0010 000X	0x20, 0x21	-	1 MHz	Front Cam

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
I2C4	PP1V8_IO_CAM	1 MHz	Juliet	0x18	0011 000X	0x30, 0x31	-	1 MHz	Top Board
			Romeo	0x66	1100 110X	0xCC, 0xCD	1 MHz	Top Board	

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C0	PP1V8_I0	1 MHz	Rigel	0x55	1010 101X	0xAA, 0xAB	-	1 MHz	Top Board

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C3	PP1V8_IO	1 MHz	Yeti	0x65	1100 101X	0xCA, 0xCB	-	1 MHz	Bot Board





11	
VFE_LO_1V2	VFE_LO_1V2
VFE_HI_3V15	VFE_HI_3V15
VPA_ET_HB	VPA_ET_HB
VPA_ET_UHB	VPA_ET_UHB
VPA_2G	VPA_2G
VIO_RFFE_TX	VIO_RFFE_TX
VIO_RFFE_RX	VIO_RFFE_RX
RFFE_XCVR_TO_LBTX_CLK	RFFE_XCVR_TO_LBTX_CLK
RFFE_XCVR_TO_LBTX_DATA	RFFE_XCVR_TO_LBTX_DATA
RFFE_XCVR_TO_HBTX_CLK	RFFE_XCVR_TO_HBTX_CLK
RFFE_XCVR_TO_HBTX_DATA	RFFE_XCVR_TO_HBTX_DATA
RFFE_XCVR_TO_FE_RX_CLK	RFFE_XCVR_TO_FE_RX_CLK
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LAST_MODIFICATION=Sun Apr 14 18:20:31 2019

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60	3	UHB MLB SPAD		04/06/2018

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60	57	10	VFE_HI_3V15
59	57	10	VPA_ET_HB
60	57	10	VPA_ET_UHB
60	57	10	VPA_2G
60	57	10	VIO_RFFE_TX
60	57	10	VIO_RFFE_RX
60	57	10	RFFE_XCVR_TO_LBTX_CLK
60	57	10	RFFE_XCVR_TO_LBTX_DATA
60	57	10	RFFE_XCVR_TO_HBTX_CLK
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60	57	10	50_XCVR_TX0_MB3
60	57	10	50_XCVR_TX0_LB2
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59	57	10	50_XCVR_RX6_MB_UHB_PRX
60	57	10	50_XCVR_RX8_MLB_HB_PRX
59	57	10	50_XCVR_RX9_MLB_HB_PRX
59	57	10	50_XCVR_RX10_MLB_HB_PRX
60	57	10	50_XCVR_RX14_LB_MLB_PRX
59	57	10	50_HBPA_ANT1
59	57	10	50_HBPA_ANT3
59	57	10	50_HBPA_DRX_OUT
60	57	10	50_UHBPA_ANT2
60	57	10	50_UHBPA_UHB
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等效信号线

CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
50_WIDE	50_WIDE	A_DIELECTRIC_4XV_50_WIDE_SE
50_WIDE	50_THIN	A_DIELECTRIC_4XD_50_WIDE_SE
50_THIN	50_THIN	A_DIELECTRIC_4XD_50_THIN_SE
50_WIDE_L1_THIN	50_WIDE	A_DIELECTRIC_4XV_50_WIDE_SE
50_WIDE_L1_THIN	50_THIN	A_DIELECTRIC_4XD_50_WIDE_L1_THIN_SE
RF_SHIELD	GND	DEFAULT
RF_SHIELD	RF_SHIELD	A_DIELECTRIC_2XD
RFFE_SHIELD	GND	DEFAULT
RFFE_SHIELD	RFFE_SHIELD	DEFAULT

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SCH, RADIO MLB ICE LOFT			
SCH, MLB, TOP, N104			
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HB SPAD

D

C

B

A

CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	CLASS COMMENT
CLASS NAME	CONSTRAINT SET		
RFFE_SHIELD	S	A_DIELECTRIC_2X	Y

CONSTRAINT SET	NET RULE ASSIGNMENT
P	PWR_100UM
P	PWR_200UM
P	PWR_SHAPE

CONSTRAINT SET	NET RULE ASSIGNMENT
P	A_50_THIN_SE
P	A_50_WIDE_SE
P	A_50_WIDE_L1_THIN_SE

CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	CLASS COMMENT
CLASS NAME	CONSTRAINT SET		
50_THIN	S	A_DIELECTRIC_2X	Y
50_WIDE	S	A_DIELECTRIC_2X_50_WIDE_SE	Y
50_WIDE_L1_THIN	S	A_DIELECTRIC_2X_50_WIDE_L1_THIN_SE	Y

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USID RX=0X0E
USID TX=0X02

高频功放

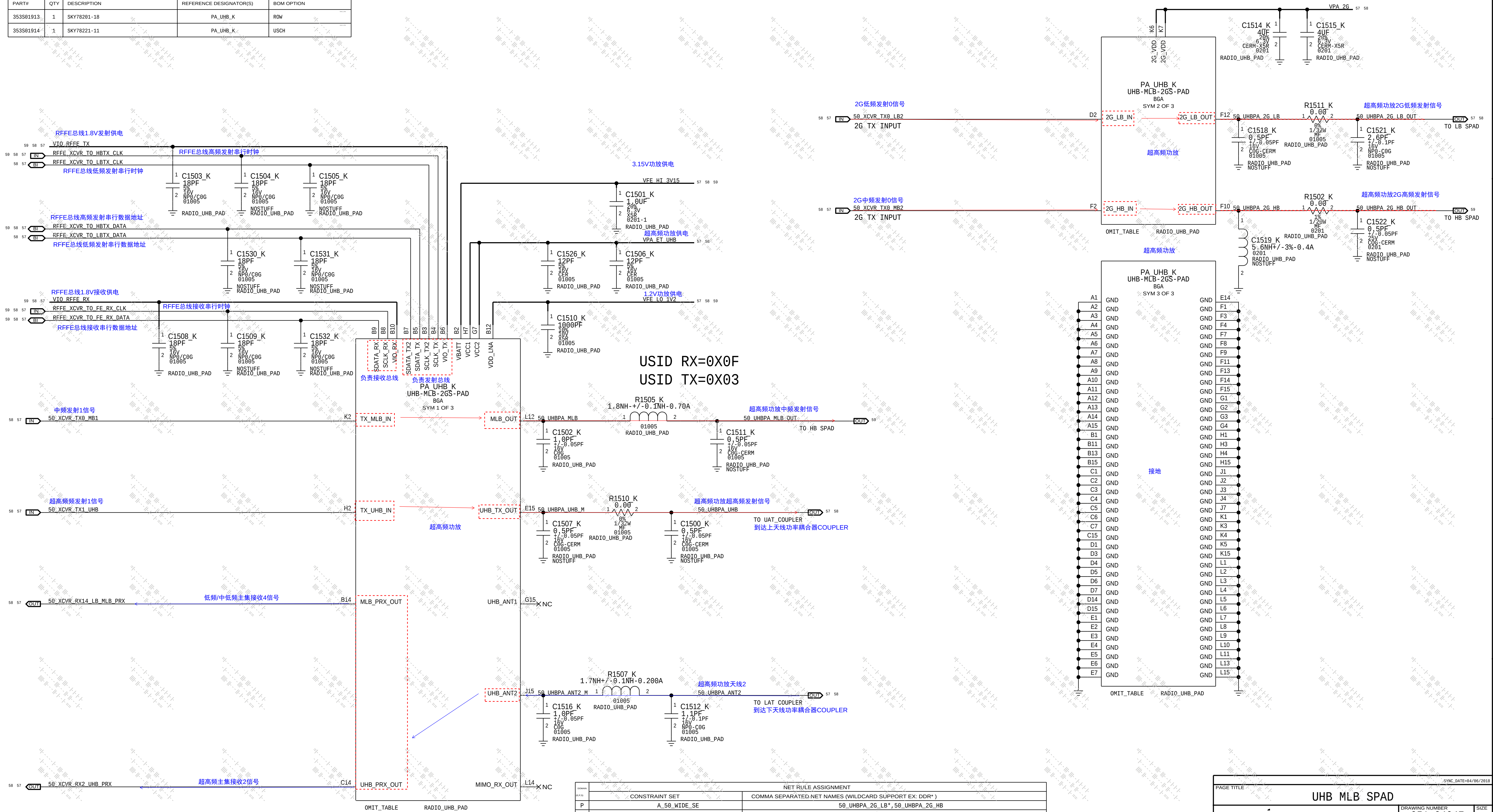
PA_HB_K
AFEM-8100-AP1
DSBGA
SYM 2 OF 2

接地

PAGE TITLE			HB SPAD		DRAWING NUMBER		SIZE	
					051-04047		D	
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UHB MLB SPAD

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S01913	1	SKY78201-18	PA_UHB_K	ROW
353S01914	1	SKY78221-11	PA_UHB_K	USCH



ID	NAME	NET RULE ASSIGNMENT	
		CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)
P	A 50_WIDE_SE		50_UHBP*_2G_LB*, 50_UHBP*_2G_HB
P	A 50_THIN_SE		50_UHBP*_MLB*, 50_UHBP*_2G_HB_OUT
P	A 50_WIDE_L1_THIN_SE		50_UHBP*_UHB*, 50_UHBP*_ANT*

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*		CLASS CHARACTER
CLASS NAME	UNIT	CONSTRAINT SET	DP NAMES EX: DP_DP_AA*,DP_BB* (LINE STARTS WITH FLAG DP:)		Y/N
50_WIDE	S	A DIELECTRIC_2X 50_WIDE_5E	50_UHPBA_2G_LB*	50_UHPBA_2G_HB	
50_THIN	S	A DIELECTRIC_2X	50_UHPBA_MLB*	50_UHPBA_2G_HB_OUT	Y
50_WIDE_L1_THIN	S	A DIELECTRIC_2X 50_WIDE_L1_THIN_5E	50_UHPBA_UHB*	50_UHPBA_ANT*	

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UHB MLB SPAD			
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Global Alternates

PART NUMBER	ALTEPART FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
132S00185	132S0316	ALT_PARTS	ALL	CAP, CER, RES, 0.1UF, 20%, 5V, SMD
138S0766	138S9739	ALT_PARTS	ALL	CAP, CER, RES, 100K, 500K, 100K, 500K, 100K, 500K
138S0049	138S0831	ALT_PARTS	ALL	CAP, CER, RES, 0.1UF, 20%, 5V, SMD, 100K, 500K
138S00133	138S00128	ALT_PARTS	ALL	CAP, CER, RES, 0.1UF, 20%, 5V, SMD, 100K, 500K
138S00269	138S00128	ALT_PARTS	ALL	CAP, CER, RES, 0.1UF, 20%, 5V, SMD, 100K, 500K
132S00232	132S00014	ALT_PARTS	ALL	CAP, CER, RES, 0.1UF, 20%, 5V, SMD, 100K, 500K
132S00233	132S00014	ALT_PARTS	ALL	CAP, CER, RES, 0.1UF, 20%, 5V, SMD, 100K, 500K
155S00414	155S0876	ALT_PARTS	ALL	FEER 80, 1500V, 20%, 100MA, 0.000C, 500MS, 10K
138S00246	138S00185	ALT_PARTS	ALL	FEER 80, 1500V, 20%, 100MA, 0.000C, 500MS, 10K
371S00108	371S00062	ALT_PARTS	ALL	FEER 80, 1500V, 20%, 100MA, 0.000C, 500MS, 10K
155S00200	155S00400	ALT_PARTS	ALL	FEER 80, 1500V, 20%, 100MA, 0.000C, 500MS, 10K
155S00194	155S00400	ALT_PARTS	ALL	FEER 80, 1500V, 20%, 100MA, 0.000C, 500MS, 10K
155S00400	155S00200	ALT_PARTS	ALL	FEER 80, 1500V, 20%, 100MA, 0.000C, 500MS, 10K
155S00194	155S00200	ALT_PARTS	ALL	FEER 80, 1500V, 20%, 100MA, 0.000C, 500MS, 10K
138S1103	138S0719	ALT_PARTS	ALL	CAP, CER, RES, 0.1UF, 20%, 5V, SMD, 100K, 500K

PARAMETER		NET RULE ASSIGNMENT	
CONSTRAINT SET		COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)	
P	A_50_THIN_SE	50_WLAN_G_TRX0_VOID_DTCK	
P	A_50_THIN_SE	50_ANT5_WLAN_G	
P	A_50_THIN_SE	50_UWB_ANT2_6G	

SYSTEM WIDE				
CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP:DP_AA*,DP_BB* (LINE STARTS WITH FLAG DP:)	
CLASS NAME	SEA	CONSTRAINT SET		
GND		S DEFAULT	GND	?
98_OHM	P	A_98_OHM_DIFF	DP:DP_*98*	?

SYSTEM WIDE					CLASS
CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT; NET NAMES EX: DDR*		Y/N
CLASS NAME	REF	CONSTRAINT SET	DP NAMES EX: DP_DP_AA*, DP_BB* (LINE STARTS WITH FLAG DP:)		
PWR_SHAPE	P	PWR_SHAPE	PP_VDD_MAIN	?	
PWR_SHAPE	P	PWR_50UM	PP1V0_S4, PP1V2_S2	?	
PWR_SHAPE	P	PWR_80UM	PP1V8_S4, PP1V8_S2	?	

SYSTEM OVERRIDES

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR* DP NAMES EX: DP-DP_AA*.DP_BB* (LINE STARTS WITH FLAG DP:)		CLASS OVERVIEW
CLASS NAME		CONSTRAINT SET			Y/N
50_THIN	S	A_DIELECTRIC_2X	50_WLAN_G_TRX0_V0ID_DTCK	Y	
50_THIN	S	A_DIELECTRIC_2X	50_ANT5_WLAN_G	Y	
50_THIN	S	A_DIELECTRIC_2X	50_UWB_ANT2_6G	Y	

CONSTRAINTS

DIELECTRIC BASED SPACING RULES		
RULE DEFINITION	LIST OF VALUES	
A_DIELECTRIC_INX (dielectric constant for inductor)	Constant 1.5, 1.6, 1.8, 2.0	2
A_DIELECTRIC_INXDX_VXX_X (dielectric constant for inductor and vias)	Constant 1.5, 1.6, 1.8, 2.0, 2.2, 2.4, 2.6, 2.8	2, 2D, 4D, 4V
A_DIELECTRIC_INXIN_OUTOUT (dielectric constant for inductor and vias)	Constant 1.5, 1.6, 1.8	?

RADIOS

HYBRID IMPEDANCE RULE			
TRACE LAYER	REFERENCE LAYER(S)	REQUIRED IMPEDANCE	TRACE WIDTH (OPTIONAL)
RULE NAME= 50_WIDE		ZONE NAME= PRIMARY	
TOP	ISL3	50	0.169
ISL6	ISL4,BOTTOM	50	0.095
ISL3	TOP, ISL4	50	0.44

HYBRID IMPEDANCE RULE			
TRACE LAYER	REFERENCE LAYER(S)	REQUIRED IMPEDANCE	TRACE WIDTH (OPTIONAL)
RULE NAME= 50 THIN		ZONE NAME= PRIMARY	
TOP	ISL2	50	0.061
ISL3	ISL2, ISL4	50	0.029
ISL5	ISL4, ISL6	50	0.044
ISL7	BOTTOM, ISL6	50	0.029

CAPPED RUL	LAYER	VALUE (MM)	RULE NAME(S)
EXAMPLE: SMD2L, SMD, TO, SMD, MOUNT, SHAPESAL	EXAMPLE: 2D, 2V, 3D, 3V, 3D, 3V, 3V		
LINE2SMD	?	0.070	2, 2D, 4D, 4V
MVIA2MWIA	?	=0.065	2, 2D, 4D, 4V
MVIA2SMD	?	=0.070	2, 2D, 4D, 4V
MVIA2SHAPE	?	=0.070	2, 2D, 4D, 4V
SMD2SMD	?	0.070	2, 2D, 4D, 4V
SMD2SHAPE	?	0.070	2, 2D, 4D, 4V
LINE2SHAPE	?	0.165	2
LINE2MWIA	?	0.165	2

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*		CLASS OVERVIEW
CLASS NAME	...	CONSTRAINT SET	DP NAMES EX: DP:DP_AA*,DP_BB* (LINE STARTS WITH FLAG DP:)		Y/N
E_PCIE_WLAN	E	PCIE_B0T	DP:DP_*	"90_PCIE*" WLAN	?
E_PCIE_BB	E	PCIE_B0T	DP:DP_*	"90_PCIE*" BB*	?
E_GENERIC	E	GENERIC_DP	DP:DP_*	"NTC", DP_50_ET*	?
E_DIGRF	E	DIGRF_DP	DP:DP_90	DIGRF*	?

Global Capacitors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138500138	138500139	ALT_PARTS	ALL	CAP, 4.7UF, 20%, 4V, KYO, 8201
138500164	138500139	ALT_PARTS	ALL	CAP, 4UF, 20%, 4V, TY, 8201

CRITICAL PART#	COMMENT
138S00139	CAP, X5R, 4.0UF, 20%, 4V, MUR, 0201

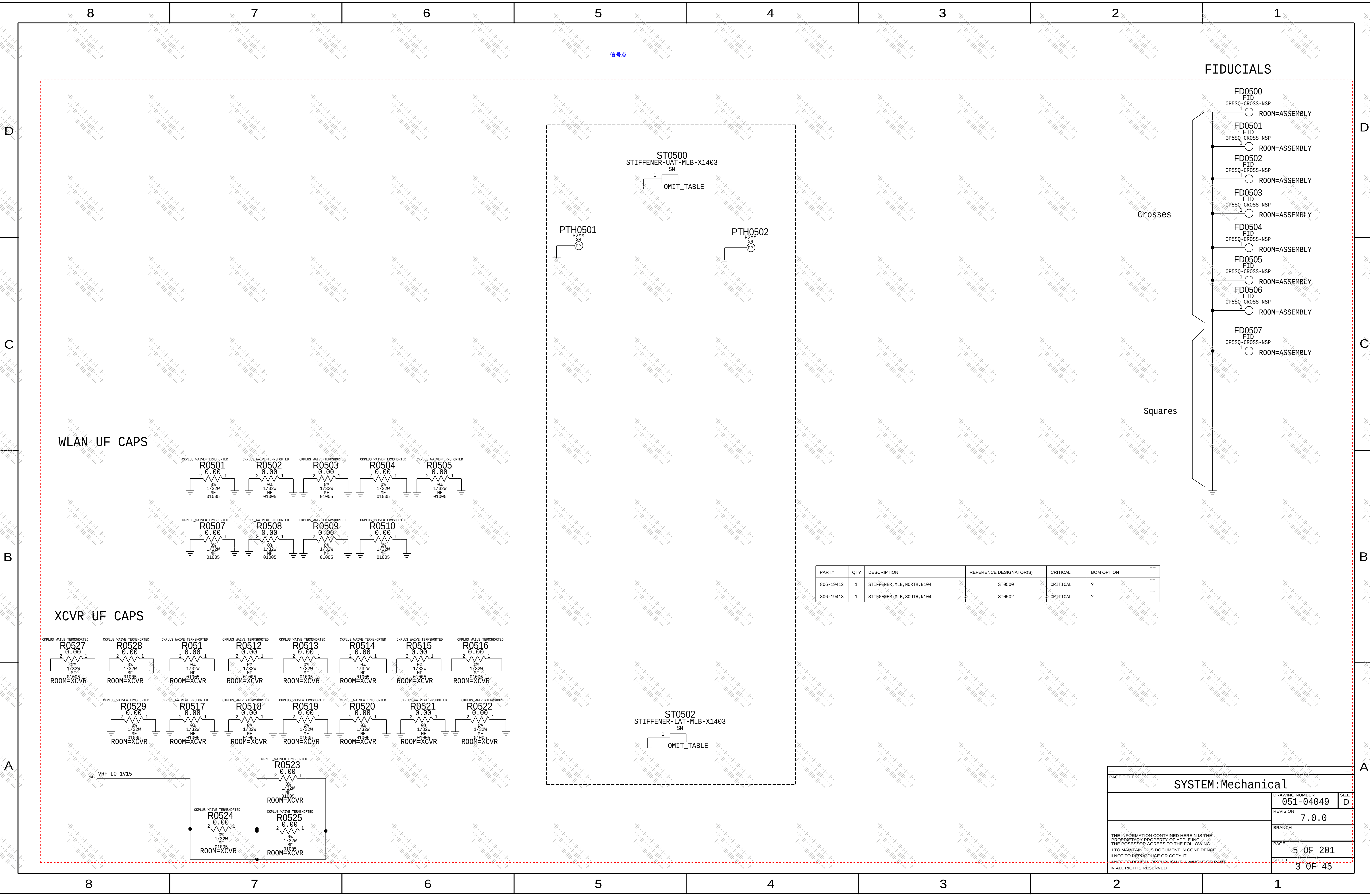
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS
138S00048	138S00003	ALT_PARTS	ALL	0402, 150F, 6.3V, Kyocera

CRITICAL PART#	COMMENT
138S00003	0402, 15uF, 6.3

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00128	138S00133	ALT_PARTS	ALL	01005, 0.470F, 6.3V, Murata
138S00269	138S00133	ALT_PARTS	ALL	01005, 0.470uF, 6.3V, TY

CRITICAL PART#	COMMENT
138S00133	01005, 0.47uF, 6.3V

PAGE TITLE		SYSTEM:BOM Tables	
		DRAWING NUMBER	SIZE
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		7.0.0	
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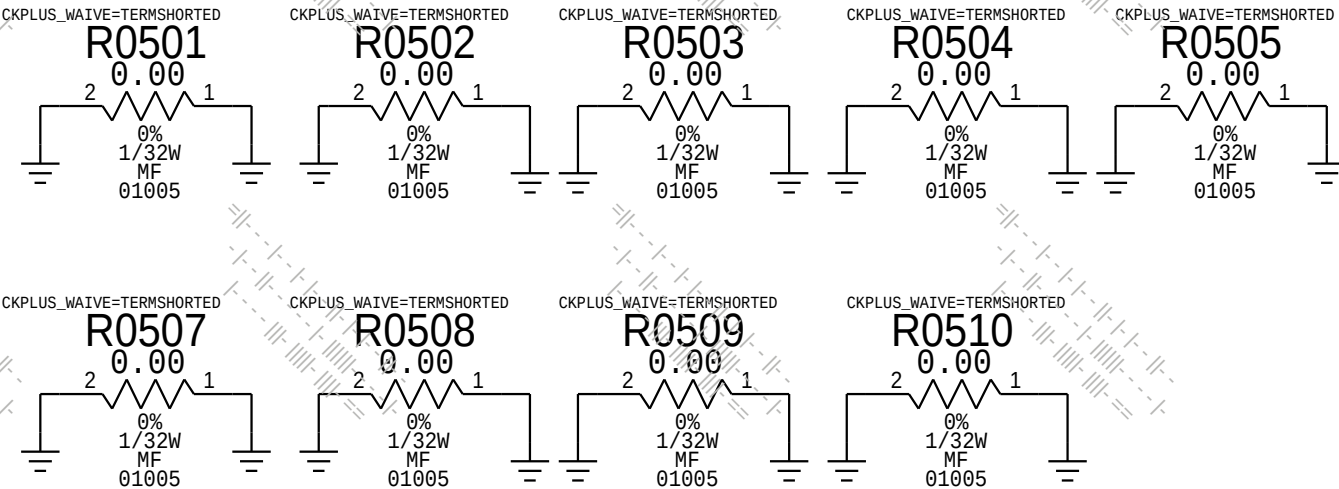
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FIDUCIALS

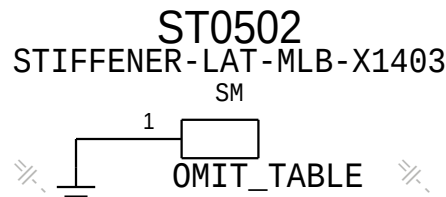
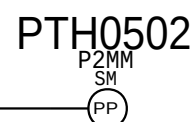
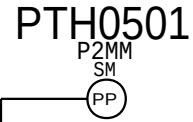
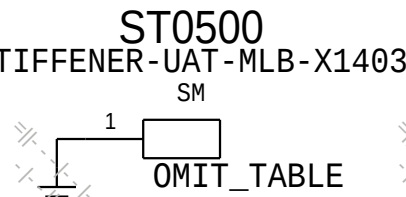
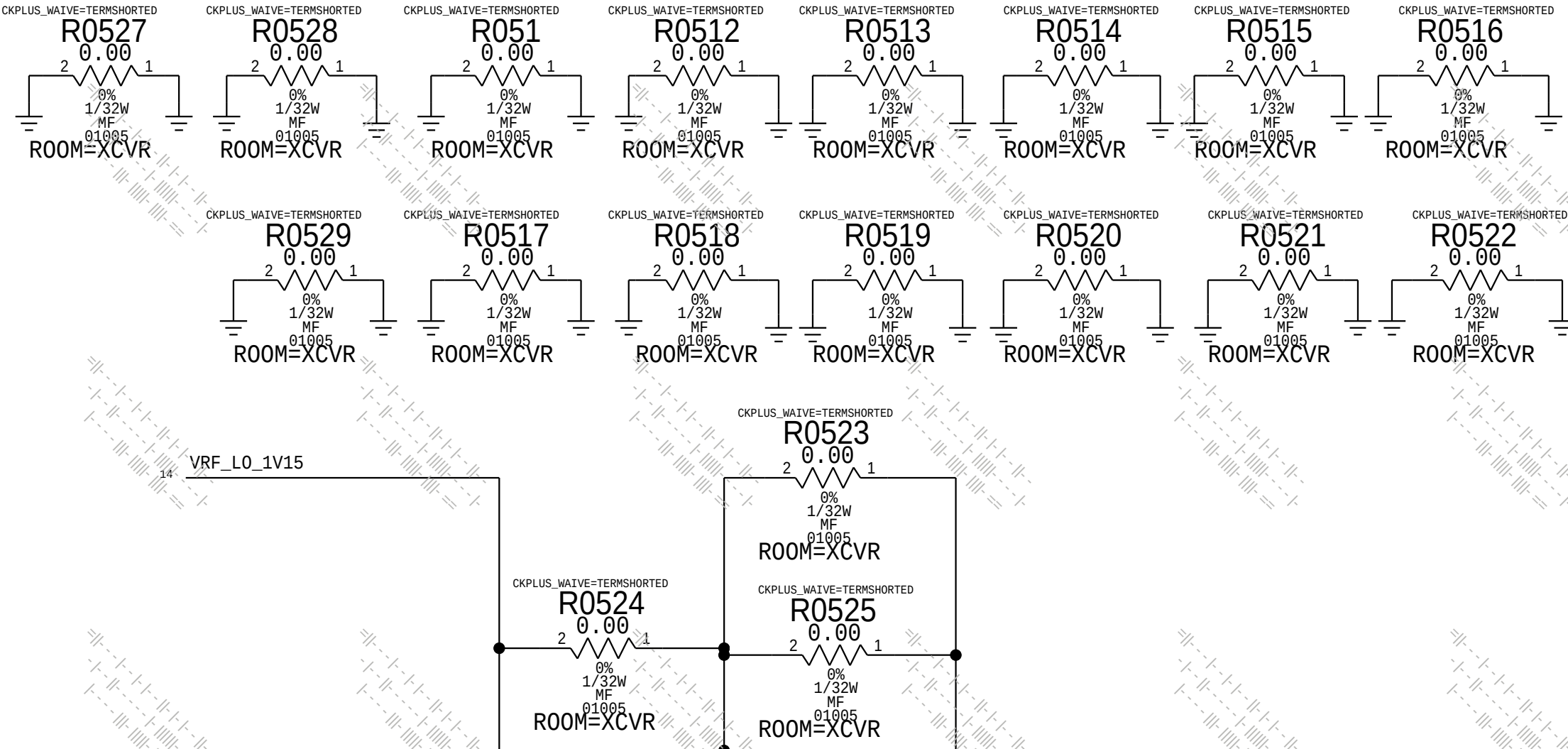
Crosses

Squares

WLAN UF CAPS



XCVR UF CAPS



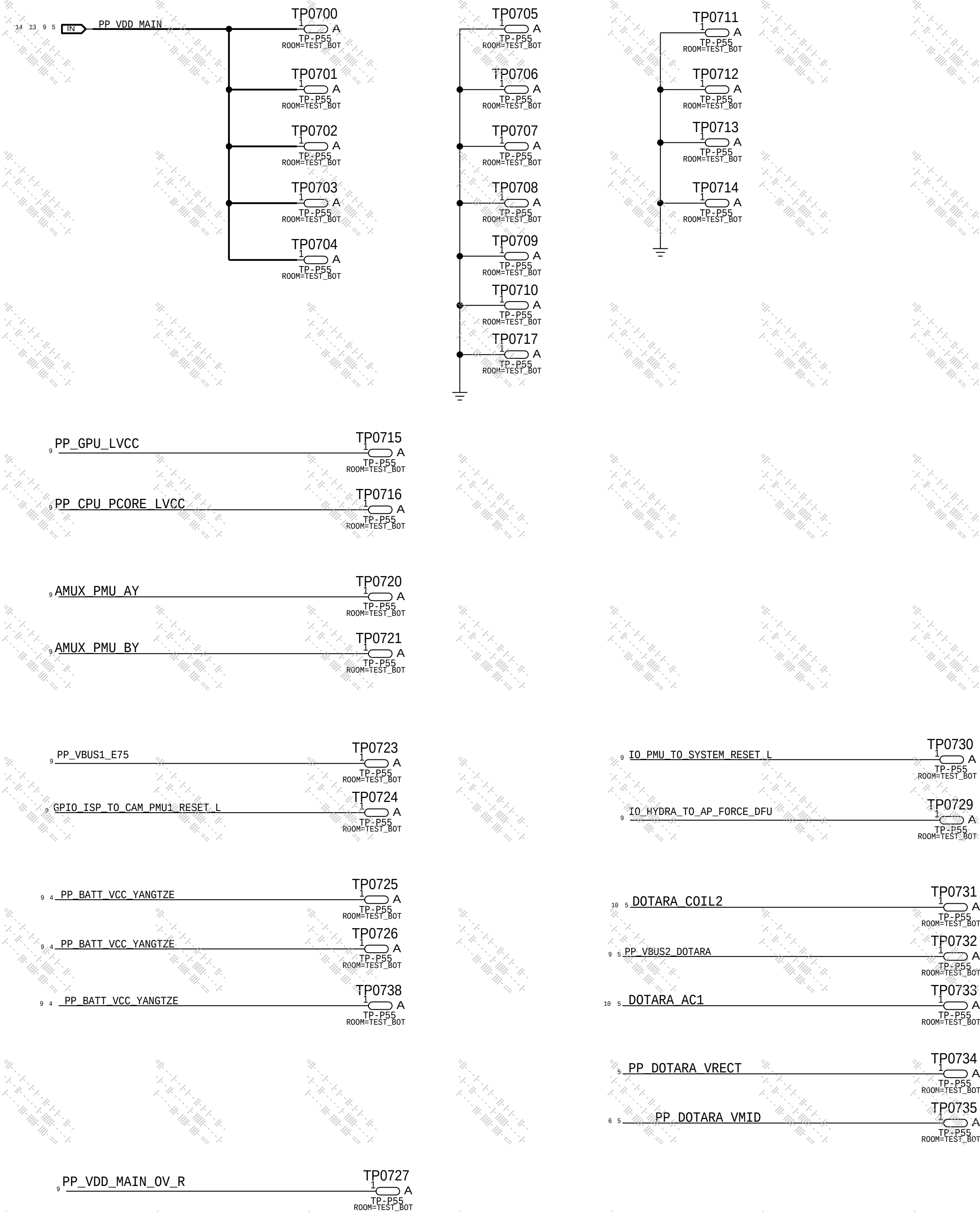
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
806-19412	1	STIFFENER, MLB, NORTH, N104	ST0500	CRITICAL	?
806-19413	1	STIFFENER, MLB, SOUTH, N104	ST0502	CRITICAL	?

PAGE TITLE		
SYSTEM:Mechanical		
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Test/Probe Points - BOTTOM

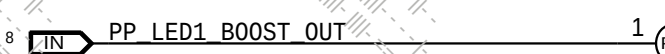
测试点

Test Points

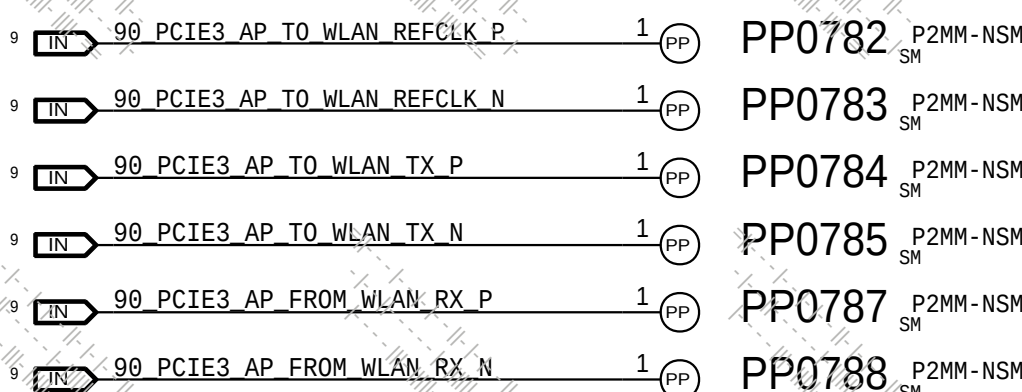


PROBE POINTS

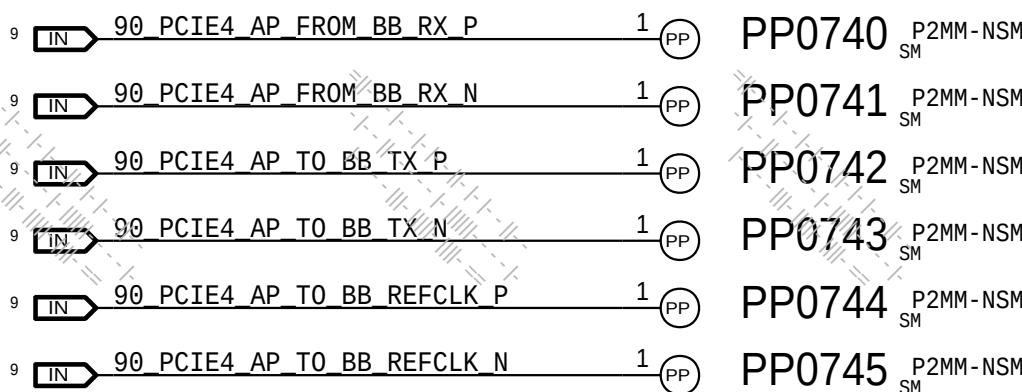
YETI Boost



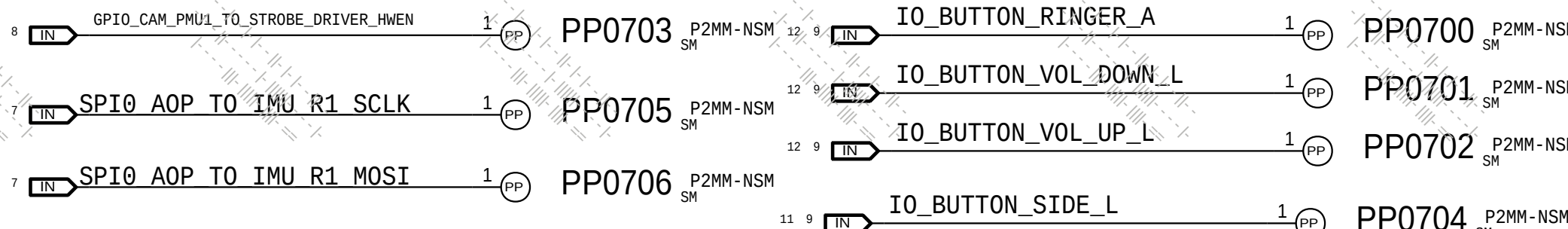
WLAN PCIe



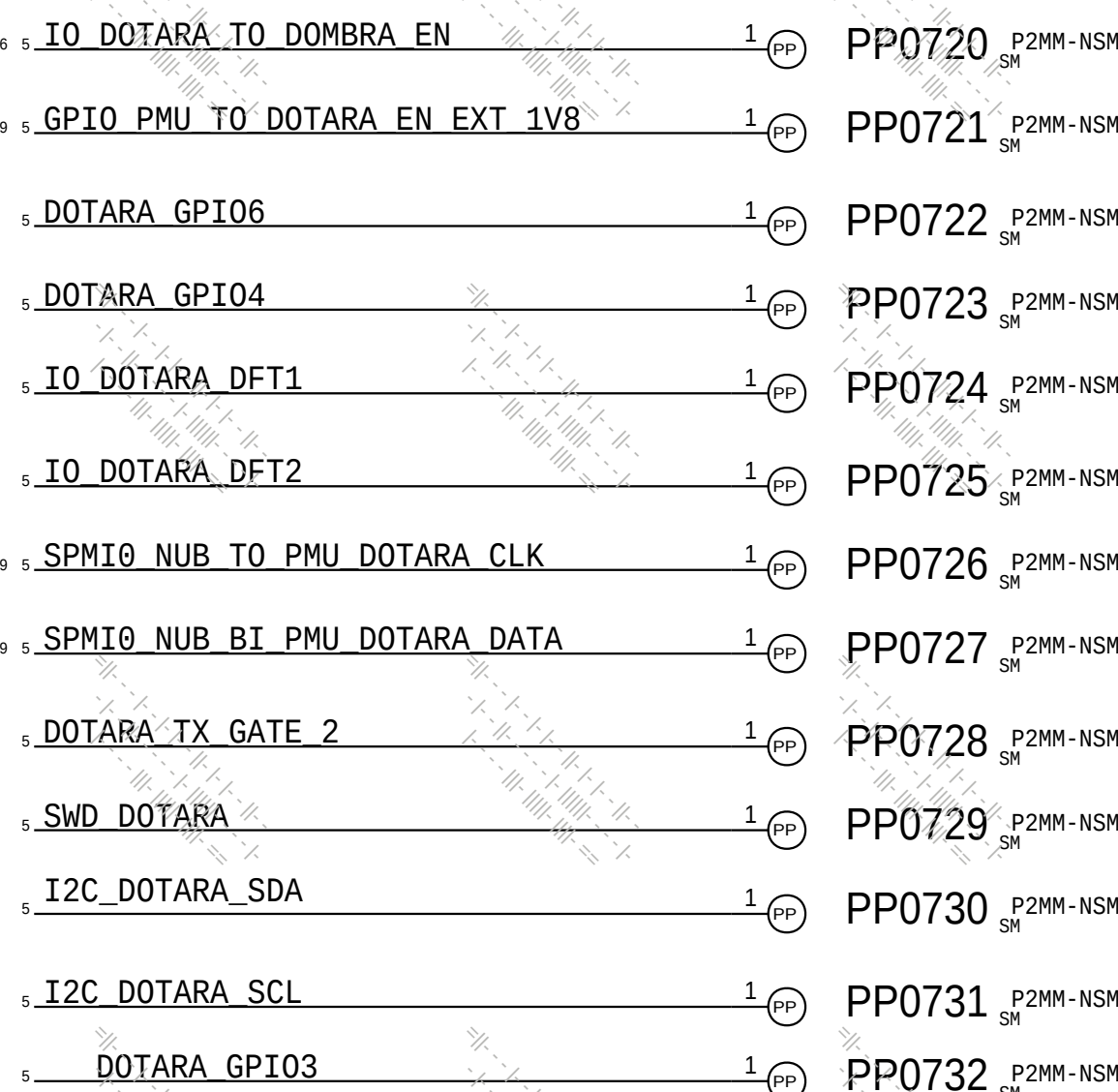
BB PCIe



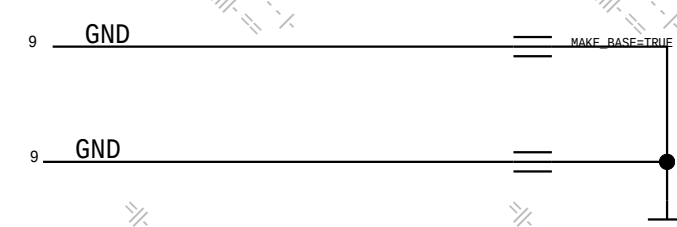
VALIDATION



Dotara



GROUND CANARIES



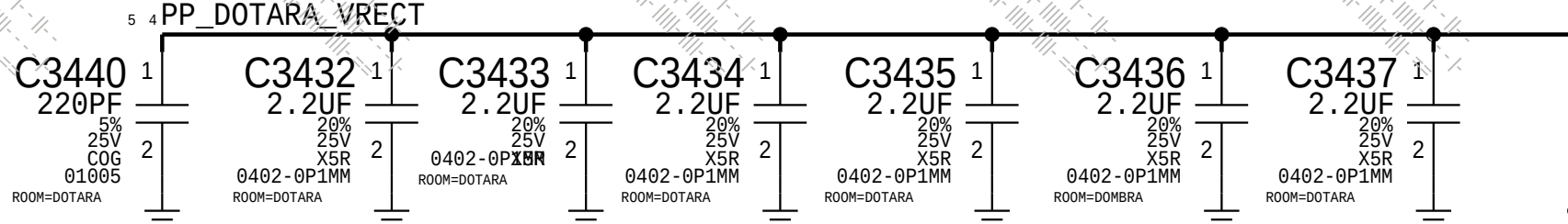
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DRAWING NUMBER	051-04049	SIZE
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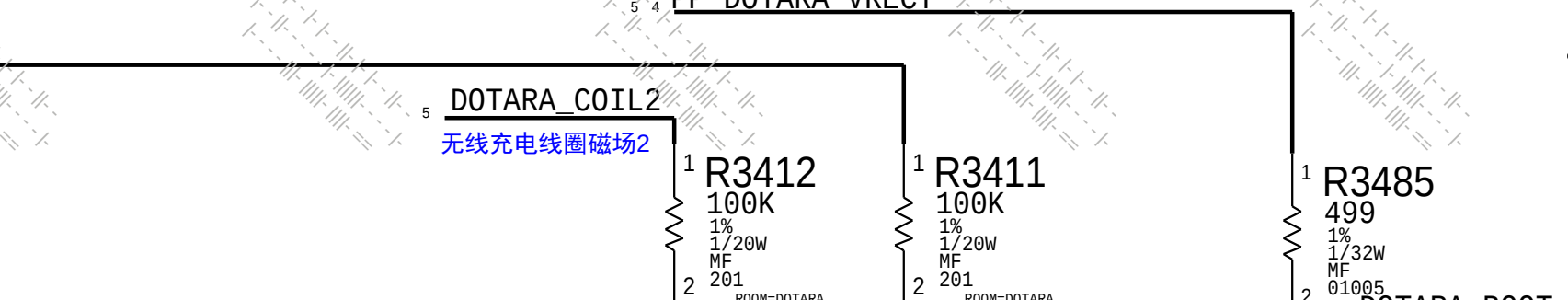
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138S00246	138S00185	ALT_PARTS	ALL	CAP, X5R, 2.2UF, 20%25V, 0402
376S00362	376S00295	ALT_PARTS	ALL	-DNSEMI, TX FET

CRITICAL PART#	COMMENT
138S00185	CAP, X5R, 14UF, 20%, 4V, MUR, 0402

19V无线充电VRECT供电



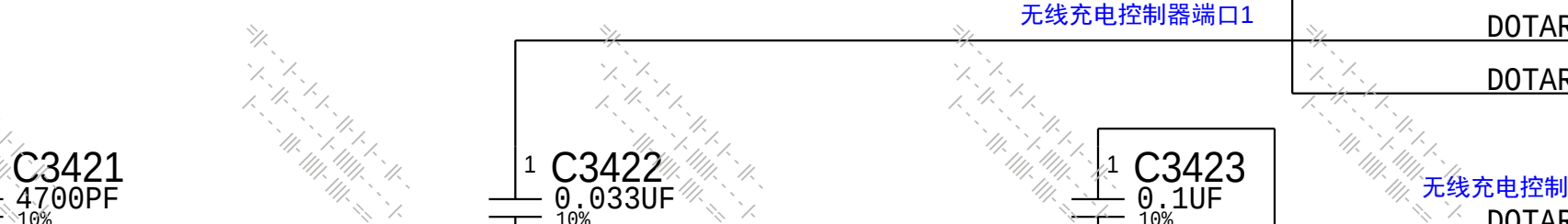
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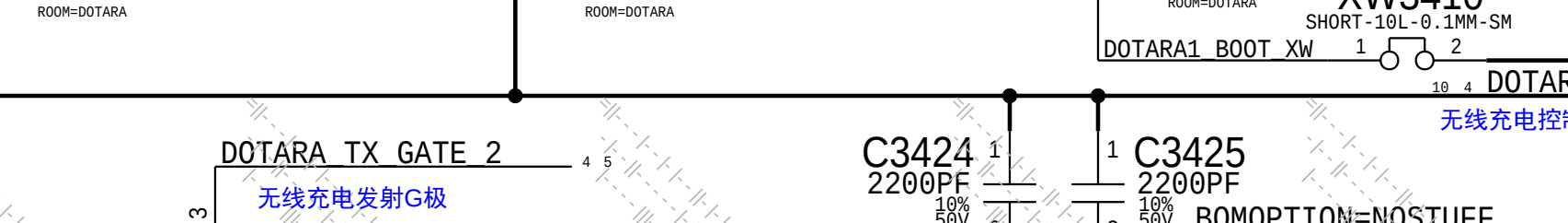
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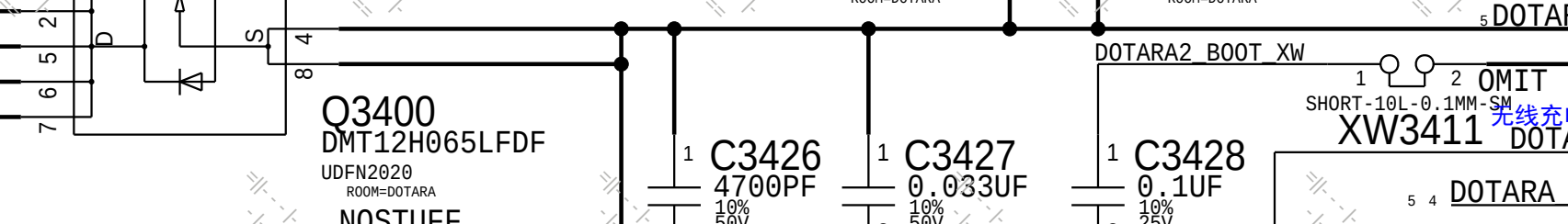
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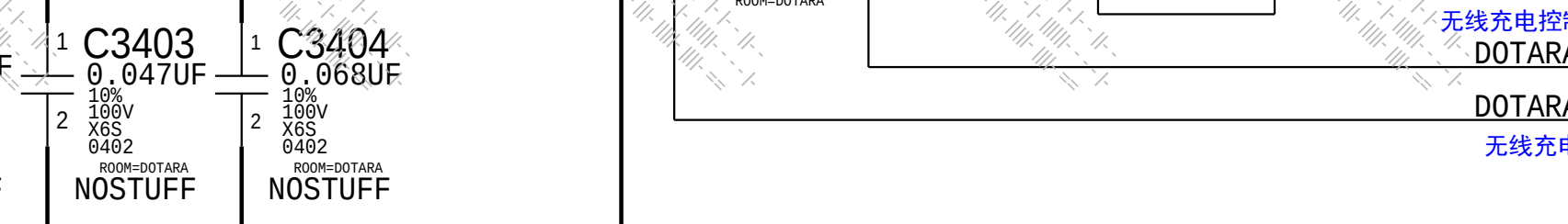
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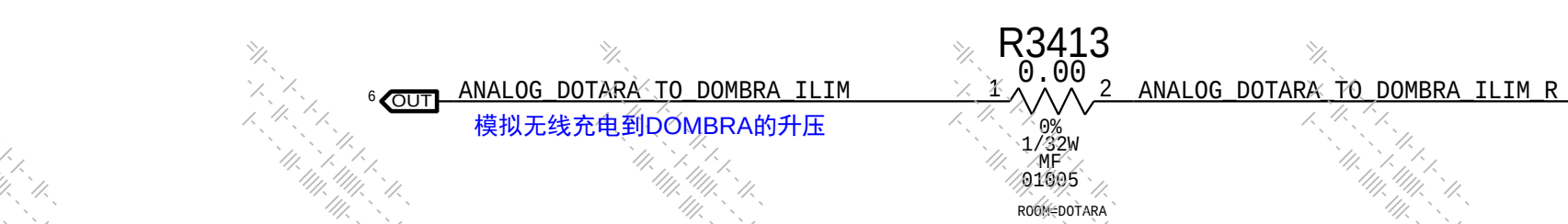
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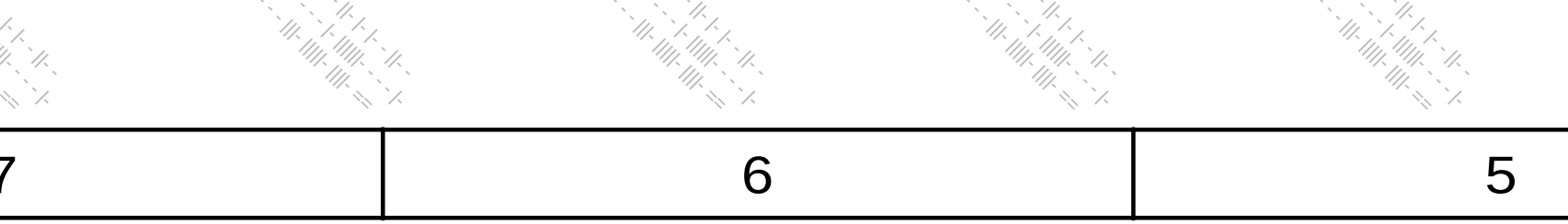
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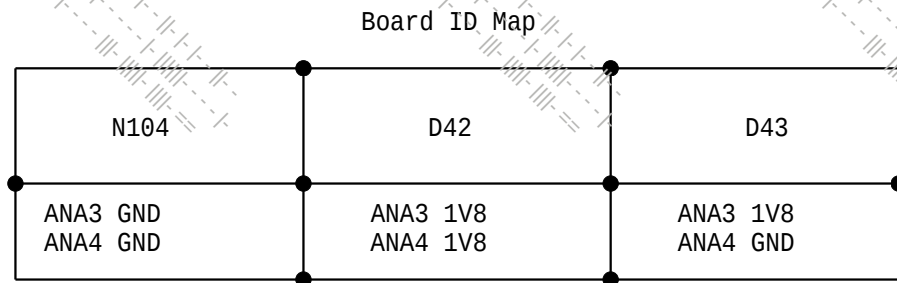
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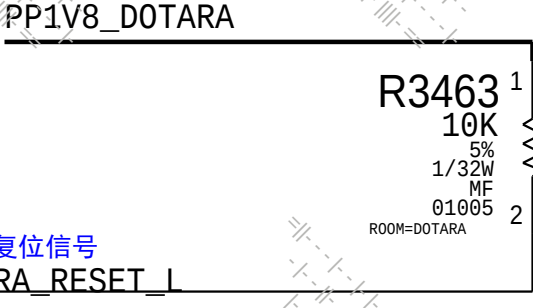
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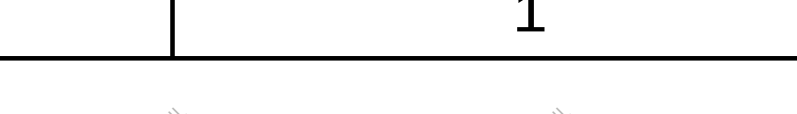
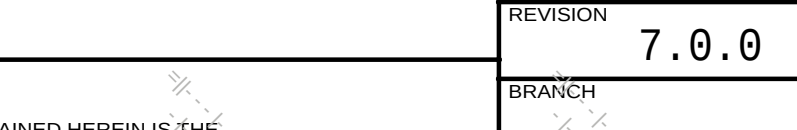
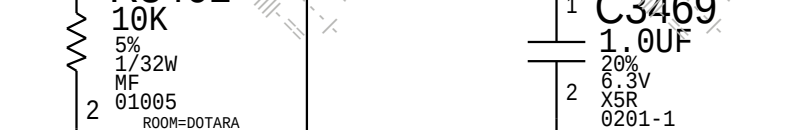
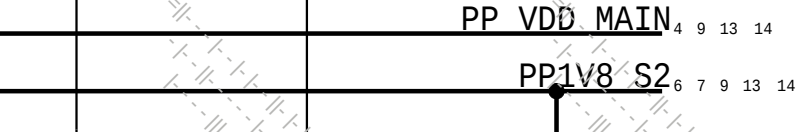
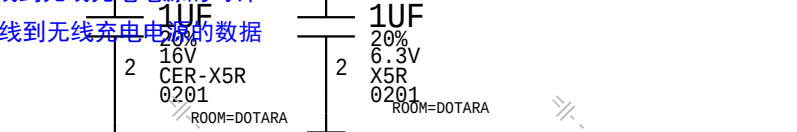
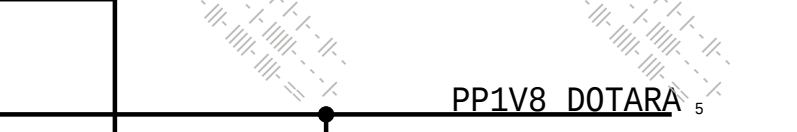
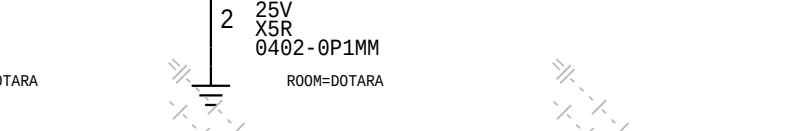
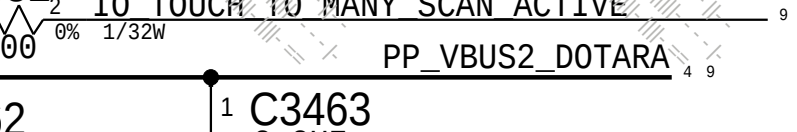
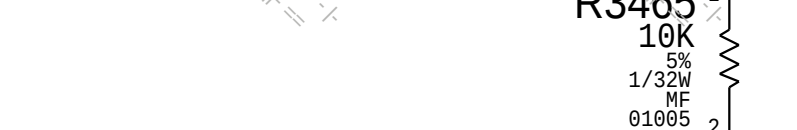
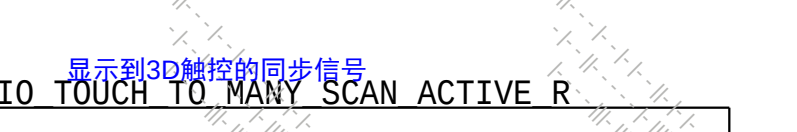
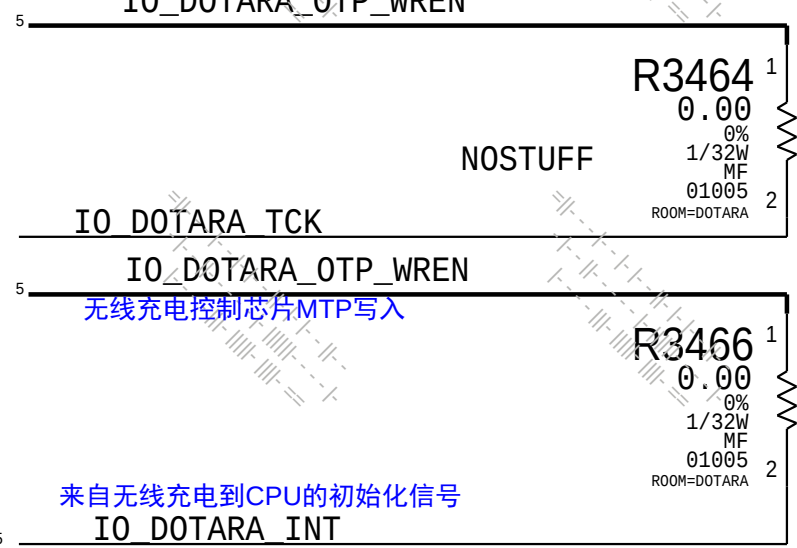
19V无线充电VRECT供电



Pull-Ups



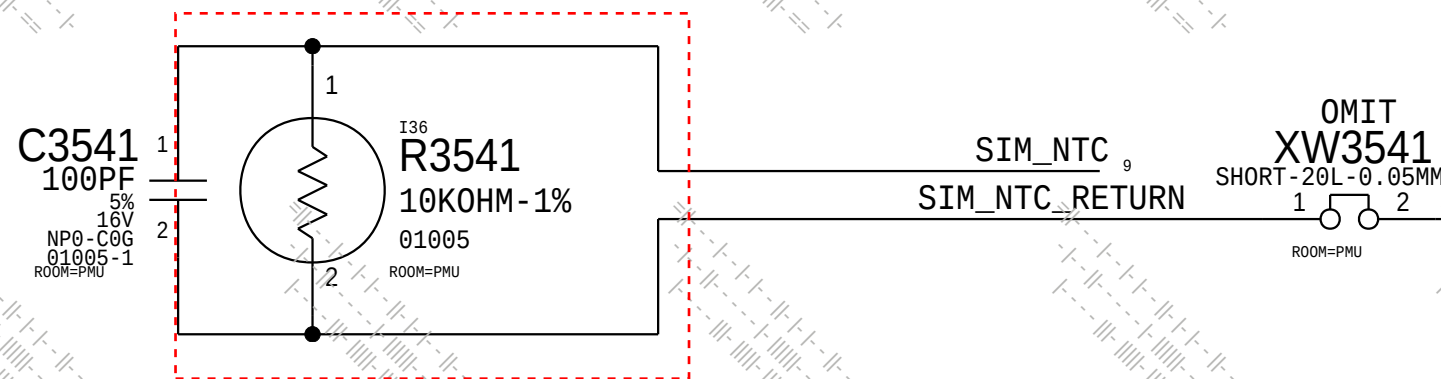
MTP



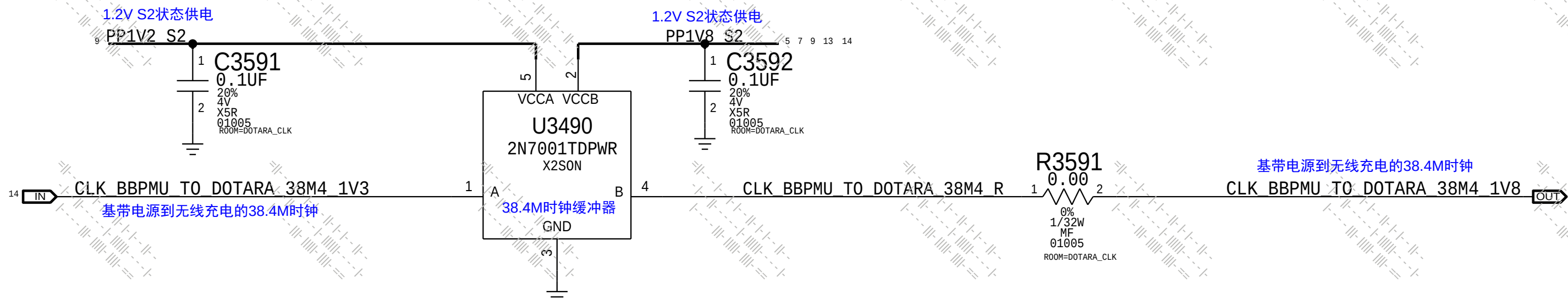
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SIM NTC

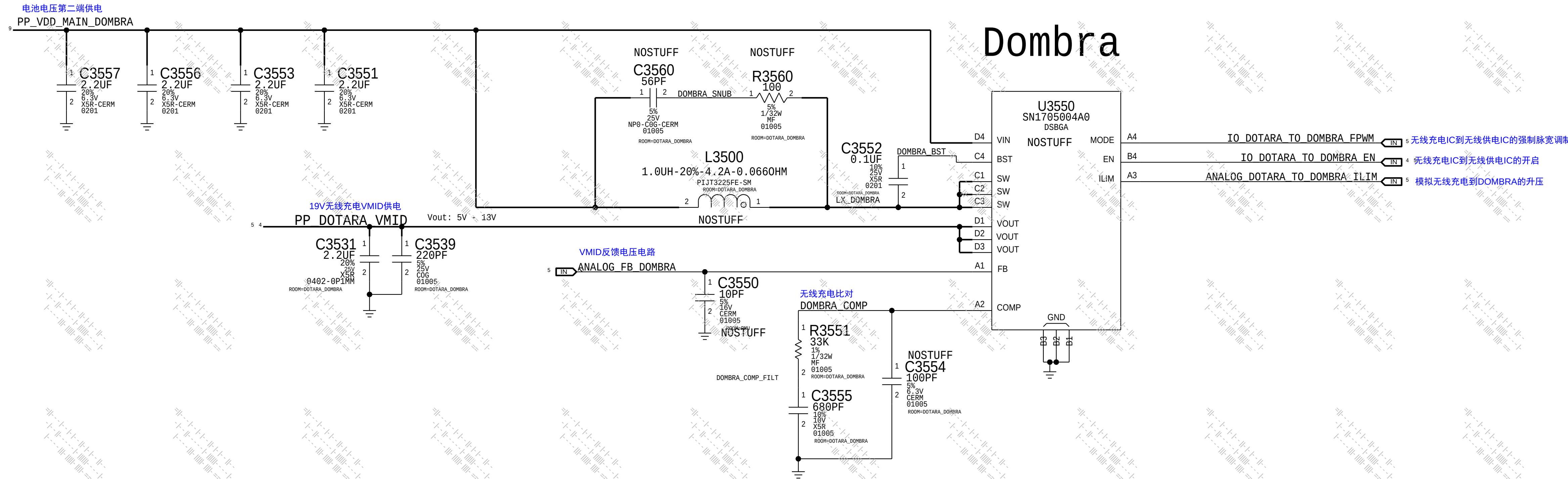
SIM卡温度检测



Dotara Clock Buffer



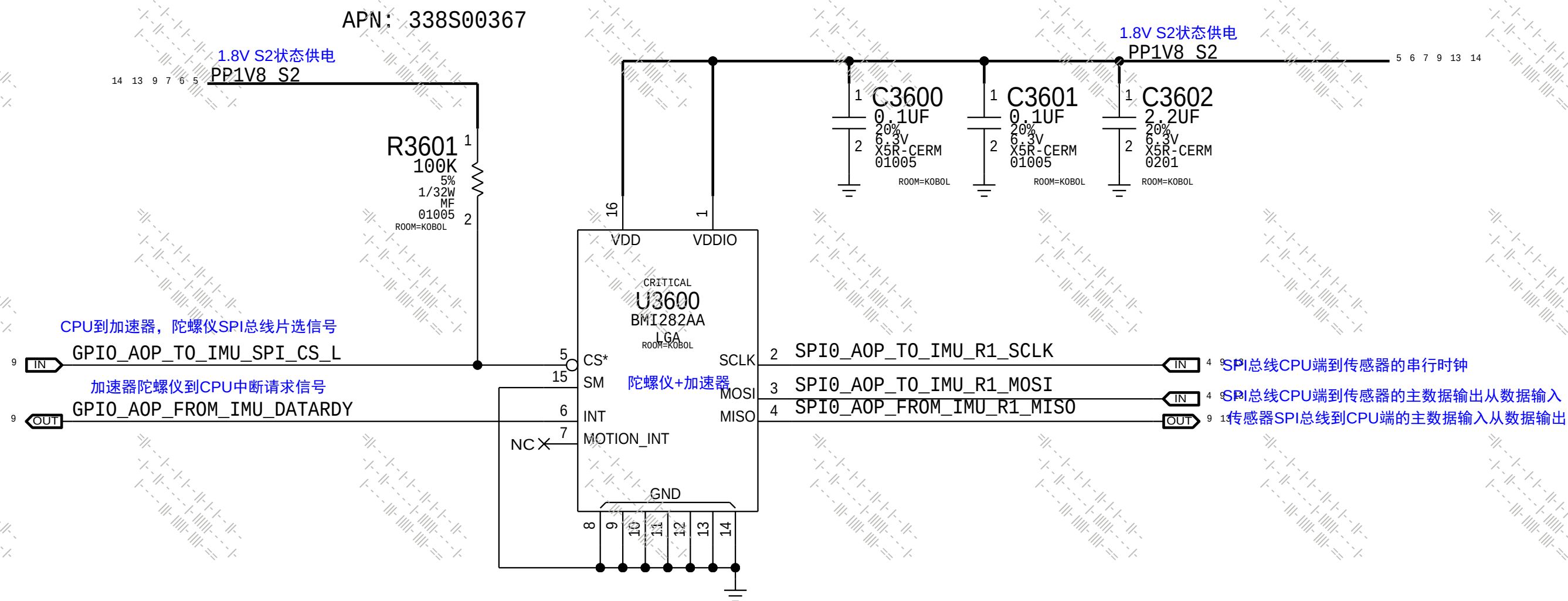
Dombra



Need 0 ohm R for termination on C3560,R3560,C3552,R3551,C3555
To be controlled by build matrix

PAGE TITLE			SYNC DATE=03/20/2018		
SYSTEM POWER: DOTARA CLK BUFF & MISC					
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Kobol - Accel & Gyro



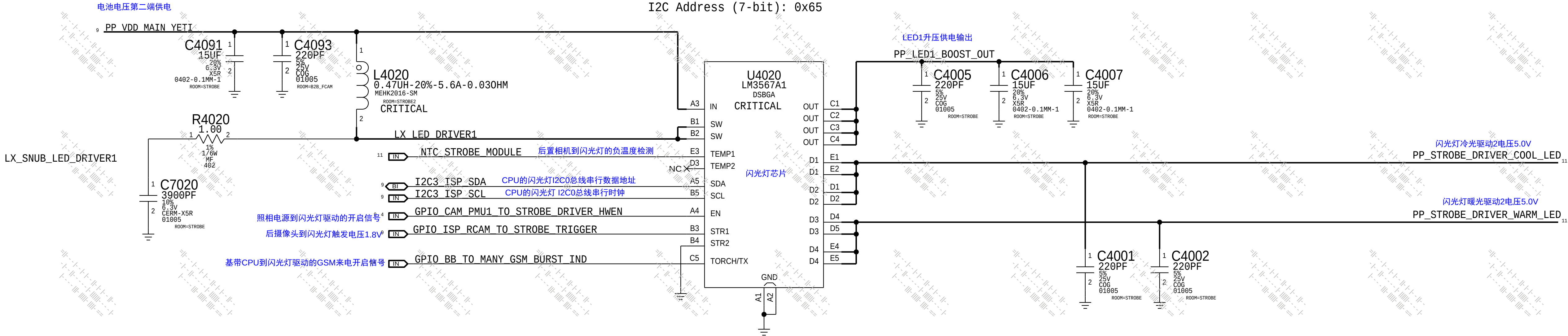
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	REVISION	7.0.0
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Yeti Inductor Alt

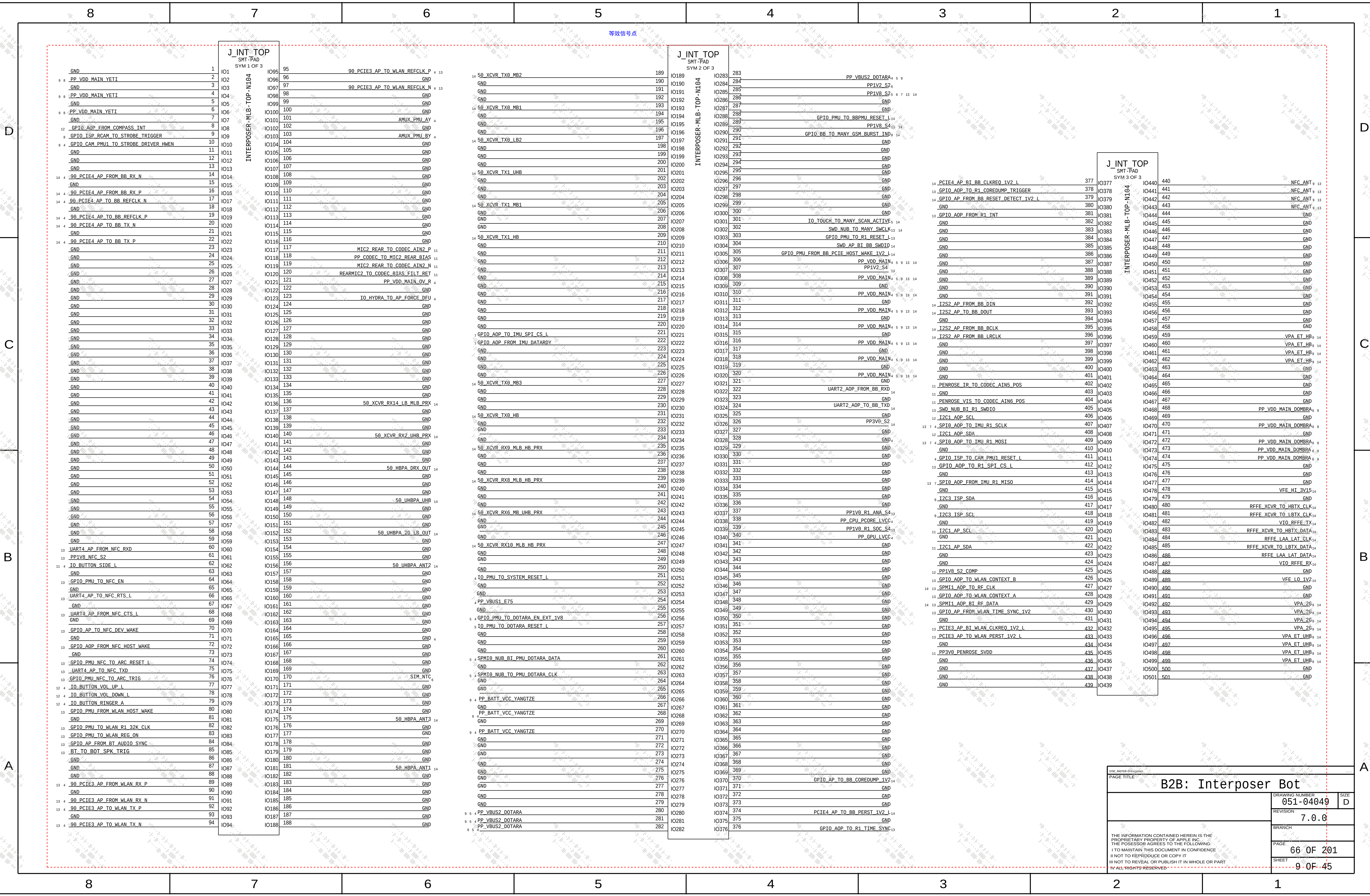
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152S08972	152S08918	ALT_PARTS	ALL	06/29/2013, 9:47:01, 1, 61, 2090, 2013
152S08947	152S08918	ALT_PARTS	ALL	CN102, 06/16/13, 9:47:01, 1, 61, 2090, 2013

LED STROBE DRIVER (YETI)

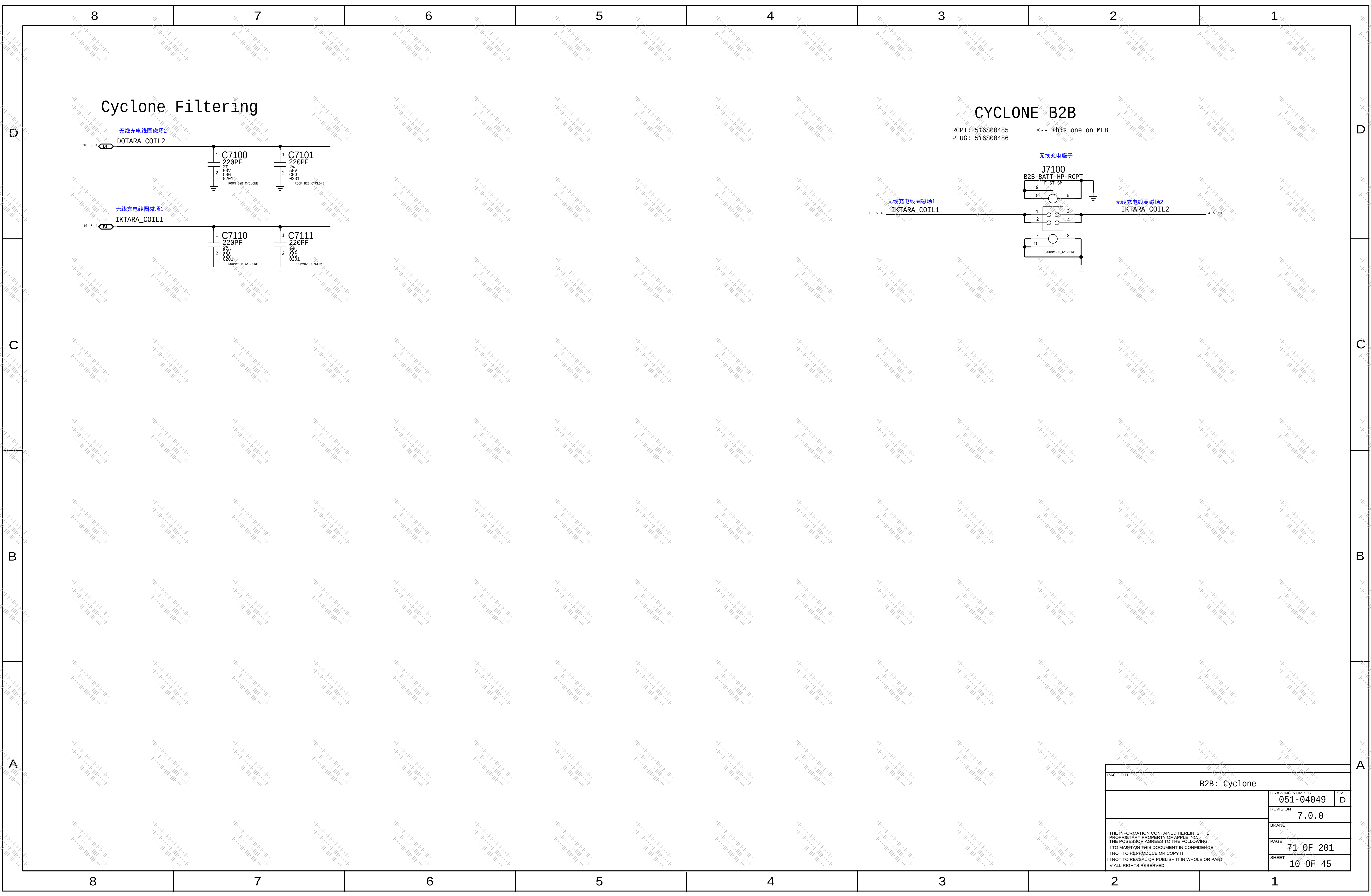
APN:353S01818
I2C Address (7-bit): 0x65



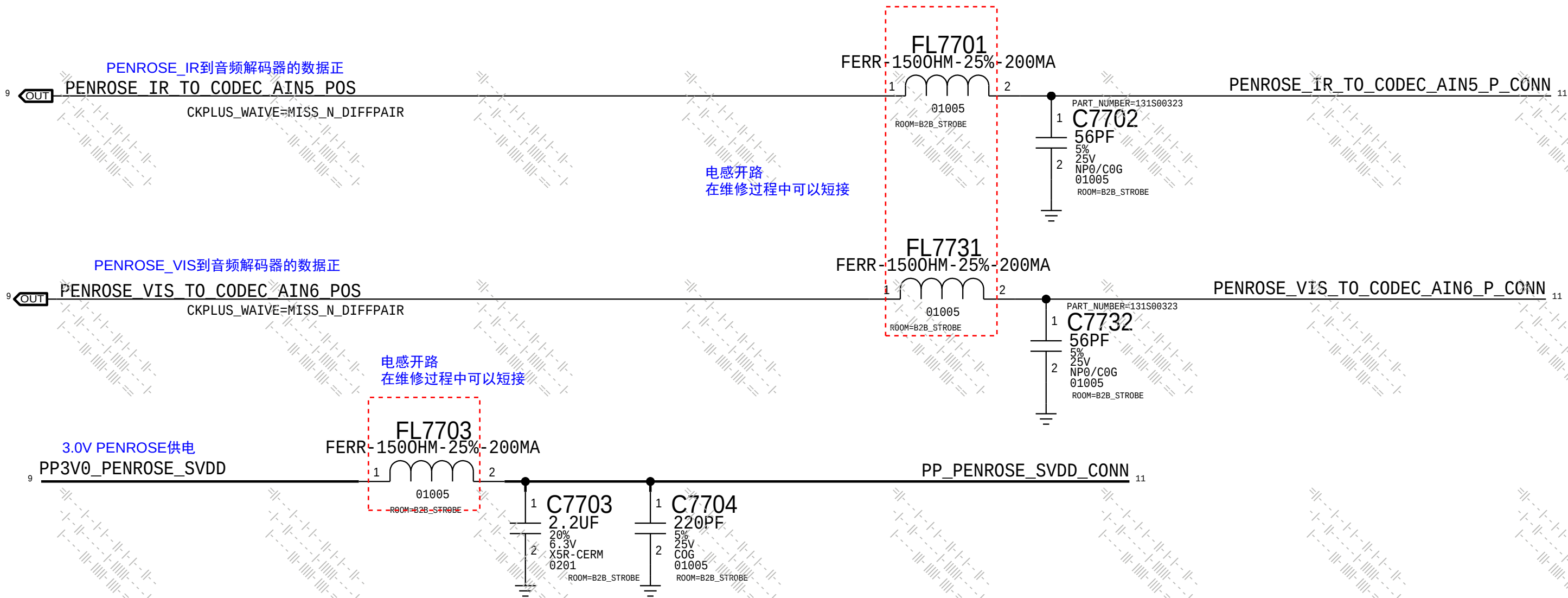
PAGE TITLE		
CAMERA: Strobe Driver		
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	REVISION	7.0.0
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	PAGE	40 OF 201
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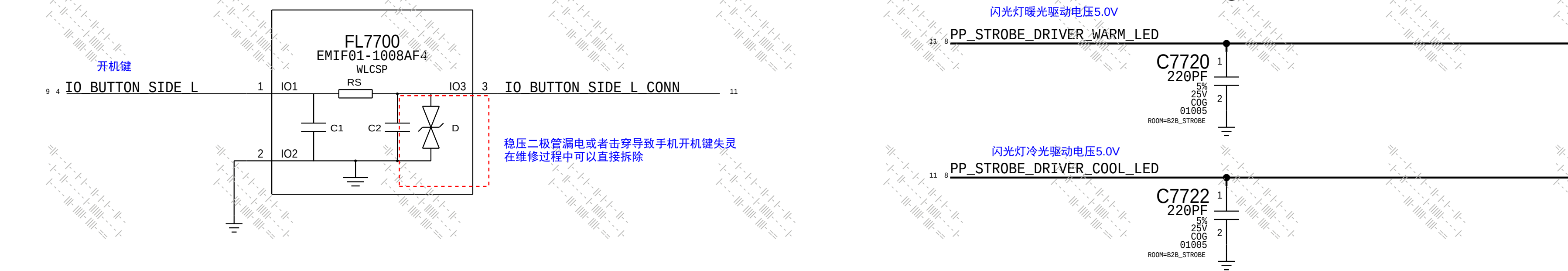
SYNC MASTER-Interposer	
PAGE TITLE	
B2B: Interposer Bot	
DRAWING NUMBER	051-04049
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PENROSE

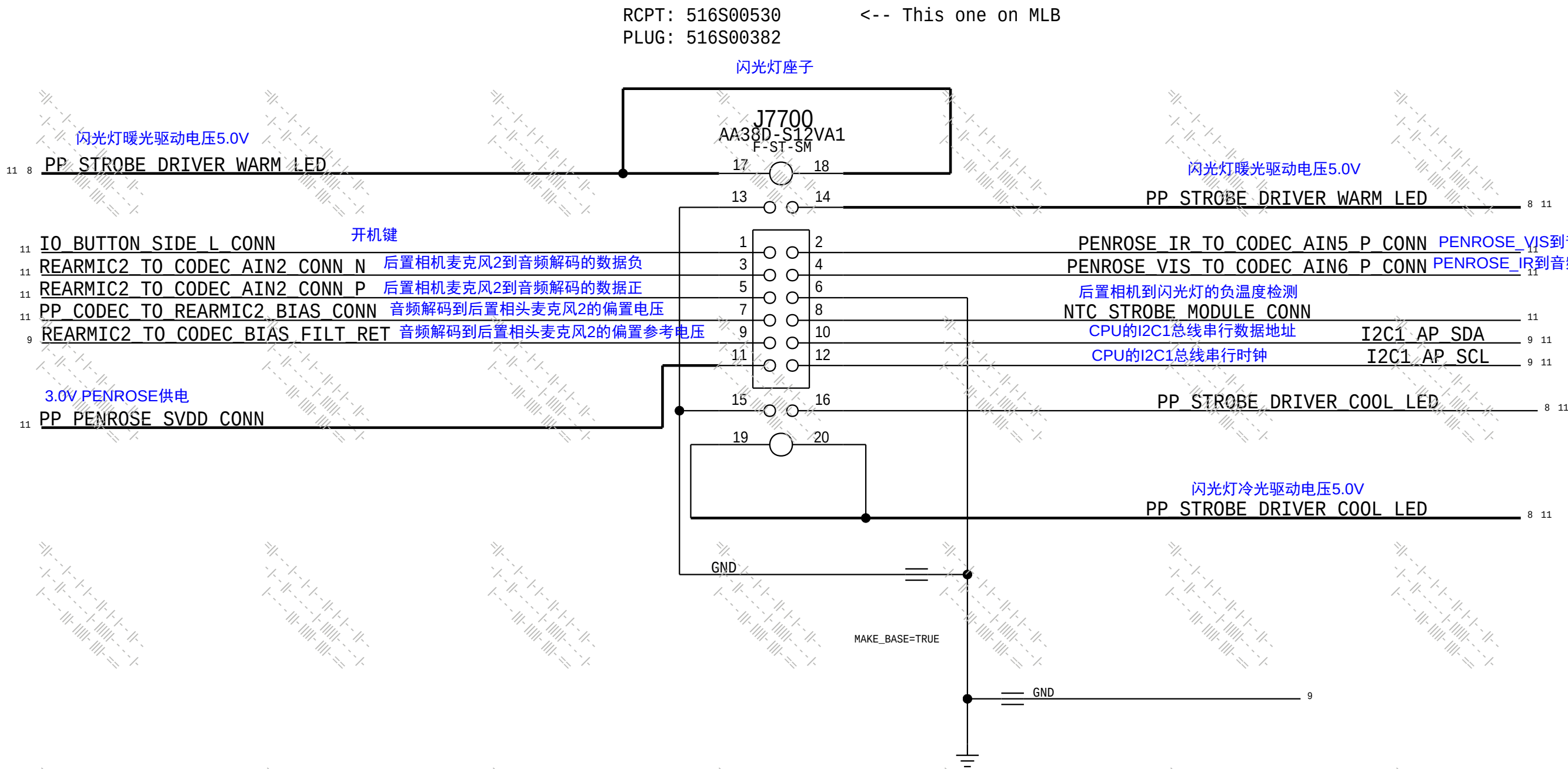


Strobe Filtering

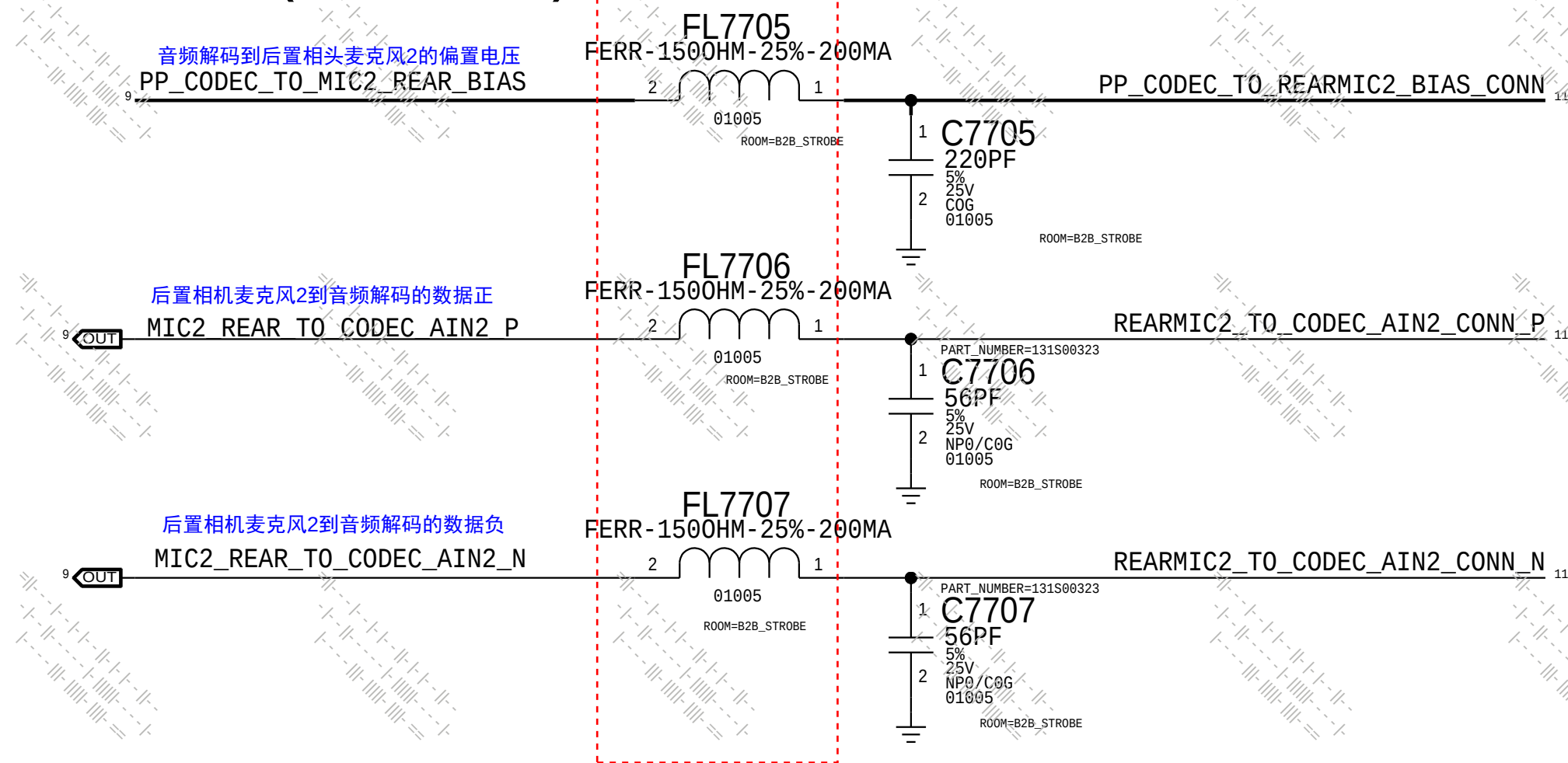


R7730 value may
change for Yeti

Strobe Connector



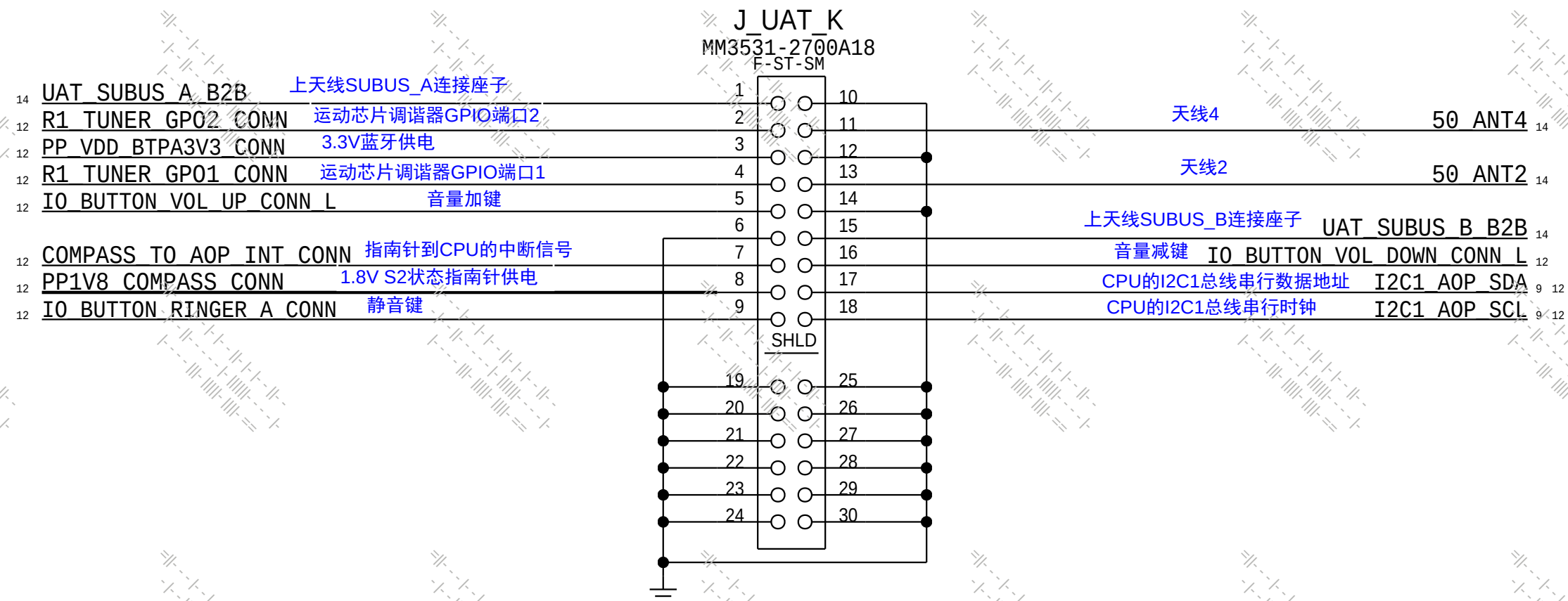
MIC2 (ANC REF)



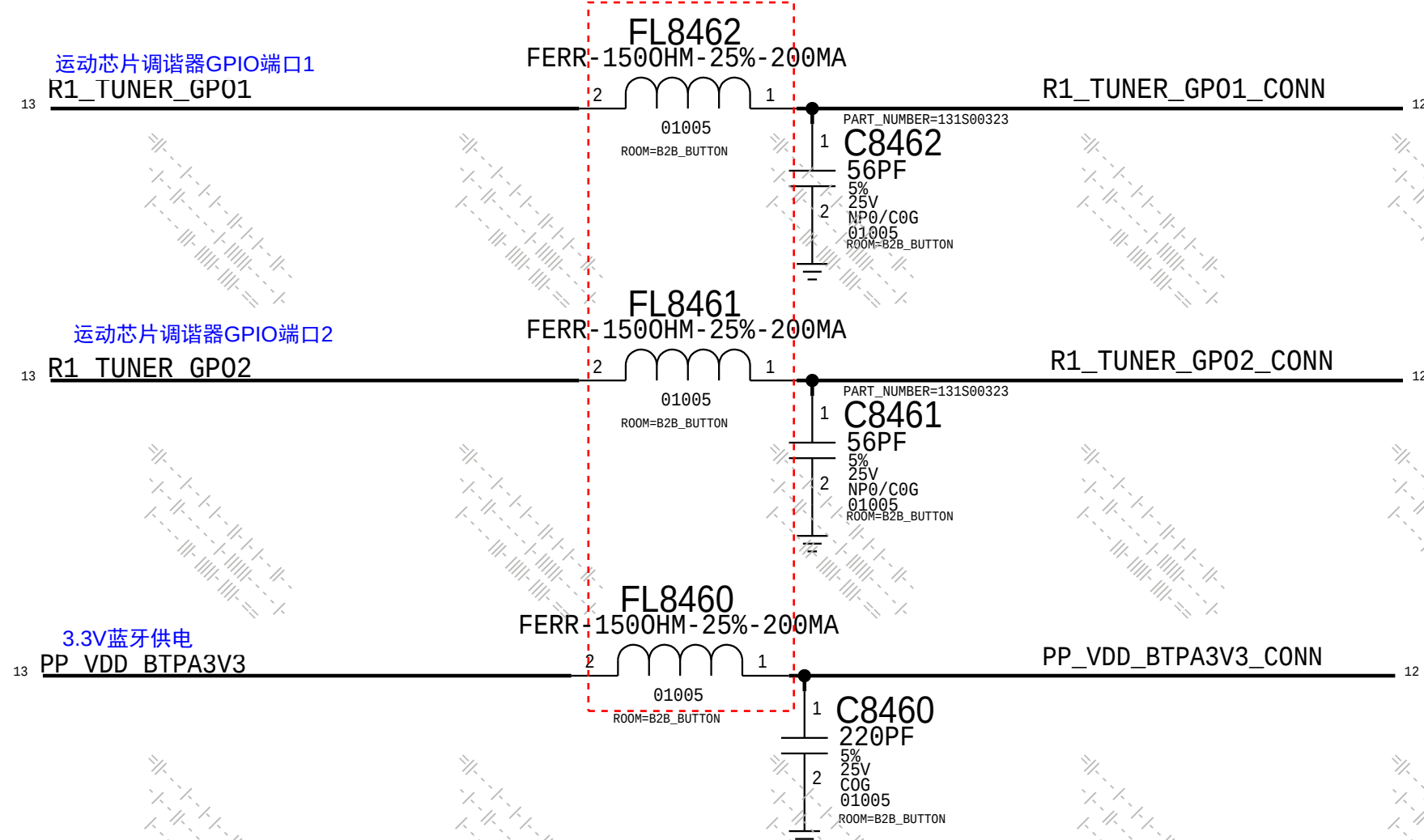
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B2B:Strobe		
DRAWING NUMBER	051-04049	SIZE
REVISION	7.0.0	D
BRANCH		
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UAT CONNECTOR

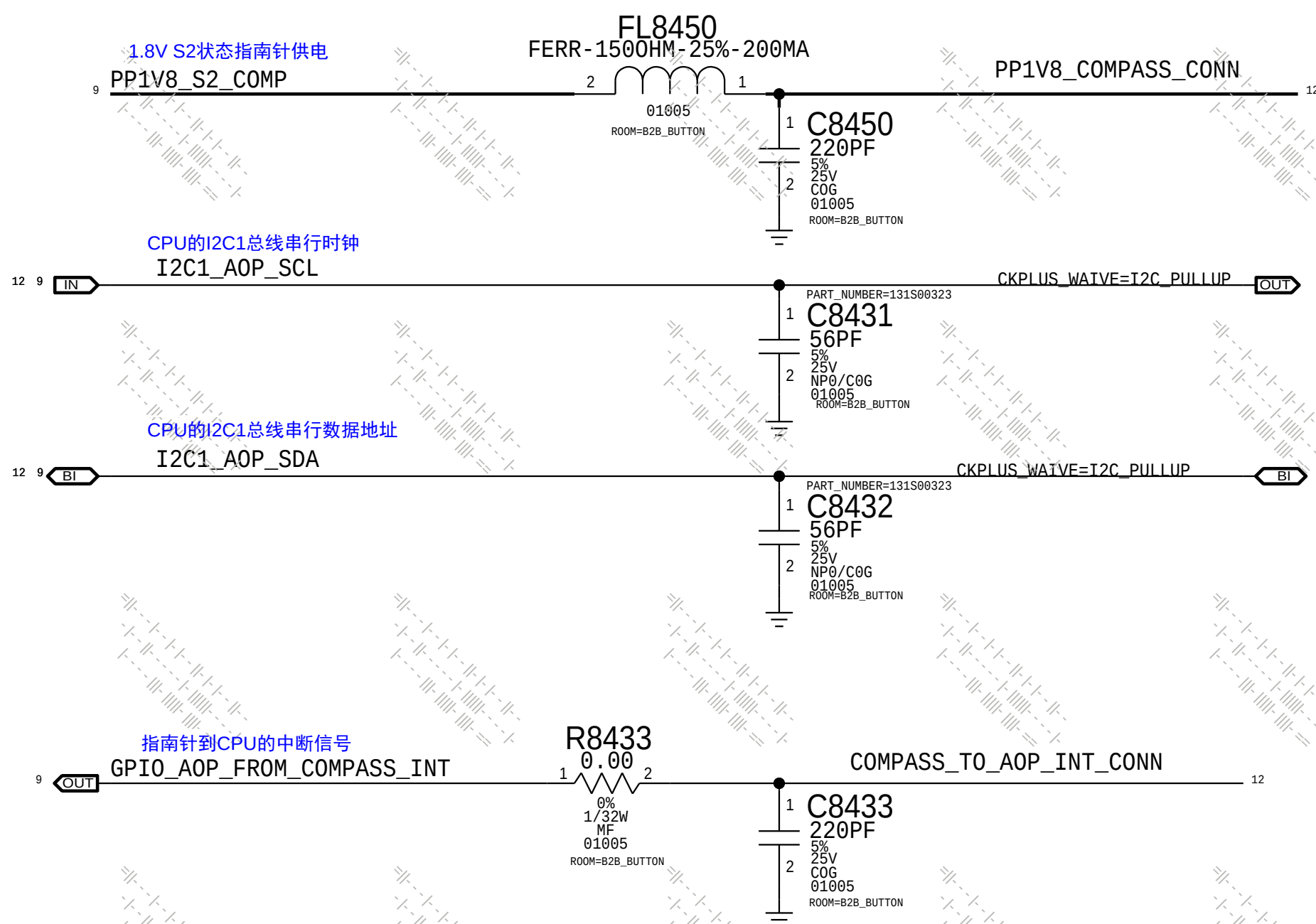
RCPT: 516S00532 <-- This one on MLB
PLUG: 516S00533 上天线座子



保险电感开了导致手机运动芯片不工作
在维修过程中可以直接拆除

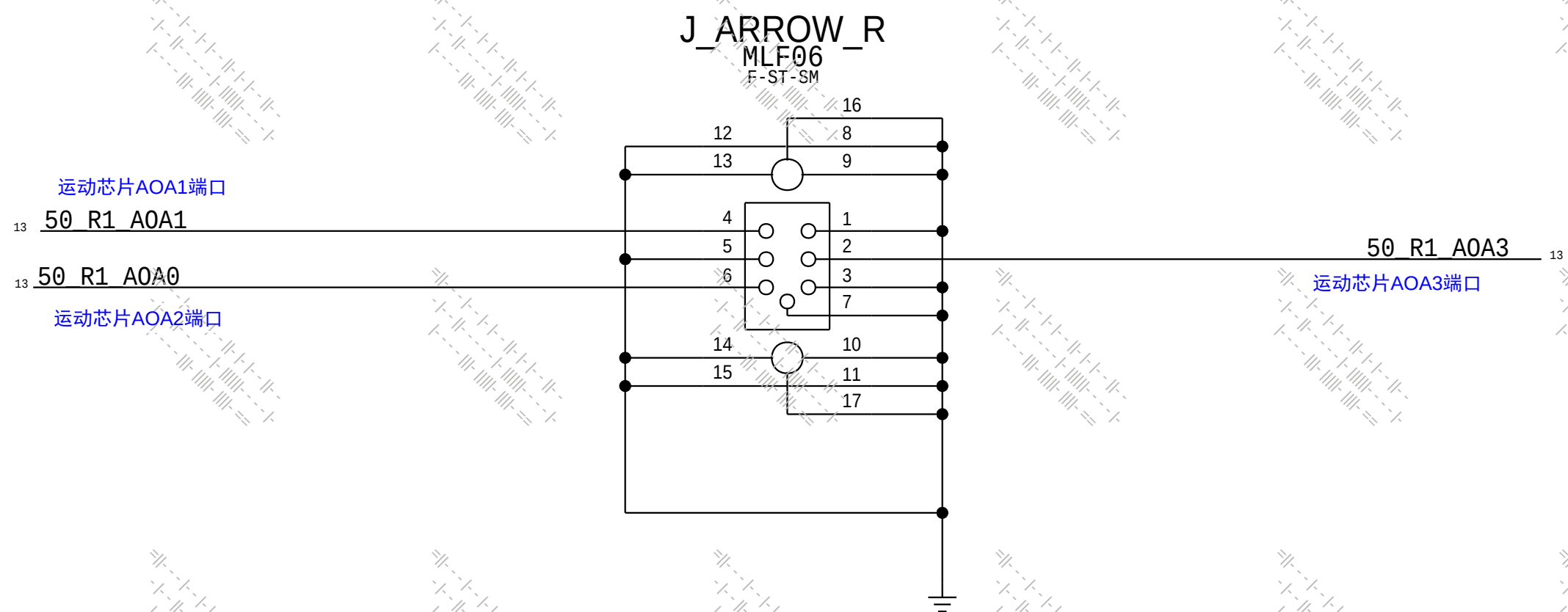


Compass

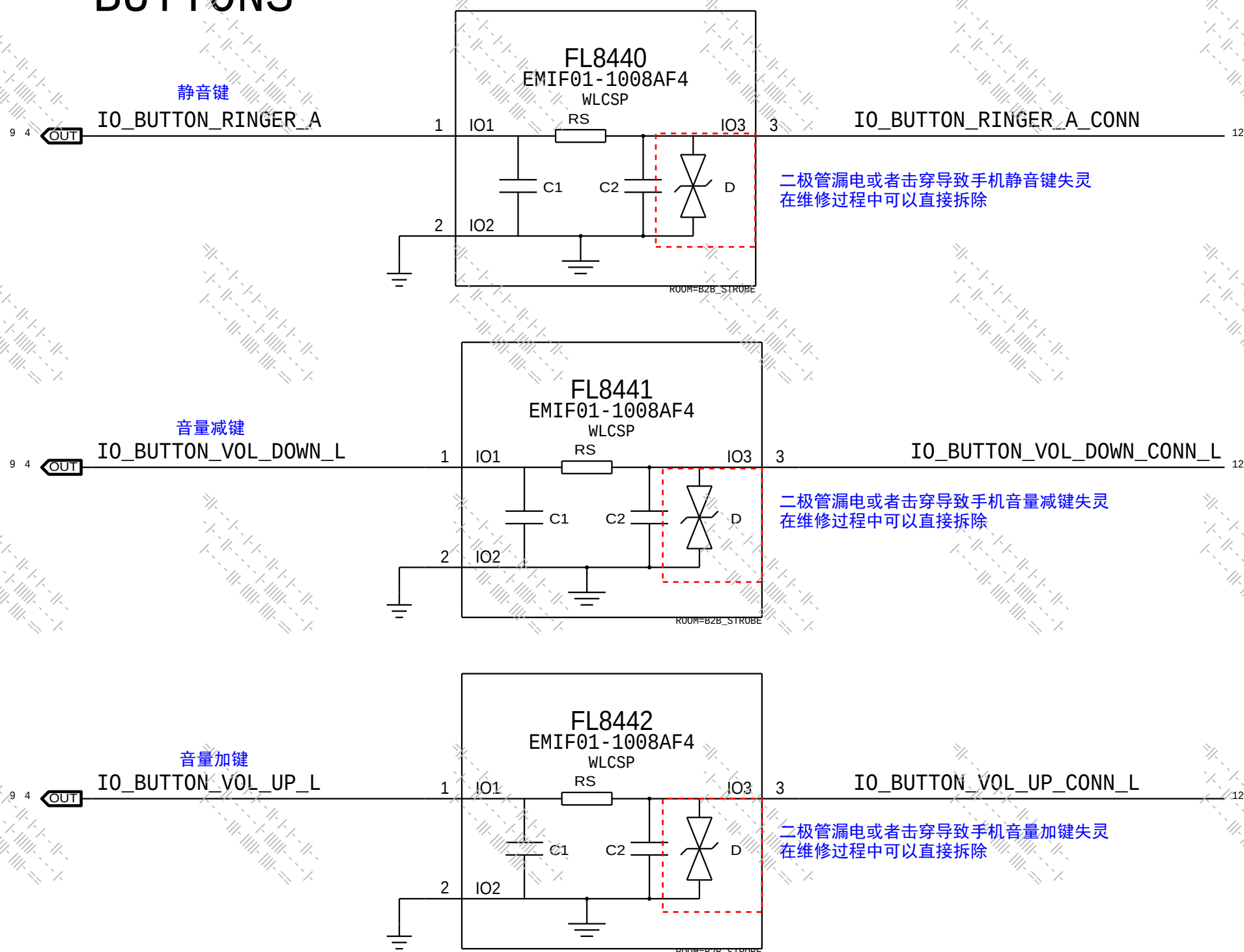


ARROW CONNECTOR

RCPT: 516S00434 <-- This one on MLB
PLUG: 516S00435

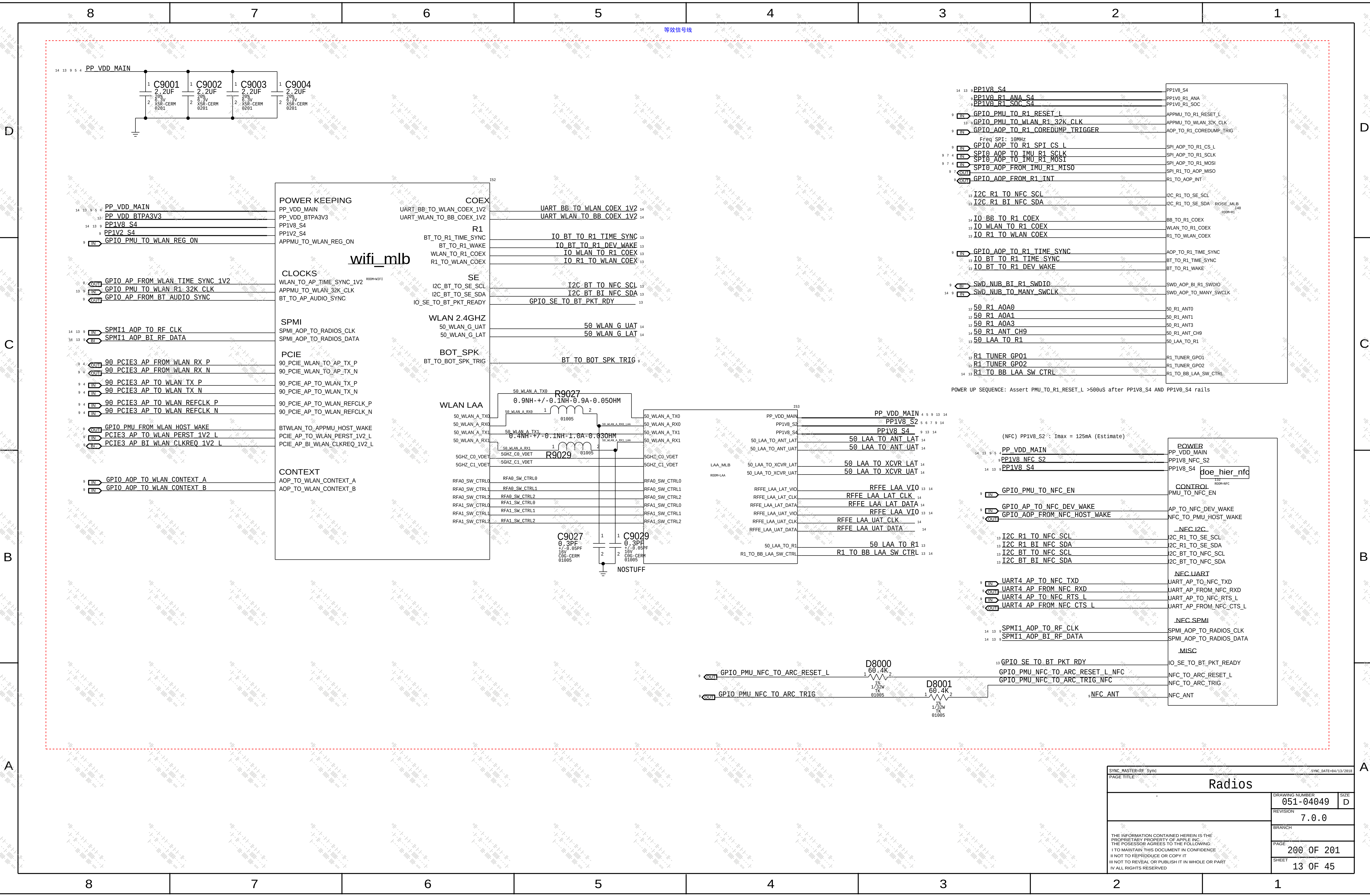


BUTTONS



UAT B2B - 02

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SYNC_MASTER=RF_Sync		SYNC_DATE=04/13/2018	
PAGE TITLE		Radios	
		DRAWING NUMBER	051-04049
		REVISION	7.0.0
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		PAGE	200 OF 201
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STOCKHOLM

NFC CONTROLLER

NET RULE ASSIGNMENT	
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)
P	PWR_SHAPE NFC_ANT*, NFC_TX*, NFC_BAL*, NFC_PROBE* PP_NFC*
P	PWR_300UM
P	PWR_200UM NFC_RX*, NFC_VCAS* VUP
P	PWR_SHAPE

D

D

C

C

B

B

A

A

NFC FRONT END

NFC

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POWER

PP1V8_S4

PP1V8_R1_ANA_S4

PP1V8_R1_SOC_S4

HOST_IPC

SPI0_AOP_TO_IMU_R1_SCLK

GPIO_AOP_TO_R1_SPI_CS_L

SPI0_AOP_FROM_IMU_R1_MISO

SPI0_AOP_TO_IMU_R1_MOSI

GPIO_AOP_TO_R1_COREDUMP_TRIGGER

GPIO_AOP_FROM_R1_INT

RESET/CLOCK

GPIO_PMU_TO_R1_RESET_L

GPIO_PMU_TO_WLAN_R1_32K_CLK

GPIO_AOP_TO_R1_TIME_SYNC

DEBUG

SWD_NUB_TO_MANY_SWCLK

SWD_NUB_B1_R1_SWDIO

SPMI/COEX_GPIOS

IO_R1_TO_WLAN_COEX

IO_BB_TO_R1_COEX

IO_WLAN_TO_R1_COEX

SE_IPC

I2C_R1_TO_NFC_SCL

I2C_R1_B1_NFC_SDA

BT_TIME_SYNC_WAKE

IO_BT_TO_R1_TIME_SYNC

IO_BT_TO_R1_DEV_WAKE

ANTENNA_SWITCH_CTRL

R1_TUNER_GP01

R1_TUNER_GP02

R1_TO_BB_LAA_SW_CTRL

ANTENNA

50_R1_A0A0

50_R1_A0A1

50_R1_A0A3

50_R1_ANT_CH9

50_LAA_TO_R1

SCH #:

951-07279

PCB #:

920-05369

BOM #:

939-05686

FRONT PAGE

DRAWING TITLE

SCH, MLB, BOT, ICE, N104

REVISION

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BRANCH

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MODULE

SPI0总线CPU端到传感器的串行时钟1.8V_{S4}
SPI0总线CPU端到传感器的片选信号₁₃
SPI0总线前往传感器的主数据输入从数据输出
SPI0总线前往传感器的主数据输出从数据输入

运动芯片I2C总线到NFC的串行时钟₁₃
运动芯片I2C总线到NFC的串行数据地址₁₃

运动定位跟踪芯片与无线网卡共存协议配置
基带CPU与运动定位跟踪芯片共存协议配置

无线网卡与运动定位跟踪芯片共存协议配置

CPU开关数据总线到基带CPU的开关时钟
CPU开关数据总线到运动芯片的开关数据交换

CPU到wifi、运动芯片的32K时钟

CPU到运动芯片的充电配置₃

CPU到运动芯片的复位信号

来自运动芯片到CPU的中断信号

CPU到运动芯片的时间同步

蓝牙到运动芯片的时间同步

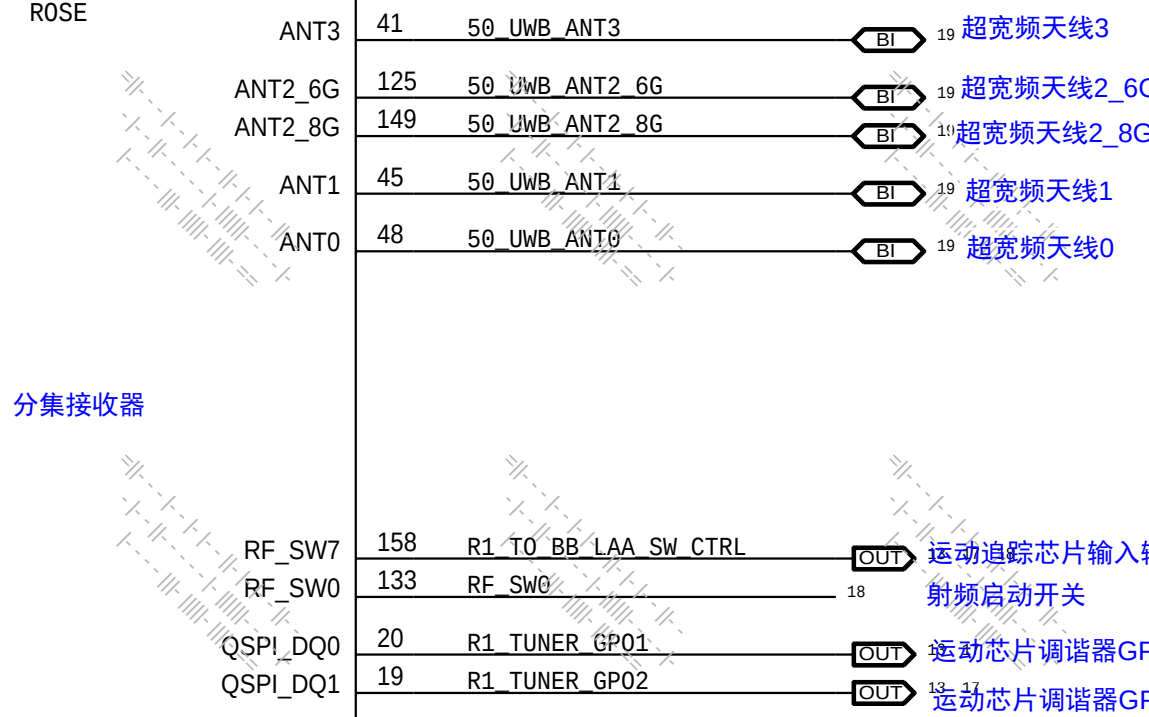
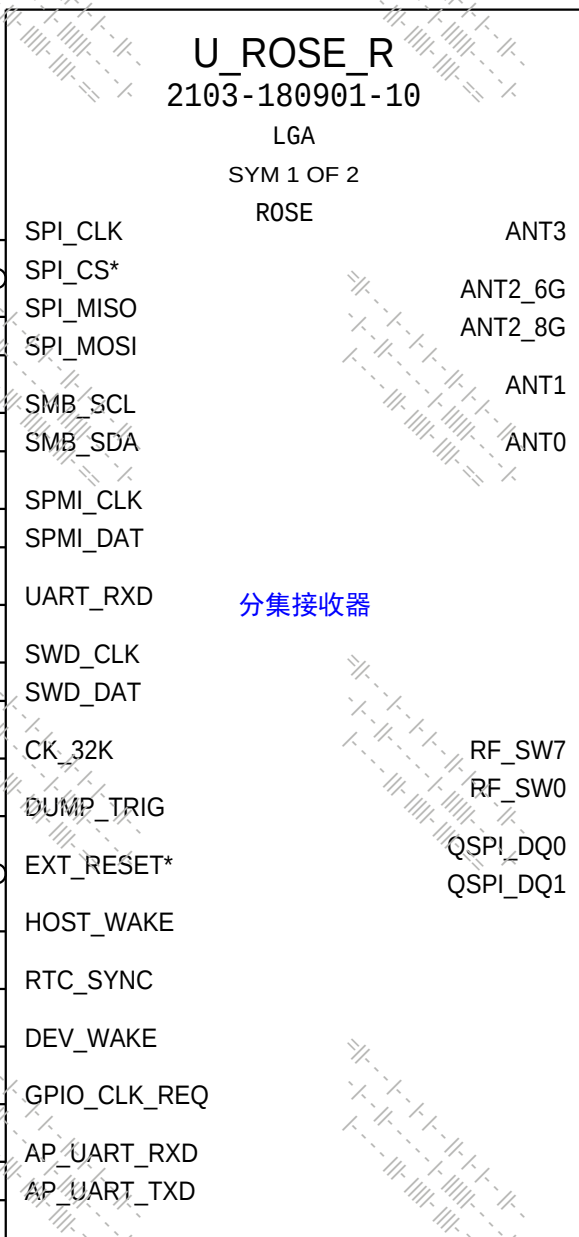
蓝牙到运动芯片的唤醒信号

CPU到运动芯片的数据接收信号₁₈

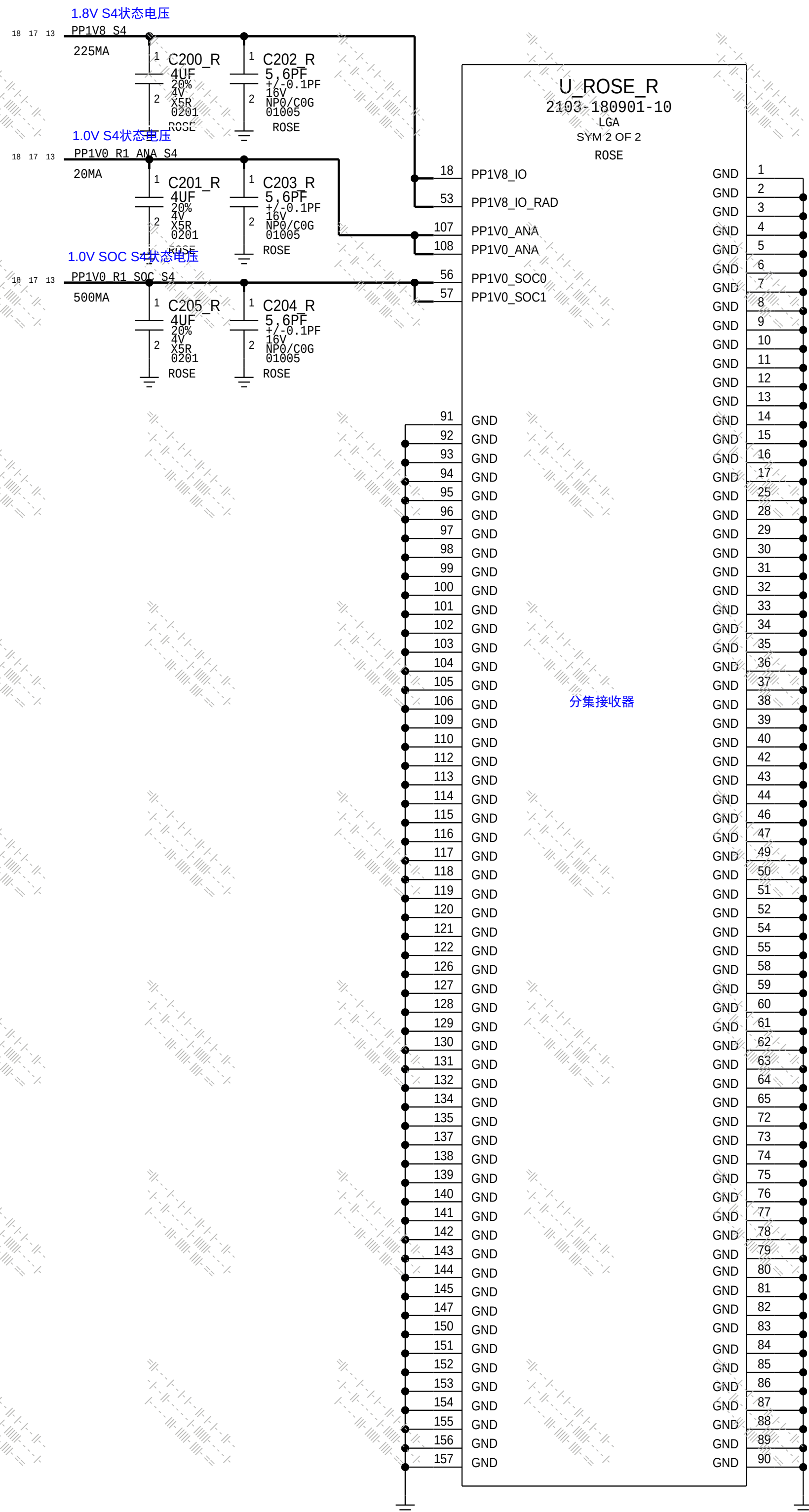
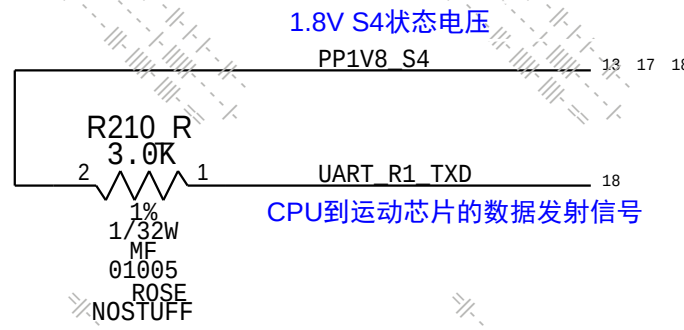
CPU到运动芯片的数据发射信号₁₈

测试点

1.8V S4状态电压
R200_R为上拉电阻
作用：使得CS片选信号持续为高电平
维修当中不可以短接，也不能丢失，但可以将其阻值改小。



UNSECURE BOOT BOOTSTRAP PULLUP

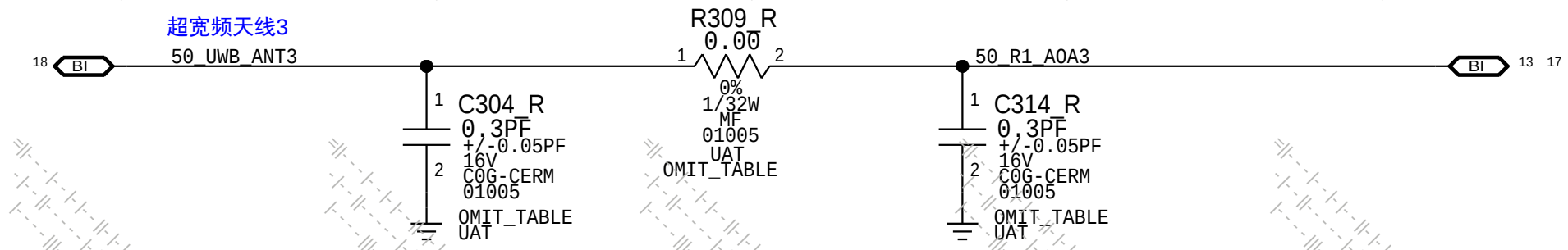
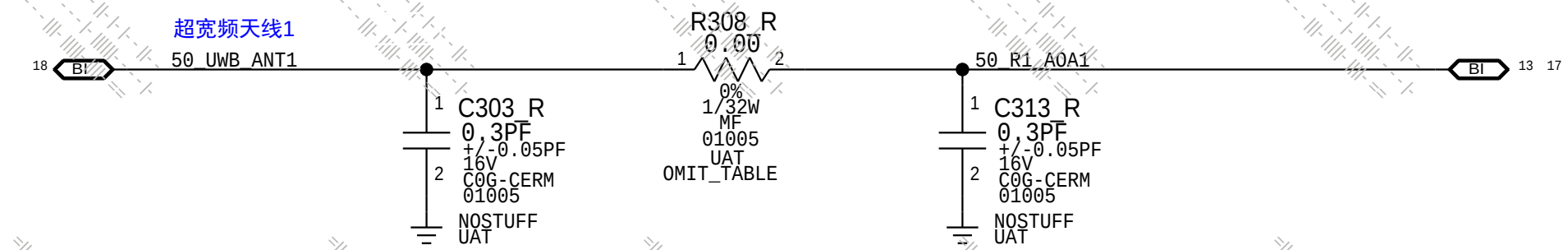
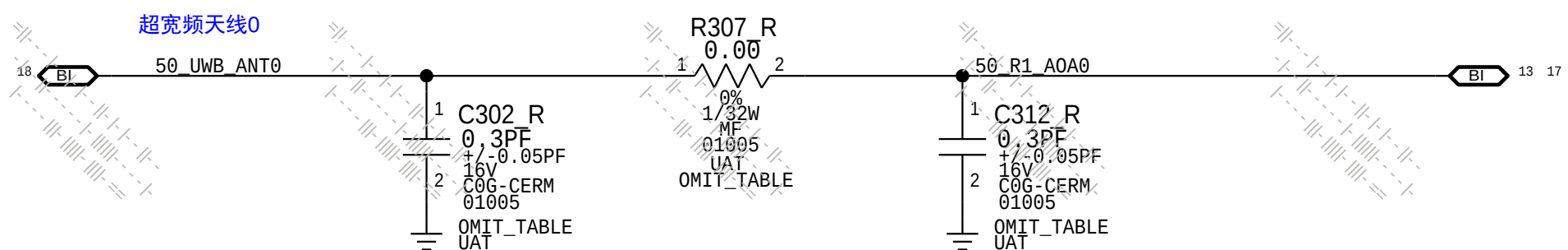
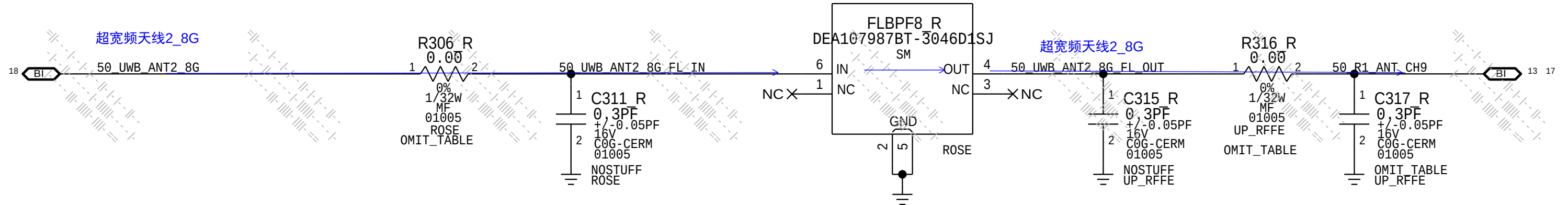
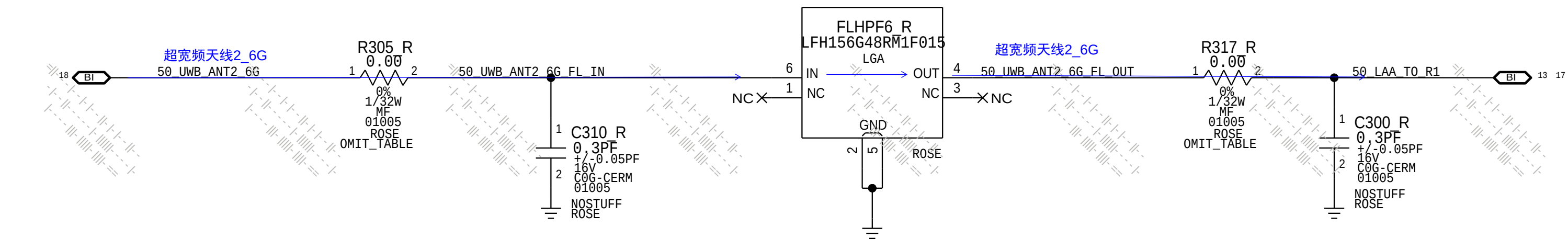


MODULE

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FILTERS/MATCHING AND ANTENNA



NET RULE ASSIGNMENT		
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)	
A_50_WIDE_SE	50_R1*, 50_UWB*, 50_LAA_TO_R1*	

CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*		CLASS
CLASS NAME	CONSTRAINT SET	DP NAMES EX: DP:DP_AA*DP_BB* (LINE STARTS WITH FLAG DP-)		YIN
50_WIDE	S A_DIELECTRIC_2X_50_WIDE_SE	50_R1*, 50_UWB*, 50_LAA_TO_R1*		?

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
11750161	1	RES,MF,0 OHM,1/32W,01005	R305_R	ROSE.X
11750161	1	RES,MF,0 OHM,1/32W,01005	R317_R	ROSE.X
11750161	1	RES,MF,0 OHM,1/32W,01005	R306_R	ROSE.X
152501109	1	IND,FILM,0.6NH,950MA,01005	R316_R	ROSE.X
131500030	1	CAP,CER,C06,0.4PF,16V,01005	C317_R	ROSE.X
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C304_R	ROSE.X
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C314_R	ROSE.X
152501109	1	IND,FILM,0.6NH,950MA,01005	R309_R	ROSE.X
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C303_R	ROSE.X
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C313_R	ROSE.X
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C302_R	ROSE.X
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C312_R	ROSE.X
152500494	1	IND,FILM,0.8NH,900MA,01005	R307_R	ROSE.X
152500494	1	IND,FILM,0.8NH,900MA,01005	R308_R	ROSE.X
13150309	1	CAP,CER,C06,0.5PF,16V,01005	C315_R	ROSE.X

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
152500497	1	IND,FILM,0.5NH,1000MA,01005	R305_R	ROSE.0
152500497	1	IND,FILM,0.5NH,1000MA,01005	R317_R	ROSE.0
152500498	1	IND,FILM,0.4NH,1000MA,01005	R306_R	ROSE.0
152500492	1	IND,FILM,1:0NH,900MA,01005	R316_R	ROSE.0
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C317_R	ROSE.0
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C304_R	ROSE.0
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C314_R	ROSE.0
152500496	1	IND,FILM,0.6NH,950MA,01005	R309_R	ROSE.0
152500496	1	IND,FILM,0.6NH,950MA,01005	R308_R	ROSE.0
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C302_R	ROSE.0
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C312_R	ROSE.0
152500496	1	IND,FILM,0.6NH,950MA,01005	R307_R	ROSE.0
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C303_R	ROSE.0
131500893	1	CAP,CER,C06,0.2PF,16V,01005	C313_R	ROSE.0
131500030	1	CAP,CER,C06,0.4PF,16V,01005	C315_R	ROSE.0

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339500664	339500665	ALT_PARTS	U_ROSE_R	MODULE,STRIDER,ES5.1

PAGE TITLE		
FILTERS MATCHING		
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D

C

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LB SPAD

D

C

B

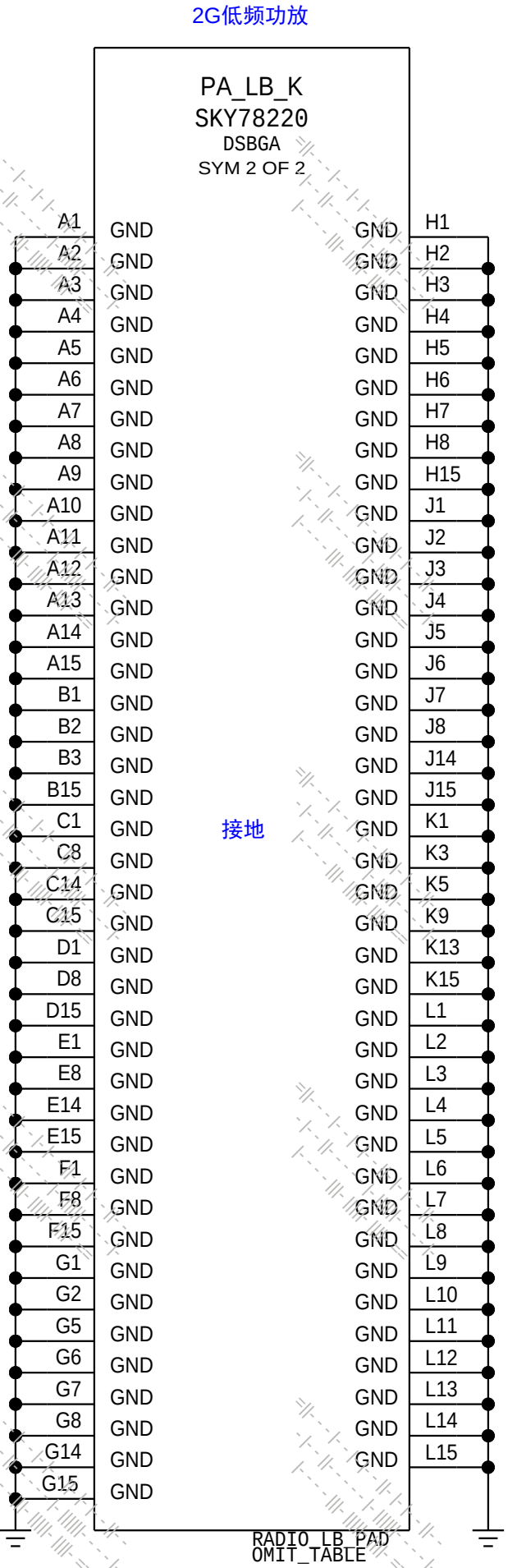
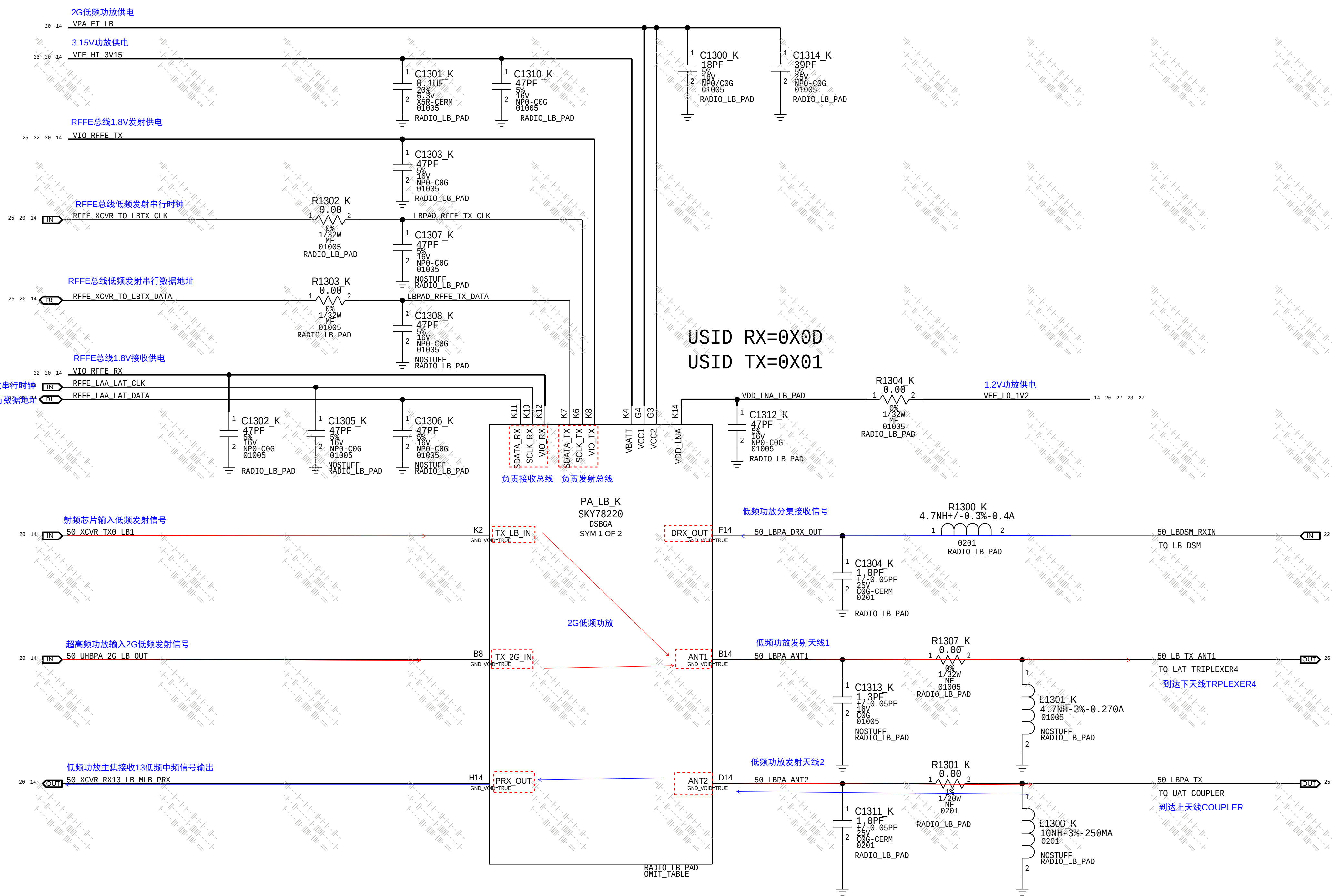
A

D

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B

A



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S01907	1	SKY78220-11	PA_LB_K	ROW
353S01908	1	SKY78223-11	PA_LB_K	USCH

NET RULE ASSIGNMENT			
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)		
A_50_THIN_SE	50_UHBPA_2G_LB_OUT*, 50_LBPA_DRX_OUT		
A_50_THIN_SE	50_LBPA_ANT*, 50_LB_TX_ANT1*		
A_50_WIDE_SE	50_LBPA_TX		
CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	
CLASS NAME	CONSTRAINT SET	DP NAMES EX: DP-DP_AA*DP_BB* (LINE STARTS WITH FLAG DP:)	CLASS DESIGNATOR
50_THIN	A_DIELECTRIC_2X	50_UHBPA_2G_LB_OUT*, 50_LBPA_DRX_OUT	Y
50_THIN	A_DIELECTRIC_2X	50_LBPA_ANT*, 50_LB_TX_ANT1*	Y
50_WIDE	A_DIELECTRIC_2X_50_WIDE_SE	50_LBPA_TX	Y
NET RULE ASSIGNMENT			
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)		
PWR_200UM	VDD_LNA_LB_PAD		

PAGE TITLE			
LB SPAD			
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LB DIVERSITY RECEIVE LNA

RFFE FILTERING

USID RX=0X0A

低频分集接收器

DSM_LB_K
LMRFRJB-K57
LGA

接地

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353501922	1	MURATA, LMRFRJB-K57 ES1.0	DSM_LB_K	ROW
353501923	1	MURATA, LMRFRJB-K69 ES1.0	DSM_LB_K	USCH

NET RULE ASSIGNMENT	
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)
A_50_WIDE_SE	50_LBDSM_RXIN
A_50_THIN_SE	50_LBDSM_TX*, 50_LBDSM_RXOUT*, 50_LBDSM_ANT*
A_50_THIN_SE	50_LBDSM_RXIN_M

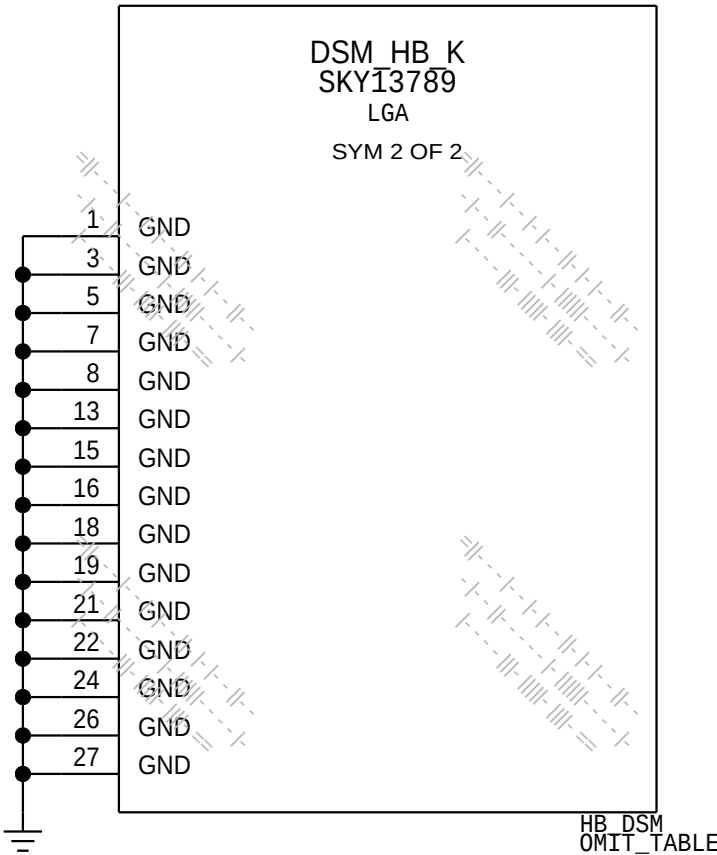
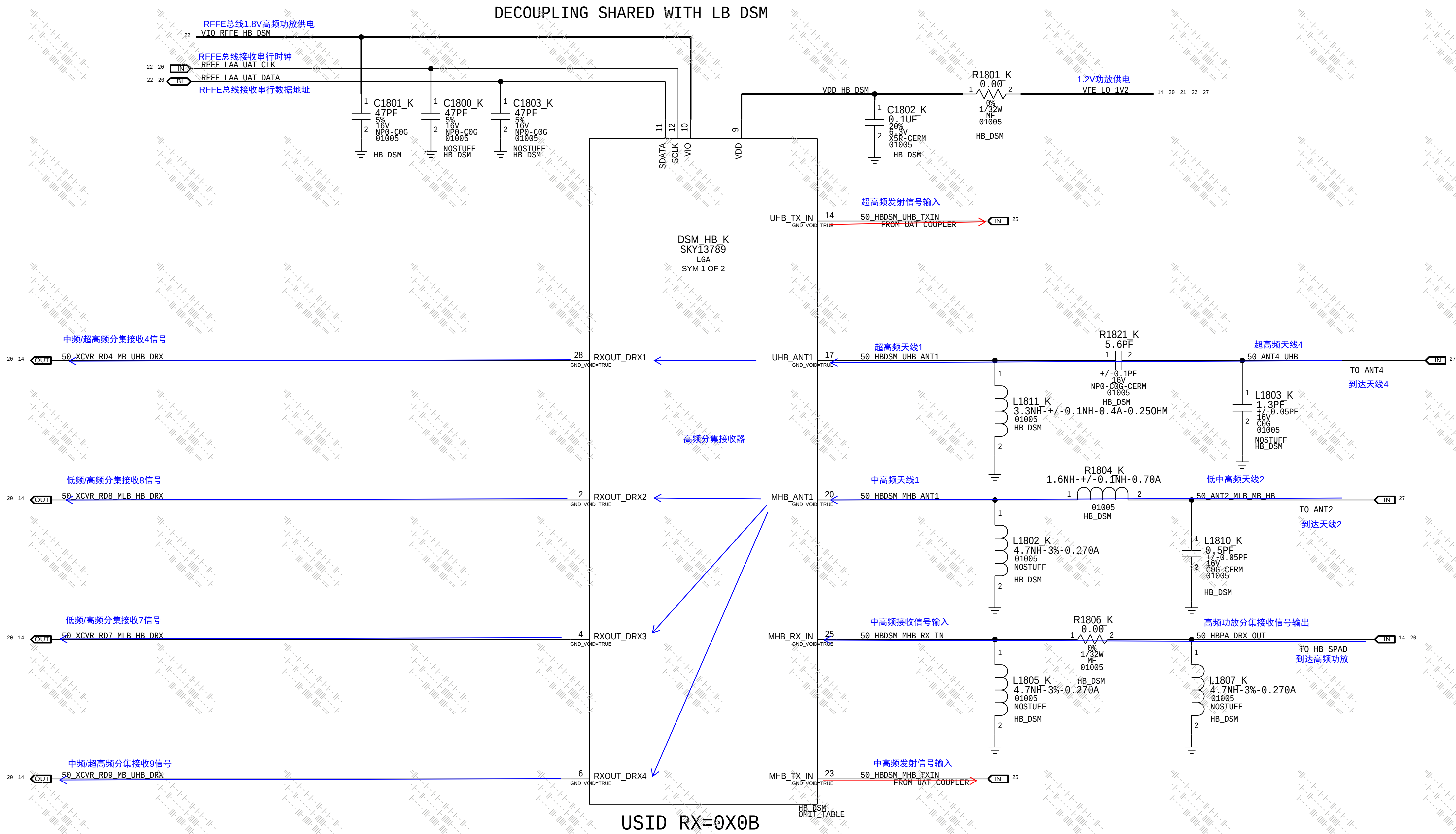
CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*		CLEAR
CLASS NAME	CONSTRAINT SET	DP NAMES EX: DP_DP_AA*DP_BB* (LINE STARTS WITH FLAG DP)		Y/N
50_WIDE	S	A_DIELECTRIC_2X_50_WIDE_SE	50_LBDSM_RXIN	Y
50_THIN	S	A_DIELECTRIC_2X	50_LBDSM_TX*, 50_LBDSM_RXOUT*, 50_LBDSM_ANT*	Y
50_THIN	S	A_DIELECTRIC_2X	50_LBDSM_RXIN_M	Y

NET RULE ASSIGNMENT	
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)
PWR_200UM	VDD_LB_DSM

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HB DIVERSITY RECEIVE LNA



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S01916	1	SKY13789-13	DSM_HB_K	ROW
353S01909	1	SKY13799-12	DSM_HB_K	USCH

NET RULE ASSIGNMENT	
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)
A_50_THIN_SE	50_HBDSM_RXOUT*, 50_HBDSM_MHB_RX_IN
A_50_THIN_SE	50_HBDSM_MHB_TXIN*, 50_HBDSM_UHB_TXIN*
A_50_WIDE_SE	50_HBPA_DRX_OUT*
A_50_THIN_SE	50_HBDSM_MHB_ANT*, 50_HBDSM_UHB_ANT*

CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*		CLASS
CLASS NAME	CONSTRAINT SET	DP NAMES EX: DP-DP_AA*DP_BB* (LINE STARTS WITH FLAG DP:)		CLASS
50_THIN	A_DIELECTRIC_2X	50_HBDSM_RXOUT*, 50_HBDSM_MHB_RX_IN		Y
50_THIN	A_DIELECTRIC_2X	50_HBDSM_MHB_TXIN*, 50_HBDSM_UHB_TXIN*		Y
50_WIDE	A_DIELECTRIC_2X_50_WIDE_SE	50_HBPA_DRX_OUT*		Y
50_THIN	A_DIELECTRIC_2X	50_HBDSM_MHB_ANT*, 50_HBDSM_UHB_ANT*		Y

NET RULE ASSIGNMENT	
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)
PWR_200UM	VDD_HB_DSM

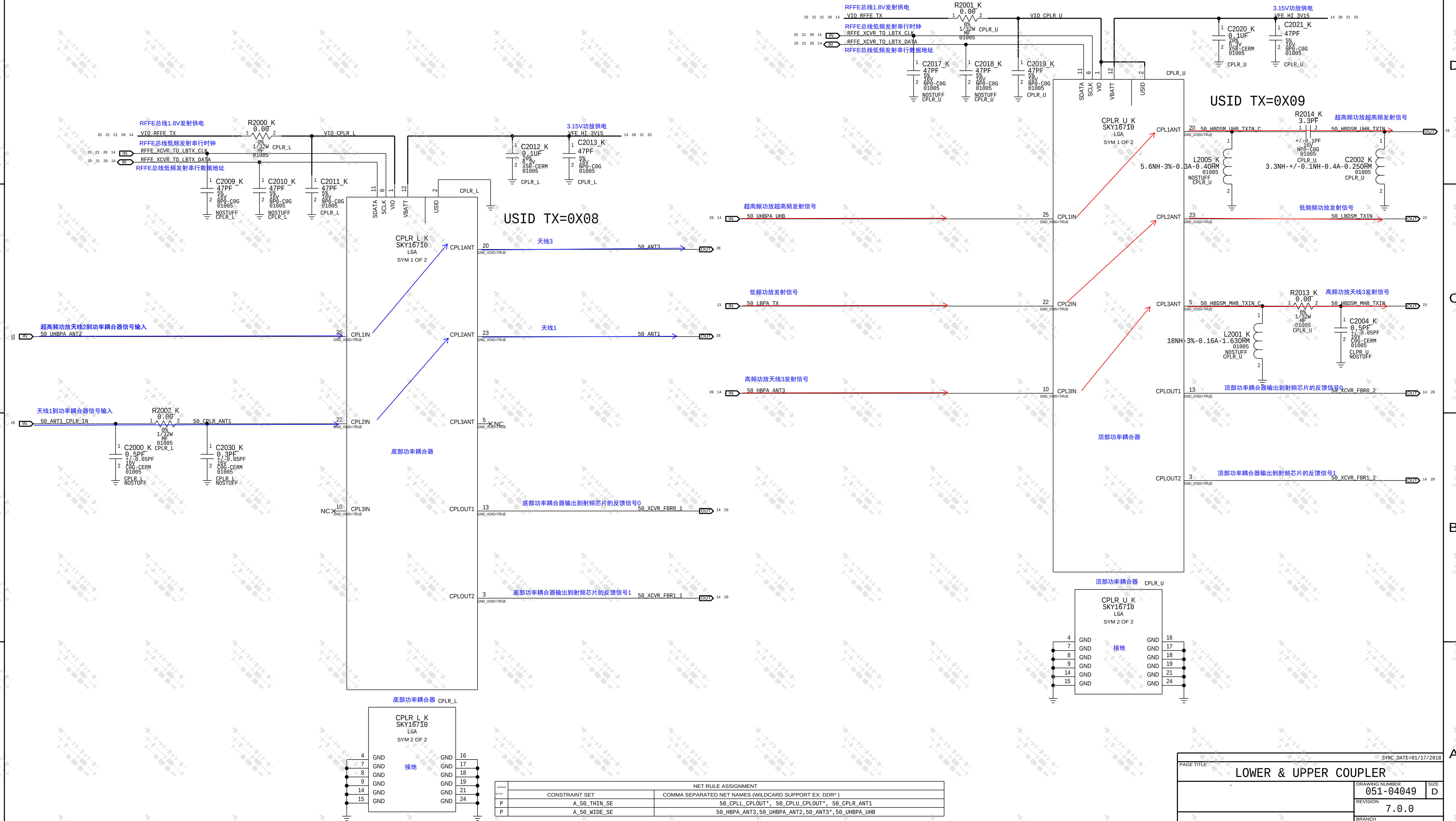
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PAGE TITLE			MIMO RECIEVE LNA			SYNC DATE=01/17/2018		
			DRAWING NUMBER		051-04049		SIZE	
			REVISION		7.0.0		D	
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			PAGE		5 OF 10			
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8 7

LOWER/UPPER COUPLER

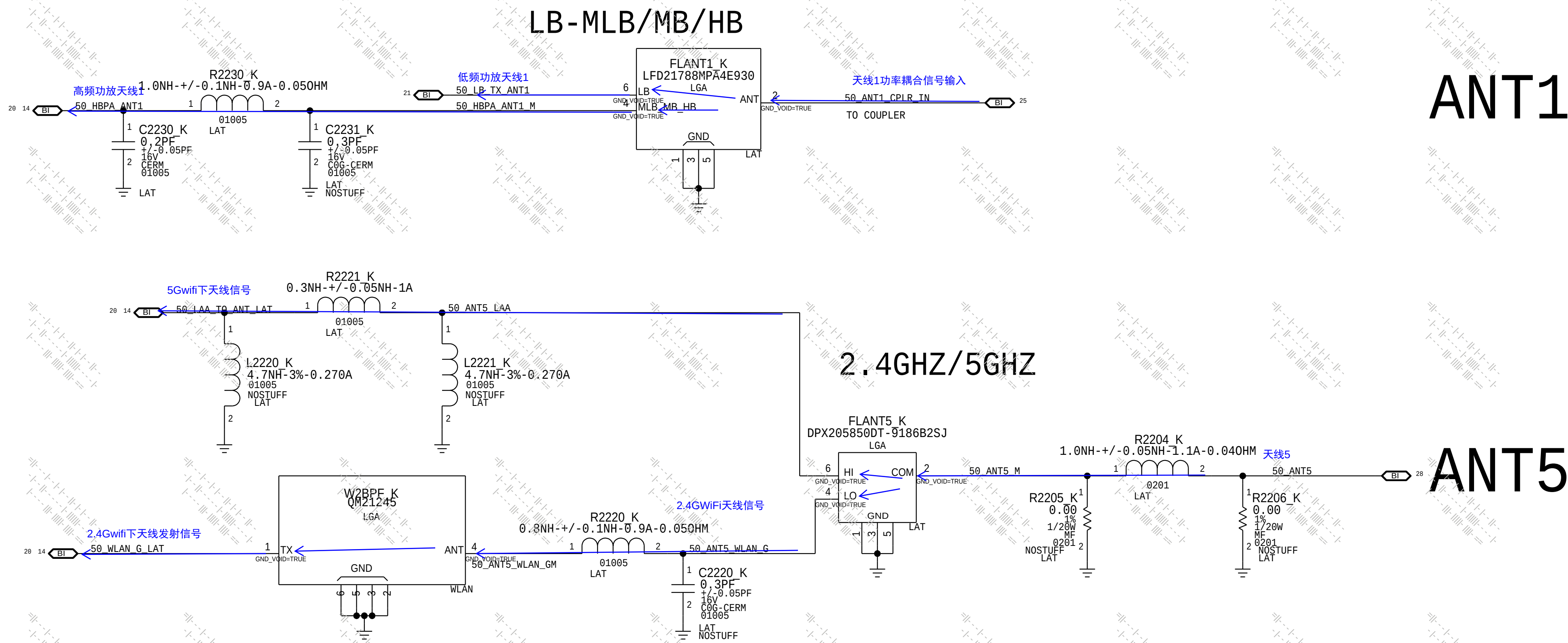


EQUATION		NET RULE ASSIGNMENT	
EXP-33	CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)	
P	PWR_100UM		VIO_CPLR*

CLASS DEFINITIONS			COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	CLASS OUTSIDE
CLASS NAME	MAX	CONSTRAINT SET	DP NAMES EX: DP-DP_AA* DP_BB* (LINE STARTS WITH FLAG DP)	Y/N
50_THIN	S	A_DIELECTRIC_2X	50_CPLL_CPL0UT*, 50_CPLU_CPL0UT*, 50_CPLR_ANT1	?
50_WIDE	S	A_DIELECTRIC_2X_50_WIDE_50	50_UHPA_ANT3, 50_UHPA_ANT2, 50_ANT3*, 50_UHPA_UHB	?

PAGE TITLE		SYNCH DATE=01/17/2018	
LOWER & UPPER COUPLER			
		DRAWING NUMBER	SIZE
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		25 OF 45	

LOWER ANTENNA FEEDS



UAT ANTENNAS

ANT2
LB
LMB/MB/HB
L1 GNSS
2.4GHZ

ANT4
R1
5GHZ
UHB

LAT ANTENNAS

ANT1
LB
LMB/MB/HB

ANT3
UHB

ANT5
2.4GHZ
5GHZ

NET RULE ASSIGNMENT		
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)	
P	A_50_THIN_SE	50_ANT1_CPLR_IN, 50_HBPA_ANT1*
P	A_50_WIDE_SE	50_ANT1, 50_ANT5*

CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*		GENERATED
CLASS NAME	CONSTRAINT SET	DP NAMES EX: DP-DP_AA*DP-BB* (LINE STARTS WITH FLAG DP)		Y/N
50_THIN	S	A_DIELECTRIC_2X		Y
50_WIDE	S	A_DIELECTRIC_2X_50_WIDE_SE		Y

PAGE TITLE			DRAWING NUMBER		SIZE
LOWER ANTENNA FEEDS			051-04049		D
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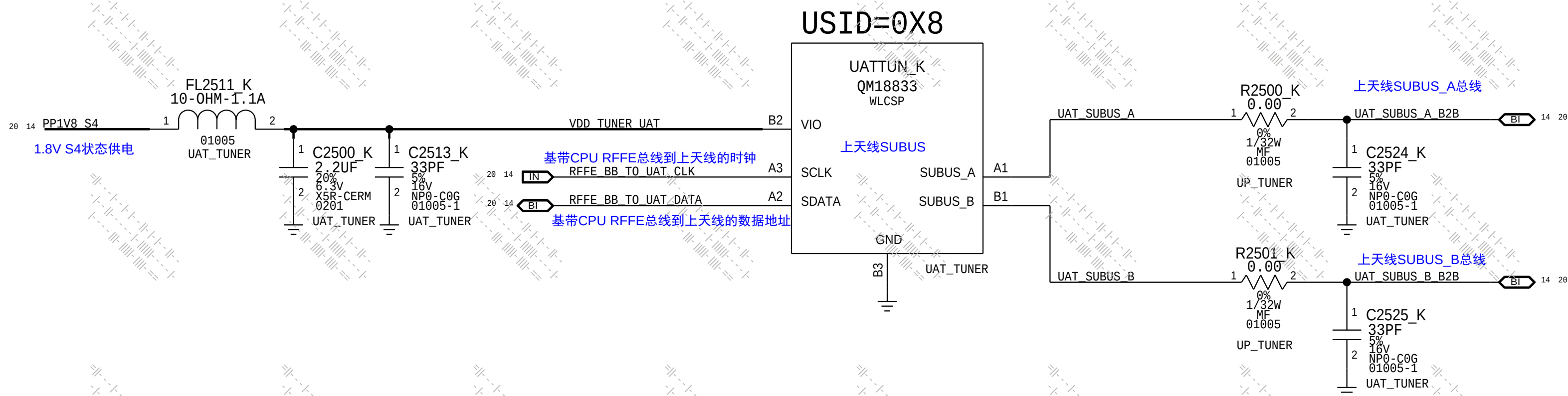


NET RULE ASSIGNMENT	
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)
A_50_THIN_SE	50_L1*, 50_GNSS_L1*, 50_LAA_TO_ANT_UAT, 50_ANT2_MLB_MB_HB, 50_ANT4_UHB
A_50_WIDE_SE	50_ANT2, 50_ANT2_M, 50_ANT2_WLAN, 50_ANT4, 50_ANT4_LAA

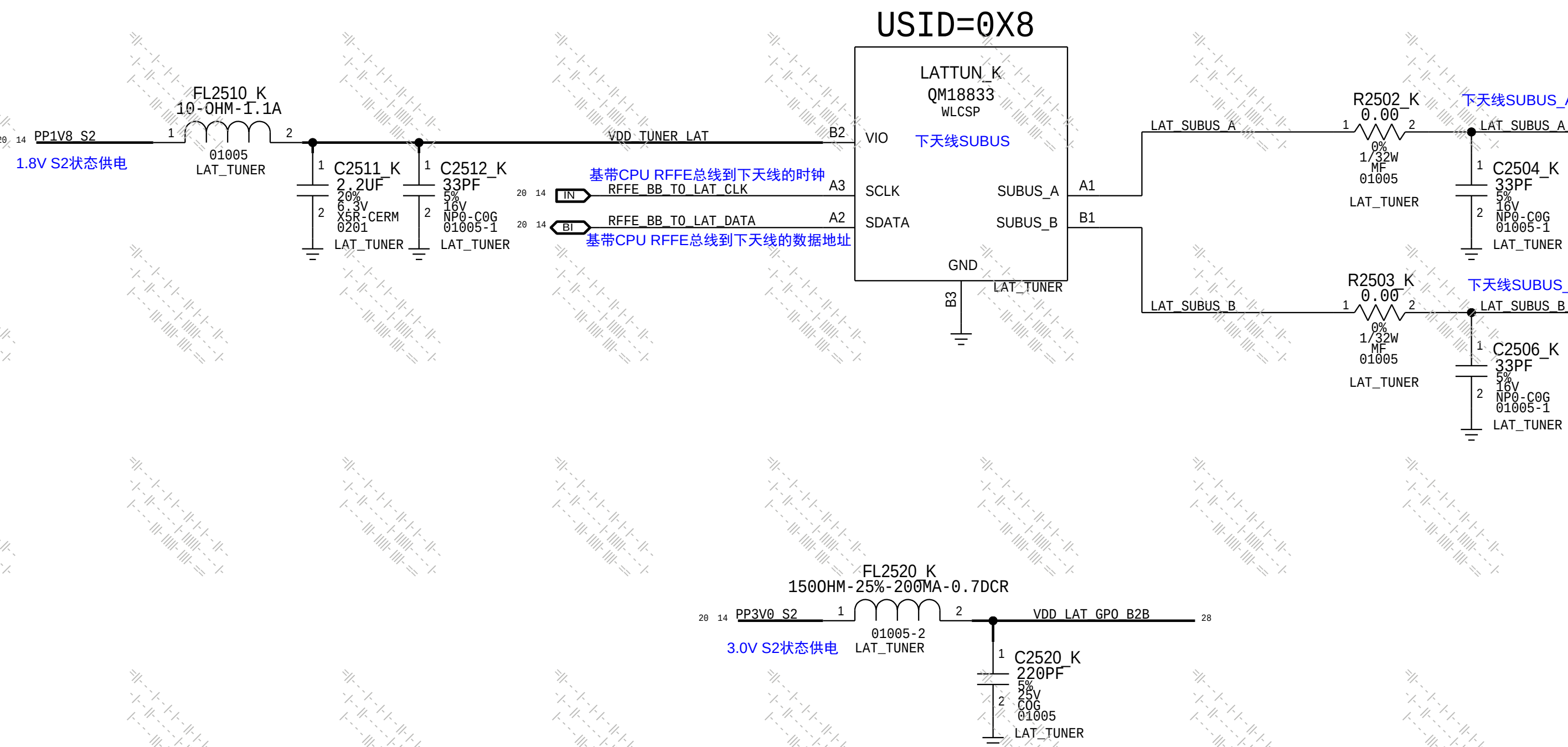
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UPPER ANTENNA FEEDS			
		DRAWING NUMBER	SIZE
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ANTENNA SYSTEM

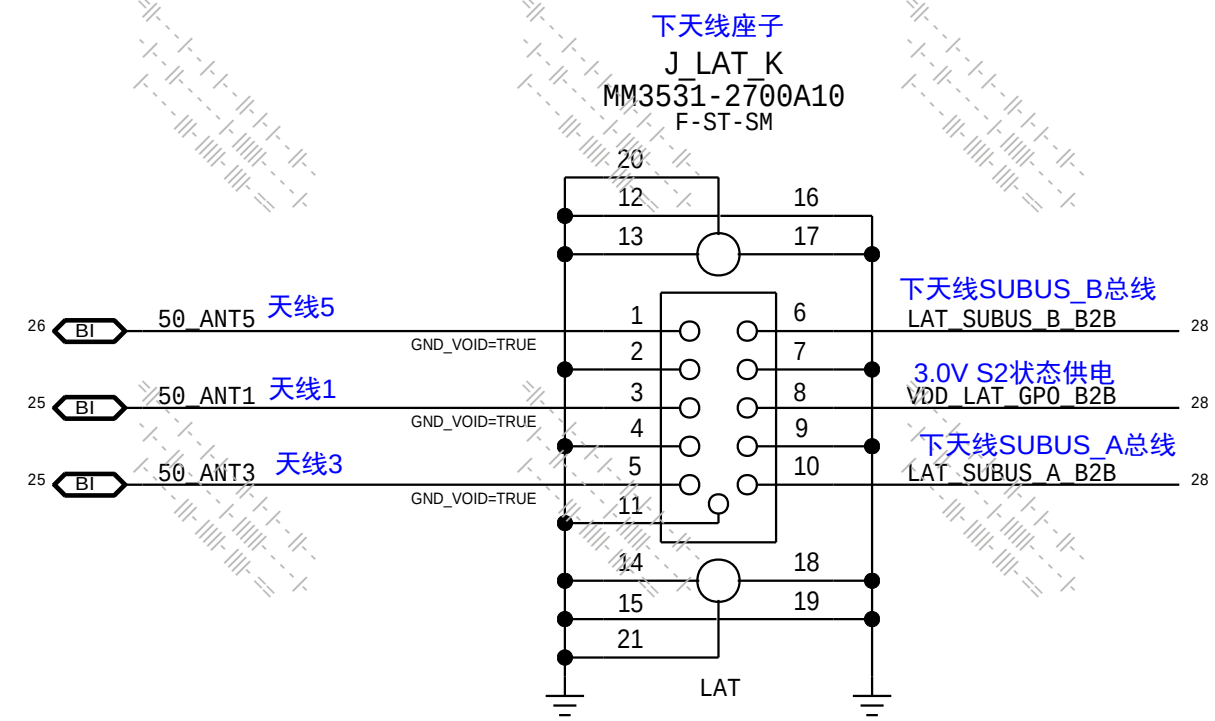
UAT SUBUS



LAT SUBUS



LAT



CLASS DEFINITIONS				COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	CLEAR OVERVIEW
CLASS NAME	TYPE	CONSTRAINT SET	DP NAMES EX: DP:DP_AA*, DP_BB* (LINE STARTS WITH FLAG DP:)		
RF_SHIELD	S	A_DIELECTRIC_2X	LAT_SUBUS*, UAT_SUBUS*	Y/N	Y

DOMAIN	NET RULE ASSIGNMENT	
DEFINITION	CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)
P	PWR_200UM	VDD_TUNER*, VDD_LAT_GPO*
P	PWR_100UM	LAT_SUBUS*, UAT_SUBUS*

PAGE TITLE		DRAWING NUMBER		SIZE
ANTENNA SYSTEM		051-04049		D
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14

IO

UART2_AOP_FROM_BB_RXD

41

34

14

IO

UART2_AOP_TO_BB_TXD

41

34

14

IO

UART_BB_TO_WLAN_COEX_1V2

41

34

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IO

UART_WLAN_TO_BB_COEX_1V2

34

34

14

IO

SPMI1_AOP_TO_RF_CLK

34

34

14

IO

SPMI1_AOP_BI_RF_DATA

41

34

14

IO

GPIO_BB_TO_MANY_GSM_BURST_IND

41

32

14

IO

GPIO_DOTARA_TO_BBPMU_CLK_EN

41

32

14

IO

CLK_BBPMU_TO_DOTARA_38M4_1V3

IO

RFFE_LAA_VIO

14

38

38

41

RFFE_LAA_VIO

IO

RFFE_LAA_LAT_DATA

14

38

38

41

RFFE_LAA_LAT_DATA

IO

RFFE_LAA_LAT_CLK

14

38

38

41

RFFE_LAA_LAT_CLK

39

14

IO

50_LAA_TO_XCVR_UAT

39

14

IO

50_LAA_TO_XCVR_LAT

41

34

14

IO

I0_BB_TO_R1_COEX

34

14

IO

R1_TO_BB_LAA_SW_CTRL

41

34

34

14

IO

I2S2_AP_FROM_BB_BCLK

41

34

34

14

IO

I2S2_AP_FROM_BB_LRCLK

41

34

14

IO

I2S2_AP_FROM_BB_DIN

41

34

14

IO

I2S2_AP_TO_BB_DOUT

41

38

33

14

IO

VRF_L0_1V15

41

38

33

14

IO

VRF_DIG_0V9

33

14

IO

VFE_L0_1V2

33

14

IO

VFE_HT_3V15

40

14

IO

VPA_ET_LB

40

14

IO

VPA_ET_HB

40

14

IO

VPA_ET_UHB

40

14

IO

VPA_2G

41

37

14

IO

GNSS_LNA_EN_L1

41

40

38

30

14

IO

RFFE_LAA_VIO

41

40

38

14

IO

RFFE_XCVR_TO_LBTX_CLK

41

40

38

14

IO

RFFE_XCVR_TO_LBTX_DATA

41

40

38

14

IO

RFFE_XCVR_TO_HBTX_CLK

41

40

38

14

IO

RFFE_XCVR_TO_HBTX_DATA

41

38

30

14

IO

RFFE_LAA_LAT_CLK

41

38

30

14

IO

RFFE_LAA_LAT_DATA

41

34

14

IO

RFFE_BB_TO_LAT_CLK

41

34

14

IO

RFFE_BB_TO_LAT_DATA

41

34

14

IO

RFFE_BB_TO_UAT_CLK

41

34

14

IO

RFFE_BB_TO_UAT_DATA

37

14

IO

50_XCVR_TX0_HB

37

14

IO

50_XCVR_TX0_LB1

37

14

IO

50_XCVR_TX0_LB2

37

14

IO

50_XCVR_TX0_MB1

37

14

IO

50_XCVR_TX0_MB2

37

14

IO

50_XCVR_TX0_MB3

37

14

IO

50_XCVR_TX1_UHB

37

14

IO

50_XCVR_TX1_HB

37

14

IO

50_XCVR_TX1_MB1

37

14

IO

50_XCVR_FBR0_1

37

14

IO

50_XCVR_FBR1_1

37

14

IO

50_XCVR_FBR0_2

37

14

IO

50_XCVR_FBR1_2

37

14

IO

50_GNSS_L1

39

14

IO

50_XCVR_RX2_UHB_PRX

39

14

IO

50_XCVR_RX6_MB_UHB_PRX

39

14

IO

GND

39

14

IO

50_XCVR_RX8_MLB_HB_PRX

39

14

IO

50_XCVR_RX9_MLB_HB_PRX

39

14

IO

50_XCVR_RX10_MLB_HB_PRX

39

14

IO

GND

39

14

IO

GND

39

14

IO

50_XCVR_RX13_LB_MLB_PRX

39

14

IO

50_XCVR_RX14_LB_MLB_PRX

39

14

IO

GND

39

14

IO

50_XCVR_RD4_MB_UHB_DRX

39

14

IO

GND

39

14

IO

GND

39

14

IO

50_XCVR_RD7_MLB_HB_DRX

39

14

IO

50_XCVR_RD8_MLB_HB_DRX

39

14

IO

50_XCVR_RD9_MB_UHB_DRX

39

14

IO

50_XCVR_RD10_LB_MLB_DRX

41

36

33

14

IO

VDD_SIM1

41

34

14

IO

SIM1_I0

41

34

14

IO

SIM1_CLK

41

34

14

IO

SIM1_RST

41

34

14

IO

SIM1_DETECT

41

36

33

14

IO

VDD_SIM2

41

34

14

IO

SIM2_I0

41

34

14

IO

SIM2_CLK

41

34

14

IO

SIM2_RST

41

34

14

IO

SIM2_DETECT

36

35

34

33

14

IO

VDD_I0_1V2

等效信号线

SCH, RADIO, KAR00

SYNC_DATE=10/09/2017

DRAWING TITLE

SCH, MLB, BOT, ICE, N104

DRAWING NUMBER

051-04049

SIZE

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REVISION

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87654321

BOM TABLES

XTAL ALTS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS
197S00156	197S00155	ALT_PARTS	Y301_K	XTAL, 38.4MHZ, TXC
197S00179	197S00155	ALT_PARTS	Y301_K	XTAL, 38.4MHZ, NDK

EEPROM ALTS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS
335S00013	335S00894	ALT_PARTS	EEPROM_K	EEPROM,3KBIT,12C

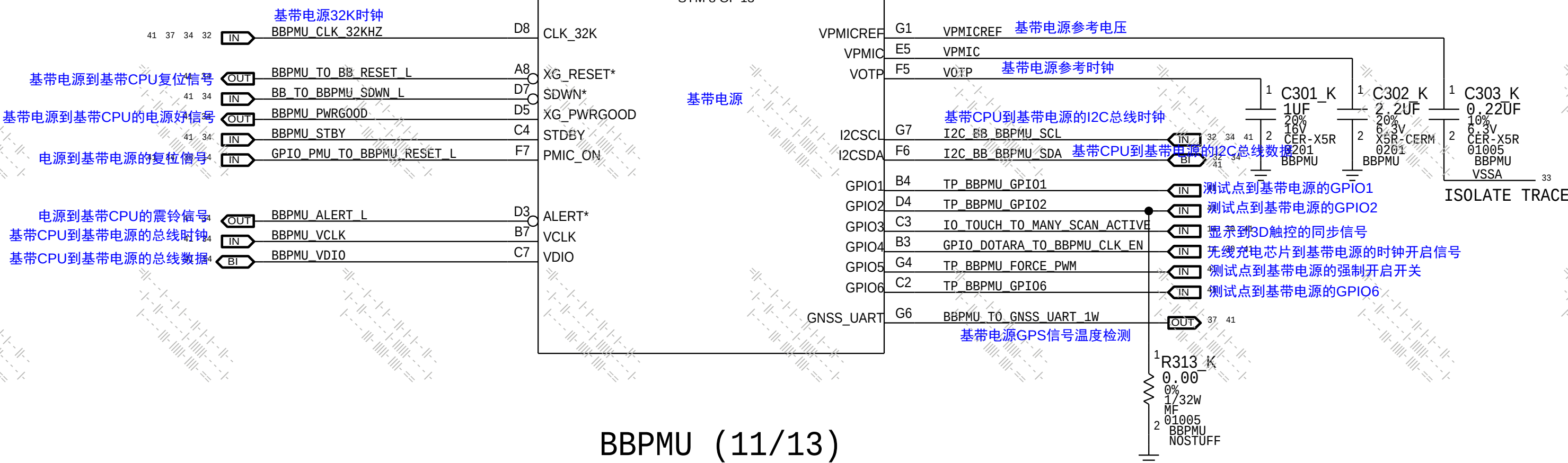
BBPMU ALTS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS
152S01141	152S01045	ALT_PARTS	L401_K, L402_K	TAIYO
152S01142	152S01043	ALT_PARTS	L405_K	TAIYO

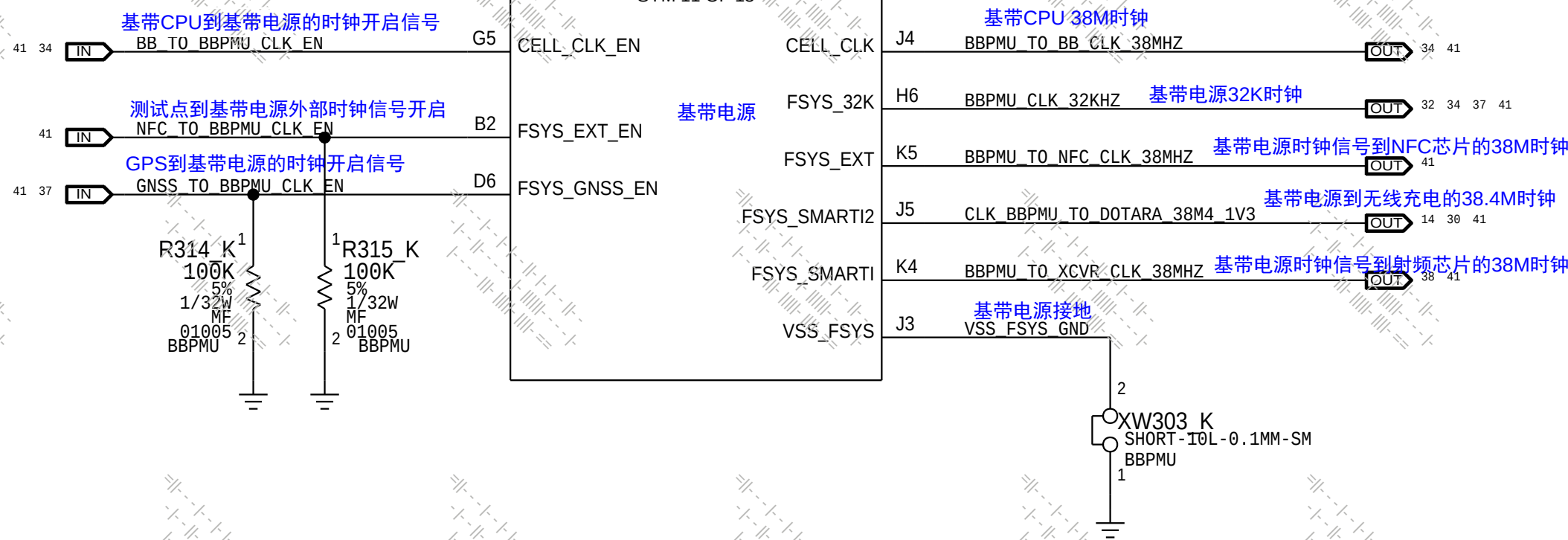
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BOM TABLES			051-04049		D	
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BBPMU: CONTROL (1/2)

BBPMU (8/13)

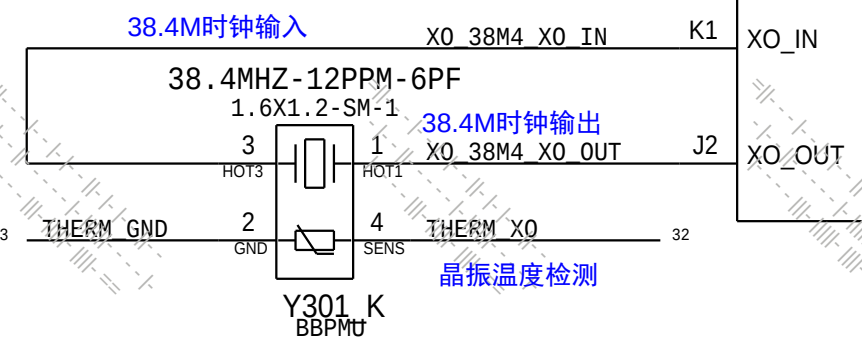


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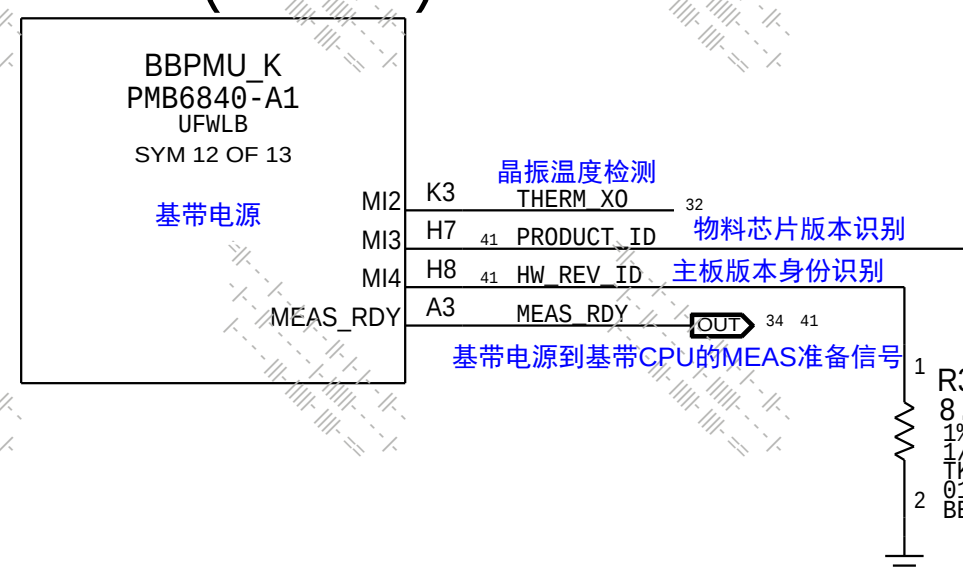


BBPMU (13/13)

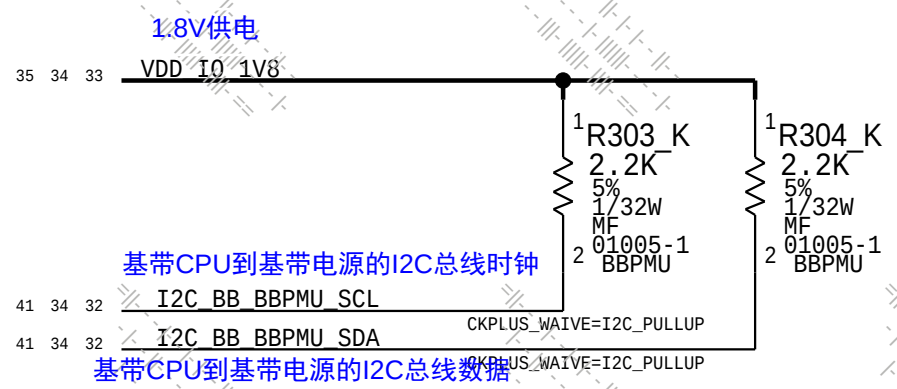
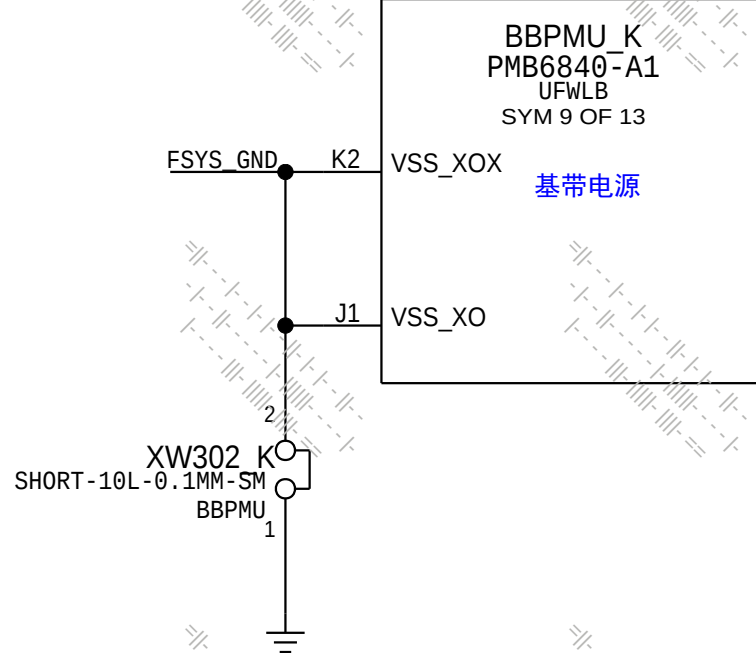
Y301_K APN=197S00155
6PF LOAD CAPACITANCE



BBPMU (12/13)



BBPMU (9/13)



BBPMU ADC TABLES

REVISION

匹配电阻对照表

PRODUCT

REVISION			PRODUCT		
R311_K	MLB	RF_DEV	R312_K	PRODUCT_ID	X-CODE
0.0	PROT00	0.1	0.0	RFDEV 19.1/2	
1.2K	PROT01	1.0	1.2K	RFDEV 19.0	
2.2K	PROT02	2.0	2.2K	ICE19.0 MLB	X1403
3.3K	PROT02 V2	2.1	3.3K	ICE19.1 MLB	X1368
4.7K	EVT	3.0	4.7K	ICE19.2 MLB	X1369
6.8K	CARRIER	RESERVED	6.8K		
8.2K	DVT	4.0	8.2K		
10K	PVT	5.0	10K		
12K			12K		
15K			15K		
18K			18K		
22K			22K		
27K			27K		
33K			33K		
39K			39K		
47K			47K		
56K			56K		
82K			82K		
100K			100K		
120K			120K		
150K			150K		

HWID "TABLE" R0AR:7/36363220

CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	
CLASS NAME	CONSTRAINT SET	DP NAMES EX: DP_DP_AA* DP_BB* (LINE STARTS WITH FLAG DP)	Y/N
RF_SHIELD	S	A_DIELECTRIC_2X	Y
RF_SHIELD	S	A_DIELECTRIC_2X	Y

CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
RF_SHIELD	GND	DEFAULT
RF_SHIELD	RF_SHIELD	A_DIELECTRIC_2X0

NET RULE ASSIGNMENT	
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)
P	PWR_100UM VPMICREF, VPMIC, VOTP

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
118500118	1	2.2K 01005 1%	R312_K	ICE19.0
118500119	1	3.3K 01005 1%	R312_K	ICE19.1
11850636	1	4.7K 01005 1%	R312_K	ICE19.2

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PAGE TITLE		SYNC_DATE=01/08/2018	
BBPMU: RAILS (2/2)			
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CLASS-TO-CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
RFFE_SHIELD	RFFE_SHIELD	DEFAULT
RFFE_SHIELD	GND	DEFAULT

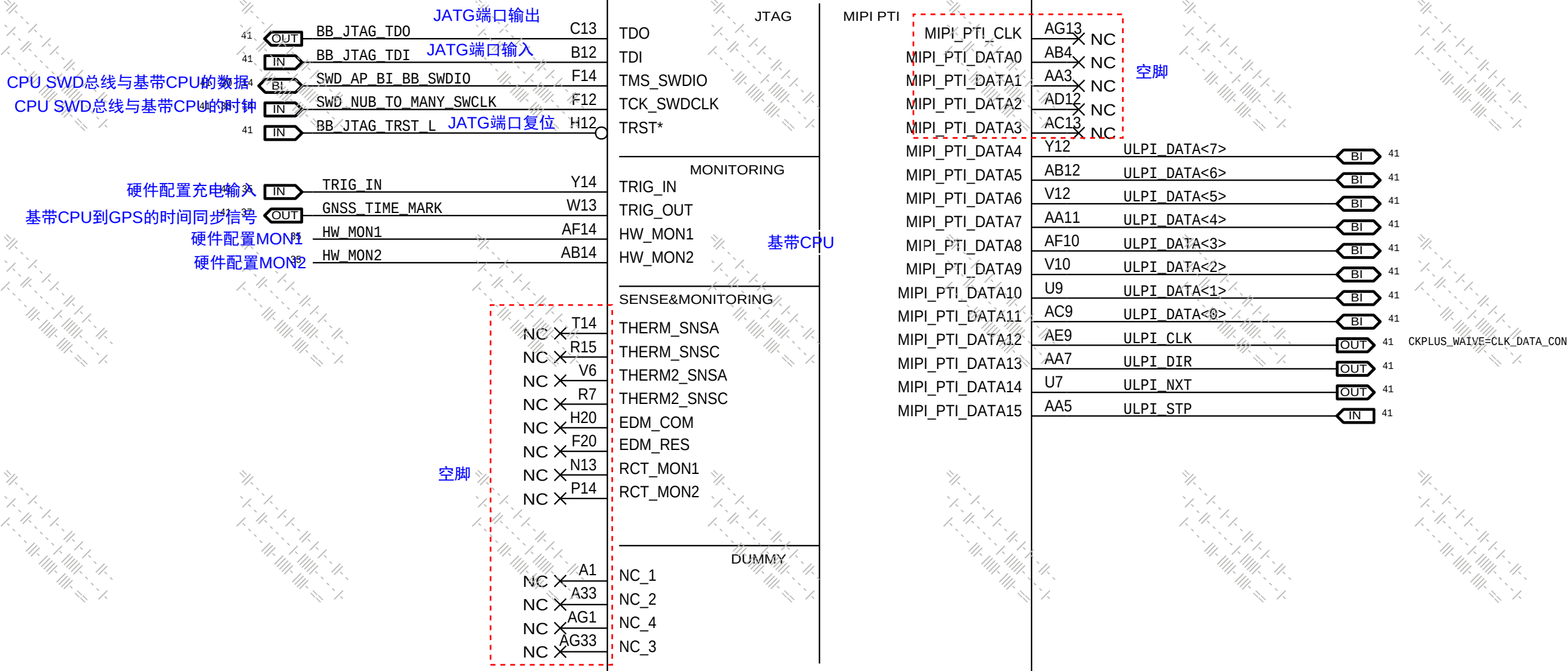
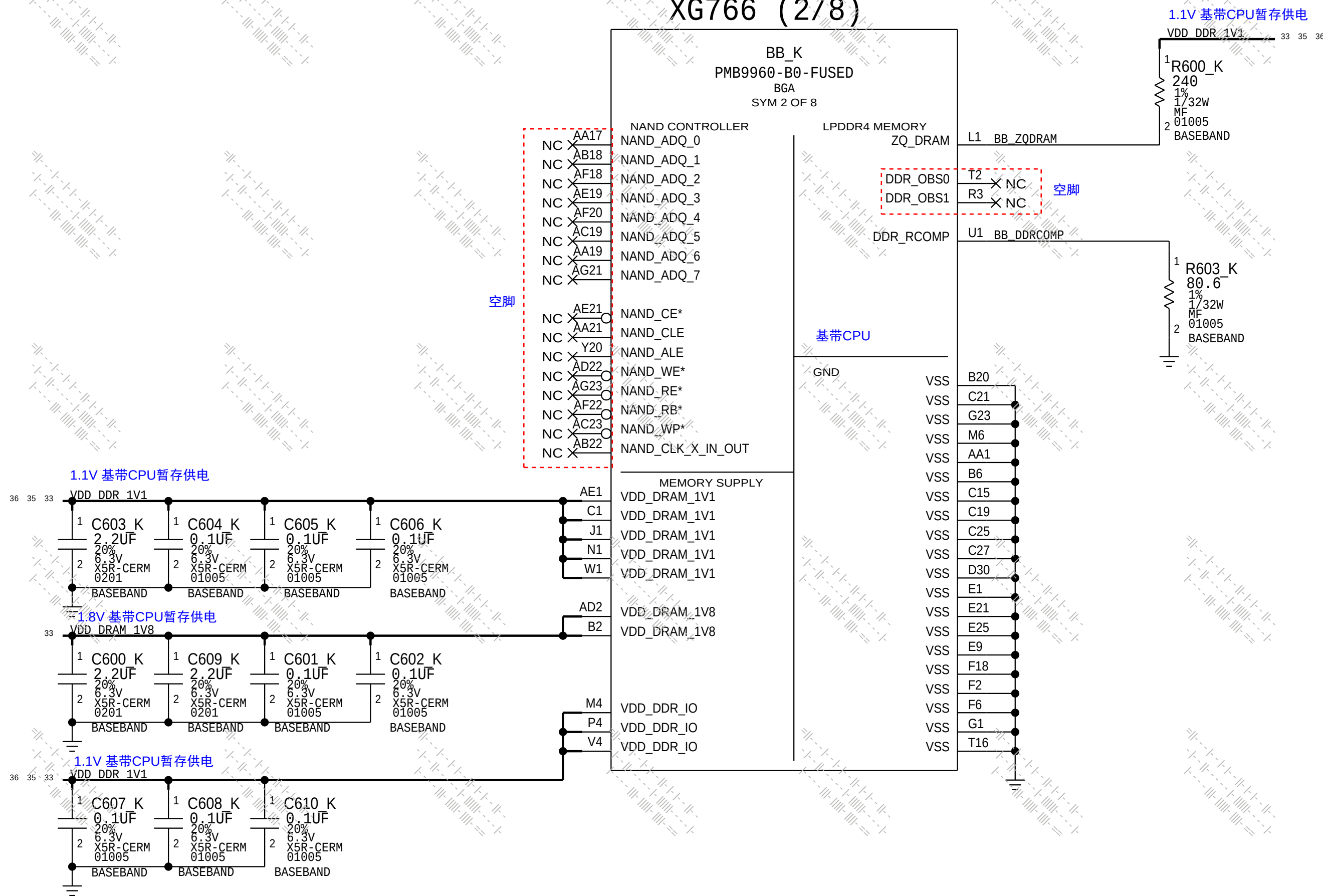


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PAGE TITLE		BB: INTERFACE (1/3)	
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BB: MEMORY & TRACING (2/3)

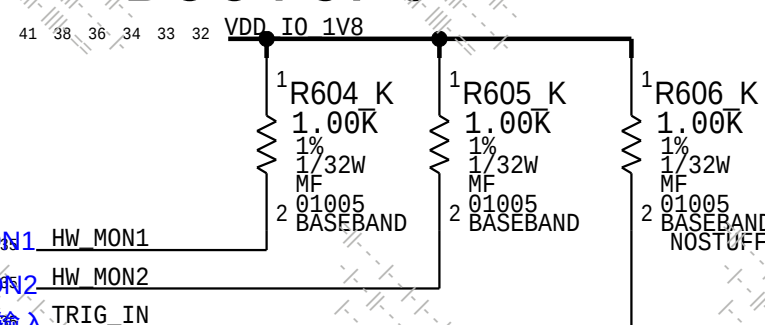
XG766 (2/8)

XG766 (7/8)



NAND	0	0	1
FLASHLESS	1	0	1
FLASHLESS WDOG DIS	1	1	0
NAND (NON-SECURE)	1	1	1
BOOTSTRAPS ARE INTERNALLY PULLED DOWN			

BOOTCFG



PERST_L IS PULLED DOWN ON AP DEV.
BB REQUIRES PERST_L BE PULLED UP WHEN PCIE IS NOT USED
ASSUME THAT IN NAND BOOT, PCIE IS NOT USED

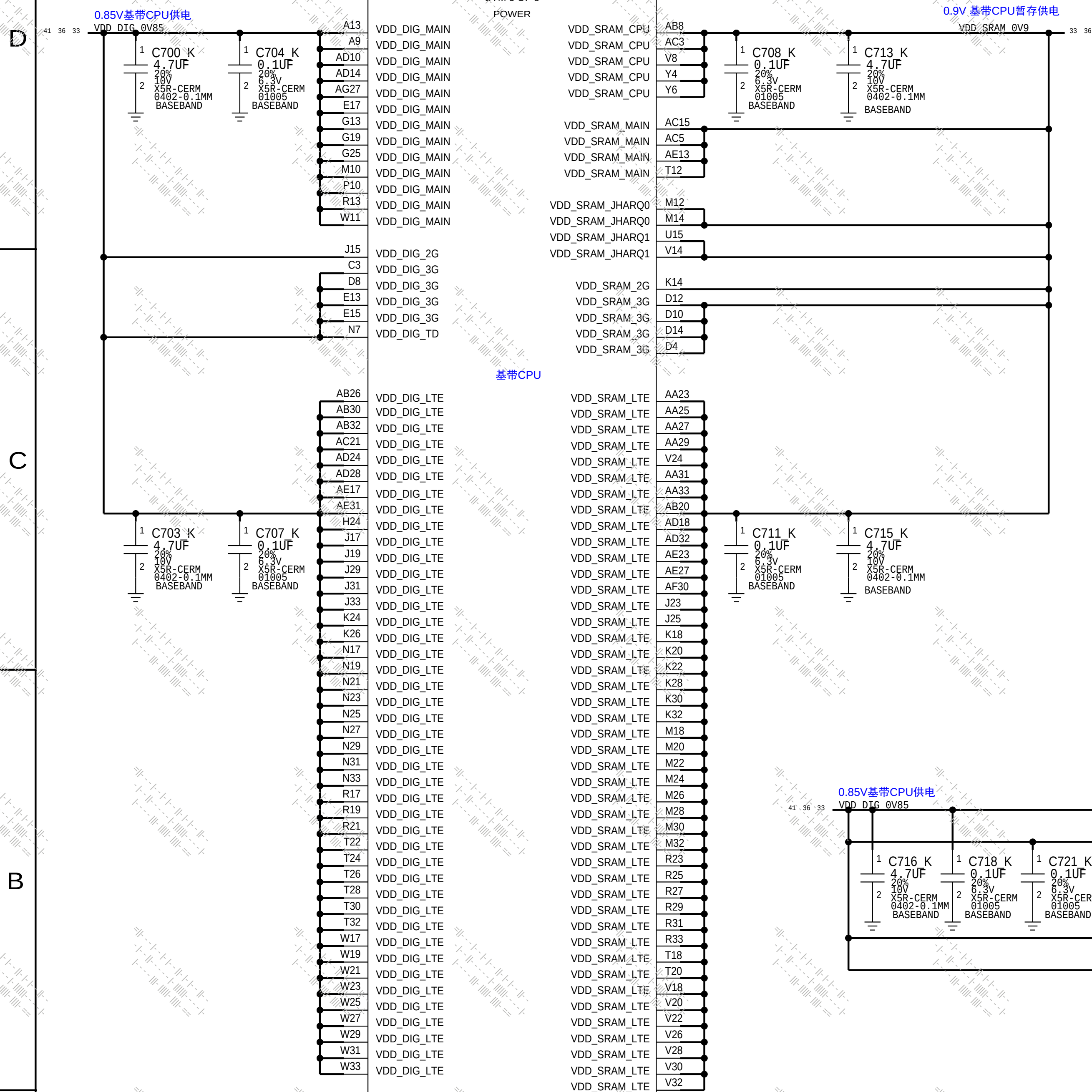
CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*	CLASS CONSTRN Y/N
CLASS NAME	CONSTRAINT SET		
RF_SHIELD	S	BB_ZQDRAM, BB_DDRCOMP	Y

			SYNC_DATE=01/08/2018		
PAGE TITLE					
BB: MEMORY & TRACING (2/3)					
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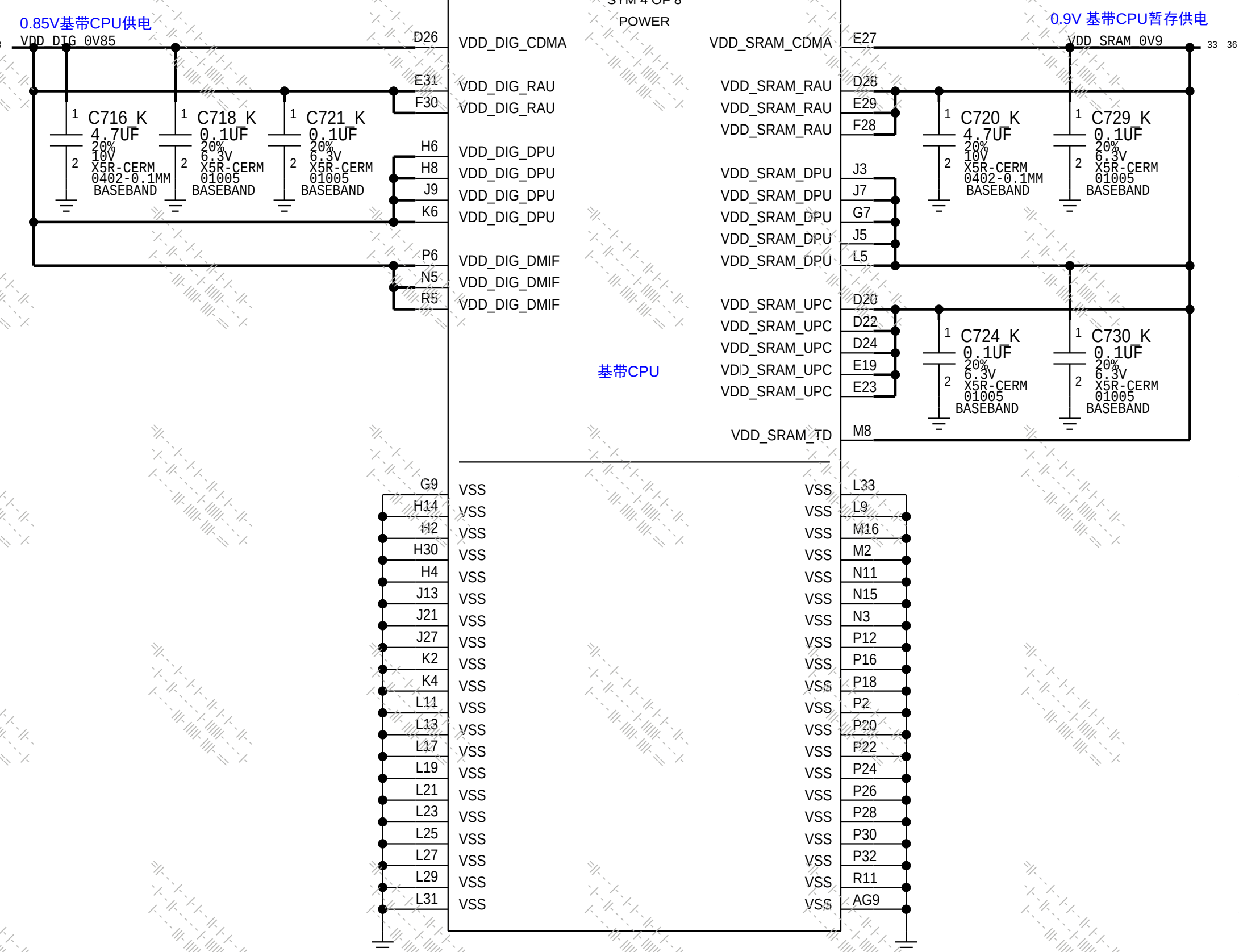
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BB: POWER (3/3)

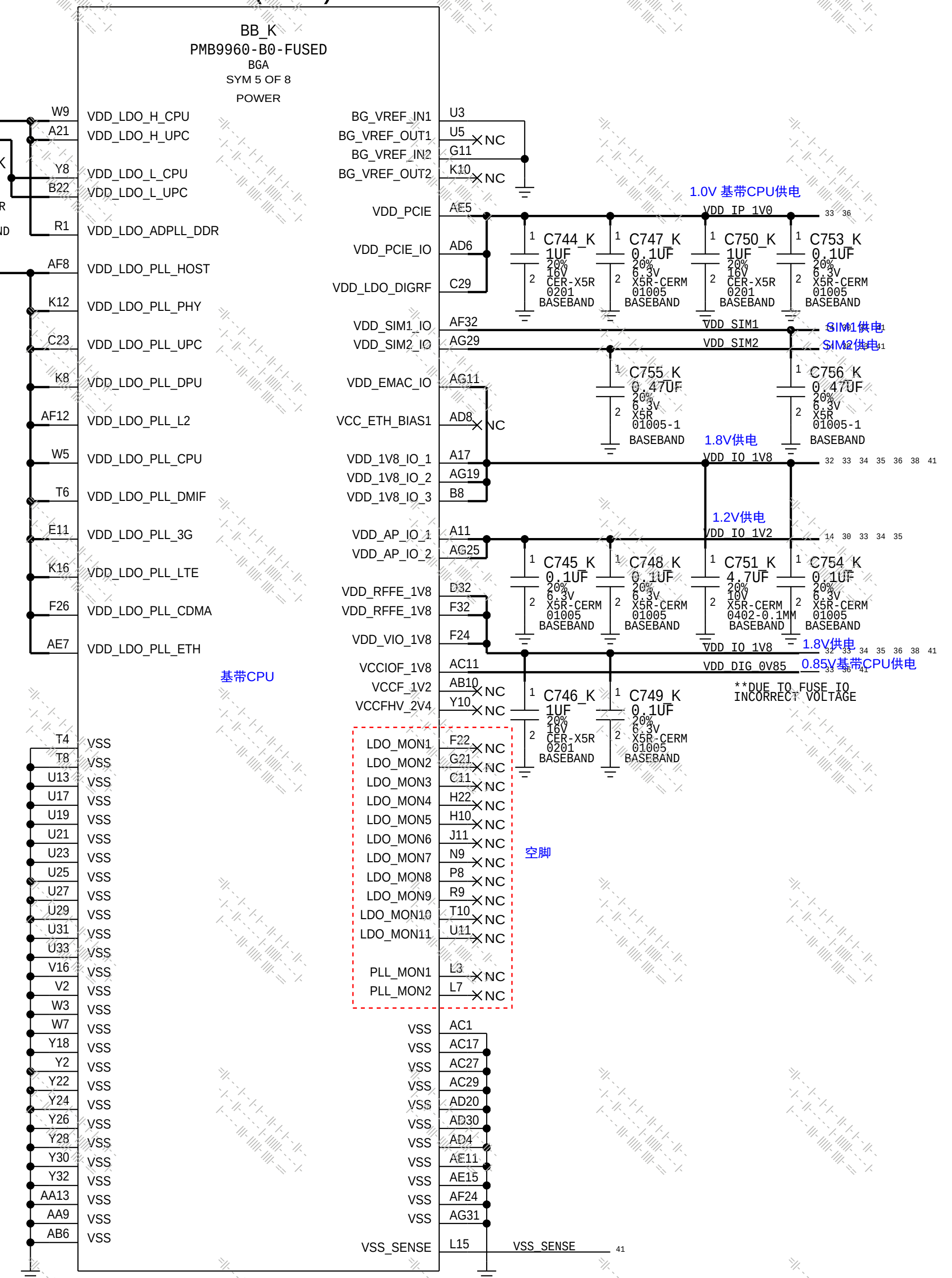
XG766 (3/8



XG766 (4/8



XG766 (5/8

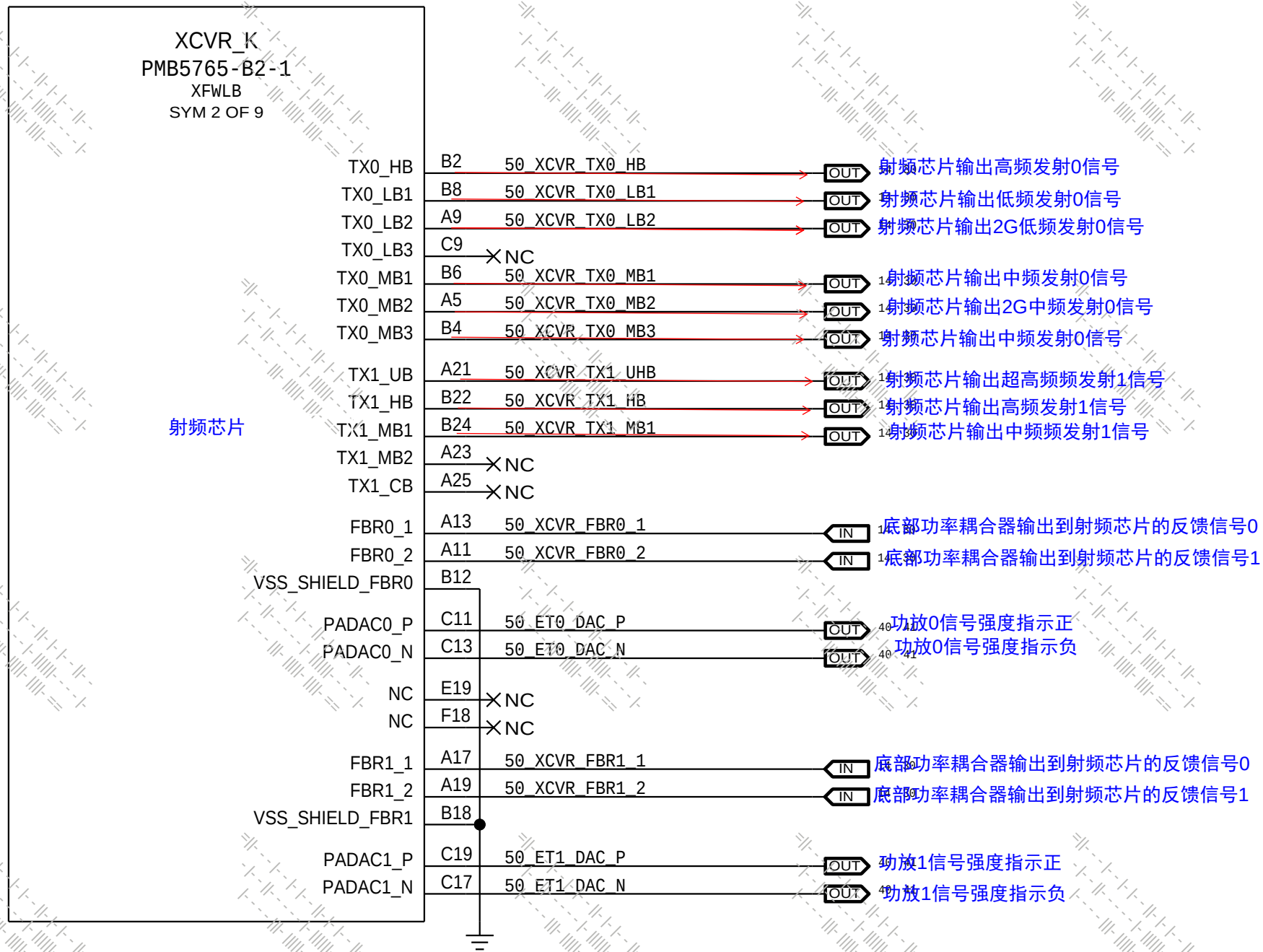


VDD AP IO = 1.2V

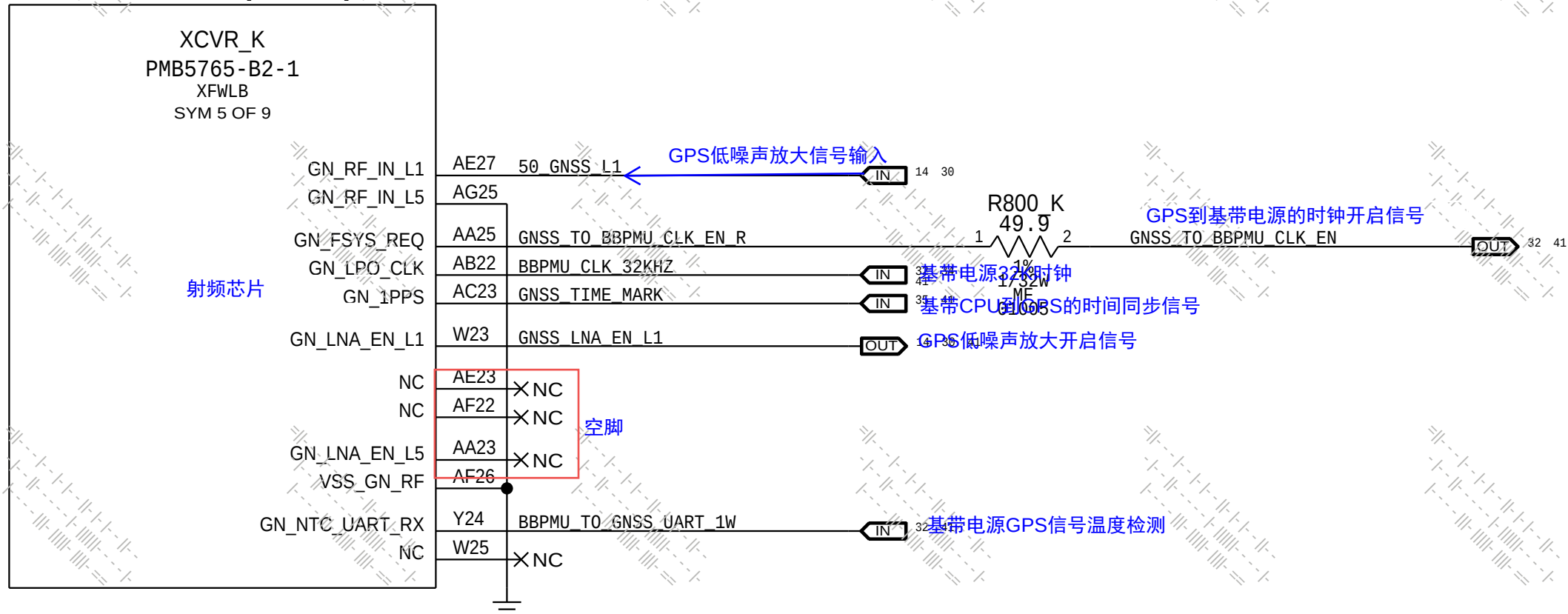
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XCVR: TRANSMIT & GNSS (1/3)

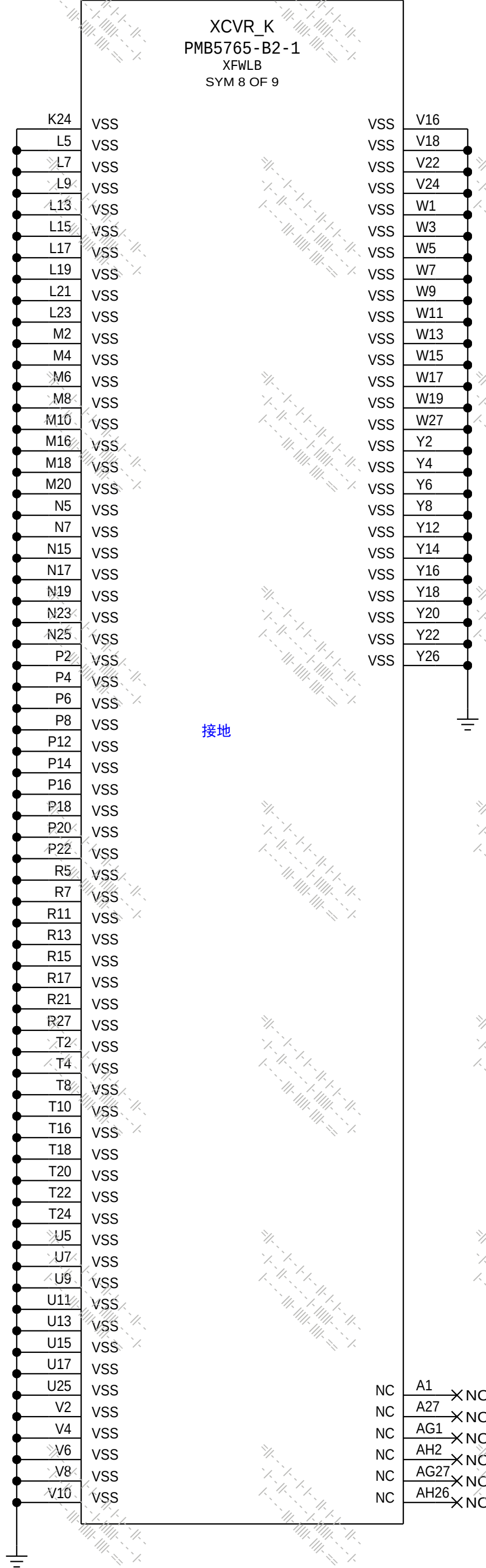
SMARTI8 (2/9)



SMARTI8 (5/9)



SMARTI8 (8/9)



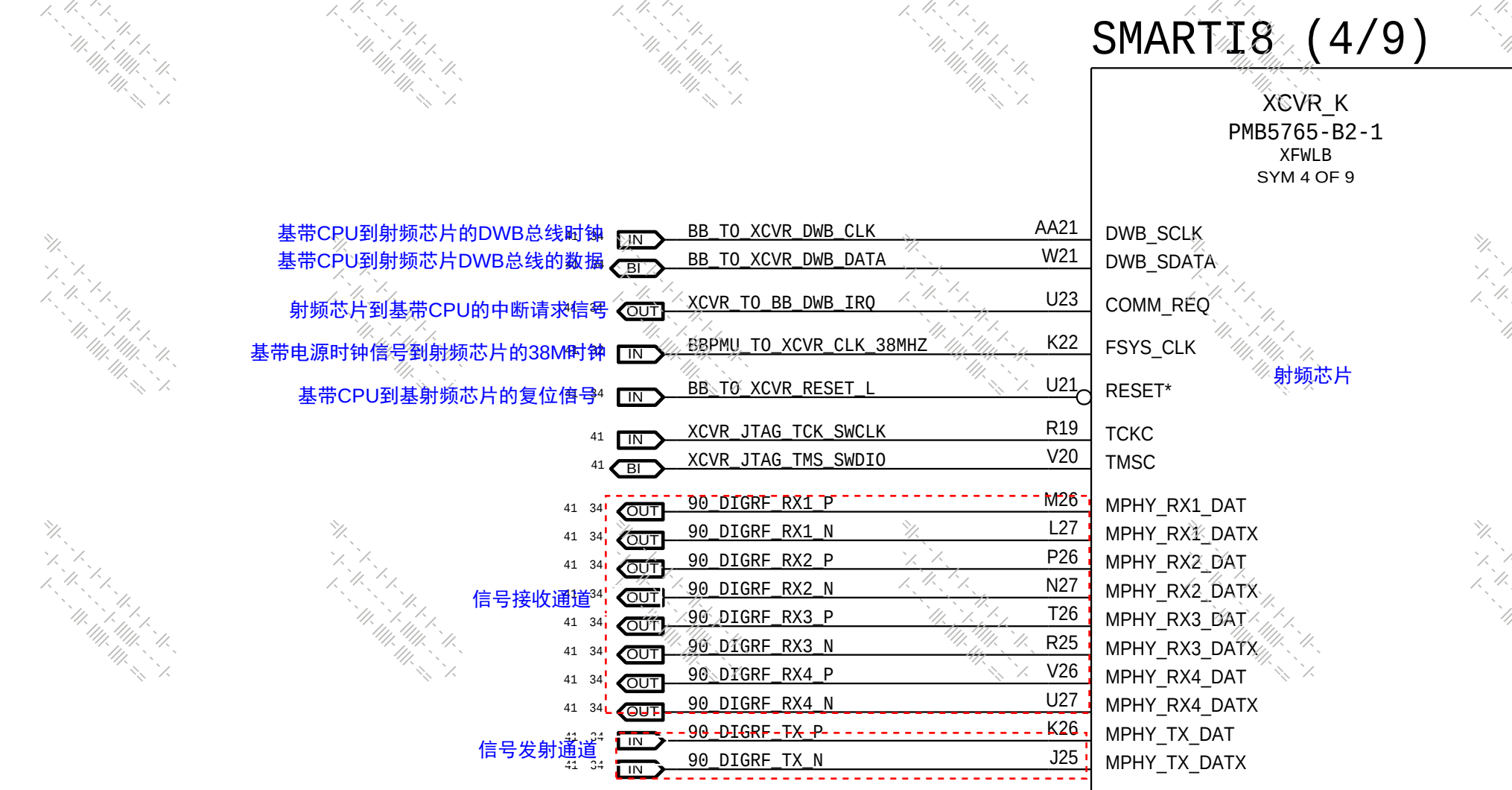
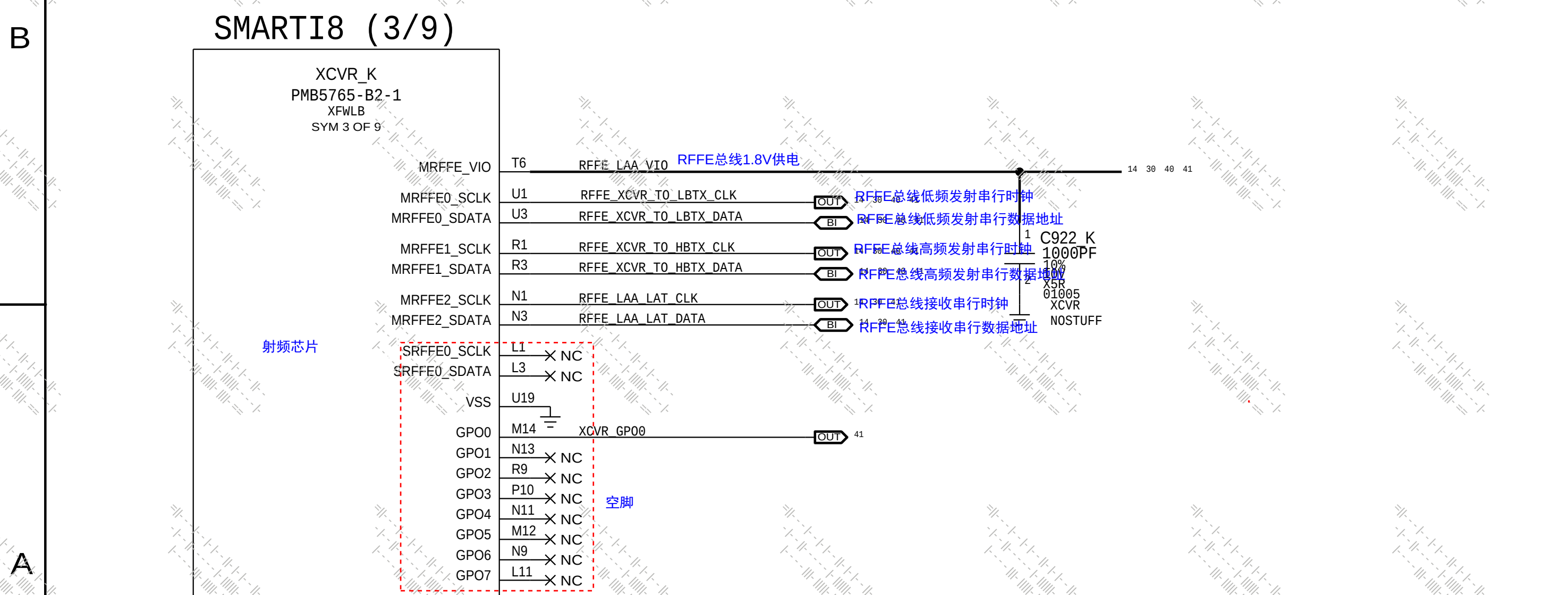
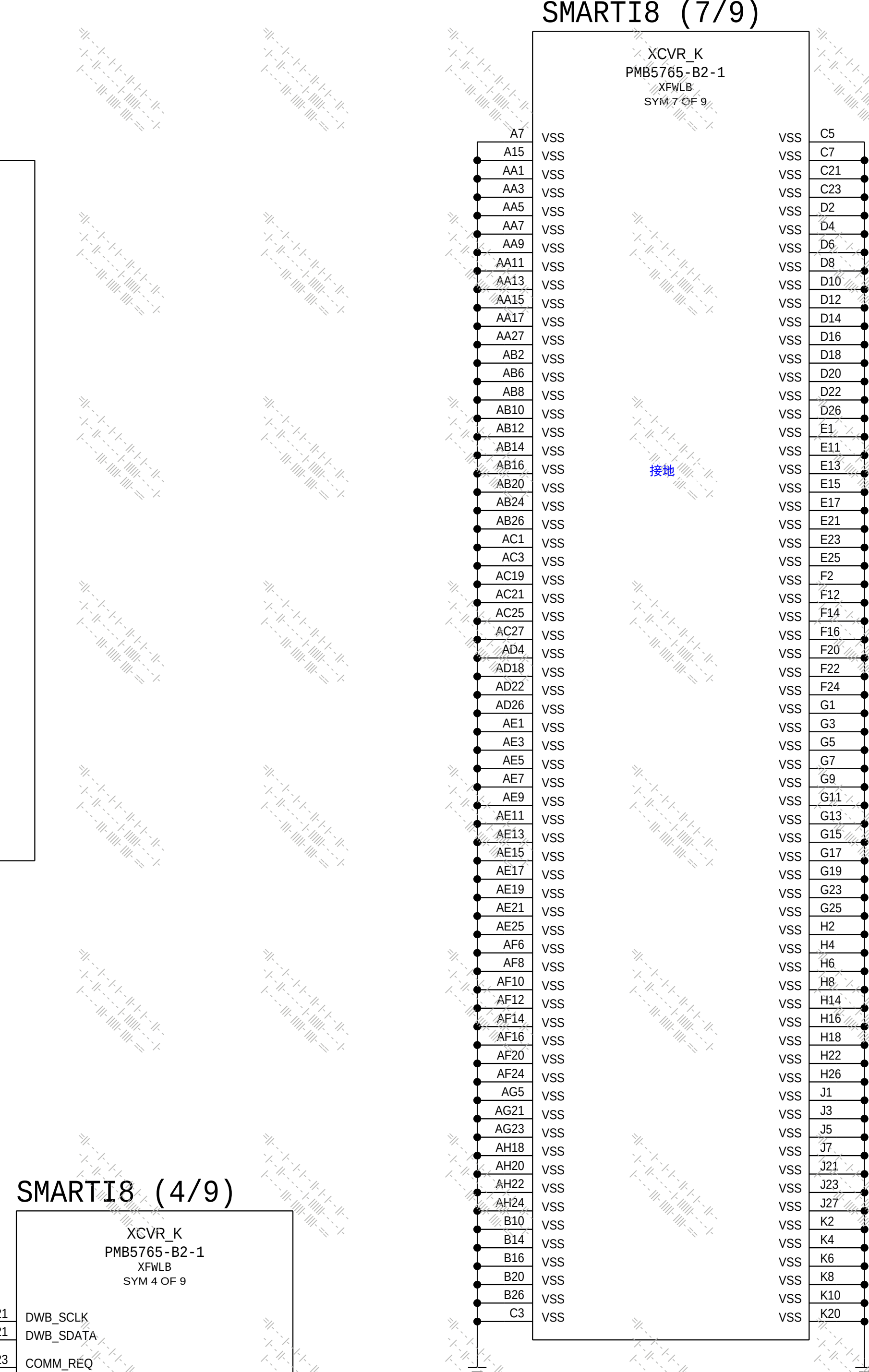
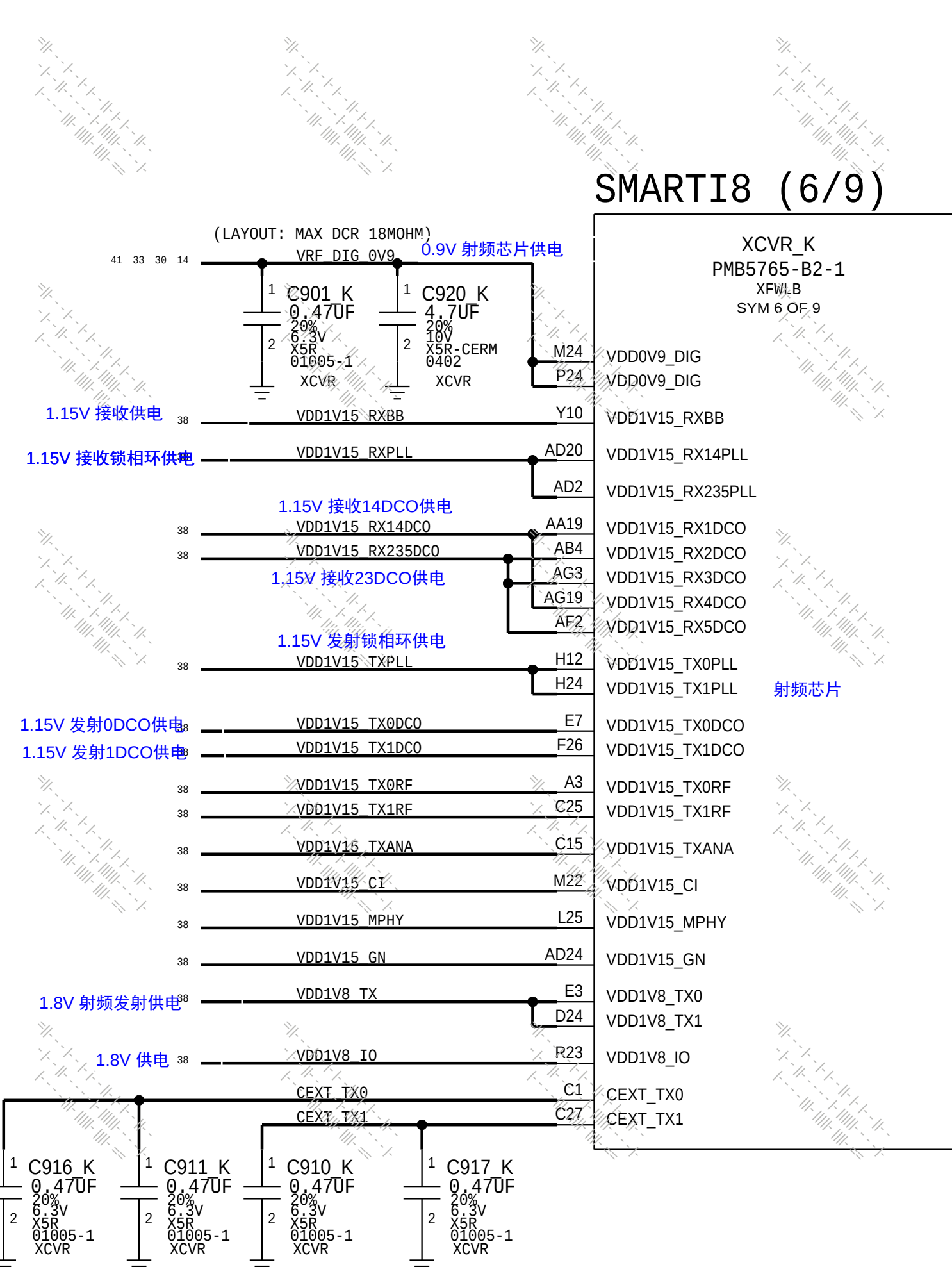
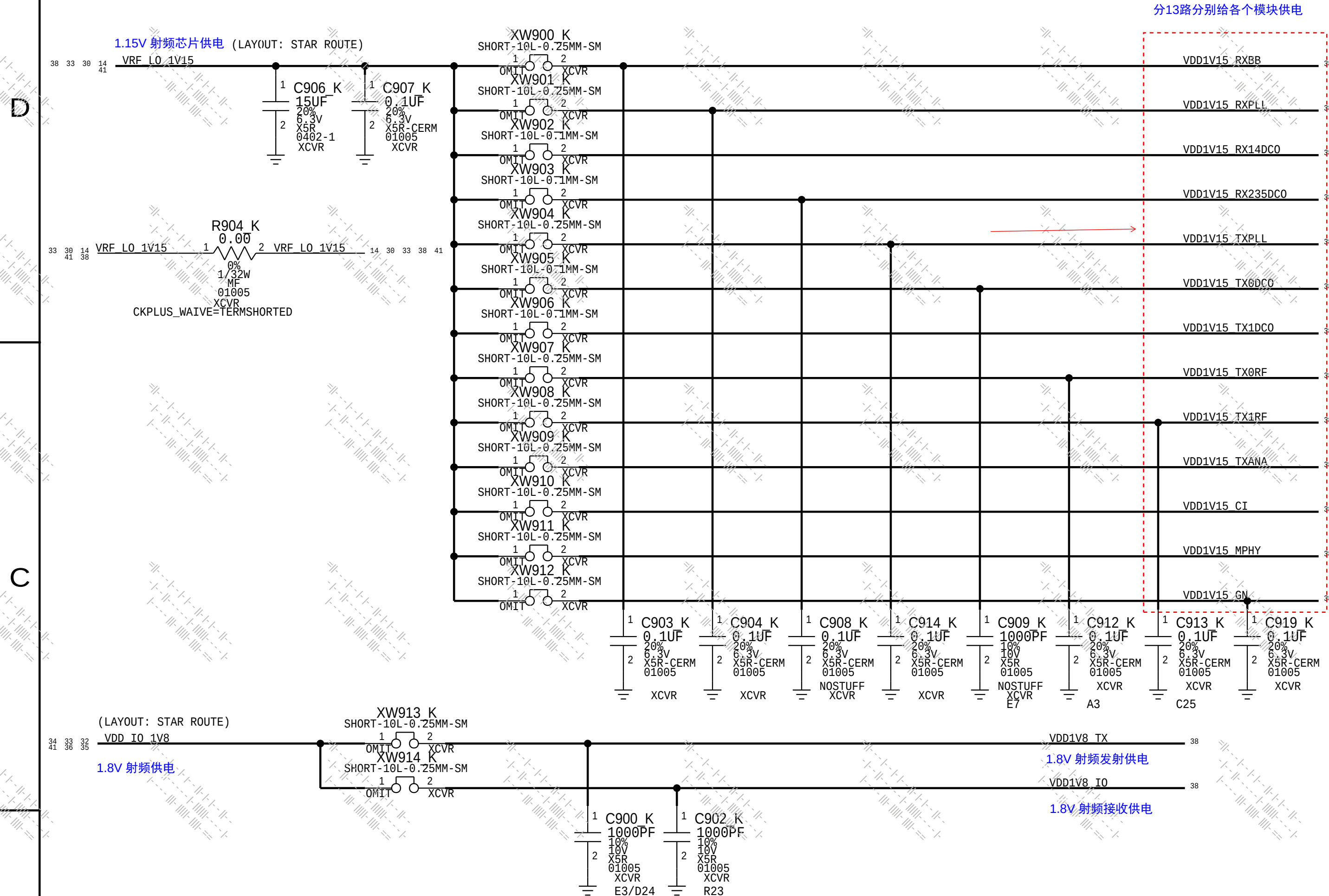
CLASS TO CLASS SPACING		
CLASS NAME	CLASS NAME	CONSTRAINT SET
50_WIDE	50_WIDE	A_DIELECTRIC_4XV_50_WIDE_SE
50_WIDE	50_THIN	A_DIELECTRIC_4XD_50_WIDE_SE
50_WIDE	50_THIN_UNSHIELDED	A_DIELECTRIC_4XD_50_WIDE_SE
50_THIN	50_THIN	A_DIELECTRIC_4XD_50_THIN_SE
50_THIN	50_THIN_UNSHIELDED	A_DIELECTRIC_4XD_50_THIN_SE
50_THIN_UNSHIELDED	50_THIN_UNSHIELDED	A_DIELECTRIC_2X

NET RULE ASSIGNMENT		
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)	
A_50_THIN_SE	50_XCVR*	
P_GENERIC_DP	50_ET*	
E_GENERIC_DP	50_ET*	

CLASS DEFINITIONS		COMMA SEPARATED WITH WILDCARD SUPPORT: NET NAMES EX: DDR*		CLASS
CLASS NAME	CONSTRAINT SET	DP NAMES EX: DP:DP_AA*,DP_BB* (LINE STARTS WITH FLAG DP)		Y/N
50_THIN_UNSHIELDED	S	A_DIELECTRIC_2X	50_XCVR_TX0_LB2, 50_XCVR_TX0_MB2	Y
50_THIN	S	A_DIELECTRIC_2X	50_XCVR_TX0_HB, 50_XCVR_TX0_LB1, 50_XCVR_TX0_MB1	Y
50_THIN	S	A_DIELECTRIC_2X	50_XCVR_TX0_MB3	Y
50_THIN	S	A_DIELECTRIC_2X	50_XCVR_TX1*, 50_XCVR_FBR*	Y
RF_SHIELD	S	A_DIELECTRIC_2X	50_ET*	Y
GENERIC_DP	P	GENERIC_DP	DP:DP_50_ET*	Y

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XCVR: TX & GNSS (1/3)					
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XCVR: INTERFACE & PWR (2/3)



CONSTRAINT	NET RULE ASSIGNMENT	
SET ID	CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)
P	PWR_300UM	VDD1V15*, VDD1V8_TX, VDD1V8_I0
P	PWR_200UM	CEXT_TX*
P	PWR_100UM	VIO_RFFE*

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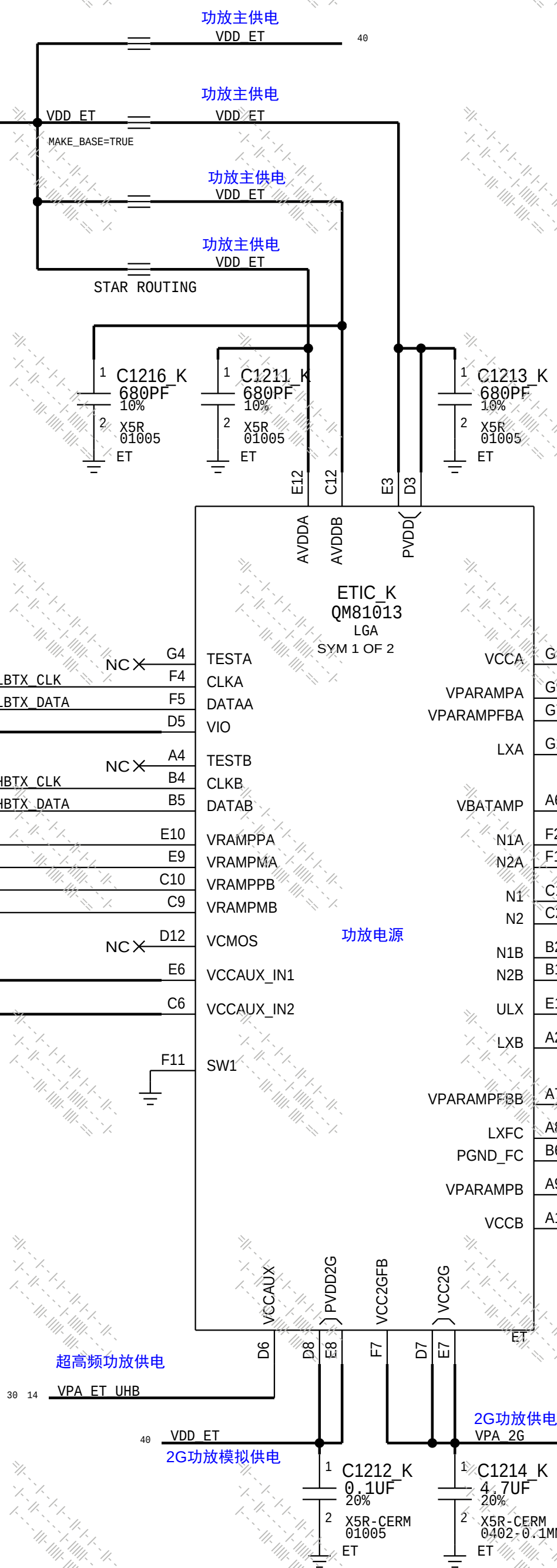
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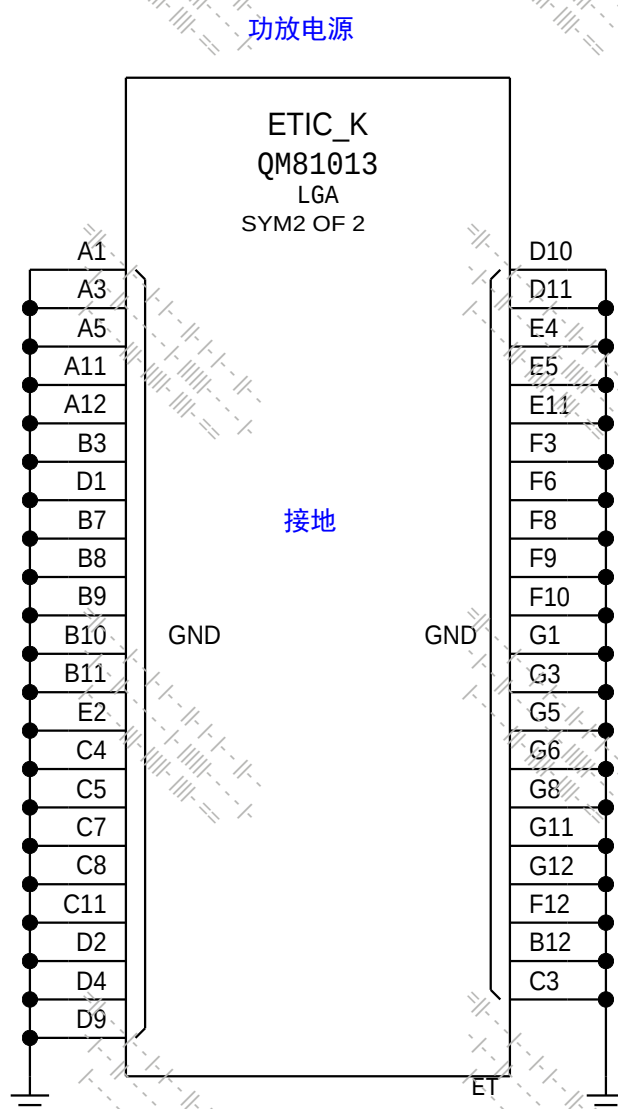
NET RULE ASSIGNMENT		
CONSTRAINT SET	COMMA SEPARATED NET NAMES (WILDCARD SUPPORT EX: DDR*)	
PWR_SHAPE	VPA_ET_LB, VPA_ET_HB, VPA_ET_UHB, VPA_2G	
PWR_SHAPE	VDD_ET*, V*_ET, VET_LX*, VET_ULX	
PWR_SHAPE	VET_N*	

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USID=0XC

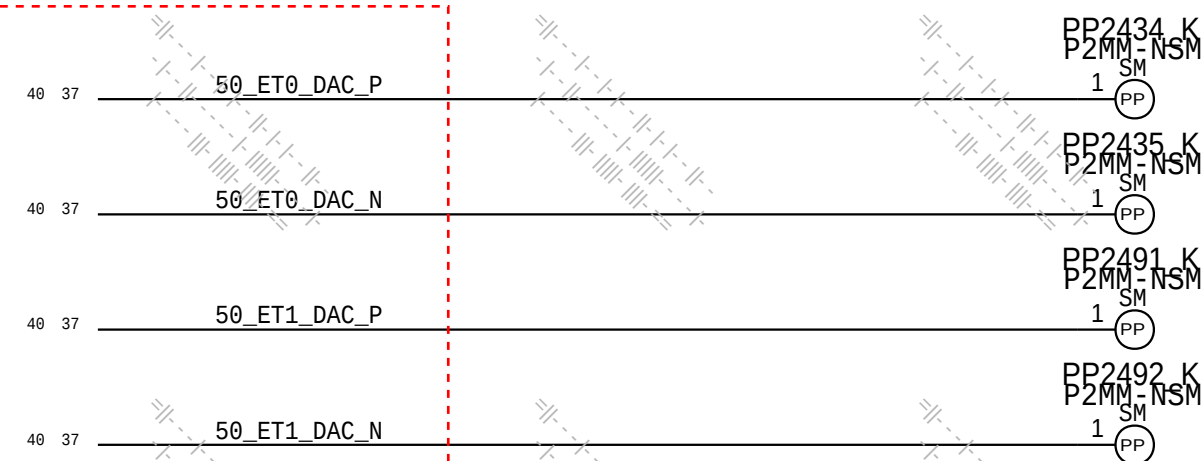
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
152590881	1	2.2UH 2012 20% 1.0MM	L1201_K	ICE19.0
152590882	1	2.2UH 2012 20% 0.8MM	L1201_K	ICE19.1
152590882	1	2.2UH 2012 20% 0.8MM	L1201_K	ICE19.2
152590811	1	2.2UH 2012 20% 1.0MM	L1202_K	ICE19.0
152590882	1	2.2UH 2012 20% 0.8MM	L1202_K	ICE19.1
152590882	1	2.2UH 2012 20% 0.8MM	L1202_K	ICE19.2
152590750	1	2.2UH 1608 20% 0.8MM	L1205_K	ICE19.0
152590750	1	2.2UH 1608 20% 0.8MM	L1205_K	ICE19.1
152590750	1	2.2UH 1608 20% 0.8MM	L1205_K	ICE19.2
131590382	1	5.6PF 01005	C1251_K	ICE19.0



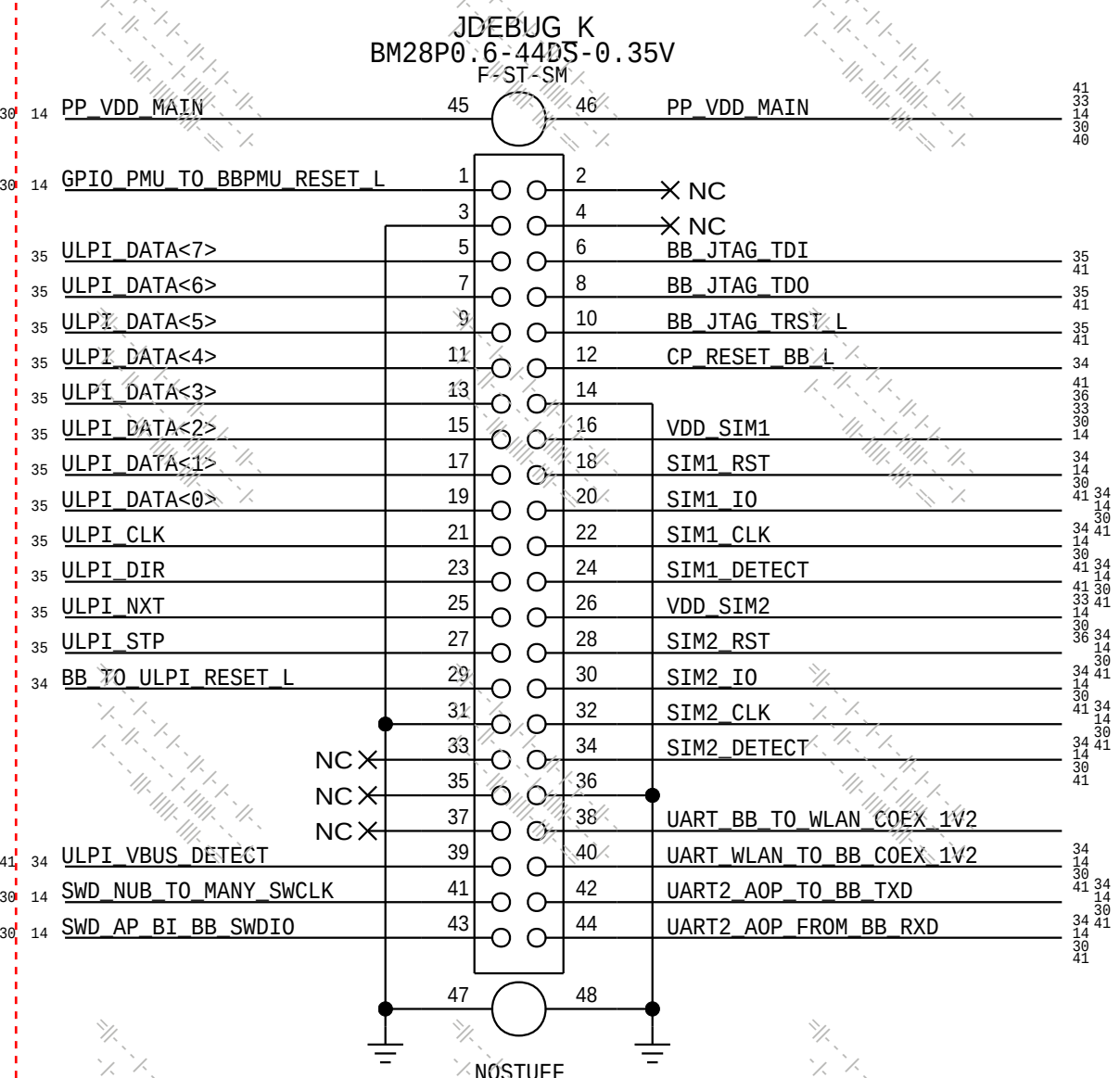
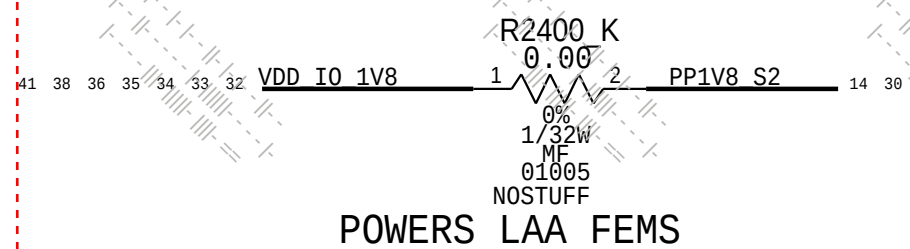
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测试点



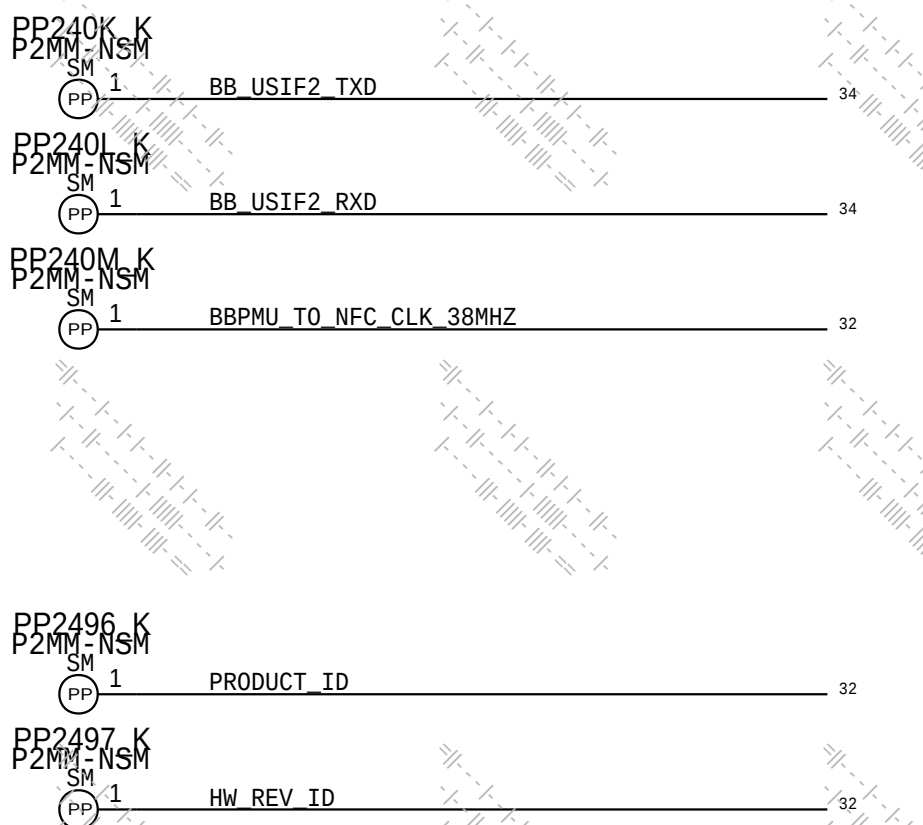
STUFF ONLY FOR VENDOR CONFIG



PP2M481 K
 P2M481 NSM
 PP 1 90_DIGRF_RX3_P 38 38
 PP2M482 K
 P2M482 NSM
 PP 1 90_DIGRF_RX3_N 38 38
 PP2M424 K
 P2M424 NSM
 PP 1 90_DIGRF_RX4_P 38 38
 PP2M493 K
 P2M493 NSM
 PP 1 90_DIGRF_RX4_N 38 38
 41
 PP2M488 K
 P2M488 NSM
 PP 1 BB_TO_XCVR_DWB_DATA 38 38
 PP2M489 K
 P2M489 NSM
 PP 1 BB_TO_XCVR_DWB_CLK 38 38
 PP2M490 K
 P2M490 NSM
 PP 1 XCVR_TO_BB_DWB_IRQ 38 38
 PP2M40P K
 P2M40P NSM
 PP 1 XCVR_GP00 38

Diagram illustrating SMPS Loading for various components:

- PP2417 K (PP, NSM) connected to EINT2 (34)
- PP2430 K (PP, NSM) connected to ULPI_VBUS_DETECT (34, 41)
- PP2436 K (PP, NSM) connected to EINT5 (34)
- PP24S1 K (PP, NSM) connected to VDD_DIG_0V85 (33, 36)
- PP24S2 K (PP, NSM) connected to VRF_DIG_0V9 (14, 38, 33, 38)
- PP24S3 K (PP, NSM) connected to VRF_L0_1V15 (14, 38, 33, 38)
- PP24S4 K (PP, NSM) connected to VRF_MED_1V3 (33)
- PP24S5 K (PP, NSM) connected to VDD_IO_1V8 (32, 34, 33, 36)



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87654321

SCHEMATIC APN: 951-08430

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[illegible]

存协议配置

存协议配置

的参考时钟差分信号负

的参考时钟差分信号正

IE总线数据接收差分信号负

IE总线数据接收差分信号正

的数据发射差分信号负

的数据发射差分信号正

的时钟请求信号交换

PP1V8_S4

43 42 13 PP1V2_S4

WAKE

1 R259 W

100K

32W

43 JTAG_WLAN_SEL

1 R257 W

10K

32W

2 01005

NOSTUFF

WL_GPIO_21_IV2

1 R258 W

10K

32W

2 01005

NOSTUFF

NOSTUFF= JTAG DISABLED

NOSTUFF=BT OVER PCIe

174	275
175	256
176	257
177	258
178	259
179	260
180	261
181	262
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184	265
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GODFATHER

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1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
7	0017939155	ENGINEERING RELEASED	2019-05-28

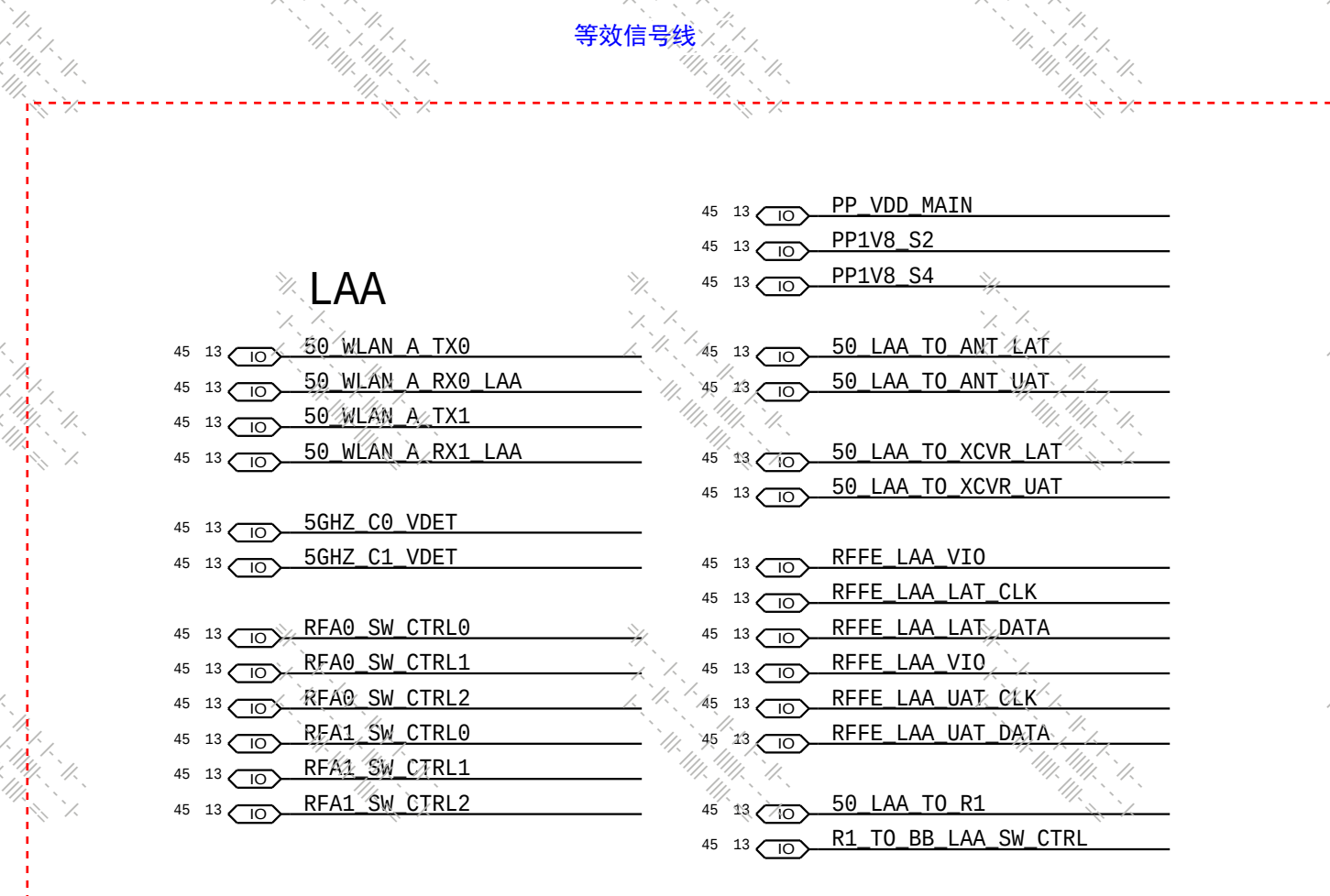
LAA_WIFI (GODFATHER)

LAST_MODIFICATION= Tue May 28 17:48:01 2019

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44	1	FEM MODULES		05/08/2018
45	2	FEM MODULES		05/08/2018

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
152S08492	1	IND, FILM, 1.0NH, 900MA, 01005	R362_W	WIFI_0
152S08498	1	IND, FILM, 0.4NH, 1000MA, 01005	R363_W	WIFI_0
152S08498	1	IND, FILM, 0.4NH, 1000MA, 01005	R372_W	WIFI_0
152S08496	1	IND, FILM, 0.5NH, 950MA, 01005	R373_W	WIFI_0
131S0648	1	CAP, CER, C0G, 0.3PF, 10V, 01005	C361_W	WIFI_0

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
152S08419	1	IND, FLUM, 1.1NH, 01005	R362_W	WIFI.X
117S0161	1	RES, MF, 0 OHM, 1/32W, 01005	R363_W	WIFI.X
117S0161	1	RES, MF, 0 OHM, 1/32W, 01005	R372_W	WIFI.X
117S0161	1	RES, MF, 0 OHM, 1/32W, 01005	R373_W	WIFI.X
131S00893	1	CAP, CER, C0G, 0.2PF, 10V, 01005	C361_W	WIFI.X



SCHEMATIC APN: 951-08431

FFM_MODULES			SYNC_DATE=05/08/2018	
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5GHZ LAT FEED



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